



Transportation Management (TM)

Generated on: 2026-09-01 12:16:50 GMT+0000

SAP S/4HANA | 2025 FPS01 (Feb 2026)

Public

Original content: https://help.sap.com/docs/SAP_S4HANA_ON-PREMISE/e3dc5400c1cc41d1bc0ae0e7fd9aa5a2?locale=en-US&state=PRODUCTION&version=2025.001

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Agreement Management

Use

As a shipper or logistics service provider (LSP), you use Agreement Management to do the following:

- Forecast and assess your transportation requirements.
- Request bids from carriers for your transportation requirements.
- Negotiate contracts (agreements) with carriers.

These features help keep your transportation costs down and efficiently manage contracts with carriers.

As a carrier, you use Agreement Management to respond to shippers or LSPs with your rates for transportation services.

Carriers, shippers, and LSPs can also use Agreement Management to set up and maintain agreements, the basis for calculating transportation charges.

Integration

Agreement Management is a component of Transportation Management (TM). Agreement Management is integrated with the Charge Management and Service Product Catalogs component because TM calculates transportation charges based on the master data stored in calculation sheets, rate tables, and scales that are created locally or that are created independently of the agreement. For more information, see [Charge Management and Service Product Catalogs](#).

Features

- Strategic freight procurement

As a shipper or LSP, you can detail your transportation requirements, request bids from carriers, and compare the bids you receive. You can then negotiate and award the transportation business to carriers and negotiate contracts with carriers (see [Strategic Freight Procurement](#)). You can also collaborate with carriers even if the carrier doesn't use TM (see [Carrier Collaboration](#)).

- Rate quotation

As a carrier, you can respond to shippers or LSPs with your rates for transportation services (see [Forwarding Agreement Quotation](#)).

- Agreements

As a shipper, LSP, or carrier, you can create long-term contracts with your business partners as a starting point for enabling charge calculation (see [Agreement Maintenance](#)).

Strategic Freight Procurement

Use

As a shipper or logistics service provider (LSP), you need to keep your transportation costs low while still ensuring reliable transportation services from carriers. Strategic freight procurement helps you to investigate the best rates from carriers to

transport items between specific locations (trade lanes) and then to agree a contract (agreement) with the carriers.

Transportation Management (TM) supplies the tools you need to request rate quotations, evaluate responses, and award the transportation business to carriers.

Prerequisites

- You have defined number ranges in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Number Range Intervals**.
- You have defined freight agreement RFQ types in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Freight Agreement RFQ Types**.
- You have defined target share strategies and strategy conditions in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Target Share Strategies**.
- If you are a carrier responding to RFQs, you must define forwarding agreement quotation types in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Forwarding Agreement Quotation Types**.
- You have configured the prerequisites for agreements (see [Agreement Maintenance](#)).
- You have configured the prerequisites for output management and business-to-business communication (see [Output Management](#)).

Process

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

1. You create a freight agreement RFQ master to detail your transportation requirements.

In the freight agreement RFQ master, you specify the trade lanes, carriers, RFQ items, and logistical parameters, for example:

- Requested capacities
- Commodity codes
- Service levels
- Movement types
- Periodicities
- Rating structure

The rating structure of each RFQ item can consist of calculation sheets and rate tables. The system uses calculation sheets to specify which transportation charges to calculate and how to calculate them, and uses rate tables to define and maintain rates directly in TM. For more information, see [Calculation Sheet](#) and [Rate Table](#).

i Note

- In the rate table of the calculation sheet, you can select the **Request Rate** checkbox for rate records. This indicates the rate records that the system downloads when you download an individual freight agreement RFQ as a spreadsheet file. When you send this file to a carrier, the carrier responds with rates only for these rate records.
- In the calculation sheet or the rate table of a calculation sheet, you can define a target rate or target amount. This specifies the maximum price you are willing to pay for a carrier's service. You also specify a tolerance as a percentage. You can take informed decisions about the carrier if the rate is beyond the target amount or target

rate after applying the tolerance percentage. For example, you specify a target amount or target rate of 1000 USD. You also specify a tolerance of 10%. If the rate the carrier has proposed is higher than 1100 USD, you do not have to take the carrier into consideration while awarding the transportation service.

You can choose to display the target rate to your carriers. To enable this feature, select the **Publish Target Rate** checkbox in Customizing for **Transportation Management** under **Master Data** **Agreement RFQs and Quotations** **Define Freight Agreement RFQ Types**.

2. You publish the freight agreement RFQ master.

As part of the publication process, the system automatically creates and sends individual freight agreement RFQs to the carriers based on your output management settings and using business-to-business communication. For more information, see [Freight Agreement Quotation Processing](#). Note that you can create individual freight agreement RFQs for multiple freight agreement RFQ masters and publish the freight agreement RFQ masters using the program **Mass Publication of Freight Agreement RFQ Masters** (/SCMTMS/RFQ_PUBLISH) in transaction SE38.

You can separate the creation of the individual freight agreement RFQs and the publication of the freight agreement RFQ master into 2 steps. This separation allows you to edit individual freight agreement RFQs before sending them to carriers. You separate the creation and publication in the Customizing activity **Define Freight Agreement RFQ Types**. Note that you cannot selectively send individual freight agreement RFQs to carriers – the system sends all the individual freight agreement RFQs to the carriers when you publish the freight agreement RFQ master.

You can enable an approval workflow so you have to get approval from an approver before publishing the freight agreement RFQ master. When an approver approves the publication of the freight agreement RFQ master, the system automatically publishes the freight agreement RFQ master and creates and sends the individual freight agreement RFQs to the carriers. If you separated the creation and the publication into 2 steps, and an approver approves the publication, you can then separately trigger the creation of the individual freight agreement RFQs and the publication of the freight agreement RFQ master.

i Note

If the system cannot publish the freight agreement RFQ master, the status of the freight agreement RFQ master changes to **Publication in Process** and does not change to **Published**. You can check the freight agreement RFQ masters that are not published in transaction SLG1. You can subsequently correct the errors in the unpublished freight agreement RFQ masters and publish the freight agreement RFQ masters again using the report **Publish Freight Agreement RFQ Master** (/SCMTMS/RFQ_RE_PUBLISH) in transaction SE38.

3. The system notifies each carrier about the individual RFQ.

The system also reminds each carrier if they have not responded and the response deadline is approaching. For more information about system notifications, see [Requesting of Quotations for Transportation Services](#).

4. The carriers respond to the individual RFQs.

The carriers evaluate and respond to their individual RFQ with their rates for the transportation services. Carriers submit their rates and promised capacity for each charge type listed in the calculation sheet of each RFQ item. If a charge type has a rate table associated with it, carriers are expected to provide rate values for each of the combinations listed in the rate table. Carriers cannot change the structure of the calculation sheets or rate tables (see [Calculation Sheet Maintenance](#) and [Rate Definition and Maintenance](#)).

For more information about responding to freight agreement RFQ masters, see [Forwarding Agreement Quotation](#) and [Forwarding Agreement Quotation Processing](#). Carriers can also respond to individual RFQs even if they don't use TM. For more information, see [Carrier Collaboration](#).

5. You evaluate the responses.

→ Recommendation

If you already know which carriers you want to award the transportation business to, skip the following 3 steps in the process. Instead, directly create or update freight agreements for the carriers you are awarding the business to. Therefore you do not need to do the following:

- a. Evaluate the responses.
- b. Review the results of the response comparison and strategy spend estimation.
- c. Accept your preferred target share strategy.

After you receive the responses from carriers, you evaluate which carrier quoted the best rate by comparing the responses by RFQ item in the comparison cockpit. You compare the responses by using the optimizer for carrier selection. The optimizer is especially useful when the comparison is complex and multiple constraints must be taken into account. You compare the responses in the following way:

- a. You insert the target share strategy and the strategy conditions that you want the system to apply when proposing target shares for carriers.

You define target share strategies and strategy conditions in the Customizing activity [Define Target Share Strategies](#).

- b. You run the optimizer to propose target shares based on the target share strategy.

The optimizer proposes the most ideal target share distribution to produce the most cost effective solution. You can run the optimizer repeatedly with different target share strategies and strategy versions.

i Note

You can run the optimizer for specific target share strategies or for all target share strategies in the freight agreement RFQ master in the back end using program [Mass Optimizer Run and Strategy Spend Estimation \(/SCMTMS/OPTIMIZER_RUN\)](#) in transaction SE38. If you do specify the target share strategies, the system runs the optimizer for all the target share strategies in the freight agreement RFQ master. The system can also estimate the strategy spend at the same time.

In the freight agreement RFQ master, when you enter a company or a corporate as an organizational unit, the system sorts all the purchasing organizations under the company or corporate and picks up the first purchasing organization for strategy spend calculation.

- c. The system simulates the estimated strategy spend based on the responses you received from the carriers and the target share strategies.
- d. You select your preferred target share strategy.

When you select your preferred target share strategy, the system automatically calculates the confirmed capacity in the freight agreement RFQ master.

i Note

If the comparison is fairly simple and the rates are the only constraints that must be taken into account, you can first compare the responses manually, as follows:

- a. You compare the responses by carrier and charge type.

You can compare the responses directly in TM. Alternatively, if the charge type has a rate table, you can compare the rate tables in a spreadsheet application.

- b. You choose whether you want to enable the confirmation of awarded capacities.

You can decide whether you want the carrier to send a response for the allocated confirmed capacities in order to execute a transportation request. You can consolidate the responses from different carriers. You can analyze and

assign transportation requests to different carriers. For more information, see [Award Confirmation](#).

- c. The system ranks the carriers based on the rates the carrier quoted per charge type.
- d. The system displays the results in a response comparison graph.

If you compare the rate tables in a spreadsheet application only, the system does not display the results in a graph.

For more information, see [Post-Bid Analysis](#).

i Note

If you are not satisfied with the responses from the carriers, you can create a new negotiation round. You can then change the details of the freight agreement RFQ master and publish it again. The carriers can then resubmit their rates.

i Note

You can create a freight agreement RFQ master assignment to delegate evaluation of responses to analysts. The analysts have expertise in analyzing bids for a trade lane. The analysts determine the best bids from the carrier bids for transportation services for a particular trade lane and merge the results back to the freight agreement RFQ master. For more information, see [Freight Agreement RFQ Master Assignment](#).

- 6. You review the results of the response comparison and strategy spend estimation.

You can use the **Award Summary** tab page to get an overview of the business you plan to award to the carriers.

- 7. You accept your preferred target share strategy.

- 8. You create or update freight agreements for the carriers you are awarding the business to.

The system notifies the carriers about the creation or update of the freight agreements. The notification includes a copy of the freight agreement and the rate tables, if any. For more information about system notifications, see [Requesting of Quotations for Transportation Services](#).

You can enable an approval workflow so that you have to request approval to create the freight agreements. You enable this workflow in the Customizing activity **Define Freight Agreement RFQ Types**.

i Note

If you update the status of the freight agreement RFQ master to completed, the system automatically sets the status of the individual freight agreement RFQs to completed.

More Information

[Freight Agreement RFQ Master](#)

[Agreement](#)

[Agreement Management](#)

[Analytics](#)

[Advanced RFQs](#)

Freight Agreement RFQ Master

Definition

A freight agreement request for quotation (RFQ) master is a business document containing the transportation requirements of a shipper or logistics service provider (LSP) for a specific future period. A shipper or LSP uses a freight agreement RFQ master to select a set of carriers to fulfill the transportation requirements.

Use

Strategic freight procurement helps you to investigate the best rates from carriers to transport items between specific locations (trade lanes) and for a specific set of logistical parameters, such as shipping type, transportation mode, and movement type. You then agree a contract (agreement) with the carriers. Freight agreement RFQ masters are a key element in strategic freight procurement, as you use them to do the following:

1. Detail your transportation requirements.
2. Request capacity and rate quotations from carriers.
3. Evaluate and compare the bids you receive from carriers.
4. Distribute target share according to the carriers' bids.
5. Simulate the estimated costs based on the carriers' rates.
6. Negotiate and award the transportation business to carriers.
7. Agree the contracts with carriers.

In essence, you use freight agreement RFQ masters to request bids from carriers for transportation services and, based on those bids, select carriers to fulfill your transportation requirements.

i Note

A carrier receives an individual freight agreement RFQ in the form of a forwarding agreement quotation. Forwarding agreement quotations allow carriers to respond with their transportation rates. For more information, see [Forwarding Agreement Quotation](#). Carriers can also respond to individual RFQs even if they don't use Transportation Management (TM). For more information, see [Carrier Collaboration](#).

You can create a freight agreement RFQ master in the following ways:

- Using the [Manage Freight Agreement RFQ Masters](#) app

i Note

This app displays all existing freight agreement RFQ masters regardless of their status in the strategic freight procurement process. You can create, edit, and display freight agreement RFQ masters using this app.

- Using a freight agreement RFQ master template

i Note

- You can create, edit, or display freight agreement RFQ master templates using the freight agreement RFQ master template apps.
- You can define Customizing templates based on industry standards for freight agreement RFQ masters in Customizing for [Transportation Management](#) under [Master Data](#) > [Agreement RFQs and Quotations](#) > [Define Freight Agreement RFQ Master Templates](#). With the freight agreement RFQ master template apps on the launchpad, you can use these Customizing templates to create master data templates for freight agreement RFQ

masters. The freight agreement RFQ master templates that you create from the Customizing templates enable you to make simple changes before creating a freight agreement RFQ master.

Structure

A freight agreement RFQ master can contain the following main tabs and screen areas:

- **General Data**

The **General Data** tab page contains high-level information about the freight agreement RFQ master, for example:

- Organizational unit responsible

Note that you can only assign a purchasing organization, company, or corporate to an organizational unit. You can't assign purchasing group or a purchasing office to an organizational unit.

- Validity period
- Carrier details

i Note

You can specify multiple carriers in the carrier table. Carrier table contains details about the status of the individual RFQs, if available.

- RFQ type
- RFQ negotiation round number
- Deadline for responding to the freight agreement RFQ
- Calculation sheet template the system uses to create a calculation sheet or local calculation sheet in the freight agreement RFQ master

- **Items**

The **Items** screen area contains information about the RFQ items. On this tab page, you can access the comparison cockpit to compare the responses you receive from carriers (see [Post-Bid Analysis](#)). You can also accept your preferred strategy before awarding the transportation business to carriers.

Note that you can specify a freight agreement type in a freight agreement RFQ type in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Freight Agreement RFQ Types**. When you create a freight agreement RFQ master with a freight agreement RFQ type, the item types assigned to the associated freight agreement type are automatically available in the input help of the item type field in the freight agreement RFQ master.

This screen area can contain the following tab pages, when relevant:

- **Calculation Sheet Overview**

This tab page displays the calculation sheet(both kinds, including local) details for each RFQ item.

- **Carriers**

This tab page displays the details of the carriers that responded to the RFQ item.

- **Trade Lanes**

On this tab page, you enter the details about the trade lanes for each RFQ item.

- **Capacities**

This tab page contains the capacity required from carriers, the capacity carriers can commit to, and the capacity awarded to carriers. For more information, see [Capacities in Rate Tables for Freight Agreement RFQ Masters](#).

- **Commodity Codes**

On this tab page, you specify the commodity codes for each RFQ item.

- **Default Route**

On this tab page, you specify the route between a location pair for delivering the RFQ item.

- **Notes**

On this tab page, you can add any notes relevant to the RFQ item.

- **Attachments**

On this tab page, you can upload files relevant to the RFQ item.

- **Document Flow**

The **Document Flow** tab page displays the business documents that were created based on the freight agreement RFQ master, with status information. For example, the individual freight agreement RFQs and the responses received from carriers.

- **Output Management**

On the **Output Management** tab page, you can print RFQ details and generate emails and faxes and send them directly to the intended recipient. You can also see a record of the documents that were already printed or sent (by e-mail or fax, for example). For more information, see [Output Management](#).

- **Excel Integration**

You can download the details of a freight agreement RFQ master. This helps you to review the details in the form of a spreadsheet.

- **Versions**

The **Versions** tab page contains a list of the RFQ versions and negotiation rounds, including the status of each version.

- **Award Summary**

The **Award Summary** tab page displays graphs summarizing all the logistical information. This summary helps you to judge how the negotiation process is going and make key decisions, such as whether to initiate a new negotiation round. You can view the award summary by trade lane, carrier, or RFQ item.

You can also check what you spent last year for transportation services with the same logistical information, that is:

- Trade lane
- Shipping type
- Transportation mode

For more information, see [Award Summary](#).

- **Notes**

On this tab page, you can add any notes relevant to the freight agreement RFQ master.

- **Attachments**

On this tab page, you can upload files relevant to the freight agreement RFQ master.

- **Capacities**

This tab page contains the total capacity required from carriers, the total capacity carriers can commit to, and the total capacity awarded to carriers.

i Note

You can activate the **Capacities** tab page or screen area in Customizing for **Transportation Management** under ► **Master Data** ► **Agreement RFQs and Quotations** ► **Define Freight Agreement RFQ Types** ►.

Integration

Freight agreement RFQ masters are integrated with forwarding agreement quotations, as carriers use forwarding agreement quotations to respond to individual freight agreement RFQs.

During strategic freight procurement, the system estimates the cost of using a carrier to provide transportation services in a trade lane. To perform this estimation, freight agreement RFQ masters are integrated with calculation sheets, local calculation sheets, rate tables, local rate tables, and ultimately charge calculation.

The system creates or updates freight agreements as part of the strategic freight procurement process.

More Information

[Strategic Freight Procurement](#)

[Requesting of Quotations for Transportation Services](#)

[Forwarding Agreement Quotation](#)

[Agreement](#)

[Calculation Sheet](#)

[Rate Table](#)

[Optimization as Part of Carrier Selection](#)

[Advanced RFQs](#)

Freight Agreement RFQ Master Assignment

Definition

A freight agreement RFQ master assignment is a business document that you as an analyst of a shipper or logistics service provider (LSP) use to evaluate rates from carriers for transportation services.

A purchasing manager can create a freight agreement RFQ master assignment from a freight agreement RFQ master item. The manager creates the freight agreement RFQ master assignment to delegate the task of bid analysis to analysts. The analysts have expertise in analyzing bids for a trade lane. The analysts determine the best bids from the carrier bids for transportation services for a particular trade lane and merge the results back to the freight agreement RFQ master item.

Use

Strategic freight procurement helps you to investigate the best rates from carriers to transport items between specific locations (trade lanes) and for a specific set of logistical parameters, such as shipping type, transportation mode, and movement type. You then agree a contract (agreement) with the carriers. Freight agreement RFQ master assignments are a key element in strategic freight procurement, as you use them to do the following:

1. Use the items that have received bids from carriers in the freight agreement RFQ assignment.
2. Evaluate and compare the bids you receive from carriers.
3. Distribute target share according to the carriers' bids.
4. Simulate the estimated costs based on the carriers' rates.
5. Determine the carriers from which you want to purchase the transportation services.
6. Merge the items back to the freight agreement RFQ master.

Structure

A freight agreement RFQ master assignment can contain the following main tab pages and screen areas:

- **General Data**

The **General Data** tab page contains high-level information about the freight agreement RFQ master assignment, for example:

- Organizational unit responsible
- Validity period
- Carrier details, including whether you already have an agreement in place with the carrier in the trade lane
- RFQ type
- RFQ negotiation round number
- Deadline for responding to the freight agreement RFQ
- Calculation sheet template the system uses to create a calculation sheet or a local calculation sheet in the freight agreement RFQ master assignment

- **Items**

The **Items** screen area contains information about the RFQ assignment items. On this tab page, you can access the comparison cockpit to compare the responses you receive from carriers (see [Post-Bid Analysis](#)). You can also accept your preferred strategy before awarding the transportation business to carriers.

This screen area can contain the following tab pages, when relevant:

- **Calculation Sheet Overview**

This tab page displays the calculation sheet (both kinds, including local) details for each RFQ assignment item.

- **Carriers**

This tab page displays the details of the carriers that responded to the RFQ assignment item.

- **Trade Lanes**

On this tab page, you display the details about the trade lanes for each RFQ assignment item.

- **Capacities**

This tab page contains the capacity required from carriers, the capacity carriers can commit to, and the capacity awarded to carriers. For more information, see [Capacities in Rate Tables for Freight Agreement RFQ Masters](#).

- **Commodity Codes**

On this tab page, you display the commodity codes for each RFQ assignment item.

- **Default Route**

On this tab page, you display the route between a location pair for delivering the RFQ assignment item.

- **Notes**

On this tab page, you can display any notes relevant to the RFQ assignment item.

- **Attachments**

On this tab page, you can display files relevant to the RFQ assignment item.

- **Business Partner**

The Business Partner tab page displays the business partners associated with the business document. For example, an invoicing party and carrier.

i Note

- You can display the business partner hierarchy in the business partner tab page.

In a freight agreement RFQ master assignment, it displays all the business partners associated with the carrier.

- You can restrict the party roles available in the business partner tab page. For more information, see [Business Partner Determination](#).
- You can display additional party roles, such as notify party and communication party, for information purposes in addition to primary party roles, such as carrier.

- **Document Flow**

The **Document Flow** tab page displays the business documents that exist based on the freight agreement RFQ master assignment, with status information. For example, the freight agreement master RFQ.

- **Output Management**

On the **Output Management** tab page, you can print RFQ assignment details and generate e-mails and faxes and send them directly to the intended recipient. You can also see a record of the documents that were already printed or sent (by e-mail or fax, for example). For more information, see [Output Management](#).

- **Excel Integration**

You can download the details of a freight agreement RFQ master. This helps you to review the details in the form of a spreadsheet.

- **Versions**

The **Versions** tab page contains a list of the RFQ assignment versions and negotiation rounds, including the status of each version.

- **Award Summary**

The **Award Summary** tab page displays graphs summarizing all the logistical information. This summary helps you to judge how the negotiation process is going and make key decisions, such as whether to initiate a new negotiation round. You can view the award summary by trade lane, carrier, or RFQ assignment item.

You can also check what you spent last year for transportation services with the same logistical information, that is:

- Trade lane
- Shipping type
- Transportation mode

For more information, see [Award Summary](#).

- **Notes**

On this tab page, you can display any notes relevant to the freight agreement RFQ master assignment.

- **Attachments**

On this tab page, you can display files relevant to the freight agreement RFQ master assignment.

- **Capacities**

This tab page contains the total capacity required from carriers, the total capacity carriers can commit to, and the total capacity awarded to carriers.

i Note

You can activate the **Capacities** tab page or screen area in Customizing for **Transportation Management** under ► **Master Data** ► **Agreement RFQs and Quotations** ► **Define Freight Agreement RFQ Types** ►.

More Information

[Freight Agreement RFQ Master](#)

[Strategic Freight Procurement](#)

[Requesting of Quotations for Transportation Services](#)

[Forwarding Agreement Quotation](#)

[Agreement](#)

[Calculation Sheet](#)

[Rate Table](#)

[Optimization as Part of Carrier Selection](#)

Award Confirmation

Use

Strategic freight procurement helps you to investigate and negotiate the best rates and optimal capacities with carriers to transport items between specific locations (trade lanes) and for specific logistical parameters, and then to create a contract (agreement) with the selected carriers.

As a shipper or a logistics service provider, Transportation Management (TM) enables you to choose whether you want the carrier to send a soft confirmation to the allocated confirmed capacities before awarding the capacities or business share through a

freight agreement. You can consolidate the responses from different carriers. When you choose to enable confirmation of awarded capacities, you can perform the following tasks:

- Define an award confirmation deadline
- Notify the carrier of award confirmation
- View the confirmation reason from each carrier for each response item in the freight agreement RFQ master
- Consolidate the confirmations from the carriers
- Upload the confirmations from carriers into the respective individual freight agreement RFQs
- View the statuses of award confirmations

Prerequisites

You've enabled award confirmation in Customizing for Transportation Management under [Master Data](#) [Agreement RFQs and Quotations](#) [Define Freight Agreement RFQ Types](#).

You've configured the prerequisites for strategic freight procurement (see [Strategic Freight Procurement](#)).

Features

Award Confirmation Deadline

In a freight agreement RFQ master, you can define a confirmation deadline before to sending notification to carriers for award confirmation. The carriers confirm the allocation of awarded capacities and forward the responses to the freight agreement RFQ master within this deadline.

Note that this confirmation is only a soft confirmation. The system doesn't reject the carriers if they don't respond within the deadline. The shipper can still communicate with the carriers offline in order to confirm the allocated capacities.

Carrier Notification

When you choose the [Notify the Carrier for Award Confirmation](#) button in a freight agreement RFQ master, the system notifies the carriers of the allotted confirmed capacities and the award confirmation deadline.

Consolidation of Confirmations

A carrier can send a response for award confirmation to the shipper through an email. You can send a spreadsheet that contains all the information, for example, items, rate tables, local rate tables, calculation sheets, local calculation sheets, transportation modes, and service levels. The carrier enters a response and a confirmation reason in the spreadsheet and sends the file back to you in an email attachment. If the carrier doesn't use TM, the carrier can also use the [Manage Freight Agreement RFQs](#) app to send you a response for award confirmation. For more information, see [Carrier Collaboration](#).

i Note

The system automatically updates the award confirmation status for master RFQ once the confirmation for an individual RFQ is available.

Optionally, you can also run the report /SCMTMS/RESPONSE_AWARD_CONFIRM manually after the award confirmation deadline. The system retrieves all the freight agreement RFQ masters with the status [Pending Confirmation from Carrier](#). Alternatively, you can also retrieve a specific freight agreement RFQ master by entering the RFQ ID.

Display of Award Confirmation Statuses

The system displays the status of award confirmation from the carrier in the following areas:

- Freight agreement RFQ master
- Freight agreement RFQ item
- Response
- Optimizer strategy

For more information on award confirmation statuses, see [Statuses in Award Confirmation](#).

Activities

To create or update freight agreement RFQ masters, in the [Create Freight Agreement RFQ Master](#) app on the fiori launchpad.

More Information

[Strategic Freight Procurement](#)

Statuses in Award Confirmation

Optimizer Strategy Statuses

Status	Appears When
Award Confirmation - Pending from One or More Carriers	Before the award confirmation deadline when one or more carriers do not send any response to the shipper or logistics service provider (LSP) for the allotted confirmed capacities for one or more RFQ items.
Rejected by One or More Carriers	One or more carriers reject the confirmed capacities allotted by the shipper for a transportation requirement (RFQ item).
No Response from One or More Carriers	After the award confirmation deadline when one or more carriers do not send any response to the shipper or LSP for the allotted confirmed capacities for one or more RFQ items.
Awarded by Shipper	The shipper or the LSP awards one or more RFQ items to one or more carriers.
Rejected by All Carriers	All carriers reject the confirmed capacities allotted by the shipper for a transportation requirement.

Freight Agreement RFQ Master Statuses or RFQ/Quotation Statuses

Status	Appears When
No Response from one or More Carriers	One or more carriers do not send a confirmation to the shipper or LSP for the allotted confirmed capacities for one or more RFQ items.
Rejected by Carriers	All carriers reject all the confirmed capacities allotted by the shipper or LSP for all RFQ items.

Status	Appears When
One or More Items Rejected by Carriers	One or more carriers reject the confirmed capacities allotted by the shipper or LSP for one or more RFQ items.
Awarded by Shipper	The shipper or LSP awards all the transportation requirements to one or more carriers.

Individual Freight Agreement RFQ Root Statuses

Status	Appears When
Award Confirmation - Pending from Carrier	The carrier does not send a confirmation to the shipper or LSP for the allotted confirmed capacities for one or more RFQ items.
Award Confirmation - Received by Shipper	The carrier sends a response to the shipper or LSP for the allotted confirmed capacities for one or more RFQ items.
Award Confirmation - Uploaded by Carrier	The carrier has sent the spreadsheet file that contains the response for award confirmation.
Award Confirmation - Submitted with Errors by Carrier	The background processing function /SCMTMS/RESPONSE_AWARD_CONFIRM chooses a file containing errors. After the award confirmation deadline, you run a recurring background processing function /SCMTMS/RESPONSE_AWARD_CONFIRM, to consolidate the responses in spreadsheet files submitted by carriers. For example, this status appears if the system chooses a file without any responses for all the RFQ items.
No Response from One or More Carriers	Before the award confirmation deadline when the carrier does not send a response to the shipper or LSP for the allotted confirmed capacities for one or more RFQ items.
Rejected by Carriers	The carrier rejects the confirmed capacities allotted by the shipper for all RFQ items.
One or More Items Rejected by Carriers	The carrier rejects the confirmed capacities allotted by the shipper for one or more RFQ items.
Accepted by Carrier	The carrier accepts the confirmed capacities allotted by the shipper for all RFQ items.

Freight Agreement RFQ Item Statuses

Status	Appears When
Preferred Strategy – Accepted by Shipper	The shipper accepts the preferred strategy for a specific transportation requirement by choosing the Accept Preferred Strategy button in the award confirmation process.
Award Confirmation – Pending from Carrier	After the shipper requests a confirmation from the carriers for a specific RFQ item, the system sets this status before the award confirmation deadline.

Status	Appears When
Rejected by One or More Carriers	One or more carriers reject the confirmed capacities allotted by the shipper or LSP.
No Response from One or More Carriers	One or more carriers do not send a response to the shipper or LSP for the allotted confirmed capacities.
Rejected by All Carriers	All carriers reject the confirmed capacities allotted by the shipper or LSP.

Response Statuses

Status	Appears When
Preferred Strategy - Accepted by Shipper	The shipper accepts the preferred strategy for a specific transportation requirement by choosing the Accept Preferred Strategy button in the award confirmation process.
Award Confirmation - Pending from Carrier	After the shipper requests a confirmation from the carrier for a specific RFQ item, this status is set on the response item before the award confirmation deadline.
Accepted by Carrier	The carrier accepts the confirmed capacities allotted by the shipper for a transportation requirement.
Rejected by Carrier	The carrier rejects the confirmed capacities allotted by the shipper for a transportation requirement.

Optimization at Individual Capacity Lines

Transportation Management enables you as a shipper to optimize the business share allocation process to evaluate the carriers at a granular level. You can specify business share conditions for a strategy at an individual capacity line-level in a freight agreement RFQ. By providing different business share constraints for different individual capacity lines, you can influence the business share output of the optimizer at a granular level. These capacity lines can be modeled as the rate table lines.

Prerequisites

1. You have defined a flat view profile in Customizing for Transportation Management under [Master Data > Agreement RFQs and Quotations > Define Excel and Flat View Profiles](#).
2. You have defined a freight agreement RFQ type in Customizing for Transportation Management under [Master Data > Agreement RFQs and Quotations > Define Freight Agreement RFQ Types](#). You have assigned the flat view profile to the freight agreement type in the same Customizing activity.

Features

Location Pair Description

You can choose the [Show Descriptions](#) or [Hide Descriptions](#) buttons to show or hide the descriptions of the source and destination locations respectively.

Add or Remove Carriers

You can specify a set of carriers for a specific capacity line. You can add or delete more carriers during the process. You can also specify whether a carrier is incumbent or non-incumbent.

You can select an individual capacity line and choose [Carrier Assignment > Add Carrier](#). The system displays all the carriers associated with the capacity line in the Carriers tab. You can add a new carrier and choose [Add Selected Carrier](#) button.

You can select an individual capacity line and choose [Carrier Assignment > Remove Carrier](#). The system displays all the carriers associated with the capacity line in the Carriers tab. You can remove a carrier and choose [Remove Selected Carrier](#) button.

Strategy Creation for Individual Capacity Lines

You can select individual capacity lines in the flat view and choose the Create Strategy button.

i Note

If you have assigned target share strategies in the Customizing for Transportation Management under [Master Data > Agreement RFQs and Quotations > Define Freight Agreement RFQ Types](#), the system displays the details of the strategy. If you have not assigned target share strategies in Customizing, you can enter the strategy details and create the strategy

Update Carrier Strategy Conditions

You can update strategy conditions for a specific carrier by selecting the corresponding response item and choosing [Update Carrier Strategy Conditions](#) button. A dialog box appears in which you can enter the condition values for the following and choose update:

- Minimum business share
- Maximum business share

Note that in the dialog box, only when you select the [Update](#) checkbox, the optimizer considers the new value during optimizer execution.

Update Master Strategy Conditions

You can update the conditions specific to an entire RFQ item by selecting the corresponding RFQ item and choosing the [Update Master Strategy Conditions](#) button. A dialog box appears in which you can enter the condition values for the following and choose update:

- Minimum number of carriers
- Maximum number of carriers
- Enable bonus-malus
- Enable transit time

Note that in the dialog box, only when you select the [Update](#) checkbox, the optimizer considers the new value during optimizer execution.

Strategy Conditions

Optimizer Constraints Based Solution Spend Calculation

Once the strategy is created, the system displays the strategies when you navigate to **Views > Strategy View**. When you select a strategy, and choose **Insert > Conditions**, a **Strategy Conditions** dialog box appears. Here you can specify optimizer strategy conditions such as the following:

- Minimum number of carriers
- Maximum number of carriers
- Bonus-malus
- Transit time

Historical Data Based Solution Spend Calculation

You can also choose to perform historical data-based analysis. In the **Strategy Conditions** dialog box, you can specify the historical spend constraints such as the following:

- Prime charge type
- Valid-from date
- Valid-to date

Optimizer Run Execution

Once you update or enter the strategy conditions, you can choose the **Run Optimizer** button. On successful run of optimizer, the system displays the target shares in the **Target Share** column. The system also displays the corresponding capacity share, which is calculated based on the business share, in the **Proposed Capacity Distribution** column.

Note that the optimizer run may fail if the strategy conditions are invalid. You can also choose to select multiple strategy lines and run the optimizer.

Simulation of Solution Spend

To calculate estimated spend, choose **Simulate Estimated Spend**. On calculation of estimated spend, the system displays the values in the strategy view. When you select the strategy, the system displays the item and item responses along with the capacities and business shares in the flat view.

Activities

1. To enable individual capacity level optimization, in the RFQ item, select the **Individual Capacity Optimization** checkbox. When you select this checkbox, the system allows you to specify business share conditions for strategies at the individual capacity line- level.
2. Select an RFQ item and choose **Display Flat View**.

The system displays the individual capacity lines associated with the flat view profile and charge type.

Note that you can filter the flat view contents based on the rate table dimensions.
3. Specify a set of carriers for the individual capacity lines.
4. Select individual capacity lines in the flat view, create strategies, and specify valid strategy conditions.

5. Run the optimizer to get the appropriate business shares for each carrier and simulate the estimated spend to view the approximate costs incurred.

i Note

If the maximum business share is zero at individual capacity lines, the system does not allocate any share to the specific carrier. The business share becomes zero when you do not mention any condition at strategy level.

You cannot create strategy conditions for individual capacity lines in the assignment documents in the **Document Flow** tab in the freight agreement RFQ master.

More Information

- [Freight Agreement RFQ Master](#)
- [Strategic Freight Procurement](#)

Requesting of Quotations for Transportation Services

Strategic freight procurement helps you to investigate and negotiate the best rates from carriers to transport items between specific locations (trade lanes) and for specific logistical parameters, and then to agree a contract (agreement) with the selected carriers.

Requesting quotations for transportation services using freight agreement request for quotation (RFQ) masters is a key element in strategic freight procurement. This is because you use freight agreement RFQ masters to request bids from carriers for transportation services and, based on those bids, select carriers to fulfill your transportation requirements. When requesting quotations for transportation services, you can do the following:

- Detail your transportation requirements.
- Request and compare the bids you receive from carriers.
- Distribute target share according to the carriers' bids.
- Simulate the estimated costs based on the carriers' rates.
- Negotiate and award the transportation business to carriers.
- Agree the contracts with carriers.

Prerequisites

You've configured the prerequisites for strategic freight procurement (see [Strategic Freight Procurement](#)).

Features

Templates

You can use templates to help simplify the process of creating freight agreement RFQ masters.

You can define Customizing templates in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Freight Agreement RFQ Master Template**. You can transfer Customizing templates across different TM systems.

You can define master data templates directly from the launchpad. If necessary, you can define a master data template based on a Customizing template. To define, view, and update master data templates for freight agreement RFQ masters, open the [Create FA RFQ Master Templates](#) app.

Transportation Requirement Details per RFQ Item

You specify your transportation requirements per RFQ item. For each RFQ item, you can specify the following, for example:

- The carrier details that are specific to the RFQ item
- The trade lanes where you want the transportation services
- The capacity that you need the carrier to be able to fulfill, the capacity the carrier can commit to, and the capacity you award to the carrier (see [Capacities in Rate Tables for Freight Agreement RFQ Masters](#))
- The commodity codes for the RFQ item
- The calculation sheet and rate tables (both kinds, including local) for carriers to use when submitting their rates for the transportation services (see [Calculation Sheet](#) and [Rate Table](#))
- Logistical parameters such as shipping type, movement type, and transportation mode

Note that if you change the validity of the agreement in the freight agreement RFQ master, you can update the validity of all or selected RFQ items and service products with the same validity period.

Approval Workflow

You can enable approval workflows so that you have to get approval from an approver before publishing a freight agreement RFQ master and before creating a freight agreement during strategic freight procurement. When an approver approves the publication of the freight agreement RFQ master, the system automatically publishes the freight agreement RFQ master and creates and sends the individual freight agreement RFQs to the carriers. You enable these approval workflows in the Customizing activity [Define Freight Agreement RFQ Types](#). You can also specify that you want the system to notify an approver when you request approval. For more information, see [Approval Check](#).

You can also use an offline approval workflow to send work items for freight agreement RFQ masters by e-mail to a user, such as a customer agent. The user can accept or reject a work item and update the system from their inbox. For more information, see [Offline Workflow for Sending and Approving Work Items](#).

Comparison Cockpit

After you receive responses from carriers during strategic freight procurement, you use the comparison cockpit to evaluate which carrier quoted the best rate by comparing and analyzing some or all of the responses per RFQ item or across RFQ items. The comparison cockpit supports you in the vital post-bid analysis step, which in turn supports your decision-making process. The comparison cockpit helps you to decide which carriers to award the transportation business to.

In the comparison cockpit, you compare responses by using the optimizer for carrier selection. The optimizer is especially useful when the comparison is complex and multiple constraints must be taken into account. You can also compare responses manually directly in TM or using spreadsheets.

For more information, see [Post-Bid Analysis](#).

Periodicity

On the Capacities tab of a freight agreement RFQ item, you can specify the periodicity. This specification enables the shipper to specify the frequency at which the carrier can expect a volume of goods to be shipped. Note that the freight agreement RFQ item must have just one trade lane. For more information, see [Check Against Transportation Allocations](#).

The system can specify the periodicity of an item if you've performed the following activities:

- Define a transportation allocation type with a specific periodicity in the Customizing activity [Define Transportation Allocation Types](#) in Customizing for [Transportation Management](#) under [►Planning►General Settings►](#)
- Assign this allocation type to a freight agreement type in the Customizing activity [Define Freight Agreement Types](#) in Customizing for [Transportation Management](#) under [►Master Data►Agreements and Service Products►](#)
- Assign this freight agreement type to the freight agreement RFQ type in the Customizing activity [Define Freight Agreement RFQ Types](#) in Customizing for [Transportation Management](#) under [►Master Data►Agreement RFQs and Quotations►](#)

When you create a freight agreement from the freight agreement RFQ master, the system uses the allocation period you've specified in [Define Transportation Allocation Types](#) Customizing activity, as the periodicity in the freight agreement RFQ item. You can't change this periodicity. If you haven't specified allocation period, the system leaves the periodicity field blank.

Negotiation Rounds and RFQ Versions

If you aren't satisfied with the responses from carriers, you can create a new negotiation round. You can then change the details of the freight agreement RFQ master and publish it again. The carriers can then resubmit their rates. The system records the number of the RFQ negotiation round, indicating how many times you published a freight agreement RFQ master to carriers. The freight agreement RFQ master must be published before you can create a new negotiation round. When you create a new negotiation round, the system updates the number of negotiation rounds by 1 and resets the version number to zero. This ensures that the latest round you publish always has an associated clean version.

You can create a new version of the freight agreement RFQ master without creating a new negotiation round. A new version enables you to make changes without publishing the freight agreement RFQ master to your carriers. You can use versions for internal changes before you publish, and you can use negotiation rounds for the changes that you send to your carriers. The freight agreement RFQ master must have the [In Process](#) status to create a new version. If you have created individual freight agreement RFQs or published the freight agreement RFQ master, you cannot create a new version of the freight agreement RFQ master. When you create a new version, the system updates the version number by 1 and the negotiation round number remains the same. When you publish the freight agreement RFQ master, the negotiation round number is updated by 1.

To create a negotiation round, in the freight agreement RFQ master, select [►Follow Up►Create Negotiation Round►](#).

To create a version, in the freight agreement RFQ masters choose the [Generate New Version](#) button.

Award Summary

You can display a summary and graphs of all the logistical information in a freight agreement RFQ master. This summary helps you to judge how the negotiation process is progressing and make key decisions, such as whether to create a new negotiation round. You can also check what you spent last year for transportation services with the same logistical information, such as trade lane, shipping type, and transportation mode. For more information, see [Award Summary](#).

Mass Individual RFQ Creation and Publication

You can perform the following activities using the program [Mass Publication of Freight Agreement RFQ Masters](#) (/SCMTMS/RFQ_PUBLISH) in transaction SE38.

- Create individual freight agreement RFQs for the carriers in multiple freight agreement RFQ masters
- Create and publish individual freight agreement RFQs for the carriers in multiple freight agreement RFQ masters

i Note

The system doesn't automatically publish the freight agreement RFQ masters if you separated the creation of individual freight agreement RFQs and the publication of the freight agreement RFQ master into 2 steps. You separate the creation and publication in Customizing for [Transportation Management](#) under [►Master Data►Agreement RFQs and Quotations►Define](#)

Freight Agreement RFQ Types . If you separated the creation and publication, you can also use this program to publish multiple freight agreement RFQ masters for existing individual freight agreement RFQs.

System Notifications

TM integrates with output management (see [Output Management](#)) to allow you to automatically notify carriers at various stages of the strategic freight procurement process as follows:

1. After publication of the freight agreement RFQ master

The system notifies carriers that you published a freight agreement RFQ, and includes the freight agreement RFQ as an attachment. The system also attaches XLS files for rate tables and local rate tables used for any charge types.

2. When the response deadline is approaching

The system sends a reminder to carriers who haven't responded to the freight agreement RFQ. You define when to send this reminder in the Customizing activity [Define Freight Agreement RFQ Types](#).

3. On the response deadline

On the last day of the response, the system sends a reminder to carriers who have still not responded to the freight agreement RFQ.

4. When the freight agreement RFQ is canceled

The system notifies carriers that the freight agreement RFQ has been withdrawn.

5. After creation of the agreement

The system notifies carriers that you created the agreement, and includes the freight agreement, rate tables, and local rate tables, if any, as attachments.

6. When the agreement is expiring

The system notifies carriers that the freight agreement is expiring and includes the freight agreement both as an attachment and as a link.

TM uses business-to-business communication to exchange this information electronically between business partners. For more information, see [Freight Agreement Quotation Processing](#) and [Forwarding Agreement Quotation Processing](#).

Business Share Integration

When you create or update an agreement from an agreement RFQ, the system proposes a target share. The target share is the output from the optimizer run that you perform in the comparison cockpit in the freight agreement RFQ master. The system copies the share into the item details of the agreement. For more information, see [Agreement Maintenance](#).

The business share is automatically included in the agreement from the agreement RFQ. The system includes the target share in the business share, and uses the business share in the carrier selection settings in planning. For more information, see [Business Share](#).

Maintenance of RFQ Masters Using Spreadsheets

You can download the RFQ master data into a spreadsheet for editing. The spreadsheet contains the following information:

- RFQ header details
- RFQ items
- Requested capacities

- Offered capacities
- Confirmed capacities
- Calculation sheets and local calculation sheets
- Calculation sheet items

The spreadsheet also contains a worksheet for each rate table or local rate table. You can't edit the details in the worksheet. However, you can download the rate tables or local rate tables as an XLS file and then upload the edited XLS file back into the RFQ master in your TM system. The system updates the RFQ master with your changes.

To download or upload a freight agreement RFQ master, choose **Freight Agreements** app from the launchpad. From the freight agreement work list, choose **Freight Agreement RFQ Masters**. Open the freight agreement RFQ master and go to the **Excel Integration** tab page. You can also download or upload a freight agreement RFQ master using program **Excel Integration with Freight Agreement RFQs** (/SCMTMS/CARR_RFQ_EXCEL) in transaction SE38.

i Note

Your carriers can also download their individual freight agreement RFQs for editing when responding to you with their rates for the transportation services. For more information, see [Responses from Carriers Using Spreadsheets](#).

If one or more carriers have responded to the freight agreement RFQ, you can download a spreadsheet that enables you to compare the responses of the carriers in the freight agreement RFQ master.

Carrier Collaboration

If your carriers don't use TM, you can allow them to view and respond to the individual freight agreement RFQs that they receive from you using a carrier worklist. Your carriers contact your purchasing organization for logon details to this personal object worklist.

Carriers can use the **Manage Freight Agreement RFQs** app to view the individual freight agreement RFQs specific to the carrier organization. For more information, see [Carrier Collaboration](#).

Activities

To create or update freight agreement RFQ masters, choose the **Freight Agreements** app from the launchpad.

To create a freight agreement RFQ master and to start the process of updating a freight agreement, choose the **Freight Agreements** app from the launchpad. Open the freight agreement and choose the **Create RFQ Master** button.

More Information

[Strategic Freight Procurement](#)

[Archiving Agreements and RFQ Masters](#)

Responses from Carriers Using Spreadsheets

When responding to you with their rates for transportation services, carriers can download an individual freight agreement RFQ into a spreadsheet for editing. You can choose a structured format, a compatible format, or a flat view format for your carriers to download.

Prerequisites

You've specified the format of the spreadsheet that you want carriers to download in Customizing for [Transportation Management](#) under [►Master Data► Agreement RFQs and Quotations► Define Freight Agreement RFQ Types ►](#).

Features

Spreadsheet Formats

- Structured format

The data shown in this format is structured so that you don't have to scroll too much to get to the most important information. There's a separate sheet for the RFQ header details and for each RFQ item. We recommend using the structured format when there's a low number of RFQ items.

The address of the location used in the trade lanes or rate tables (both kinds, including local) is also included.

- Compatible format

This format displays the following details in one row in one sheet:

- RFQ header details
- RFQ items
- Requested capacities
- Offered capacities
- Confirmed capacities
- Calculation sheet items

There are then separate sheets for each rate table (both kinds, including local). This format is available for compatibility reasons only.

- Flat View Format

This format displays the capacity demand and rates as a consolidated view consisting of all the charges and rates that are relevant for negotiating a freight agreement RFQ.

Editing of the Spreadsheet by Carrier

Carriers can only edit certain pieces of information such as the following:

- RFQ Items: Offered capacities, **Responded** column, and **Confirmed** column
- Calculation sheet items: Amounts for flat charges
- Local rate table: Offered amounts, offered capacities, and offered transit times

You can also download or upload a freight agreement RFQ using program [Excel Integration with Freight Agreement RFQs](#) (/SCMTMS/CARR_RFQ_EXCEL) in transaction SE38.

Related Information

[Requesting of Quotations for Transportation Services](#)

Capacities in Rate Tables and Local Rate Tables for Freight Agreement RFQ Masters

Use

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You can specify capacities in the rate tables of freight agreement RFQ masters. Capacities enable you to publish the amount of business you require from a carrier. Note that the business share feature also enables you to specify the amount of business you want to allocate to a carrier. However, you use the business share for your own analysis and business decisions. You use capacities to seek a response from carriers and as an output of business share decisions. For more information about business shares, see [Business Share](#).

In the rate table of a freight agreement RFQ master, the system displays the following fields for a scale dimension in the **►Dates and Validity ► Rates ►** screen area:

- **Requested Capacity**

The capacity you require from a carrier. You enter this information before publishing the freight agreement RFQ master.

- **Offered Capacity**

The capacity the carrier offers to you. The carrier enters this information when responding to the freight agreement RFQ master.

- **Confirmed Capacity**

The actual capacity you award to a carrier. After you compare the responses from the carriers and select a carrier, the system enters the confirmed capacity based on the target share proposed by the optimizer. You can manually change the confirmed capacity.

- **Average Load**

The average capacity per shipment that you transport in a trade lane during the validity period of the freight agreement RFQ master.

- **Target Share**

The capacity as a percent of the total business for an item in a trade lane that you award to a carrier.

i Note

You can create rate tables or local rate tables from within a freight agreement RFQ master. However, you can't specify rate tables. Also, rate tables only display the confirmed capacity.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Capacity Drivers and Capacity Dimensions

- Capacity drivers

Capacity drivers are the dimensions in a rate table that determine your capacity requirements. For example, when you specify source location A and destination location B as capacity drivers, you specify that you need the specified capacity from location A to location B. When you specify the capacity drivers, the system doesn't take any of the other dimensions into account for capacity.

In most cases, the source location and destination location are the capacity drivers (given that you're working in a trade lane for a freight agreement RFQ master). However, you can also use other drivers, such as equipment type and commodity code. You must specify at least one capacity driver in a rate table and you can have multiple capacity drivers.

Note that if you specify source location and destination location as the capacity drivers, you can't select source location and destination location as column scales in the **Dates and Validity** > **Rates** screen area.

i Note

You must have the same requested capacity for all instances of a combination of capacity drivers in a rate table.

For example, you have 3 dimensions in a rate table: source location, destination location, and gross weight. The capacity drivers are source location and destination location. The rate table appears as follows:

Source Location	Destination Location	Gross Weight >= (TON)	Requested Capacity
A	B	100	100
A	B	500	100
A	C	100	200
A	D	100	300
A	D	500	300

The requested capacity is the same for all the gross weight values for the location pairs. The system copies the requested capacity of the first instance of the combination of capacity drivers throughout the rate table for all instances of that combination of capacity drivers.

- Capacity dimension

You can specify the dimension that you use to enter your capacity requirements for an item in a trade lane. You can only use a number-based dimension. Therefore, you must use a number-based calculation base as the capacity dimension. This in turn means you must use a calculation base that is associated with one of the following scale bases:

- **Number**
- **Quantity**
- **Volume**
- **Weight**

For example, you can't specify a commodity code as a capacity dimension because it isn't number-based. Note that the system doesn't use dimensional weight when estimating the strategy spend. Therefore, you can't use the following calculation bases:

- **Chargeable Weight** (CHRG_WEIGHT)
- **Chargeable Weight Based on Net Weight** (CHRG_WEIGHT_NET)
- **Customs Value** (CUSTOMS_VALUE)

You also can't use an amount-based dimension, for example, you can't use the following calculation bases:

- **Goods Value** (GOODS_VALUE)
- **Insurable Value** (INSURABLE_VALUE)
- **Customs Value** (CUSTOMS_VALUE)

Note that you can specify only one capacity dimension in a rate table.

You can't specify the same dimension as both a capacity driver and a capacity dimension.

i Note

If you don't specify a capacity dimension, the system assumes that you're working with full container loads (FCLs) or full truck loads (FTLs). The capacity of the truck or container is then considered as full.

Creating Freight Agreement RFQ Masters from Freight Agreements

You can create a freight agreement RFQ master from a freight agreement. If there's a rate table in the freight agreement, the system adds a copy of the table as a rate table for the RFQ master. The system keeps the dimensions in the rate table, but clears the amounts for the rates and capacities.

Full Container Loads or Full Truck Loads

When working with FCLs or FTLs, the capacity of the truck or container is considered as full and you don't need to specify capacities. In an FCL or FTL scenario, you also can't edit the average load. You can't edit the average load because you work with full loads, and the average load is therefore not a factor in your capacity requirements. The system doesn't validate the shipping type against the capacity dimension.

Deleting Rates

If you delete the rate for a selected line, the system deletes only the relevant rate in the **►Dates and Validity►Rates►** screen area. The system doesn't delete the amounts of other related values for the same scale dimension combination. For example, you have 3 rates for a source location and destination location, with 3 different gross volumes. If you delete the rate in the first row, the system doesn't delete the rate in the 2 remaining rows. The following table illustrates this behavior:

Source Location	Destination Location	Gross Volume	>=1,000	Requested Capacity
CSI-APORT-AMS	CSI-APORT-HKG	1000	None	100
CSI-APORT-AMS	CSI-APORT-HKG	2000	110	100
CSI-APORT-AMS	CSI-APORT-HKG	3000	115	100

i Note

If you delete the capacity for a selected line, the system deletes the capacity for all other lines with the same combination of capacity drivers. This occurs because you must have the same requested capacity for all instances of a combination of capacity drivers.

Activities

To display the capacities, on the **►Dates and Validity►Rates►** tab page of the rate table or local rate table choose the **Show More Info** button. The system only displays capacities if you've selected at least one item as the capacity driver in the **►General Data►Scales►** screen area.

You can only enter the requested capacity and the average load. The carrier can enter only the offered capacity, in a response.

The system displays the confirmed capacity in the rate table or the local rate table when you award transportation business to a carrier and create an agreement.

To enable FCL or FTL, select **FCL** or **FTL** as the shipping type for the item in the **Items** screen area of the freight agreement RFQ master.

More Information

[Agreement RFQ](#)

[Requesting of Quotations for Transportation Services](#)

[Rate Table](#)

[Rate Definition and Maintenance](#)

Post-Bid Analysis

Use

After you receive responses from carriers for the quotations you requested, you use the comparison cockpit to evaluate which carrier quoted the best rate for the specified periodicity. You compare and analyze some or all of the responses per RFQ item or across RFQ items. The comparison cockpit supports you in the vital post-bid analysis step, which in turn supports your decision-making process. The comparison cockpit helps you to decide which carriers to award the transportation business to and in what proportion.

i Note

The system simulates an estimation of the cost; the system doesn't calculate the actual charges.

If you know which carriers you want to award the transportation business to, you can directly create or update the freight agreements.

Integration

You compare responses during strategic freight procurement (see [Strategic Freight Procurement](#)).

Prerequisites

- A freight agreement RFQ master has been published.
- Carriers have responded to the individual freight agreement RFQs with their rates for the transportation services.
- You have defined target share strategies in Customizing for **Transportation Management** under **Master Data** **Agreement RFQs and Quotations** **Define Target Share Strategies**.
- You've defined a calculation profile for the organizational unit in Customizing for **Transportation Management** under **Basic Functions** **Charge Calculation** **Basic Settings** **Define Calculation Profiles**.
- You've defined a charges profile for the organizational unit, including specifying a local currency, and assigned the calculation profile defined above.

You define the charges profile in Customizing for [Transportation Management](#) under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Define Charges Profiles](#).

- You've assigned the charges profile defined above to the organizational unit using transaction PPOME.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Optimization-Based Response Comparison

You compare responses by using the optimizer for carrier selection. The optimizer is especially useful when the comparison is complex and multiple constraints must be taken into account. With the optimizer, you can simulate and analyze the potential strategy spend depending on your strategy conditions. The system can then display the results by amount and strategy version in a strategy spend comparison graph. The system displays results for the periodicity specified in the freight agreement RFQ master.

The optimizer uses target share strategies and strategy conditions to propose target shares for carriers. You define target share strategies and strategy conditions in the Customizing activity [Define Target Share Strategies](#). You can define strategy conditions such as the minimum and maximum target share percentage that the optimizer should respect when proposing target shares for carriers and the minimum and maximum number of carriers to include in the target share strategy. You can specify strategy conditions for incumbent carriers and strategy conditions for all other carriers. Depending on your business process, you might want to award a higher percentage of business to an incumbent carrier. Note that the optimizer also takes the capacity promised by the carriers into account. This ensures that the optimizer proposes a target share for a carrier that is within the carrier's confirmed capacity.

In the comparison cockpit, you can use the [Copy](#) button to create a new strategy by copying an existing strategy. You can manually change the attributes of the copied strategy. You can either copy all the attributes of a strategy or you can copy a strategy without the target shares. When you copy a strategy without the target shares, the system copies only the strategy conditions and doesn't copy all other attributes such as, business shares, target shares, and proposed capacities. You can manually allocate target shares for the carriers and run the optimizer.

i Note

- After specifying target share strategies and strategy conditions in a freight agreement RFQ master, you can run the optimizer in the back end for specific target share strategies or for all target share strategies in the freight agreement RFQ master. The system can also estimate the strategy spend at the same time.

You can run the optimizer in the back end using program [Mass Optimizer Run and Strategy Spend Estimation](#) (/SCMTMS/OPTIMIZER_RUN) in transaction SE38. You can use this program for a single freight agreement RFQ master or for multiple freight agreement RFQ masters.

If you don't specify the target share strategies when using this program for a single freight agreement RFQ master, the system runs the optimizer for all the target share strategies in the freight agreement RFQ master. When using this program for multiple freight agreement RFQ masters, you can run the optimizer only for all target share strategies in the freight agreement RFQ masters.

- You can add additional constraints to the optimizer (for example, in relation to risks or bonuses) when comparing responses to freight agreement RFQs.

For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BADIs\) for Transportation Management](#) [Master Data](#) [Agreement RFQs](#) [BAdI: Additional Optimizer Constraints](#).

- You can add additional parameters and costs before or after estimating the strategy spend.

For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BADIs\) for Transportation Management](#) [Master Data](#) [Agreement RFQs](#) [BAdI: RFQ Strategy Spend Estimation](#).

- You can replace the standard strategy spend estimation with your own customer-specific strategy spend estimation.

For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) > [Master Data](#) > [Agreement RFQs](#) > [BAdI: Customer-Specific Strategy Spend Estimation](#).

- You can reward or penalize a carrier because of the past performance of a carrier in a trade lane in the optimizer.

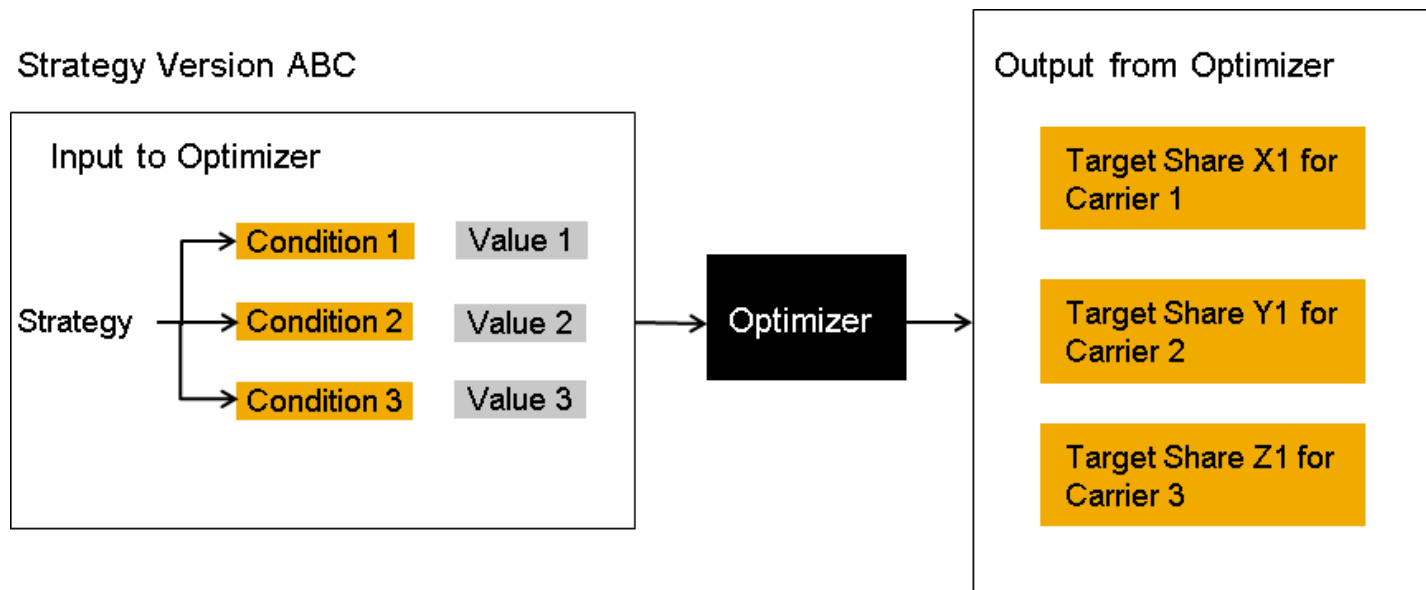
For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) > [Master Data](#) > [Agreement RFQs](#) > [BAdI: Specification of Historical Parameters for Optimizer](#).

- You can reward or penalize a carrier based on the difference between the requested transit time and the offered transit time in the optimizer. For example, if the difference is of less value, you can award a higher reward to the carrier.

For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) > [Master Data](#) > [Agreement RFQs](#) > [BAdI: Specification of Transit Time for Optimizer](#).

The optimizer proposes the most ideal target share distribution to produce the most cost effective solution. The system can then simulate the estimated strategy spend based on the responses you received from the carriers and the target share strategies proposed by the optimizer. This simulation gives you an indication of how much you'll potentially spend if you choose the target share distribution proposed by the optimizer. Note that you can run the optimizer repeatedly with different target share strategies and strategy versions. This allows you to simulate the impact of strategy changes and compare spend variations across different strategy versions. For information about how to use the optimizer to compare responses, see [Strategic Freight Procurement](#). For information about how the system estimates the strategy spend, see [Strategy Spend Estimation](#).

An optimization-based response comparison with two versions of the same strategy mentioned above is depicted in the following figures:



i Note

- The system can compare only one rate table at a time. The system can't compare charge types with different rate tables.
- Rate tables must not contain the calculation bases [Transportation Zone of Source Location](#) (SOURCELOC_ZONE) or [Transportation Zone of Destination Location](#) (DESTLOC_ZONE).
- The system takes into account the minimum and maximum rate values that you can specify in rate tables for the line items of RFQ items.

For more information, see [Calculation Sheet Maintenance](#).

- A local rate table can have just one validity period.
- The system takes into account the rates for all the capacity drivers.

For more information about capacity drivers, see [Capacities in Rate Tables for Freight Agreement RFQ Masters](#).

- If you manually change the target share for a carrier after the optimizer run, the system estimates the strategy spend based on the carrier-level target share.

The system no longer bases the estimation on the rate-table target share.

- If the optimizer proposes a target share at rate-table level for a location pair, the system applies the same target share for all the charge types in the calculation sheets for the location pair.

If the optimizer doesn't propose a target share at the rate-table level, the system bases the estimation for all the charge types on the carrier-level target share.

- If the optimizer can't fill the total requested capacity because of the minimum or maximum target share you specified for the carriers, the optimizer will dismiss the minimum or maximum target share and accept the capacity offered by the carriers.

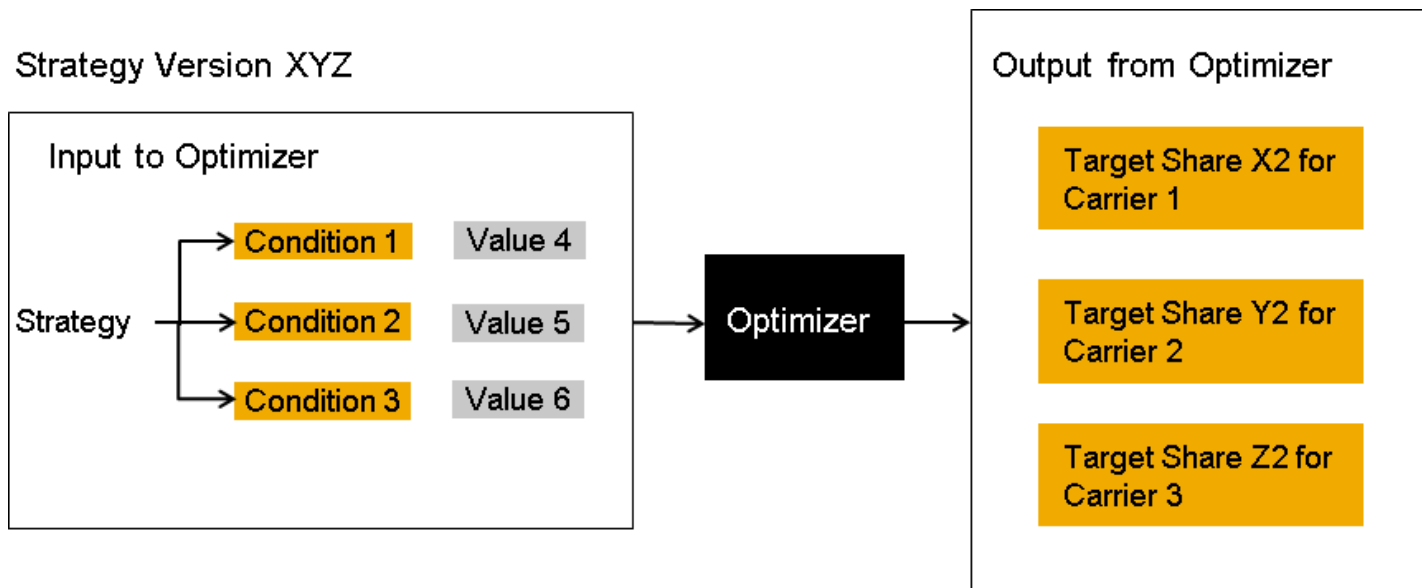
❖ Example

You request a capacity of 100 kg. Carrier 1 offers 60 kg and carrier 2 offers 100 kg.

You specified a maximum target share of 70% for carrier 1 so accept the offered capacity of 60 kg from that carrier.

You specified a maximum target share of 30% for carrier 2. However, this doesn't fill the total requested capacity of 100 kg. Therefore, the optimizer dismisses the maximum target share of 30% for carrier 2 and instead accepts 40 kg of the capacity offered by carrier 2. The total requested capacity is filled.

Strategy Version XYZ



To insert a target share strategy into a freight agreement RFQ master, in the [Response Comparison](#) screen area of the comparison cockpit you must first select the charge types that you want to compare using the optimizer.

Manual Response Comparison

If the comparison is fairly simple and the rates are the only constraints that must be taken into account, you can compare responses manually by carrier and charge type directly in Transportation Management (TM). Note that you can only compare the responses manually directly in TM for charge types with flat amounts. The system ranks the carriers according to the lowest quoted rate per charge type. The system also displays the results in a response comparison graph.

Alternatively, if the charge type has a rate table or a local rate table associated with it, you can compare the rate tables in a spreadsheet. If you compare the rate tables in a spreadsheet, the system doesn't display the results in a response comparison graph.

Note that if you compare responses manually, you manually enter the target share for the carriers. As there's no optimizer involvement, the system estimates the strategy spend based on the carrier-level target share. The system doesn't base the estimation on the rate-table target share.

Historical Spend Simulation

In the optimizer conditions, you can specify a prime charge type and validity periods. The system uses the structure of the rate table or the local rate table of the prime charge type to find previous orders with the same attributes within the validity periods. The system then calculates the charges for these orders with the rates that the carriers have proposed in the individual freight agreement RFQs. The system uses the total charge for all the previous orders to determine the target share when you run the optimizer. Since the input to the optimizer are charges and not rates, the accuracy of the optimizer increases. Note that the system uses the report [Historical Spend Simulation](#) (/SCMTMS/SOL_SPEND_SIMULATE) in transaction SE38 to find the previous orders.

Currency Conversion

If you include multiple currencies when comparing responses, the system converts the currencies into a single currency as follows:

- If the RFQ items contain multiple currencies, the system converts the currencies and displays the amounts in the currency of the freight agreement RFQ master.
- If the RFQ items contain a single currency that is different to the currency of the freight agreement RFQ master, the system displays the amounts in the currency of the RFQ items.
- If the RFQ items contain a single currency but the charge types contain a different currency, the system converts the currencies and displays the amounts in the currency of the RFQ items.

You can specify the exchange rates in Customizing for SAP NetWeaver under [General Settings](#) > [Currencies](#) > [Enter Exchange Rates](#).

Graphs

You can use the response comparison graph and strategy spend comparison graph to display the comparisons visually. There's a range of graph types available, for example, stacked and horizontal graphs, pie charts, line charts, and area charts. The default graph type is a 100% stacked column graph.

You can use the comparison graph to view the system proposed capacities. You can also view the amount that is spent if you accept the capacity proposal.

Preferred Target Share Strategy

After comparing the responses using the optimizer for carrier selection, you select your preferred target share strategy. You must then accept your preferred strategy before awarding the transportation business to carriers. When you accept your preferred target share strategy, the system automatically calculates the confirmed capacity in the freight agreement RFQ master based on the target share proposed by the optimizer. You can manually change this confirmed capacity. You can then create or update freight agreements for the carriers you're awarding the business to.

More Information

[Award Summary](#)

Strategy Spend Estimation

Use

As part of the post-bid analysis during the strategic freight procurement process, you can compare responses from carriers in the comparison cockpit. Comparing responses helps you to evaluate which carrier quoted the best rate for the specified periodicity. In Transportation Management, you can either compare the responses manually by carrier and charge type directly or by using the optimizer for carrier selection. For more information about how to compare responses, see [Post-Bid Analysis](#).

With optimization-based response comparison, the system proposes the most ideal target share distribution across carriers, taking into account your strategy conditions and the capacity promised by the carriers. The system then estimates the total cost of using a particular carrier or carriers to provide the transportation business for the selected RFQ items in a freight agreement RFQ master for the specified periodicity. The system bases the estimated cost on the target share proposed by the optimizer. The system estimates the strategy spend in different ways depending on the responses you select in the [Response Comparison](#) screen area of the comparison cockpit.

i Note

- The system estimates the cost; the system doesn't calculate the actual charges.
- If you compare responses manually, you enter the target share for the carriers. Since there's no optimizer involvement, the system estimates the strategy spend based on the carrier-level target share. The system doesn't base the estimation on the rate-table target share.
- Rate tables (both kinds, including local) must not contain the calculation bases [Transportation Zone of Source Location](#) (SOURCELOC_ZONE) or [Transportation Zone of Destination Location](#) (DESTLOC_ZONE).

Prerequisites

- You've configured the prerequisite settings for the post-bid analysis of the responses you receive from carriers (see [Post-Bid Analysis](#)).
- If a line item has a flat amount, you must not specify a calculation resolution base.

Estimation Considerations

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

- The system estimates the strategy spend for the full validity period of each RFQ item.

If the periodicity of the RFQ item is different to the validity period, the system adjusts the calculation to take this into account. For example, the validity period for an RFQ item is 12 months and the periodicity is monthly. Therefore, the system estimates the strategy spend for the full 12 months by multiplying the estimated strategy spend by the periodicity value of 12.

- The system takes into account the minimum and maximum rate values that you can specify in rate tables for the line items of RFQ items.

For more information about minimum and maximum rate values, see [Calculation Sheet Maintenance](#).

- The system can estimate the strategy spend for a single rate table only.

The system can't compare charge types with different rate tables at the same time.

- A local rate table can have just one validity period.
- The system takes into account the rates for all the capacity drivers.

For more information about capacity drivers, see [Capacities in Rate Tables for Freight Agreement RFQ Masters](#).

- If you manually change the target share for a carrier after the optimizer run, the system estimates the strategy spend based on the carrier-level target share.

The system no longer bases the estimation on the rate-table target share.

- If the optimizer proposes a target share at rate-table level for a location pair, the system applies the same target share for all the charge types in the calculation sheets for the location pair.

If the optimizer doesn't propose a target share at the rate-table level, the system bases the estimation for all the charge types on the carrier-level target share.

- You can specify only one of the following capacity dimensions when you run the optimizer:
 - Gross weight
 - Gross volume
 - Container count
- You can add additional parameters and costs before or after estimating the strategy spend.

For more information, see the following business add-ins in Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#) [Master Data](#) [Agreement RFQs](#) [BAI: Customer-Specific Strategy Spend Estimation](#):

- [BAI: RFQ Strategy Spend Estimation](#)
- [BAI: Additional Optimizer Constraints](#)
- [BAI: Specification of Historical Parameters for Optimizer](#)
- You can replace the standard strategy spend estimation with a strategy spend estimation that suits your specific requirements.

For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#) [Master Data](#) [Agreement RFQs](#) [BAI: Customer-Specific Strategy Spend Estimation](#).

Estimation Examples

i Note

In the examples below, the validity period for RFQ items is 12 months and the periodicity is monthly. Therefore, the periodicity value is 12.

Example Data

You have a freight agreement RFQ master for the US East Zone to the US West Zone listing the following carriers:

- SON_CAR001
- SON_CAR002
- SON_CAR003
- SON_CAR004

You have the following RFQ items in the freight agreement RFQ master:

RFQ Item	Transportation Mode	Movement Type	Requested Capacity	Yearly Budget (USD)	Shipping Type	Validity Period
Item 100: US East Zone to Dallas	Road	Door to port	5,200 kg per month	200,000	Less than truck load (LTL)	01.02.2014-31.01.2015
Item 200: Dallas to Denver	Road	Port to port	30 containers per month	450,000	Full truck load (FTL)	01.02.2014-31.01.2015
Item 300: Denver to US West Zone	Road	Port to door	5,200 kg per month	150,000	LTL	01.02.2014-31.01.2015
Item 400: US East Zone to US West Zone	Road	Door to door	30 containers per month	750,000	FTL	01.02.2014-31.01.2015

The RFQ items have the following capacity details:

Details	Requested Capacity	Offered Capacity	Confirmed Capacity
Gross Weight	5,200 kg	Not applicable at this stage	Not applicable at this stage
Equipment Group	Pallets	Pallets	Pallets
Equipment Type	IG2	IG2	IG2
Periodicity	Monthly	Not applicable	Not applicable
Capacity Dimension	Gross Weight	Not applicable	Not applicable

Based on the details above, you can estimate the strategy spend for the following scenarios:

1. For items that divide the end to end transportation into stages
For each stage with each carrier for RFQ items 100, 200, and 300
2. For an Individual Item that covers all stages of end to end transportation
For door-to-door FTL with each carrier for RFQ item 400

Scenario 1: Estimate for Each Stage with Each Carrier for RFQ Items 100, 200, and 300

For this scenario, you can run the optimizer by specifying constraints across RFQ items 100, 200, and 300. Alternatively, you can run the optimizer separately per item with the same constraints for RFQ items 100, 200, and 300 for the same result.

For scenario 1, the optimizer proposes the following target share strategy for the freight agreement RFQ master at the carrier level **across the RFQ items**:

Carrier	Target Share (%)	Estimated Spend (USD)
SON_CAR001	20	5,592.60
SON_CAR002	20	5,856
SON_CAR003	40	10,812
SON_CAR004	20	5,642

You indicate your preferred target share strategy, before you create or update agreements with the carriers you're awarding the business to.

If you run the optimizer **per item**, the optimizer proposes the following target share strategy per item:

- RFQ item 100 (strategy spend estimated as USD 16,525.60):

Carrier	Target Share (%)	Estimated Spend (USD)
SON_CAR001	20	3,286.40
SON_CAR002	20	3,452.80
SON_CAR003	40	6,406.40
SON_CAR004	20	3,380

- RFQ item 200 (strategy spend estimated as USD 2,940):

Carrier	Target Share (%)	Estimated Spend (USD)
SON_CAR001	20	585
SON_CAR002	20	630
SON_CAR003	40	1,140
SON_CAR004	20	585

- RFQ item 300 (strategy spend estimated as USD 8,437):

Carrier	Target Share (%)	Estimated Spend (USD)
SON_CAR001	20	1,721.20
SON_CAR002	20	1,773.20
SON_CAR003	40	3,265.50
SON_CAR004	20	1,677

The system also proposes the target strategy share at each RFQ item level. If there's a rate table (both kinds, including local) included in the RFQ items you're estimating, then the system stores the proposed target strategy share for the trade lane using the rate table (both kinds, including local) in the **Target Share** column of the rate table (both kinds, including local).

Scenario 2: Estimate for Door-to-Door Full Truck Load with Each Carrier for RFQ Item 400

For this scenario, you can estimate the strategy spend by specifying constraints and running the optimizer for RFQ item 400. The optimizer proposes the following target share strategy:

- Version 1 (strategy spend estimated as USD 13,998) with the following target share distribution:

Carrier	Target Share (%)	Estimated Spend (USD)
SON_CAR002	20	2,856
SON_CAR003	20	2,730
SON_CAR004	20	2,796
SON_CAR001	40	15,616

- Version 2 (strategy spend estimated as USD 13,930.50) with the following target share distribution:

Carrier	Target Share (%)	Estimated Spend (USD)
SON_CAR002	20	2,856
SON_CAR003	35	4,777.50
SON_CAR004	35	4,893
SON_CAR001	10	1,404

Version 2 would be your preferred target share strategy if the decision is based solely on the strategy that yields the lowest costs.

Based on these analyses, you can decide if you want to use scenario 1 or scenario 2 for your transportation needs.

More Information

[Strategic Freight Procurement](#)

Award Summary

Use

You can display a summary and graphs of all the logistical information in a freight agreement RFQ master. This summary helps you to judge how the negotiation process is progressing and make key decisions, such as whether to create a new negotiation round.

To access the award summary, in the freight agreement RFQ master choose the **Award Summary** tab page.

Integration

The **Award Summary** tab page is available in the freight agreement RFQ master (see [Freight Agreement RFQ Master](#)). You use the award summary during strategic freight procurement (see [Strategic Freight Procurement](#)).

Prerequisites

- A freight agreement RFQ master has been published.
- Carriers have responded to the individual freight agreement RFQs with their rates for the transportation services.

Features

Views

You can view the award summary using the following views:

Trade Lane

This view provides a summary of all RFQ items across the trade lanes of the freight agreement RFQ master. You can view important logistical parameters associated with trade lanes, such as the following:

- Equipment type
- Transportation mode
- Estimated strategy spend for the trade lane
- Budget allocated for the trade lane for the specified periodicity

On the **Items** tab page, you can view further information, for example:

- Number of carriers operating in the trade lane
- Target share distribution across various carriers
- Agreement-related information

You can see how the negotiation process with carriers is progressing because you can see how the estimated spend for the trade lane compares to the planned periodicity budget. For example, if you can easily see that the estimated spend is greater than the planned periodicity budget, you can consider creating a new negotiation round.

The graphs indicate the comparison of planned periodicity budget, estimated spend, and historical spend. On the **Items** tab page, the graph displays the information by estimated spend and by capacity.

Carrier

This view provides an award summary of all the carriers, including the target share for each carrier and the carriers that you awarded the transportation business to. In this view, you find information related to important logistical parameters that the carrier has agreed to provide or operate within, such as:

- Transportation mode
- Equipment type
- Shipping type
- Strategy spend version
- Amount

In the **Estimated Spend** view, the graph indicates the spend variation per RFQ item for all carriers. In the **Capacity** view, the graph indicates the capacity per RFQ item for all carriers and the target strategy spend assigned to them.

RFQ Item

This view provides an award summary by RFQ item. The graphs display a comparison of the periodicity budget versus the estimated spend per item and the spend per carrier for the chosen item. On the [Trade Lanes](#) tab page, you can view further information, for example, carrier and target share.

Historic Spend

In the [Trade Lane](#) and [RFQ Item](#) view, the system can display what you spent historically for transportation services with the same logistical information, that is, with the same trade lane, transportation mode, and shipping type. The system sends a query to SAP NetWeaver Business Warehouse for the historical data and displays it in graph format with the periodicity budget and the estimated strategy spend. Note that if the currency from SAP NetWeaver Business Warehouse differs to the currency in the RFQ master, the system converts the currency to match the currency of the RFQ master.

To check what you spent historically for transportation services with the same logistical information, choose the [Display Historic Spend](#) button in the [Trade Lane](#) or [RFQ Item](#) view on the [Award Summary](#) tab page.

More Information

[Requesting of Quotations for Transportation Services](#)

Advanced RFQs

Allows you to work with the improved usability and features of freight agreement requests for quotation (RFQs).

Prerequisites

You've ensured that the checkbox to use advanced RFQs is selected for the freight agreement RFQ type. You define freight agreement RFQ types in Customizing for [Transportation Management](#) under [Master Data](#) [Agreement RFQs and Quotations](#) [Define Freight Agreement RFQ Types](#).

i Note

This option is selected by default when you define new freight agreement RFQ types.

If you don't keep this option selected, note the following points:

- Improved usability and features aren't available when working with freight agreement RFQ masters.
- You continue to work with freight agreement RFQ masters as before.

We recommend that you don't select this checkbox for existing freight agreement RFQ types that are used in ongoing RFQ negotiations. This is to ensure that there are no compatibility issues with your existing documents. However, when defining new freight agreement RFQ types for new documents, we recommend that you keep this checkbox selected.

Features

- New Web Dynpro application configuration `/SCMTMS/TCM_EFA_RFQ_ADV` so you can work with advanced freight agreement RFQ masters
- The option to maintain only one purchasing organization in freight agreement RFQ masters (selected by default)

- The option to create only one RFQ item in freight agreement RFQ masters (selected by default)
- Maintaining trade lanes for freight agreement RFQ items is now optional when publishing freight agreement RFQ masters.
- Capacity details can be specified using the quantity dimension
- Option to edit certain details of a freight agreement RFQ master after it is published

You can edit the response deadline, update the planned agreement validity period, responsible person, freight agreement RFQ master description, and add notes and attachments.

- Work with a single user interface for capacity demand and rates when negotiating freight agreement RFQs with carriers

For more information, see [Capacity Demand and Rates](#).

- Option to reject participation in a freight agreement RFQ on behalf of a carrier
- Option to reopen a freight agreement RFQ for negotiation with a carrier based on the following conditions:
 - The response deadline is not in the past.
 - The freight agreement RFQ has one of the following statuses: **Submitted by Carrier**, **Submitted with Errors by Carrier**, **Participation Rejected by Carrier**.
 - The status of the freight agreement RFQ master is either **Published** or **Updated After Publication**.
- Option to use capacity demand and rates when a freight agreement RFQ is created from a freight agreement
- Option to implement a business add-in (BAI) to perform business-specific validations when a carrier uploads a response for a freight agreement RFQ

For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management > Master Data > Agreement RFQs and Quotations > BAI: Validate Carrier Response](#).

Capacity Demand and Rates

Allows you as a shipper or logistics service provider to work with a single user interface (UI) for capacity demand and rates when negotiating freight agreement requests for quotation (RFQs) with carriers.

Prerequisites

- You have defined a freight agreement RFQ type in Customizing for [Transportation Management](#) under [Master Data > Agreement RFQs and Quotations > Define Freight Agreement RFQ Types](#).
- Ensure that you have selected the checkbox to use advanced RFQs in the freight agreement RFQ type.
- Ensure that you have selected the checkbox to enable capacity demand and rates for the freight agreement RFQ type.
- You have defined a calculation sheet template in Customizing for [Transportation Management](#) under [Basic Functions > Charge Calculation > Basic Settings > Templates > Define Calculation Sheet Templates](#).

Features

You can use the following features of capacity demand and rates to help simplify working with freight agreement RFQs:

- Capacity Demand and Rates Profile

Use a capacity demand and rates profile to create a calculation sheet, calculation sheet line items, and rate tables. You can assign a capacity demand and rates profile to an RFQ item in an Advanced RFQ. (For more information, see [Advanced](#)

[RFQs.\)](#)

You define a capacity demand and rates profile in Customizing for **Transportation Management** under **Master Data** **Agreement RFQs and Quotations** **Define Capacity Demand and Rates Profiles**.

You can use a capacity demand and rates profile to do the following:

- Specify a calculation sheet template in the profile.
- Based on the calculation sheet template, the system proposes fields that you can select and assign to the profile. These fields are displayed in the capacity demand and rates flat view (see Capacity Demand and Rates Flat View).
- Each field that you assign to the profile is displayed as a column in the capacity demand and rates flat view (flat view). You can specify the relative position of a field, which is then used to display information in the flat view.
- Allow fields to be displayed as search criteria for the flat view.
- Allow additional related fields to be displayed in the flat view. The following is an example of additional related fields.

❖ Example

If you select **Source Location** as a field, and select the checkbox to show related fields, some of the related fields that are displayed in the flat view are as follows: **Source Location City**, **Source Location Country/Region**, **Source Location Zip Code** etc.

- Pivot the flat view on a particular field by selecting a field to be a pivot column. This option transposes the existing data of the selected field from several rows to one row by adding new columns to the flat view.
- Capacity Demand and Rates Flat View (Flat View)

The flat view is a consolidated view of all the charge items and rate tables that are relevant for negotiating freight agreement RFQs. The flat view displays information using the capacity demand and rates profile that you have assigned to an RFQ item. The following example explains what information is displayed in the flat view:

❖ Example

The following table displays a calculation sheet in a freight agreement RFQ that is under negotiation:

Item Hierarchy	Line No.	Instruction Type	Charge Type	Amount	Currency	Rate Table	Target Amount	Target Currency
Basic rate	10	Standard	FB00		EUR	18859		EUR
Loading Charges	20	Standard	TM7-LOAD		EUR	18860		EUR
Base Charges	30	Standard	TM7-BASE		EUR			EUR

The following table displays the scales that are configured for the rate table **18859**:

Scale Dimension	Calculation Base	Calculation Base Description	Scale Type	Scale Unit of Measure
01	SOURCELOC	Source Location	Same Scale (=)	
02	DESTLOC	Destination Location	Same Scale (=)	
03	QUANTITY_VAL	Quantity	To Scale (<=)	PAL

The following table displays the scales that are configured for the rate table **18860**:

Scale Dimension	Calculation Base	Calculation Base Description	Scale Type	Scale Unit of Measure
01	SOURCELOC	Source Location	Same Scale (=)	
02	EQUIP_TYPE	Equipment Type	Same Scale (=)	

If you use capacity demand and rates, the flat view displays the following fields:

Source Location (=)	Destination Location (=)	Equipment Type (=)	Quantity (PAL) (<=)	Basic rate – Target Rate (EUR per 1 PAL)	Basic rate – Tolerance (%)	Loading charges – Target Rate (EUR)	Loading charges – Tolerance (%)	Base Charges – Target Amount (EUR)	Base Charges – Tolerance (%)
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i Note

When using capacity demand and rates, the [Calculation Sheet Overview](#) tab is not visible.

- Search Criteria

Apply search criteria to the flat view to view only specific records. You can also limit the number of records to be retrieved and displayed in the flat view. In the capacity demand and rates profile, you can select the fields to be displayed as search criteria on the flat view.

- File Download and Upload

Download the capacity demand and rates records as a spreadsheet, which you can then use to update. You can choose to download the records with or without applying the search criteria.

You can upload a spreadsheet to update the capacity demand and rates. For example, the spreadsheet can contain your updated lanes, break-weights, target rates, and target amounts for flat charges.

i Note

For information about carrier spreadsheet formats, see [Responses from Carriers Using Spreadsheets](#).

- Capacity Demand and Rates Flat View for Carriers

Enable your carriers to use the flat view to submit their responses to freight agreement RFQs. Select the flat view format under Carrier Settings in the freight agreement RFQ type that you have defined in Customizing for [Transportation Management](#) under [Master Data](#) [Agreement RFQs and Quotations](#) [Define Freight Agreement RFQ Types](#).

- Capacity Demand and Rates in Freight Agreement RFQs Created from Freight Agreements

Use capacity demand and rates in a freight agreement RFQ created from a freight agreement. The system determines the capacity demand and rates profile based on the calculation sheet in the freight agreement items. If the system could not determine a capacity demand and rates profile, you can manually add the profile in the freight agreement RFQ.

Evaluations and Awarding

Allows you as a shipper to work with a single user interface (UI) for evaluating and awarding capacities to carriers or logistics service providers.

Prerequisites

- You have defined a capacity demand and rates profile in Customizing for [Transportation Management](#) under [Master Data > Agreement RFQs and Quotations > Define Capacity Demand and Rates Profiles](#).

You have selected [Dimensions](#) or [RFQ Item](#) for the award level in the capacity demand and rates profile.

If you have selected [Dimensions](#) as the award level, in [Assign Fields and Configure Properties](#), you have also selected the capacity dimensions as the award level dimensions.

- Carriers or logistics service providers have responded to your request for quotations.

Features

- Capacity Demand and Rates Flat View for Evaluations

After you publish a freight agreement RFQ master, the flat view displays the individual responses from your carriers. You can use this information to evaluate the responses by comparing offered rates, offered capacities, and offered transit times.

You can apply search criteria to evaluate specific records. You can also download and view the records as a spreadsheet.

- Evaluations

The evaluations UI is displayed after you publish the freight agreement RFQ master. You can use this UI to insert or edit simulations to view and evaluate target share strategies based on the responses received from the carriers.

You can use simulations to evaluate using the optimizer or perform a manual evaluation.

- In an optimizer-based evaluation, the simulation uses the optimizer to allocate the target shares based on the target share strategy that you have defined in Customizing for [Transport Management](#) under [Master Data > Agreement RFQs and Quotations > Define Target Share Strategies](#).

In the simulation, you can specify an optimizer strategy and the conditions to use when comparing carrier responses.

- In a manual evaluation, you must manually allocate the target shares to carriers based on the award level dimensions or at the RFQ Item level.

You can use a spreadsheet to award the carriers based on percentages or capacities on the award level dimensions.

- Simulation Estimated Spends and Awarding

- Simulate the estimated spend to view the estimated spends based on offered capacity, requested capacity, and missing capacity.

Missing capacity represents the capacity that was not committed by a carrier.

- Insert multiple simulations and compare the responses from carriers based on estimated spends.
- Set one of the simulations to be your preferred simulation. This simulation represents your preferred target share allocations to carriers.
- Accept the preferred strategy for a freight agreement RFQ item for awarding.

i Note

When using capacity demand and rates, the comparison cockpit is not available. You can use the evaluations and awarding feature of capacity demand and rates to compare carrier responses.

Reverse Auction

Allows you as a shipper to enable carriers to revise their bids and view the current best rates for the lanes.

Prerequisites

You've defined the following in the Customizing for **Transportation Management**, under **►►Master Data ► Agreement RFQs and Quotations ►**:

- You've selected **Advanced RFQs**
- You've selected **Enable Capacity Demand and Rates**
- You've selected **Enable Reverse Auction**
- (Optional) You've selected **Allow Carriers to Increase their Bids** if you want to enable your carrier to quote higher than the previous bid. You can select this checkbox, only if you've selected **Enable Reverse Auction** for the RFQ

Features

- As a shipper, you can:
 - Allow your logistic service providers to revise their bids
 - View the current best rates by your logistics service providers on the user interface as well as spreadsheets
- Logistics service providers (LSPs) can:
 - View the current best rates available for any given lane
 - Revise their bids that could be higher or lower

i Note

- LSPs can submit higher bids only if the shipper has enabled the following:

In the Customizing for **Transportation Management**, under **►►Master Data ► Agreement RFQs and Quotations ► ►Allow Carriers to Increase their Bids ►**

- By default, LSPs can submit only lower bids

- Submit bids multiple times until the response deadline is or the negotiation is completed or cancelled
- Use spreadsheets to upload latest bids for resubmission

Related Information

[Capacity Demand and Rates](#)

Strategic Freight Sales

Use

Strategic freight sales helps you as a carrier or logistics service provider (LSP) to determine the best rates you can provide to transport items between specific locations (trade lanes) and then to agree a contract (agreement) with shippers or other LSPs. Transportation Management (TM) supplies the tools you need to receive requests for rate quotations, evaluate your responses, and also proactively send your rates to shippers or other LSPs.

This feature enables you to quickly and conveniently determine the prices you can provide for various services based on the prices you have quoted previously. This gives you higher productivity and better decision making.

Process

1. You create a forwarding agreement (FWA) quotation

A shipper or LSP can communicate requests for you to provide rates to transport specific items as follows:

- o Business-to-business (B2B)

In a freight agreement RFQ master, a shipper or LSP that uses TM can specify transportation items along with attributes such as transportation mode and service type. Subsequently, the shipper or LSP sends the freight agreement RFQ master to you using business-to-business communication. For more information, see [Freight Agreement Processing](#). The system displays the freight agreement RFQ as a forwarding agreement quotation in your worklist. You can see the worklist in the **Forwarding Agreements** app on the fiori launchpad. For more information, see [Strategic Freight Procurement](#).

- o E-mail

A shipper or LSP uses e-mail to send you all the details of a request. Details can include information such as, items, rate tables, local rate tables, calculation sheets, local calculation sheets, transportation modes, and service levels. You perform the following actions to use this information in a forwarding agreement quotation:

- a. Download a new forwarding agreement quotation as a spreadsheet file. For more information, see [Forwarding Agreement Quotation](#)
- b. Enter the information the shipper or LSP has sent in to the spreadsheet.
- c. Upload the spreadsheet in the forwarding agreement quotation. For more information, see [Forwarding Agreement Quotation](#).

The system automatically fills all the fields in the forwarding agreement quotation according to the details specified in the spreadsheet. Subsequently, you enter rates for the various transportation items and download the details of the forwarding agreement quotation in another spreadsheet in the **Excel Integration** tab page of the forwarding agreement quotation. You can then send this spreadsheet file to the shipper or LSP.

If a shipper or LSP sends you a spreadsheet file with the request details in the format of a downloaded Microsoft Excel file, you can directly upload the file in the **Excel Integration** tab page of a new forwarding agreement quotation.

i Note

You can also change an existing forwarding agreement quotation using a Microsoft Excel file. You upload the Microsoft Excel file in the **Excel Integration** tab page of the forwarding agreement quotation. The system updates the information in the forwarding agreement quotation.

- o Microsoft Excel

For more information, see [Creation of FWA Quotation from Non-TM Formatted Excel](#)

A shipper or LSP can contact you with a request for rates for items by email, telephone, and so on. You can create a forwarding agreement quotation and enter the items the shipper or LSP has specified. For more information, see

[Forwarding Agreement Quotation](#)

To proactively send rates for items to shippers and LSPs, you can create a forwarding agreement quotation from any of the following documents or the document items:

- Service product catalogs

You can open a service product catalog and select the service products from which you create a forwarding agreement quotation. For more information on service product catalogs, see [Service Product Catalogs](#).

- Forwarding agreements

You can open a forwarding agreement and select the agreement items you want to offer to shippers or LSP. Subsequently, you can create a forwarding agreement quotation from these items. For more information on forwarding agreements, see [Forwarding Agreement Maintenance](#).

- Forwarding agreement quotation template

- Opportunity in an SAP Customer Relationship Management (CRM) system

You can use the service products you define from the TM system in an **Opportunity** business document in the SAP CRM system. Subsequently, you can create a forwarding agreement quotation to send rates for these service products.

i Note

After you have created the forwarding agreement quotation, you can create forwarding agreement quotation assignments. You create forwarding agreement quotation assignments to delegate the task of rate specification for transportation services for a particular trade to analysts who has expertise in providing rates for the trade lane. The analysts perform step 2 and step 3 and proposes rates. Subsequently, the analysts merge the rates back to the forwarding agreement quotation. For more information, see [Forwarding Agreement Quotation Assignment](#).

2. You specify the rate for each forwarding agreement quotation item

For more information, see [Rate Determination for Forwarding Agreement Quotation Items](#)

3. You request for approval of rate from the approver

You can enter a rate proposal for an item in the item or in a response to the item. You send the item or response to your approver for approval. The approver receives the request for approval in the workflow inbox. The approver can decide whether to accept or reject your rate proposal in the response or item. If the approver accepts the proposal for a response, the system automatically copies the rates you have proposed in the response to the calculation sheet (both kinds, including local) in the quotation item and the status of the response and quotation item changes to **Approved**. Note that the system does not copy all the rates but only the relevant rates for similar dimensions in the quotation item. If the approver accepts the rate in the item, the system changes the status of the item to **Approved**. If the approver rejects the rate proposal in the response or item, the status of the response or item changes to rejected. You can subsequently change the rate proposal and send it to the approver again.

i Note

You must select the checkbox **Publication Workflow** in Customizing for Transportation Management under **Master Data > Agreement RFQs and Quotations > Define Forwarding Agreement Quotation Types** to enable the approver to receive the quotation items for approval. If you do not select this checkbox, you must manually set the status of the forwarding agreement quotation item to **Approved** before you can send the rate proposal to the shipper or LSP.

4. You send the forwarding agreement quotation to a shipper or an LSP

After the status of a forwarding agreement quotation changes to **Approved**, you can send the quotation to the shipper or LSP. The system includes the rates you have proposed in the quotation.

You choose the **Submit** button to send the forwarding agreement quotation to the shipper or LSP. You can send the forwarding agreement quotation using business-to-business communication. For more information, see [Forwarding Agreement Processing](#). The forwarding agreement quotation is sent through the B2B (business to business) communication. The status of the forwarding agreement quotation changes to **Submitted**. You can also download the FWAQ as a Microsoft Excel file and send it as an email attachment.

If the shipper or LSP accepts your quotation item, you can create a forwarding agreement from the quotation item. However, if the shipper or LSP wants to renegotiate the prices or wants additional services, the system increments the customer negotiation cycle by one or creates a new version of the document with the changes the shipper or LSP has requested. Subsequently, you repeat the process from step 2 and propose new rates.

i Note

If you want to create a new version of the forwarding agreement quotation when a shipper or LSP wants to renegotiate the prices or wants additional services, you must select the checkbox **Create New Version** in Customizing for Transportation Management under **Master Data > Agreement RFQs and Quotations > Define Forwarding Agreement Quotation Types**. The system creates the new version with the changes the shipper or LSP has requested. If you do not select this checkbox, the system increments the customer negotiation cycle by one.

When you use B2B communication to send the forwarding agreement quotation, the system creates a freight agreement RFQ in the TM system of the shipper or LSP.

5. You create a forwarding agreement

When the shipper or LSP awards you with a quotation item, you can create a new forwarding agreement or update an existing forwarding agreement for the shipper or LSP.

Once one or more of the items have been awarded by the shipper or LSP, you can select the items and create or update a forwarding agreement with the shipper or LSP.

i Note

The forwarding agreement can also be sent to the shipper through B2B. The system creates a freight agreement in the TM system of the shipper.

Forwarding Agreement Quotation

Definition

A forwarding agreement quotation is an individual business document that you as a carrier or logistics service provider (LSP) proactively send to a shipper or another LSP to bid for the provision of future transportation services in a trade lane for a defined period of time.

You can also use a forwarding agreement quotation to respond to a request for a quotation from a shipper or an LSP with your rates for transportation services. For more information, see [Strategic Freight Sales](#).

You can create, edit, or display forwarding agreement quotations in the **Create Forwarding Agreement Quotation** or **Edit Forwarding Agreement Quotation** app on the fiori launchpad.

i Note

- You must define forwarding agreement quotation types before you create a forwarding agreement quotation. You can create forwarding agreement quotation types in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Forwarding Agreement Quotation Types**.

- You can use a forwarding agreement quotation template to create a forwarding agreement quotation. You can create, edit, or display forwarding agreement quotation templates in the [Forwarding Agreement Quotation Template](#) app on the fiori launchpad.

Use

In a forwarding agreement quotation, you can perform the following actions on a forwarding agreement quotation item:

- Business context viewer

You can review the transportation history of shippers and logistics service providers (LSPs) to which you send rate proposals for forwarding agreement quotation items. For more information, see [Customer Analysis](#).

- Build a rate proposal for an item and response, manually or automatically

For more information, see [Rate Determination for Forwarding Agreement Quotation Items](#).

- Create a forwarding agreement quotation assignment

A sales manager can create an FWAQ assignment from a forwarding agreement quotation. The manager creates the FWAQ assignment to delegate the task of rate specification for transportation services for a particular trade to an analyst who has expertise in providing rates for the trade lane. For more information, see [Forwarding Agreement Quotation Assignment](#)

- Display as flat view

You can view the entire forwarding agreement quotation (FWAQ) or an FWAQ item in one screen. This enables you to make quick and efficient decisions about rates based on logistical information. For more information, see [Flat View](#).

- Send item for approval

After you've decided the rates for a particular item or response, you send the item for approval to your approver. For more information, see [Strategic Freight Sales](#)

- Estimate forwarding charges

You can estimate forwarding charges for a forwarding agreement quotation item. The sales organization, sold-to party, movement type, traffic direction, transportation mode, and shipping type are copied from the forwarding agreement quotation to the estimate forwarding charges application. When you calculate charges, the system uses the forwarding agreement quotation item as an agreement item. It also uses the calculation sheet (both kinds, including local) and the rates of the forwarding agreement quotation item to calculate the charges.

- Display duplicate documents

The system checks and displays other business documents with matching attributes when you save or check the business document.

Structure

A forwarding agreement quotation can contain the following main tab pages and screen areas:

- **General Data**

The **General Data** tab page contains high-level information about the transportation services the shipper or LSP wants you to bid for, for example:

- Organizational unit responsible
- Validity period

- Your details, including whether the shipper or LSP already has an agreement in place with you for the trade lane
- forwarding agreement quotation type
- Customer negotiation cycle
- Deadline for responding to the shipper or LSP

- **Items**

The **Items** screen area contains information about the quotation items. This screen area can contain the following tab pages, when relevant:

- **Calculation Sheet Overview**

This tab page displays the calculation sheet (both kinds, including local) details for each quotation item.

- **Trade Lanes**

This tab page displays the details about the trade lanes for each quotation item.

- **Details**

On this tab page, you can specify the details of the payment terms and general terms. Note that when you create an agreement from the forwarding agreement quotation, the system copies the information you specify here to the details tab page of the agreement.

- **Capacities**

This tab page contains the capacity required from you, the carrier or logistics service provider, the capacity you can commit to, and the capacity awarded to you.

- **Commodity Codes**

This tab page displays the commodity codes for each quotation item.

- **Notes**

This tab page displays any notes relevant to the quotation item.

- **Attachments**

On this tab page, you can upload files relevant to the quotation item.

- **Default Route**

On this tab page, you can use default routes to specify the sequence of locations that the system uses if there's no direct connection between your locations. If there's a direct connection between your locations, you can specify a default route to predefine entries for the organization interaction status, transportation mode, and so on.

i Note

In the forwarding agreement quotation item, you can also specify an **Opportunity** business document and the corresponding item that exists in an SAP Customer Relationship Management system. This enables you to access the CRM opportunity from the forwarding agreement quotation item. This increases your response time and reduces workload.

- **Business Partner**

The Business Partner tab page displays the business partners associated with the business document. For example, a bill-to party and sold-to party.

i Note

- You can display the business partner hierarchy in the business partner tab page.

In a forwarding agreement quotation, it displays all the business partners associated with the sold-to party.

- You can restrict the partner functions available in the business partner tab page. For more information, see [Business Partner Determination](#).
- You can specify additional partner functions, such as notify party and communication party, for information purposes in addition to primary partner functions, such as sold-to party.
- The information on this tab page isn't considered when you create an agreement from the forwarding agreement quotation.

• Output Management

On the **Output Management** tab page, you can print forwarding agreement quotation details and generate e-mails and faxes and send them directly to the intended recipient. You can also see a record of the documents that were already printed or sent (by e-mail or fax, for example). For more information, see [Output Management](#).

• Notes

On this tab page, you can add any relevant notes.

• Attachments

On this tab page, you can upload relevant files.

• Change Documents

You can view all changes made to a forwarding agreement quotation on the **Change Documents** tab page or screen area. The changes are listed by date and time. You view the details of the changes at header level by selecting the relevant row in the **Transactions** screen area and at field level by selecting the relevant row in the **Record Level Changes** screen area. You must refresh the forwarding agreement quotation to view recently saved changes.

i Note

You can activate the **Change Documents** tab page or screen area in Customizing for **Transportation Management** under **►Master Data ► Agreement RFQs and Quotations ► Define Forwarding Agreement Quotation Types ►**.

• Document Flow

You can view the business documents from which the forwarding agreement quotation was created and the business documents created from the forwarding agreement quotation.

• Excel Integration

You can use a Microsoft Excel file to update details when you create a new forwarding agreement quotation or when you edit an existing forwarding agreement quotation. You can perform the following actions:

- Download the forwarding agreement quotation as a Microsoft Excel file
- Update the Microsoft Excel file that you download with the shipper's or LSP's requirements
- Upload the Microsoft Excel file in the Excel Integration tab page

The system uses the information in the uploaded Microsoft Excel file to automatically enter information in the relevant fields in the quotation. You can then propose rates for various items in the forwarding agreement quotation.

You can also perform the following activities:

- Download the forwarding agreement quotation as a customer excel

If you've created the forwarding agreement quotation from a Microsoft Excel provided by a customer, you can download the forwarding agreement quotation in the same format of the customer's Microsoft Excel file. For more information on creating a forwarding agreement quotation from a customer's Microsoft Excel file, see [Creation of FWA Quotation from Non-TM Formatted Excel](#)

- Download all items, all item responses, or all items and item responses in the forwarding agreement quotation as a Microsoft Excel file

i Note

You specify the fields of forwarding agreement quotations that you want another LSP or a shipper to be able to edit in Microsoft Excel in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Editable Fields in Excel**. Note that this Customizing isn't taken into account if you download the forwarding agreement quotation as a customer excel.

• Versions

The **Versions** tab page contains a list of the FWAQ versions, including the status of each version.

• Capacities

This tab page contains the total capacity required from you, the carrier or logistics service provider, the total capacity you can commit to, and the total capacity awarded to you.

i Note

You can activate the **Capacities** tab page or screen area in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Forwarding Agreement Quotation Types**.

More Information

[Freight Agreement RFQ Master](#)

[Agreement](#)

[Strategic Freight Sales](#)

For more information about SAP CRM opportunities, see SAP Library on SAP Help Portal at <http://help.sap.com/crm>. Choose a release and under **Application Help**, choose **SAP Customer Relationship Management > Sales > Opportunity Management > Integration with SAP Transportation Management**.

Forwarding Agreement Quotation Assignment

Definition

A forwarding agreement quotation (FWAQ) assignment is a business document that you as an analyst of a carrier or logistics service provider (LSP) use to propose rates for transportation services.

A sales manager can create an FWAQ assignment from a forwarding agreement quotation item. The manager creates the FWAQ assignment to delegate the task of rate specification for transportation services for a particular trade to an analyst who has expertise in providing rates for the trade lane.

You, as an analyst, can also create an FWAQ assignment and use the items that are relevant to you from various FWAQs and propose rates for the transportation services. For example, you have expertise in providing rates between Asia and Europe. You can

create a forwarding agreement quotation assignment and use items from various FWAQs that have routes between these locations to provide rates for transportation services.

For every item in the FWAQ assignment, the system displays the FWAQ from which the item is used.

You can create, edit, and display FWAQ assignments in the **Forwarding Agreement Quotation Assignment** app in the fiori launchpad.

Use

In an FWAQ assignment, you can perform the following actions on an FWAQ assignment item:

- Business context viewer

You can review the transportation history of shippers and logistics service providers (LSPs) to which you send rate proposals for forwarding agreement quotation items. For more information, see [Customer Analysis](#).

- Build a rate proposal for an item and response, manually or automatically

For more information, see [Rate Determination for Forwarding Agreement Quotation Items](#).

- Display as a flat view

You can view the entire forwarding agreement quotation assignment (FWAQ) or an FWAQ assignment item in one screen. This enables you to make quick and efficient decisions about rates based on logistical information. For more information, see [Flat View](#).

- Estimate forwarding charges

You can estimate forwarding charges for an FWAQ assignment item. The sales organization, sold-to party, movement type, traffic direction, transportation mode, and shipping type are copied from the FWAQ assignment to the estimate forwarding charges application. When you calculate charges, the system uses the forwarding agreement quotation assignment item as an agreement item. It also uses the calculation sheet (both kinds, including local) and the rates of the forwarding agreement quotation assignment item to calculate the charges.

- Send item for approval

After you've decided the rates for a particular item or response, you send the item for approval to your approver. For more information, see [Strategic Freight Sales](#).

- Merge items

After you've proposed the rates, you can merge the rates back into the FWAQ item. Note that the rates for the items must be approved before you can merge the rates into the FWAQ item.

- Display duplicate documents

The system checks and displays other business documents with matching attributes when you save or check the business document.

Structure

A forwarding agreement quotation assignment can contain the following main tab pages and screen areas:

- **General Data**

The **General Data** tab page contains high-level information about the transportation services the shipper or LSP wants you to bid for, for example:

- Organizational unit responsible
- Validity period
- FWAQ type
- Customer negotiation cycle
- Deadline for responding to the shipper or LSP
- Business partner hierarchy

- **Items**

The **Items** screen area contains information about the FWAQ assignment items. This screen area can contain the following tab pages, when relevant:

- **Calculation Sheet Overview**

This tab page displays the calculation sheet (both kinds, including local) details for each FWAQ assignment item.

- **Trade Lanes**

This tab page displays the details about the trade lanes for each FWAQ assignment item.

- **Capacities**

This tab page contains the capacity required from you, the carrier or logistics service provider, the capacity you can commit to, and the capacity awarded to you.

- **Commodity Codes**

This tab page displays the commodity codes for each FWAQ assignment item.

- **Notes**

This tab page displays any notes relevant to the FWAQ assignment item.

- **Attachments**

On this tab page, you can upload files relevant to the FWAQ assignment item.

- **Default Route**

On this tab page, you can use default routes to specify the sequence of locations that the system uses if there's no direct connection between your locations. If there's a direct connection between your locations, you can specify a default route to predefine entries for the organization interaction status, transportation mode, and so on.

- **Business Partner**

The Business Partner tab page displays the business partners associated with the business document. For example, a bill-to party and sold-to party.

i Note

- You can display the business partner hierarchy in the business partner tab page.

In an FWAQ assignment, it displays all the business partners associated with the sold-to party.

- You can restrict the partner functions available in the business partner tab page. For more information, see [Business Partner Determination](#).

- You can specify additional partner functions, such as notify party and communication party, for information purposes in addition to primary partner functions, such as sold-to party.

- **Output Management**

On the **Output Management** tab page, you can print the FWAQ assignment details and generate e-mails and faxes and send them directly to the intended recipient. You can also see a record of the documents that were already printed or sent (by e-mail or fax, for example). For more information, see [Output Management](#).

- **Notes**

On this tab page, you can add any relevant notes.

- **Attachments**

On this tab page, you can upload relevant files.

- **Change Documents**

You can view all changes made to an FWAQ assignment on the **Change Documents** tab page or screen area. The changes are listed by date and time. You view the details of the changes at header level by selecting the relevant row in the **Transactions** screen area and at field level by selecting the relevant row in the **Record Level Changes** screen area. You must refresh the FWAQ assignment to view recently saved changes.

i Note

You can activate the **Change Documents** tab page or screen area in Customizing for **Transportation Management** under **►Master Data ► Agreement RFQs and Quotations ► Define Forwarding Agreement Quotation Types ►**.

- **Excel Integration**

You can use a spreadsheet to update details when you create a new FWAQ assignment or when you edit an existing FWAQ assignment. You can perform the following actions:

- Download the FWAQ assignment to a spreadsheet
- Update the spreadsheet that you download with the shipper's or LSP's requirements
- Upload the spreadsheet in the **Excel Integration** tab page

The system uses the information in the uploaded spreadsheet to automatically enter information in the relevant fields in the FWAQ assignment. You can then propose rates for various items in the FWAQ assignment.

You can also download all items, all item responses, or all items and item responses in the forwarding agreement quotation to a spreadsheet

i Note

You specify the fields of an FWAQ assignment that you want another LSP or a shipper to be able to edit in a spreadsheet in Customizing for **Transportation Management** under **►Master Data ► Agreement RFQs and Quotations ► Define Editable Fields in Excel ►**.

- **Capacities**

This tab page contains the total capacity required from you, the carrier or logistics service provider, the total capacity you can commit to, and the total capacity awarded to you.

i Note

You can activate the **Capacities** tab page or screen area in Customizing for **Transportation Management** under **►Master Data ► Agreement RFQs and Quotations ► Define Forwarding Agreement Quotation Types ►**.

More Information

[Forwarding Agreement Quotation](#)

[Freight Agreement RFQ Master](#)

[Agreement](#)

[Strategic Freight Sales](#)

Customer Analysis

Use

You can use this feature to review the transportation history of shippers and logistics service providers (LSPs) to which you send rate proposals for forwarding agreement quotation items.

Prerequisites

You have performed the following tasks:

- Integrated Transportation Management (TM) with SAP NetWeaver Business Warehouse (SAP NetWeaver BW) BI CONTENT 747 – SP07. The system uses the analyzed data in SAP NetWeaver BW to display the carriers' performance in the BCV.
- Implemented SAP note [1715521](#) .
- Activated the business configuration set /SCMTMS/BCV_CUSTOMER_ANALYSIS.

Features

The system displays the following customer analysis information:

- Revenue per year

Displays the total revenue earned from a shipper or an LSP over a period of time in a graph. This enables you to determine the impact the customer has on your business profits.

You can also review the revenue per quarter. This enables you to check if the revenue you generated was uniform over a particular year.

- Demand per year

Displays the total load that you carried for a customer over a period of time in a graph. This enables you to determine whether a customer is inflating demand in the request.

You can also review the demand per quarter. This enables you to check if the load you delivered was uniform over a particular year or if the load increased or decreased during certain periods.

Note that you can also enter the sales organization so that the system displays analytics that is specific to the transportation history of the agreements between the sold-to party and sales organizations. You can also analyze data for the shipper or LSP based on trade lanes. This enables you to truncate your analysis over a trade lane.

More Information

Flat View

Use

As a logistic service provider or carrier, you can view the entire forwarding agreement quotation (FWAQ) or an FWAQ item in one screen. This enables you to see a broader view of the FWAQ to make quick and efficient decisions about rates based on logistical information.

You can display the flat view of an FWAQ or an FWAQ item when you open an FWAQ in the [Edit Forwarding Agreement Quotation](#) app on the SAP Fiori launchpad.

Prerequisites

You've performed the following activities:

- Defined a flat view profile in Customizing for [Transportation Management](#) under [Master Data](#) > [Agreement RFQs and Quotations](#) > [Define Excel and Flat View Profiles](#) > [Define Flat View Profiles for FWAQs](#) >
- Assigned the flat view profile to an FWAQ type in Customizing for [Transportation Management](#) under [Master Data](#) > [Agreement RFQs and Quotations](#) > [Define Forwarding Agreement Quotation Types](#) >

When you display the flat view from an FWAQ of this FWAQ type, the system uses the flat view profile to display a flat view of the FWAQ or the FWAQ item.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You can perform the following activities:

- Change rates

For the same charge type across items, you can simultaneously update the rates for a particular scale or for all scales. For example, you can increase the rates for basic freight with the charge type of FB00 by 20% with source location Bangalore and destination location Hamburg.

- Margin and target margin

The system displays the following:

- The rate obtained by the system by subtracting the rate in the cost charge types from the total rate in all the non-cost charge types as the margin
- The rate obtained by the system by subtracting the rate in the cost charge types from the total target rate in all the non-cost charge types as the target margin

i Note

You can select the **Cost** checkbox only if you've selected the **Statistical Charge** checkbox. You can perform these activities on the **Classification** tab page of a charge type when you edit a calculation sheet.

- Business context viewer

You can review the transportation history of shippers and logistics service providers (LSPs) to which you send rate proposals for forwarding agreement quotation items. For more information, see [Customer Analysis](#).

- Update repeated rates

A repeated rate is the rate for the same logistical combination that is displayed multiple times in flat view. The system automatically updates all the rows of a charge type that have repeated rates when you update one of the rate rows.

❖ Example

A forwarding agreement quotation item contains two charge types with the following rate tables:

- FB00 with the rate table values:

Source Location	Destination Location	Rate
Berlin	Chicago	10

- BAF with the rate table values:

Source Location	Destination Location	Gross Weight	Rate
Berlin	Chicago	100	10
Berlin	Chicago	200	30

When you display the item as a flat view, the system displays the two charge types as follows:

Source Location	Destination Location	Gross Weight	FB00	BAF
Berlin	Chicago	100	10	20
Berlin	Chicago	200	10	30

If you change the rate of FB00 from 10 to 5 in one row, the system automatically updates all the rows of the charge type with the same rate for the same logistical combination if you've enabled the repeated rates feature.

Source Location	Destination Location	Gross Weight	FB00	BAF
Berlin	Chicago	100	5	20
Berlin	Chicago	200	5	30

- Copy or delete a row

You can use the **Copy** button to copy an existing row and the **Delete** button to delete an existing row.

- Mass update

You can use the [Mass Update](#) button to update all the rates displayed in the table.

- Download as a spreadsheet

You can download the visible columns and rows in the flat view of an FWAQ to a spreadsheet file.

- Filter data based on specific attributes

You can filter data based on criteria you specify. For example, simulation date enables you to view the rates for a particular date. In the rate table, the system finds the validity period in which this date exists and displays the rate for the validity period.

- Display trade lane, commodity code, and default route for a row

You select a row. The rates in the row are present in a particular item. The system displays the trade lane, commodity code, and default route of the item.

- Change and save the view

You can edit the manner in which the system displays the flat view and can also save this view for later use.

- Filter based on values

You can filter the values in the flat view based on specific values of columns and hide the rest of the data. For example, you can display the rates that have values for an equipment type of 20G.

i Note

- The number of characters for the value of a field name can't exceed 29 characters if the orientation is [By Column](#). Instead, you can specify an orientation of [By Row](#) to accommodate more than 29 characters.
- The system doesn't display the calculation rule in the flat view.

Example

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

An FWAQ with an FWAQ type FLAT_QT contains an item with charge types with the following rate tables:

- Charge type FB00 with the rate table values:

Source Location	Destination Location	Commodity	Equipment Type	Rate	Target Rate
Berlin	Chicago	FAK1	20G	100	90
Berlin	Chicago	FAK1	40G	110	100
Berlin	Chicago	FAK2	20G	200	190
Berlin	Chicago	FAK2	40G	210	200

- Charge type BAF with the rate table values:

Source Location	Destination Location	Equipment Type	Rate	Target Rate
Berlin	Chicago	20G	50	40
Berlin	Chicago	40G	40	30

- Cost Charge type FB01 with the rate table values:

Source Location	Destination Location	Commodity	Equipment Type	Rate
Berlin	Chicago	FAK1	20G	20
Berlin	Chicago	FAK1	40G	20
Berlin	Chicago	FAK2	20G	20
Berlin	Chicago	FAK2	40G	20

In the Customizing activity [Define Flat View Profiles for FWAQs](#), you've specified the following settings for a flat view profile FLAT_DEM.

Document Hierarchy Level	Field Name	Orientation	Sequence Number
Calculation Sheet	TCET084	By Column	
Scale	SOURCELOC	By Row	1
Scale	DESTLOC	By Row	2
Scale	COMMODITY_CODE	By Row	3
Scale	EQUIP_TYPE	By Row	4
Scale	TARGET_RATE	By Row	

You've also assigned the flat view profile to the forwarding agreement quotation type FLAT_QT

When you display the FWAQ item as a flat view in the FWAQ, the system displays the following view of the item:

Source Location	Destination Location	Commodity Code	Equipment Type	FB00	Target Rate FB00	BAF	Target Rate BAF	FB01	Margin	Target Margin
Berlin	Chicago	FAK1	20G	100	90	50	40	20	130	110
Berlin	Chicago	FAK1	40G	110	100	40	30	20	130	110
Berlin	Chicago	FAK2	20G	200	190	50	40	20	230	210
Berlin	Chicago	FAK2	40G	210	200	40	30	20	230	210

Note that the target margin is the sum of the target rates in all the non-cost charge types after subtracting the rate in the cost charge type.

You can also change the values of the rates and save the FWAQ.

More Information

[Forwarding Agreement Quotation](#)

Rate Determination of Forwarding Agreement Quotation Items

Use

The system enables you to easily and efficiently determine the rates you want to provide to shippers or logistics service providers (LSPs) for forwarding agreement quotation (FWAQ) items based on the existing service products. Note that you can also use this feature in a forwarding agreement quotation assignment for the forwarding agreement quotation assignment items.

Prerequisites

- In the forwarding agreement quotation, you've selected a leading charge type in the calculation sheet (both kinds, including local) of an item. This enables the system to easily find service products based on the information, such as location pairs, in the rate table (both kinds, including local) of the leading charge type. Subsequently, the system tries to find matching rate records of the leading charge type in the rate records of the charge type in the service products.

→ Recommendation

We recommend that you choose a charge type that contains the source locations and destination locations as the leading charge type. This enables the system to find matching service products with the similar source locations and destination locations in the rate records of the charge type in the service product. This enables the system to match the most optimum rates.

We recommend that you select only one charge type as the leading charge type. This enables the system to find matching service products quickly.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Automatically Build Rate

In the forwarding agreement quotation item, you can choose one or more items and automatically build rate. The system performs the following activities for each item:

1. Tries to find service products with the attributes as the rate table in the leading charge type
2. If it finds matching service products, the system tries to find matching rate records of the leading charge type in the rate records of the charge type in the service product
3. If it finds matching rate records, the system creates a forwarding agreement quotation item response in the item

The forwarding agreement quotation item response contains not only the matching charge type and matching rate record, but all the charge types with a matching rate record.

The system creates separate forwarding agreement quotation item responses for every matching service product.

i Note

- After the system finds a match for the leading charge type and a rate record combination in a service product, the system doesn't search for more service products for the rate record.
- The system selects a rate from the rate table of the matching service product based on the middle date of the agreement validity in the forwarding agreement quotation. For example, the rates in a rate table of a matching service product are valid from April 1, 2015 to April 20, 2015 and another rate from April 21, 2015 to May 21, 2015. If the agreement validity is from April 15, 2015 to May 15, 2015, the system selects the rate that is valid on April 30, 2015, which is the middle date of the agreement validity. Therefore, the system selects the rate that is valid from April 21, 2015 to May 21, 2015.

❖ Example

Two service products exist in the system with charge type FB00 with the following rate tables:

- FB00 rate table for service product 1

Source Location	Destination Location	Equipment Type	Rate (USD)
Hamburg	Moscow	20G	100

- FB00 rate table for service product 2

Source Location	Destination Location	Rate (USD)
Hamburg	Moscow	110

The leading charge type FB00 of the item contains the following rate table:

Source Location	Destination Location	Equipment Type	Rate (USD)
Hamburg	Moscow	20G	

Both the service products match the leading charge type. However, the system uses the one it finds first. It doesn't continue to search after it has found one of the two service products.

Rate Builder Cockpit

For one or more items, you can open the rate builder cockpit to manually build rate from matching service products. When you open the rate builder cockpit for an item, the system automatically finds matching service products for the leading charge type and the corresponding rate records in the item. Subsequently, you can perform the following activities:

- Copy the service product as your response

You choose one or more matching service products and copy the service product as a response. The system automatically copies the service product along with the calculation sheet as a response to the forwarding agreement quotation item.

- Copy charge type

You choose the forwarding agreement quotation item or a forwarding agreement quotation item response. Subsequently, you can choose a matching service product and copy the charge type in the service product to your forwarding agreement quotation item or the forwarding agreement quotation item response.

i Note

You can copy a charge type with or without currency conversion. When you choose to copy a charge type with currency conversion, the system uses the currency specified in the calculation sheet item for currency conversion. If you haven't maintained any currency in the calculation sheet item, the system chooses the document currency you've specified in the forwarding agreement quotation for currency conversion. You specify the exchange rates in Customizing for SAP NetWeaver under ► [General Settings](#) ► [Currencies](#) ► [Enter Exchange Rates](#) ►.

- Build rate in the forwarding agreement quotation item

You select one or more service products and build rate within the item. The system performs the following steps:

1. Tries to find matching rate records of the leading charge type in the rate records of the charge type in the service product
2. If it finds matching rate records, the system copies all the matching rate records for only the matching charge types.

i Note

You can build rate in the forwarding agreement quotation item with or without currency conversion. When you choose to build rates with currency conversion, the system uses the currency specified in the calculation sheet item for currency conversion. If you haven't maintained any currency in the calculation sheet item, the system chooses the document currency you've specified in the forwarding agreement quotation for currency conversion. You specify the exchange rates in Customizing for SAP NetWeaver under ► [General Settings](#) ► [Currencies](#) ► [Enter Exchange Rates](#) ►.

- Build rate with responses

You select one or more service products and build rate as responses. The system performs the following steps:

1. Tries to find matching rate records of the leading charge type in the rate records of the charge type in the service product
2. If it finds matching rate records, the system creates a forwarding agreement quotation item response in the item

The forwarding agreement quotation item response contains not only the matching charge type and matching rate record, but all the charge types with a matching rate record.

The system creates separate forwarding agreement quotation item responses for every matching service product.

i Note

The system selects a rate from the rate table of the matching service product based on the middle date of the agreement validity in the forwarding agreement quotation. For example, the rates in a rate table of a matching service product are valid from April 1, 2015 to April 20, 2015 and another rate from April 21, 2015 to May 21, 2015. If the agreement validity is from April 15, 2015 to May 15, 2015, the system selects the rate that is valid on April 30, 2015, which is the middle date of the agreement validity. Therefore, the system selects the rate that is valid from April 21, 2015 to May 21, 2015.

- Search additional service products

You can choose to ignore the service products that the system has found and instead specify your criteria to search for service products

Porting of Rates

After you have finished building rates automatically or in the rate builder cockpit for an item, and you have approved the rates, you can perform one of the following activities:

- Send rates only for the requested services of the customer

The system merges the forwarding agreement quotation item responses that are relevant to the item before you send it to the customer. This ensures that the customer is sent the rates for only the requested transportation services.

To use this feature, you must specify a format type of **Customer** in Customizing for **Transportation Management** under **Master Data** > **Agreement RFQs and Quotations** > **Define Forwarding Agreement Quotation Types**.

- Send rates not only for the requirements of the customer but also additional service rates

If you want to send additional service products to the customer, you can build additional responses and send the item and the additional forwarding agreement quotation item responses to the customer.

To use this feature, you must specify a format type of **Carrier** in Customizing for **Transportation Management** under **Master Data** > **Agreement RFQs and Quotations** > **Define Forwarding Agreement Quotation Types**.

Example

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

A forwarding agreement quotation item contains a calculation sheet with the FB00 charge type and the following rate table:

Source Location (=)	Destination Location (=)	Equipment Type (=)	Commodity Code (=)	Value
BUSAN	ROTTERDAM	20G0	4701	0.00
BUSAN	ROTTERDAM	20G0	4701	0.00
BUSAN	ROTTERDAM	42G0	4701	0.00
BUSAN	ROTTERDAM	42G0	4701	0.00
NINGBO	HAMBURG	20G0	4701	0.00
NINGBO	HAMBURG	20G0	4701	0.00
NINGBO	HAMBURG	42G0	4701	0.00
NINGBO	HAMBURG	42G0	4701	0.00
NINGBO	ROTTERDAM	20G0	4701	0.00
NINGBO	ROTTERDAM	20G0	4701	0.00
NINGBO	ROTTERDAM	42G0	4701	0.00
NINGBO	ROTTERDAM	42G0	4701	0.00
SHANGHAI	HAMBURG	20G0	4701	0.00
SHANGHAI	HAMBURG	20G0	4701	0.00
SHANGHAI	HAMBURG	20G0	4701	0.00
SHANGHAI	HAMBURG	20G0	4701	0.00
MOSCOW	DELHI	20G0	4701	0.00
MOSCOW	DELHI	42G0	4701	0.00

The system finds one service product with the following charge type rate tables:

This is custom documentation. For more information, please visit [SAP Help Portal](#).

- FB00

Source Location (=)	Destination Location (=)	Equipment Type (=)	Commodity Code (=)	No. of Containers in TEU	Value
NINGBO	ROTTERDAM	20G0	4701	100	96
NINGBO	ROTTERDAM	20G0	4701	200	90
NINGBO	ROTTERDAM	20G0	4701	300	72
NINGBO	ROTTERDAM	20G0	4711	100	138
NINGBO	ROTTERDAM	20G0	4711	200	120
NINGBO	ROTTERDAM	20G0	4711	300	108
NINGBO	SOUTHAMPTON	20G0	4701	100	112
NINGBO	SOUTHAMPTON	20G0	4701	200	126
NINGBO	SOUTHAMPTON	20G0	4701	300	133
NINGBO	SOUTHAMPTON	20G0	4711	100	145
NINGBO	SOUTHAMPTON	20G0	4711	200	157
NINGBO	SOUTHAMPTON	20G0	4711	300	178

- THC

Transportation Zone of Destination Location (=)	Equipment Type (=)	Value
EUROPE	20G0	30
EUROPE	22B0	60
EUROPE	42G0	45
EUROPE	42H0	80
MALAYSIA	20G0	25
MALAYSIA	22B0	45
MALAYSIA	42G0	60
MALAYSIA	42H0	75

- BAF

Destination Location (=)	Equipment Type (=)	Value
HAMBURG	20G0	50
HAMBURG	42G0	80
ROTTERDAM	20G0	60

Destination Location (=)	Equipment Type (=)	Value
ROTTERDAM	42G0	90
SOUTHAMPTON	20G0	40
SOUTHAMPTON	42G0	70

When you build a forwarding agreement quotation response, the system creates a forwarding agreement quotation response with the following charge types and rate tables:

- FB00

Source Location (=)	Destination Location (=)	Equipment Type (=)	Commodity Code (=)	No. of Containers in TEU	Value
NINGBO	ROTTERDAM	20G0	4701	100	96
NINGBO	ROTTERDAM	20G0	4701	200	90
NINGBO	ROTTERDAM	20G0	4701	300	72
NINGBO	ROTTERDAM	20G0	4711	100	138
NINGBO	ROTTERDAM	20G0	4711	200	120
NINGBO	ROTTERDAM	20G0	4711	300	108

- THC

Transportation Zone of Destination Location (=)	Equipment Type (=)	Value
EUROPE	20G0	30
EUROPE	22B0	60
EUROPE	42G0	45
EUROPE	42H0	80

- BAF

Destination Location (=)	Equipment Type (=)	Value
HAMBURG	20G0	50
HAMBURG	42G0	80

The system finds a second service product with the following charge type rate tables:

- FB00

Source Location (=)	Destination Location (=)	Equipment Type (=)	Commodity Code (=)	No. of Containers in TEU	Value
MOSCOW	DELHI	20G0	4701	100	96

- THC

Transportation Zone of Destination Location (=)	Equipment Type (=)	Value
ASIA	20G0	30

When you build another forwarding agreement quotation response, the system creates a new forwarding agreement quotation response with the following charge types and rate tables:

- FB00

Source Location (=)	Destination Location (=)	Equipment Type (=)	Commodity Code (=)	No. of Containers in TEU	Value
MOSCOW	DELHI	20G0	4701	100	96

- THC

Transportation Zone of Destination Location (=)	Equipment Type (=)	Value
ASIA	20G0	30

More Information

[Strategic Freight Sales](#)

[Forwarding Agreement Quotation](#)

[Forwarding Agreement Quotation Assignment](#)

Creation of FWA Quotation from Non-TM Formatted Excel

Use

A shipper or logistics service provider can send you, the carrier or logistics service provider, the requirements for transportation services in a Microsoft Excel file that isn't in the format of a Microsoft Excel downloaded from a forwarding agreement quotation. From the Microsoft Excel file, in **SAP NetWeaver Business Client (NWBC)** under **► Forwarding Agreement Management ► Forwarding Agreement Quotations ► Create/Update Forwarding Agreement Quotation from Microsoft Excel ►**, you can perform the following activities:

- Create a new forwarding agreement quotation

- Create a new version of a forwarding agreement quotation
- Replace the items in a forwarding agreement quotation
- Add items to a forwarding agreement quotation

You can perform these activities using Microsoft Excel 2007 and higher releases.

i Note

- After you've created the forwarding agreement quotation, you must manually specify values for fields that haven't been entered automatically from the Microsoft Excel file for determining the rates you can provide. To change the values that have been entered from the Microsoft Excel file, you must manually change the values in the forwarding agreement quotation.
- You can map the rates in the Microsoft Excel file to the rate or target rate, or both in the forwarding agreement quotation.

You can download the forwarding agreement quotation in the same format of the customer's Microsoft Excel file. You can then send this file to the customer. Since the customer receives the Microsoft Excel file in a known format, it's easy for the customer to analyze the rates.

Prerequisites

To ensure that the values in the Excel are mapped to the correct field in the forwarding agreement quotation, you've performed the following activities in Customizing for [Transportation Management](#) under [►Master Data► Agreement RFQs and Quotations► Define Excel and Flat View Profiles► Map Data in Non-TM Formatted Excel to TM Fields. ►](#):

1. Defined an excel mapping profile
2. Assigned the method in which the values in the Microsoft Excel file are mapped to the fields in the forwarding agreement quotation.

In [SAP NetWeaver Business Client](#) (NWBC) under [►Forwarding Agreement Management► Forwarding Agreement Quotations► Create/Update Forwarding Agreement Quotation from Microsoft Excel ►](#), you assign excel mapping profiles to the sheets of a Microsoft Excel file to create the forwarding agreement quotation.

The values in the Microsoft Excel file must be valid Transportation Management master data values. Else, you can use the business add-in [BADl: Master Data Mapping for FWAQ](#) under [►Business Add-Ins \(BADIs\) for Transportation Management► Master Data► Agreement RFQs and Quotations ►](#) to map your master data to TM master data.

Example

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

A shipper sends you a Microsoft Excel file in the following format:

	A	B	C	D	E	F	G	H	I
1									

	A	B	C	D	E	F	G	H	I
2									
3	Shipping Type	Source Region	Source Country/Region	Source Airport	Destination Region	Destination Country/Region	Destination Airport	Transportation Mode	Shi Typ
4	10	PA_APORT_EUR	DE	BER	PA_APORT_US	US	ATL	05	10
5	10	PA_APORT_EUR	DE	BER	PA_APORT_US	US	AUS	05	10
6	10	PA_APORT_EUR	DE	BER	PA_APORT_US	US	BOS	05	10
7	10	PA_APORT_EUR	DE	BER	PA_APORT_US	US	BWI	05	10

In Customizing for [Transportation Management](#) under [Master Data](#) [Agreement RFQs and Quotations](#) [Define Excel and Flat View Profiles](#) [Map Data in Non-TM Formatted Excel to TM Fields](#), you define an [Excel Mapping Profile](#) of CUST_1 and specify the following settings in the [Assign Excel Fields to Document Fields](#) screen:

Document Hierarchy Level	Field Name	Orientation	Starting Column	End Column	Row Number
Charge		By Column	M	R	1
Header	SHIPPING_TYPE	By Row	A		
Item	DEST_ZONE	By Row	E		
Item	MOT	By Row	H		
Item	MOVEM_TYPE	By Row	J		
Item	SHIPPING_TYPE	By Row	I		
Item	SOURCE_ZONE	By Row	B		
Scale	CHRG_WEIGHT	By Column	M	R	2
Scale	DESTLOC_CNTRY	By Row	F		4
Scale	DEST_AIR_IATA	By Row	G		
Scale	SOURCELOC_CNTRY	By Row	C		
Scale	SOURCE_AIR_IATA	By Row	D		

i Note

You've specified a row number of 4 for field name DESTLOC_CNTRY to specify that the values for the all the field names for which you haven't specified a row number, and have an orientation of **By Row** start from the 4th row. Therefore, the system considers the values for following fields from the 4th row:

- SOURCE_AIR_IATA
- SOURCELOC_CNTRY
- DEST_AIR_IATA

- DESTLOC_CNTRY
- SOURCE_ZONE
- SHIPPING_TYPE
- MOVEM_TYPE
- MOT
- DEST_ZONE
- SHIPPING_TYPE

You don't have to specify a row number of 4 only for field name DESTLOC_CNTRY. You can specify a row number of 4 for any of the field names for which the value start from the 4th row, and has an orientation of **By Row**, for example MOVEM_TYPE. The system automatically assumes that the values for the all the field names for which you haven't specified a row number, and has an orientation of **By Row** begins from the 4th row.

If you don't specify a row number of 4 for any of these fields, the system assumes that the values start right after the last specified row number that has an orientation or **By Column**. Therefore, the system would assume that the values start from row 3. This would be incorrect for this example.

In **SAP NetWeaver Business Client** (NWBC) under **Forwarding Agreement Management** **Forwarding Agreement Quotations** **Create/Update Forwarding Agreement Quotation from Microsoft Excel** , you upload the Microsoft Excel file. Specify the Excel Mapping Profile of CUST_1 for the sheet that contains the relevant values and create a forwarding agreement quotation. The system automatically maps the values in the Excel to the fields in the forwarding agreement quotation in the following method:

- Specifies a shipping type of 10 in the header
- Creates an item with the following attributes:
 - A shipping type of 10
 - A movement type of DIA
 - A transportation mode of 05
 - A trade lane with a source of PA_APORT_EUR and destination of PA_APORT_US

The system also creates a calculation sheet with a charge type of FB00. The rate table of the charge type contains the following scale calculation base values:

- SOURCELOC_CNTRY
- SOURCE_AIR_IATA
- DESTLOC_CNTRY
- DEST_AIR_IATA
- CHRG_WEIGHT

The system selects the minimum value checkbox for chargeable weight and creates the following rate table:

Source Country/Region (=)	IATA Source Location (=)	Destination Country/Region (=)	IATA Destination Location (=)	Minimum Value	<= 1 KG (R)	<= 45 KG (R)	<= 100 KG (R)	<= 300 KG (R)	<= 500 KG (R)

DE	BER	US	ATL	0	0	0	0	0	0
DE	BER	US	AUS	0	0	0	0	0	0
DE	BER	US	BOS	0	0	0	0	0	0
DE	BER	US	BWI	0	0	0	0	0	0

Agreement

Definition

An agreement is a long-term contract between business partners that details their commitment to conduct business with each other in an agreed manner.

An agreement can be legally binding. Available agreement types are, for example, freight agreement, forwarding agreement, charges templates, and internal agreement. Agreement types can be mutual between the business partners.

Use

Agreements are the basis for calculating transportation charges and can involve one or more parties. You use forwarding agreements to calculate transportation charges billable to your customer and freight agreements to calculate transportation charges billable to you by your carrier. You use internal agreements to calculate transportation charges between organizations of the same company code and between organizations of different company codes in the same company. You use charges template when a valid released agreement is missing between the purchasing organization and carrier.

An agreement includes contractual data, such as the following:

- Organizational unit, for example, the sales or purchasing organization
- Involved parties, for example, the sold-to party or carrier (except charges templates)
- Terms of payment
- Validity dates

The system determines the agreement based on organization, business partner, validity period, and preconditions. Agreement determination is based mainly on the partner relationship.

Structure

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

An agreement can contain the following main tabs and screen areas:

- **General Data**

The **General Data** tab contains general information about the agreement, for example, the organizational unit responsible for the forwarding order or freight order, the validity period of the agreement, and business partner details. Note that in agreements with multiple parties, you can specify business partners and business partner hierarchy trees. In agreements with a single party, you can specify either a business partner or business partner hierarchy tree.

You can also specify the calculation sheet template the system can use when creating a calculation sheet from within the agreement.

i Note

If you don't specify either the sold-to party/carrier or the business partner hierarchy tree for an agreement, the system considers the agreement to be a standard agreement. A standard agreement is valid only if an agreement isn't identified during agreement determination for the sold-to party specified in the forwarding order or for the carrier specified in the freight order.

You can define a freight agreement or forwarding agreement type to apply to multiple organizational units and multiple business partners in Customizing for **Transportation Management** under **► Master Data ► Agreements and Service Products ► Define FWA and Service Product Catalog Types ►** or under **► Master Data ► Agreements and Service Products ► Define Freight Agreement Types ►**. You can also specify multiple organizational units and business partners for mutual agreements. While defining the freight agreement types, if you don't specify a number range, the system allows you to manually enter the agreement ID when you create an agreement using that freight agreement type.

On this tab, you can add an agreement determination exclusion rule or a BRFplus condition to exclude the agreement from agreement determination. During agreement determination, if the system finds a condition that matches the condition in the agreement determination exclusion rule, the system excludes the agreement. For more information, see [Rules and Conditions for Charge Management and SP Catalogs](#).

• Items

The **Items** screen area contains the details of the agreement items. You must define at least one item for the agreement to be relevant for charge calculation. You can assign preconditions to agreement items and each agreement item can represent any combination of them, for example:

- Service level
- Stage category
- Transportation mode
- Shipping type

The system uses these preconditions and external strategies (see [External Strategies](#)) to determine the pick-up frequency and transit duration. Service preconditions define whether a service is valid for certain freight units.

i Note

You can assign service level, stage category, transportation mode, and shipping type only to agreements with **Editable** entered for the display control setting. You make this setting in Customizing for **Transportation Management** under **► Master Data ► Agreements and Service Products ►**.

This screen area can contain the following main tabs:

- **Precondition**
- **Details**
- **Dimensions** (freight agreements only)
- **Capacities**
- **Calculation Sheet Overview**
- **Commodity Codes**
- **Business Share** (freight agreements only)

- [Default Routes](#)
- [Instructions](#) (forwarding agreements only)
- [Notes](#)
- [Attachments](#)

You must attach a calculation sheet to each agreement item and you can also specify additional preconditions, payment terms, and commodity codes for each agreement item. For more information about the [Calculation Sheet Overview](#) screen area, see [Calculation Sheet](#).

i Note

You can choose an existing calculation sheet or create a calculation sheet. If you enter the name of an existing calculation sheet, the system attaches the calculation sheet to the agreement item. If the calculation sheet doesn't already exist, the system prompts you to create it. The calculation sheet that the system creates is based on the calculation sheet template you specified on the [General Data](#) tab. If you didn't specify a calculation sheet template, the system creates the calculation sheet with one sum line at the top level. You can then add additional lines to the calculation sheet. For more information, see [Calculation Sheet Maintenance](#).

• [Business Partner](#)

The [Business Partner](#) tab displays the business partners associated with the business document. For example, a bill-to party and sold-to party is specified in a forwarding agreement.

i Note

- You can display the business partner hierarchy in the business partner tab.

In a freight agreement or freight agreement RFQ master, it displays all the business partners associated with the carrier.

In a forwarding agreement, forwarding agreement quotation, or internal agreement, it displays all the business partners associated with the sold-to party.

- You can restrict the partner functions available in the business partner tab. For more information, see [Business Partner Determination](#).
- The information on this tab isn't considered when the system determines the agreement to use in a forwarding order or freight order.
- You can specify additional partner functions, such as notify party and communication party, for information purposes in addition to primary partner functions, such as sold-to party or carrier.

• [Output Management](#)

On the [Output Management](#) tab, you can print agreement details and generate emails and faxes and send them directly to the intended recipient. You can also see a record of the documents that were already printed or sent (by email or fax, for example). For more information, see [Output Management](#).

• [Versions](#)

The [Versions](#) tab contains a list of the agreement versions, including the status of each version.

• [Change Documents](#)

You can view all changes made to an agreement on the [Change Documents](#) tab or screen area. The changes are listed by date and time. You view the details of the changes at header level by selecting the relevant row in the [Transactions](#) screen

area and at field level by selecting the relevant row in the **Record Level Changes** screen area. You must refresh the agreement to view recently saved changes.

i Note

You can activate the **Change Documents** tab or screen area in Customizing for **Transportation Management** under **Master Data** > **Agreements and Service Products** > **Define FWA and Service Product Types** or under **Master Data** > **Agreements and Service Products** > **Define Freight Agreement Types**.

Integration

All agreement types are integrated with calculation sheets, local calculation sheets, rate tables, local rate tables, scales, and local scales. For more information, see [Calculation Sheet](#), [Rate Table](#), and [Scale](#).

More Information

[Agreement Management](#)

[Agreement Maintenance](#)

[Charge Calculation](#)

[Charges Templates](#)

[Setup of MD for Charge Management and SP Catalogs](#)

[Management of Instructions](#)

Agreement Maintenance

Use

You create and maintain freight agreements, forwarding agreements, internal agreements, and mutual agreements as a starting point for enabling charge calculation.

i Note

If an agreement is a mutual agreement, the agreement usage is **Customer and Service Provider**. The agreement usage of an agreement is visible in the relevant agreement personal object worklist.

Integration

Agreement maintenance is a step in the process to set up master data in the Charge Management and Service Product Catalogs component and is integrated with calculation sheets, local calculation sheets, rate tables, local rate tables, scales, and local scales. You can create all charge management master data from within an agreement. For more information, see [Setup of MD for Charge Management and SP Catalogs](#).

You can use the master data cockpit to access an overview of the agreements and the charge management master data per organizational unit, business partner, and charge type. The system also shows the relationship between the different types of charge management master data. For more information, see [Master Data Cockpit](#).

Prerequisites

- You have defined number ranges in Customizing for **Transportation Management** under ► **Basic Functions** ► **Charge Calculation** ► **Basic Settings** ► **Number Range Intervals** ► **Assign No. Ranges to Calc. Sheets, Rate Tables, & Scales** ►.
- For freight agreements, forwarding agreements, and internal agreements, you have defined agreement item types, agreement types, and a default agreement type in Customizing for **Transportation Management** under ► **Master Data** ► **Agreements and Service Products** ►.
- If you enable approval workflow for the release of agreements, you must create conditions and configure workflow settings (see [Workflow in Transportation Management](#)).

Features

Agreement Versions

You can generate different versions of agreements, with the same validity period or with mutually exclusive validity periods. Agreement versions ensure that you do not need to create a new agreement when an agreement's validity period expires or the details of an agreement change depending on the time period. Note that you can have only one released agreement version with the same validity period or an overlapping validity period but can have multiple released agreement versions with mutually exclusive validity periods. You can revert to an earlier agreement version by deactivating the released version and releasing the earlier version. You can also delete agreement versions. The **Versions** tab page contains a list of the agreement versions, including the status of each version.

Business Partners

The system displays the business partners associated with the business document. For example, a bill-to party and sold-to party.

i Note

- You can display the business partner hierarchy in the business partner tab page.
For example, in a forwarding agreement, it displays all the business partners associated with the sold-to party.
- You can restrict the partner functions available in the business partner tab page. For more information, see [Business Partner Determination](#).
- You can specify additional partner functions, such as notify party and communication party, for information purposes in addition to primary partner functions, such as sold-to party.

Approval Workflow

You can enable an approval workflow for the release of agreements. You can specify that you want the system to notify an approver when you request approval for the release of an agreement. You can also specify whether you can edit agreements during the approval workflow. In agreements with multiple parties, only one approver must approve the release of the agreement.

Note that the approver is the person assigned to the position that is the head of the organizational unit. Therefore, at least one organizational unit must have a position that is the head of the organizational unit. For more information, see [Approval Check](#).

The approver uses Business Workplace to approve or reject the release of agreements. For more information, see SAP Library for **SAP Business Workflow** on SAP Help Portal at <http://help.sap.com/netweaver> ►. In SAP Library, choose ► **SAP NetWeaver 7.5** ► **Application Help** ► **SAP NetWeaver Library: Function-Oriented View** ► **Application Server** ► **Application Server ABAP** ► **Other Services** ► **Services for Application Developers** ► **SAP Business Workflow** ► **SAP Business Workflow: Reference Documentation** ►.

i Note

You enable the approval workflow and allow the agreement to be edited during the approval workflow when you define agreement types. You define freight agreement, forwarding agreement, and internal agreement types in Customizing for [Transportation Management](#) under [Master Data](#) > [Agreements and Service Products](#).

Offline Approval Workflow

Transportation Management (TM) can send work items for agreements by e-mail to a user, such as a customer agent. The user can accept or reject a work item and update the system by e-mail. For more information, see [Offline Workflow for Sending and Approving Work Items](#).

Calculation Sheets, Rate Tables, and Scales

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You can create calculation sheets, rate tables, and scales from within an agreement in the following ways:

- Create a calculation sheet for each agreement item in the [Items](#) screen area
- Create a rate table for each calculation sheet item on the [Calculation Sheet](#) tab page of the [Items](#) screen area
- Create a scale for a rate by going to the rate table details in the [Items](#) screen area

You can also copy and delete the calculation sheets, rate tables, and scales from within an agreement. For more information, see [Calculation Sheet](#), [Rate Table](#), and [Scale](#).

i Note

If you copy an agreement, the system copies the calculation sheets, rate tables, and scales that you created from within the agreement or that you referenced in the agreement.

If you delete an agreement, the system deletes the calculation sheets, rate tables, and scales that you created from within the agreement. However, the system does **not** delete the calculation sheets, rate tables, and scales that you referenced in the agreement.

You can edit local charge management master data objects independently of the agreement they were created within. To enable this, you select the [Edit Loc. MD Objects](#) (Edit Local Charge Management Master Data Objects) checkbox in Customizing for [Transportation Management](#) under [Basic Functions](#) > [General Settings](#) > [Define General Settings for TM](#).

Charge-Calculation-Specific Features

- Chargeable weight
- Agreement determination
- Uncontrolled transportation
- Header-level charges
- External reference number
- Resource-based internal agreements

For more information, see [Agreements and Charge Calculation](#).

Preconditions

Depending on your Customizing settings and the agreement type (freight agreement, forwarding agreement, or internal agreement), you can add preconditions at agreement and agreement-item level, for example:

- Transportation mode
- Shipping type
- Traffic direction
- Stage category
- Service level
- Contract basis
- Consolidation type

In the **Item Precondition** tab page, you can add second and subsequent preconditions to an agreement item. You can also add second and subsequent values to the same precondition. This enables you to use the same agreement item and calculation sheet or local calculation sheet for more than one precondition, and more than one precondition value. For example, you can specify a precondition of **Transportation Mode** in an agreement item that already contains a precondition of **Service Level**. You can also specify precondition values of **Sea** and **Road** for the same agreement item.

If you use the search on a precondition field to search for a precondition and you have entered an item type for the agreement item, the system displays all preconditions that are available in the system. However you can only edit the preconditions, including any default values, that you specified for the item type in the Customizing activity **Define Freight Agreement Item Types**. If you have not entered an item type for the agreement item, the system displays all the preconditions that are available in the system.

For more information, see Customizing for **Transportation Management** under **►►Master Data ► Agreements and Service Products ► Define Freight Agreement Item Types ►**.

i Note

The availability of some preconditions is dependent on the transportation mode of the agreement or agreement item. For example, you can enter the contract basis and air waybill type only if the transportation mode is **Air**. The transportation mode in an agreement also restricts the calculation method types available in calculation sheets created from the agreement.

The system also uses preconditions during agreement determination. You can use preconditions to search for agreements in the relevant agreement personal object worklist on the launchpad.

You can add or remove these preconditions for freight agreements, forwarding agreements, and internal agreements in Customizing for **Transportation Management** under **►►Master Data ► Agreements and Service Products ►**.

Commodity Codes

You can define commodity codes for agreement items in forwarding, freight, and internal agreements. Commodity codes enable you to define the categories of products that the parties in the agreements have agreed to transport between them.

You can also use a wildcard entry to represent a group of commodity codes in the agreement item. The system calculates the charges when the commodity code in the forwarding order fully or partially matches the commodity code or wildcard entry in the agreement item. On the **Charges** tab page, you can see the commodity code type and commodity item number used for charge calculation. Note that you can use commodity code-based agreement determination only for item-level calculation.

You can define a rate table or a local rate table with the rates for commodity codes. You must specify the commodity code type for the rate table or the local rate table and you can use the commodity codes belonging to the commodity code type to define rates.

For example, your customer needs a product with a set of rates to transport livestock and another product and set of rates to transport food. To differentiate between the commodities, you use a commodity type of **LF**, with commodity codes of 01*to 05* for livestock and 16*to 24* for food. When you create a forwarding agreement, you select the appropriate product, and the system calculates charges based on the product you transport.

You must have made settings in Customizing for **Transportation Management** under **Master Data > Classification of Goods**. For more information, see [Classification of Goods](#).

Default Routes

You can specify default routes for agreement items. This enables you to specify the route you take between a location pair in delivering an order. You can assign existing default routes, create new default routes, and edit default route details in the **General Data > Default Routes** tab page under the agreement item.

When you create a forwarding order from a forwarding agreement item with a default route, the system copies the default route stages to the forwarding order. The system uses the default route you specify in the agreement as the ordered route in the order. The system uses the default route stages in charge calculation if you specify the data source as **Ordered Route** in the calculation profile. You can also specify or change the actual route later in the **Stages** tab page of the order or in planning (see [Determination of the Route](#)).

You can create and edit default routes under **Master Data > Transportation Network > Default Route**, and assign these default routes to an agreement item. You must assign a default route type to a default route. For more information, see Customizing for **Transportation Management** under **Master Data > Transportation Network > Define Default Route Types**.

Duplication Check

The system checks and displays other business documents with matching attributes when you save or check the business document.

i Note

You can activate the feature for the following documents:

- Freight agreements in Customizing for **Transportation Management** under **Master Data > Agreements and Service Products > Define Freight Agreement Types**
- Forwarding agreements in Customizing for **Transportation Management** under **Master Data > Agreements and Service Products > Define FWA and Service Product Catalog Types**
- Internal agreements in Customizing for **Transportation Management** under **Master Data > Agreements and Service Products > Define Internal Agreement Types**
- Freight agreement RFQs in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Freight Agreement RFQ Types**
- Forwarding agreement quotations in Customizing for **Transportation Management** under **Master Data > Agreement RFQs and Quotations > Define Forwarding Agreement Quotation Types**

Capacities

The system displays the total capacity required from carriers, the total capacity carriers can commit to, and the total capacity awarded to carriers.

i Note

You can activate the **Capacities** tab page or screen area in Customizing for **Transportation Management** under ► **Master Data** ► **Agreements and Service Products** ► **Define Freight Agreement Types** ► for freight agreements.

You can activate the **Capacities** tab page or screen area in Customizing for **Transportation Management** under ► **Master Data** ► **Agreements and Service Products** ► **Define FWA and Service Product Catalog Types** ► for forwarding agreements.

Release of Agreements with Rate Tables

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

In freight agreements, forwarding agreements, and internal agreements, you can simultaneously release the agreement and the corresponding rate tables.

In the forwarding agreement management worklist and the freight agreement management worklist, you can select and release multiple agreements with the corresponding rate table.

i Note

If you have enabled an approval workflow for the agreement and the corresponding rate tables, the system processes the agreement release workflow and then the rate table release workflow before releasing the agreement with the corresponding rate table.

Freight-Agreement-Specific Features

- Specify service levels
- Create freight booking, freight orders, and service orders from items
- Create freight agreement RFQ masters
- Integrate business share
- Create freight agreements for general sales agents
- Create transportation allocations
- Specify dimensions and units of measure for items
- Use a spreadsheet file to update details when you create or edit a freight agreement
- Restrict the editing of released freight agreements only to super users

For more information, see [Freight Agreement Maintenance](#).

Forwarding-Agreement-Specific Features

- Specify service levels and create forwarding orders and forwarding quotations based on service levels
- Specify instructions for the processing of the forwarding agreement
- Specify to settle for transportation services on the basis of execution criteria, such as buyer's consolidation, route, or trailer unit
- Consolidate goods from different suppliers into one container

For more information, see [Forwarding Agreement Maintenance](#).

Consumed Capacity and Amount

At agreement-item level, you can calculate how much of the capacity specified in an agreement is consumed by orders and by invoices, and the charge amount this corresponds to. This helps you to check how much of the possible capacity you plan to use during a specified period, and how much is actually used. The system calculates what you plan to consume by checking freight orders or forwarding orders. The system calculates what is actually consumed by checking invoices. You can see the consumed capacity and amount for an agreement item on the **Capacities** tab page in the **Items** screen area of the agreement. The system calculates the actual amount based on the calculated rate for the actual capacity from the freight orders or forwarding orders. Note that you can recalculate the consumption if you want to ensure you have the most up-to-date consumption data visible in the agreement.

You can calculate the consumed capacity and amount for each agreement item on the launchpad by opening the agreement. In the **Items** screen area, select the item and choose **Follow-Up > Calculate Consumption**.

You can calculate the consumed capacity and amount for multiple or all agreement items in one or more agreements by using program **Calculate Consumed Capacity and Amount at Agreement-Item Level** (/SCMTMS/FAG_CONSUMPTION) in transaction SE38.

Validity Period Update

If you change the validity of an agreement, you can update the validity of all the items and service products with the same validity period. You can also update the validity of selected items and service products instead.

General

- Agreement search

In the relevant agreement POWL, you can search for agreements within a zone. You do not need to specify an exact source or destination location. If you search a zone and specify to include the location within the zone, the system also searches agreements whose location is within the zone. You can also search for agreements involving a specific carrier by using the carrier's standard carrier alpha code (SCAC).

- Printing, e-mailing, and faxing

You can print agreement details and generate e-mails and faxes and send them directly to the intended recipient. For more information, see Customizing for **Cross-Application Components** under **Processes and Tools for Enterprise Applications > Reusable Objects and Functions for BOPF Environment > PPF Adapter for Output Management**, as well as [Printing](#).

- Creation of agreements from Customizing templates

You can create Customizing templates for freight, forwarding, and internal agreements and in turn use the Customizing templates to create agreements. You can also create a mutual agreement from a Customizing forwarding or freight agreement template.

You can reuse these Customizing templates across multiple TM systems so that you don't need to recreate a Customizing template in each different system. This decreases duplication and work effort. In a Customizing template, you can predefine various attributes of an agreement such as, organization, business partner, preconditions, and commodity code.

You define a Customizing forwarding agreement template in Customizing for Transportation Management under **Basic Functions > Charge Calculation > Basic Settings > Templates > Define Forwarding Agreement Templates**.

You define a Customizing freight agreement template in Customizing for Transportation Management under **Basic Functions > Charge Calculation > Basic Settings > Templates > Define Freight Agreement Templates**.

You define a Customizing internal agreement template in Customizing for Transportation Management under **Basic Functions > Charge Calculation > Basic Settings > Templates > Define Internal Agreement Templates**.

- Creation of agreements from master data templates

You can create master data templates for freight, forwarding, and internal agreements on the launchpad and in turn use the master data templates to create agreements. You can also create a mutual agreement from a master data forwarding or freight agreement template. However, you cannot reuse these master data templates across multiple TM systems.

In a Customizing template, you can predefine the different attributes of an agreement such as, organization, business partner, preconditions, and commodity code.

Note that you can also create a master data template from an existing Customizing template.

You define a master data forwarding agreement template in the [Create Forwarding Agreement Template](#) app on the launchpad.

You define a master data freight agreement template in the [Create Freight Agreement Template](#) app on the launchpad.

You define a master data internal agreement template in the [Create Internal Agreement Template](#) app on the launchpad.

Activities

Create or update freight agreements in the [Freight Agreements](#) app on the launchpad.

Create or update forwarding agreements in the [Forwarding Agreements](#) app on the launchpad.

Create or update internal agreements, in the [Internal Agreements](#) app on the launchpad.

Create a forwarding agreement from a Customizing or a master data template in the [Create Forwarding Agreement](#) app on the launchpad. Enter the forwarding agreement template ID in the [Forwarding Agreement Template](#) field and choose [Continue](#).

Create a freight agreement from a Customizing or a master data template in the [Create Freight Agreement](#) app on the launchpad. Enter the freight agreement template ID in the [Freight Agreement Template](#) field and choose [Continue](#).

Create an internal agreement from a Customizing or a master data template in the [Create Internal Agreement](#) app on the launchpad. Enter the internal agreement template ID in the [Internal Agreement Template](#) field and choose [Continue](#).

Create a mutual agreement from a forwarding agreement Customizing or a master data template in the [Create Forwarding Agreement](#) app on the launchpad. Enter the forwarding agreement template ID in the [Forwarding Agreement Template](#) field, select the [Mutual Agreements](#) checkbox, and choose [Continue](#).

More Information

[Agreement](#)

[Agreement Management](#)

[Archiving Agreements and RFQ Masters](#)

Agreements and Charge Calculation

Use

Transportation Management (TM) calculates transportation charges on the basis of the data stored in agreements and charge management master data. You can make specific charge calculation settings as part of the creation and maintenance of agreements.

Prerequisites

You have configured the prerequisites for agreements (see [Agreement Maintenance](#)).

Features

Chargeable Weight

You can assign a dimensional weight profile at agreement level and at agreement-item level to calculate the chargeable weight of a product. By assigning a dimensional weight profile at agreement-item level, you can have a different dimensional weight profile for each stage of a freight order or forwarding order. During charge calculation, you can change the dimensional weight profile at the charge-item level in a forwarding order.

i Note

A dimensional weight profile at agreement-item level has priority over a dimensional weight profile at agreement level. If you do not assign a dimensional weight profile at agreement-item level, the system uses the dimensional weight profile assigned at agreement level.

You can also assign a dimensional weight profile for manually-changed charge lines.

You can also assign a dimensional weight profile to the calculation profile of the organizational unit. If you do not assign a dimensional weight profile at the agreement-item and agreement header level, the system uses the dimensional weight profile assigned to the calculation profile of the organizational unit. You can assign dimensional weight profiles to calculation profiles in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Define Calculation Profiles**.

You can define dimensional weight profiles in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Data Source Binding** > **Define Dimensional Weight Profiles**.

Agreement Determination

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

The system determines the agreement based on business partner and organizational unit (sales organization for forwarding agreements and purchasing organization for freight agreements). If there are multiple agreements for a carrier or customer when the system tries to calculate transportation charges, the system determines which agreement to use within the organizational hierarchy. As part of the agreement determination, if you select the **Evaluate All Agreements** checkbox in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Define Charges Profiles**, the system takes the leading charge type into account by skipping agreements that use calculation sheets without a leading charge type. The system then skips agreements if the leading charge type does not have a rate. If there are multiple leading charge types in a calculation sheet, all the leading charge types must have a rate defined for the system to consider the agreement and calculation sheet for charge calculation. We recommend that you maintain a single leading charge type in a calculation sheet.

The system also uses preconditions during agreement determination. For more information, see [Agreement and Calculation Sheet Determination](#).

Uncontrolled Transportation

You can specify whether an agreement is for controlled or uncontrolled transportation, which impacts on whether the system can calculate the charges for the main stage of an uncontrolled transportation. For more information, see [Uncontrolled Transportation](#).

Header-Level Charges

The system can calculate charges for an agreement item at header level even when calculating the rest of the charges at stage level. You can group the charges common to all stages in a separate agreement item. During charge calculation, the system calculates the header-level charges independently of the stage charges. The system calculates the stage charges as usual.

Note that when you create an internal settlement document, you can choose to settle header-level charges along with one of stage-level charges or you can also settle header-level charges independently. You must run the program /SCMTMS/CFIR_CREATE_BATCH to manage this setting. When you select the **Header Charges in Separate Settlement Document** checkbox, the system settles the head-level charges independently. Also, the system displays the **Ready for Invoicing** status at the header charges along with the charge item status.

To enable header-level charges when calculating charges at stage level, you must maintain a calculation profile with the calculation level as stage level. For more information, see Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Calculation Profiles**. For information about calculation profiles, see [Calculation Profiles](#).

External Reference Number

You can specify the reference number of a separate agreement in an external system or the reference number your business partner uses in their internal system for your same agreement in TM.

Resource-Based Internal Agreements

You can specify internal agreements as resource-based agreements to use when charging only for the use of resources. For more information, see [Internal Settlement for Resources](#).

❖ Example

A shipper wants carrier A to transport goods on its behalf. Carrier A outsources the transportation of the goods to carrier B. Carrier B uses carrier A's vehicles to transport the goods. Carrier A uses a resource-based internal agreement to charge carrier B for the use of its vehicles.

More Information

[Charge Calculation](#)

[Agreement](#)

Freight Agreement Maintenance

Use

As part of the creation and maintenance of freight agreements, you can use the following features specific to freight agreements only.

Note that you can use many other features that are not specific to freight agreements. For more information, see [Agreement Maintenance](#).

Prerequisites

You have configured the prerequisites for agreements (see [Agreement Maintenance](#)).

Features

Service Levels

The service levels you can choose for freight agreement items are service levels that are defined for the carrier. For more information, see [Definition of Business Partners](#).

Capacity Dashboard

For an item in a freight agreement, you choose **Follow Up > Calculate Consumption** and open the dashboard. You can view the quantity consumed and the amount calculated in order documents and settlement documents for the validity period of the agreement item. You can subsequently compare this data to the quantity decided in the agreement. This enables you to analyze the usage of the agreement.

You can view the quantity consumed and the amount calculated in order documents and settlement documents for specific periods of time. You can also view these individual orders and the settlement documents.

Freight Booking, Freight Orders, and Service Orders

When you create a freight document or service order from an agreement item, the system copies the services you have specified in the item to the freight document or service order. For more information, see [Definition of Carrier Service Codes](#).

Note that when you create a freight order from a multi-party freight agreement the carrier information is not copied to the freight order.

Freight Agreement RFQ Masters

You can create freight agreement RFQ masters from within freight agreements to start the process of updating the freight agreements. To create a freight agreement RFQ master from within a freight agreement, in the **Freight Agreements** worklist app on the launchpad. Open the freight agreement and choose the **Create RFQ Master** button.

You can also create freight agreements from within freight agreement RFQ masters. You can do this, in the **Create Freight Agreement RFQ Master** app on the launchpad. Open the freight agreement RFQ master and choose the **Follow Up > Create/Update Agreement** button.

For more information about freight agreement RFQ masters, see [Freight Agreement RFQ Master](#).

Business Share Integration

If you include a target share when you create or update a freight agreement from a freight agreement RFQ master, the system includes the target share in the freight agreement items. If you create a freight agreement without reference to a freight agreement RFQ master, you can manually add a target share to the freight agreement items.

If you want the system to take the target share into account in planning, you must choose the **Create Business Share** button. The system includes the target share in the business share, and uses the business share in the carrier selection settings in planning. You can specify mandatory settings by choosing the **Edit Business Share** button on the **Business Shares** tab page.

To create a business share for planning, you must have specified the following settings in the freight agreement:

- Valid-from date and valid-to date
- Carrier
- Trade lane on the **Precondition** tab page
- Periodicity on the **Capacities** tab page

When you create and edit a business share, the system displays a temporary identifier (ID) in the **Business Shares** tab page. It assigns a permanent ID when you save the freight agreement.

For more information about how you can use a business share in planning, see [Business Share](#).

General Sales Agents

You can create a freight agreement for a general sales agent (GSA), and specify the airlines to which the freight agreement applies. To enable you to create this freight agreement, you must specify the following settings in **BP – Maintain Business Partner** **Display in BP Role** **Vendor Data**:

- **Carrier Category of Agent**
- Airlines the GSA represents in the **Airline Codes** screen area

You can open the **BP – Maintain Business Partner** transaction in SAP Menu under **Logistics** **Transportation Management** **Master Data**.

In the freight agreement, when you specify the GSA carrier, you cannot specify an airline at header level. Instead you can only specify the airlines you specified in the BP master, at item level.

Transportation Allocations

In freight agreements with a single party, you can create and edit transportation allocations for freight agreement items to allocate a certain volume of business to a carrier on a specific trade lane. Note that the freight agreement item must have just one trade lane. For more information, see [Check Against Transportation Allocations](#).

Dimensions and Units of Measure

You can enter the dimensions and units of measure for a freight agreement item. You can specify the attributes, such as length, width, height, and girth, on the **Dimensions** tab page of the freight agreement. When you perform direct shipment option determination, the system compares the attributes of the parcel specified in the freight unit with that specified in the freight agreement. For the system to consider a freight agreement for charge calculation, all the attribute values specified in the freight unit must be less than or equal to the attribute values specified in the freight agreement. If there are multiple freight agreements that meet this condition, the system displays a list from which you can choose one. For more information, see [Rate Determination for Direct Shipment of Parcels](#).

❖ Example

You define dimensions for the following freight agreement items:

- Item 100 with a height of 200 cm and width of 100 cm
- Item 200 with a height of 250 cm

When you perform direct shipment option determination for a freight unit with a height of 210 cm, the system can use freight agreement item 200 to determine the charges.

When you perform direct shipment option determination for a freight unit with height of 200 cm and width of 90 cm, the system can use both freight agreement item 100 and freight agreement item 200 to determine the charges. In this case, the system displays both from which you can choose one.

Excel Integration

You can use a spreadsheet file to update details when you create a new freight agreement or when you edit an existing freight agreement. You can perform the following actions:

- Download the freight agreement as a spreadsheet file.
- Update the file that you downloaded.
- Upload the file on the [Excel Integration](#) tab page.

The system uses the information in the uploaded file to automatically enter information in the relevant fields in the freight agreement.

i Note

You specify the fields that can be edited in the downloaded spreadsheet file in Customizing for [Transportation Management](#) under [Master Data](#) > [Agreements and Service Products](#) > [Define Editable Fields in Excel](#).

Rate Change Requests

Your carriers can request rate changes for released freight agreements. Your carriers can therefore request changes to the rates previously agreed between you, without the need for a new agreement. For more information, see [Request of Rate Changes by Carriers](#).

Editing of Released Freight Agreements

You can restrict the editing of released freight agreements only to super users if you set up authorizations using authorization object T_AGR_FRS. However, you can also allow all users to edit released freight agreements in Customizing for [Transportation Management](#) under [Basic Functions](#) > [General Settings](#) > [Define General Settings for TM](#).

i Note

If you upgraded from a lower release of SAP S/4HANA to SAP S/4HANA 1909 or higher, you're using authorization object T_AGR_FR instead of T_AGR_FRS. Then as a default, all users with edit authorization can edit released freight agreements.

More Information

[Agreement](#)

Request of Rate Changes by Carriers

Your carriers can request changes to the rates previously agreed between you, without the need for a new agreement.

Use

Your carriers can request rate changes for released freight agreements with a current or future validity period. If you, the shipper, accept the request from the carrier, the carrier then submits the new rates. If you accept the new rates, then the freight agreement (rates) is adjusted according to the new rates and the new rates are used when calculating the transportation charges.

i Note

Your carriers can request rate changes only for rate tables and calculation sheets created within the freight agreement, that is, local rate tables and calculation sheets.

Prerequisites

- The freight agreement is released.
- The rates your carrier wants to change are in rate tables or local rate tables and not directly in calculation sheets or local calculation sheets.
- You've selected the **Negotiable** checkbox for the charge type of the calculation sheet line items.

Process

1. Typically, the carrier contacts you, the shipper, to request a change of rates for the charge lines in a released freight agreement.
2. In the freight agreement, you can specify from when the rate change is valid and also a deadline for you to decide about the rate change.

Note that the start date of the rate validity of the calculation sheet line items should be the same as the start date of the rate change validity.

3. You accept or reject the rate change request at the calculation sheet line item level.

If you reject the rate change request, the freight agreement then has the status **Request Rejected** and the process ends.

If you accept the rate change request, the freight agreement then has the status **Request Accepted**.

Note that if you accept or reject the rate change request at header level, this action overwrites any changes accepted or rejected at rate level.

4. You download the `Update_Rate_<Date_of_Shipper_Deadline>.xlsx` file available on the **Attachments** tab page of the freight agreement and rename it to `Updated_Rate_<Date_of_Shipper_Deadline>.xlsx` before sending it to the carrier.
5. The carrier enters the rate changes into the file and sends the file back to you.

6. Once you receive the file from the carrier, you upload it to the **Attachments** tab page of the freight agreement and then upload the rate changes to the system by choosing the **Rate Change Request** > **Submit Rates** button.

The freight agreement now has the status **Rates Uploaded**.

7. You submit the rate changes to the system using program **Update Responses in SAP TM** (/SCMTMS/UPLOAD_RFQ_RESPONSE).

Once the system is updated, the freight agreement has the status **Rates Submitted** or **Rates Submitted with Errors**. The program creates a new validity period in the rate tables (both kinds, including local) of the calculation sheet line items with the rate changes proposed by the carrier.

8. You review each successfully submitted rate change and accept or reject the changes at rate table validity-period level.

If you accept the rate changes, the system truncates the existing validity period and releases the new validity period. The calculation sheet line items have the status **Rate Change Accepted**.

Note that if you accept or reject the changes at header level, this decision overwrites any changes accepted or rejected at rate level.

Result

The new rates come into effect from the new valid-from date. The previous rates remain relevant up until the new valid-from date.

More Information

Forwarding Agreement Maintenance

Use

As part of the creation and maintenance of forwarding agreements, you can use the following features specific to forwarding agreements only.

Note that you can use many other features that are not specific to forwarding agreements. For more information, see [Agreement Maintenance](#).

Prerequisites

You have configured the prerequisites for agreements (see [Agreement Maintenance](#)).

Features

Service Levels

The service levels you can choose for forwarding agreement items are the service levels that are defined for the forwarding agreement item's transportation mode, or service levels that are defined independently of the transportation mode. You define service levels in Customizing for [Transportation Management](#) under [▶ Forwarding Order Management ▶ Define Transportation Service Level Codes ▶](#).

Capacity Dashboard

For an item in a forwarding agreement, you choose [▶ Follow Up ▶ Calculate Consumption ▶](#) and open the dashboard. You can view the quantity consumed and the amount calculated in order documents and settlement documents for the validity period of the agreement item. You can subsequently compare this data to the quantity decided in the agreement. This enables you to analyze the usage of the agreement.

You can view the quantity consumed and the amount calculated in order documents and settlement documents for specific periods of time. You can also view these individual orders and the settlement documents.

Instructions

If you have a forwarding agreement with service products that have services, the system automatically adds the instructions specified for the service type to the [Instructions](#) tab page of the forwarding agreement. You can edit or delete these instructions, and also add additional instructions. Note that if you create a forwarding order from a forwarding agreement with instructions, the system includes the instructions in the forwarding order and subsequent business documents.

You define service types in Customizing for [Transportation Management](#) under [▶ Basic Functions ▶ General Settings ▶ Define Service Types ▶](#). You define instructions and instruction sets in Customizing for [Transportation Management](#) under [▶ Basic Functions ▶ Instructions ▶ Define Instructions and Instruction Sets ▶](#). You assign instruction sets to service types in Customizing for [Transportation Management](#) under [▶ Basic Functions ▶ Instructions ▶ Assign Instruction Sets ▶](#). For more information, see [Management of Instructions](#).

Forwarding Orders and Forwarding Quotations

You can create forwarding orders and forwarding quotations based on service products.

Forwarding Agreement Quotations

You can create forwarding agreement quotations from within forwarding agreements to start the process of updating the forwarding agreements. You can create a forwarding agreement quotation from within a forwarding agreement in the [Create Forwarding Agreement](#) or [Edit Forwarding Agreement](#) app on the fiori launchpad. Open the forwarding agreement and choose the [Follow Up > Create Forwarding Agreement Quotation >](#) button.

You can also create forwarding agreements from within forwarding agreement quotations. You can do this in the [Create Forwarding Agreement Quotation](#) or [Edit Forwarding Agreement Quotation](#) app on the fiori launchpad. Open the forwarding agreement quotation and choose the [Create Agreement](#) button.

For more information about forwarding agreement quotations, see [Forwarding Agreement Quotation](#).

Buyer's Consolidation

For forwarding agreement items, you can consolidate goods from different suppliers that are intended for one ship-to party. For more information, see [Buyer's Consolidation](#).

Settlement Basis

At the agreement item level in forwarding agreements, you can specify that you bill your business partners based on certain execution criteria, such as by the trailers used to fulfill the order, or the route taken to execute the order. During settlement, the system creates a single forwarding settlement document for all the forwarding orders that meet the specified execution criterion. By default, the settlement basis is at forwarding order level. For more information, see [Execution-Based Settlement](#).

More Information

[Agreement](#)

Settlement

Use

You can use settlements to meet the following business requirements:

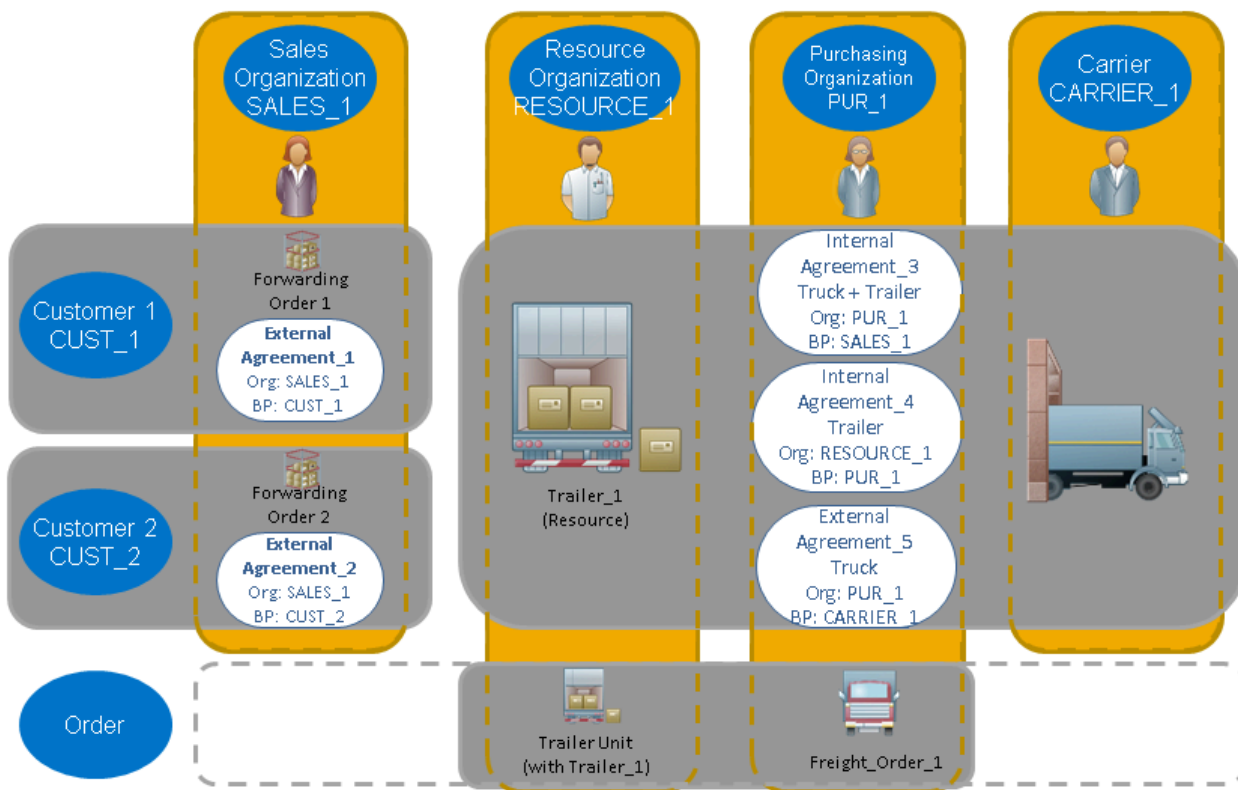
- Create a freight settlement document in Transportation Management (TM) for the transportation services provided to you by a carrier. TM transfers the freight settlement document to Materials Management (MM). You can verify the invoices you receive from your carrier against the information contained in the freight settlement document (see [Freight Settlement](#)).
- Create a forwarding settlement document in TM to bill a customer for the transportation services you provide to the customer. TM transfers the forwarding settlement document to Sales and Distribution (SD) and SD sends the invoice to the customer (see [Forwarding Settlement](#)).
- Create an internal settlement document for a transportation service provided to you by an internal organization in your company. The internal organization bills you to recover the cost of the transportation services it provided to you when you fulfilled a forwarding order to a customer (see [Internal Settlement Management](#)).
- Create an internal settlement document for the resources you use in providing a transportation service in a freight order to another internal organization in your company (see [Internal Settlement for Resources](#)).

The following functions that are described below are currently not released:

- Group logistics scenario in system internal integration

- Internal settlement for resources

The figure below illustrates a typical customer order process for you as a provider of logistics services. The figure is followed by an explanation:



To execute your forwarding orders you have the following organizations in place:

- Sales (SALES_1)

A sales organization that is customer facing.

- Resource (RESOURCE_1)

An internal sales organization that owns the resource.

You can create, change, or display resources as follows:

- On the **SAP Easy Access** screen, choose **Transportation Management** > **Master Data Resource** > **Resource**.
- In the SAP launchpad, open the **Define Resource** app

- Purchasing (PUR_1)

A purchase organization that is responsible for the procurement of the transportation service.

You have a carrier (CARRIER_1), who provides you with transportation services.

You have the following customers:

- Customer 1 (CUST_1)
- Customer 2 (CUST_2)

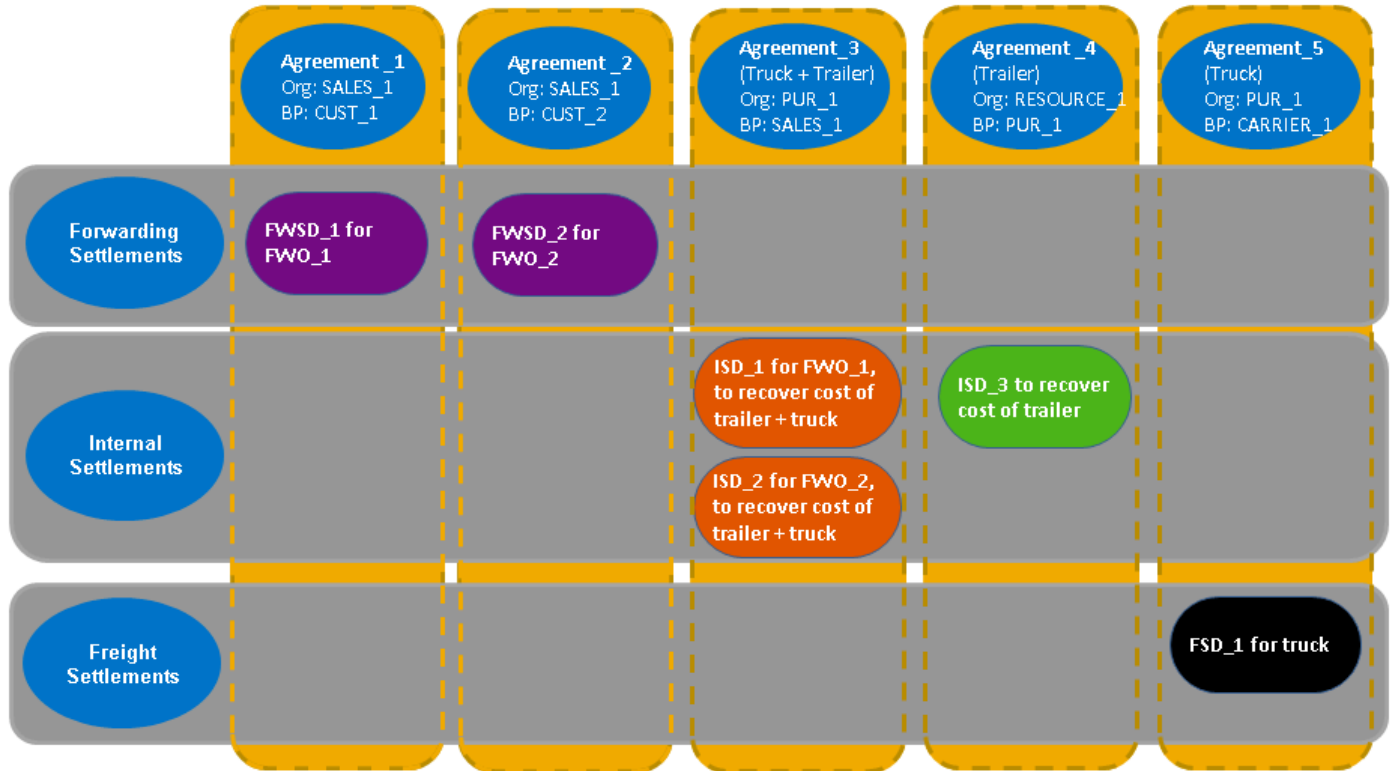
You create a forwarding order for each of customers 1 and 2, to transport goods on their behalf from a source location to a destination location. To move the goods, you need a trailer. Your internal resource organization owns a trailer, and agrees to provide you with a trailer. You need a truck to pull the trailer. Your purchasing organization buys the services of a truck from your carrier.

You create a freight order (Freight_Order_1) to fulfill the demand, and you subcontract the freight order to a carrier.

In the freight order, you provision a trailer unit that contains the trailer resource (Trailer_1). An internal organization in your company owns the trailer. The internal organization recovers the cost of providing the trailer from you.

The carrier uses his or her own truck and your trailer (Trailer_1) to move the goods from the source location to the destination location.

The figure below illustrates the settlement process for the above customer order process and is followed by an explanation:



To settle your forwarding orders, you have the following agreements in place:

- A forwarding agreement between your sales organization SALES_1 and your first customer, CUST_1 so that your sales organization gets paid for providing the transportation service. Also a forwarding agreement between your sales organization SALES_1 and your second customer, CUST_2, again so that your sales organization gets paid for providing the transportation service.
- An internal agreement between your purchasing organization (PUR_1) and your sales organization (SALES_1), so that your purchasing organization recovers the following costs:
 - Payment to the carrier for providing the truck
 - Payment to the resource organization (RESOURCE_1) for providing the trailer
- An internal agreement between your resource organization (RESOURCE_1) and your purchasing organization (PUR_1), so that your resource organization gets paid for providing the trailer.
- An external agreement between your purchasing organization (PUR_1) and your carrier, so that your carrier (CARRIER_1) provides the truck and hauls the trailer and gets paid for supplying the transportation service.

You have the following settlement documents:

- Your sales organization creates a forwarding settlement document for each of the forwarding orders it has in place with customer 1 and customer 2. This combined amount is what you get paid for the transportation service you provide to your

customers. You post the forwarding settlement documents to SD. SD sends the invoices to your customers and receives payment from your customers.

- Your purchasing organization creates an internal settlement against your sales organization for each of the forwarding orders for which it provided a truck and trailer. This enables your purchasing organization to recover the cost of providing the transportation service.

If the purchase organization belongs to a different company code in your company, you have an intercompany settlement. You post the internal settlement document to SD.

If the purchase organization belongs to the same company code as that of your sales organization, you have an intracompany posting. You post the internal settlement document directly into financial accounting.

- Your resource organization creates an internal settlement document against your purchasing organization, to recover the cost of providing the trailer. You post the internal settlement document to SD to post to financials if the purchase organization belongs to a different company code in your company. If the purchase organization belongs to the same company code as that of your sales organization, the internal settlement document posts directly into financial accounting to do an intra-company posting.
- Your purchasing organization creates a freight settlement document against your carrier. This simulates the cost of providing the truck in TM. You post the freight settlement document to MM. Later, when your carrier sends you the invoice for providing the truck and hauling the trailer, you verify the amount in the invoice against the amount you transferred in the freight settlement document to MM. If the amounts match, you pay the carrier.

Settlement for Shippers

For shippers, your carrier or logistics service provider (LSP) provides you with a freight transportation service. You follow the freight settlement process to settle the freight costs with your carrier or LSP. You can also use cost distribution to distribute the freight costs to the level of each individual order/delivery item in your freight order to allocate the freight cost to financial accounting and controlling. The system performs the cost distribution when it creates the associated freight settlement document. You post the freight settlement document with the distributed costs to MM and settlement management.

When your carrier sends you the invoice for providing the truck and hauling the trailer, you verify the amount in the invoice against the amount you transferred in the freight settlement document to MM. If the amounts match, you pay the carrier.

More Information

The following table gives more information on the settlement process for LSPs:

Subject	More Information
Overview of how the different settlement types work together in a business context for a logistics service provider	Settlement for LSPs in an Ocean Freight Scenario
Agreements, including forwarding and internal agreements	Agreement Management
Overview of forwarding order management	Forwarding Order Management
Overview of internal settlement management	Internal Settlement Management
Overview of freight order management	Freight Order Management
Overview of how the forwarding settlement process works	Forwarding Settlement

The following table gives more information on the settlement process for shippers:

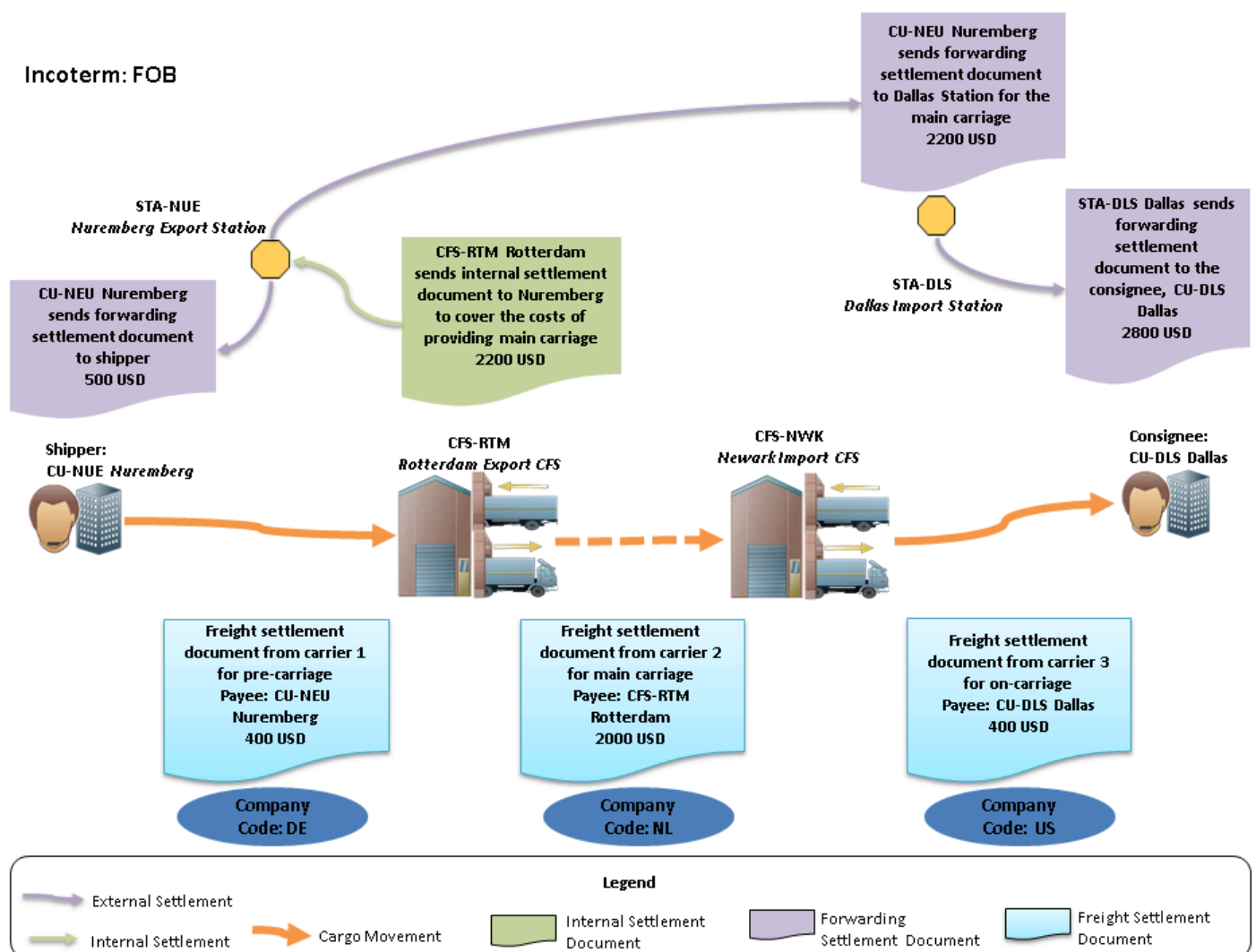
Subject	More Information
Overview of how you can create and edit agreements with transportation service providers	Agreement
Overview of how the freight settlement process works	Freight Settlement
Overview of cost distribution management	Cost Distribution Management
Overview of how the system distributes freight costs	Cost Distribution for Shippers

Settlement for LSPs in an Ocean Freight Scenario

Example

You are a logistics service provider (LSP) and you have a sales organization that takes forwarding orders from your customers. You have an internal gateway organization that manages your container freight station, consolidates shipments from different sales organizations, and creates consolidated bookings with your ocean carrier. Your gateway organization creates an intercompany internal settlement document against your sales organization to recover the cost of providing the ocean booking, and any additional costs incurred at the container freight station, for example consolidation costs. Only the sales organization is in contact with your customer.

The figure below illustrates an example of a business scenario in which you use forwarding settlement documents, freight settlement documents, and internal settlement documents. It is followed by an explanation:



Term Explanation

Some of the key terms in the graphic are explained as follows:

- STA

A station, which is usually the sales organization of an LSP.

- Gateway

A gateway consolidation operation that includes a container freight station. The gateway is both a sales organization for internal sales and a purchasing organization for the procurement of freight services from carriers. The purchasing organization usually procures for the main carriage.

- CU

A customer.

- STA-NUE

The export station at Nuremberg, and also the export sales organization.

- STA-DLS

The import station at Dallas, and also the import sales organization.

Incoterms

The customer in Dallas (not shown on the graphic) uses an Incoterm of FOB (Free On Board) to Rotterdam. The shipper CU-NUE Nuremberg pays for the pre-carriage. The ship-to party CU-DLS Dallas pays for the main carriage and the on-carriage.

In your forwarding order, you specify the shipper CU-NUE Nuremberg as the prepaid agreement party and Dallas import station STA-DLS as the collect agreement party. You specify Dallas import station STA-DLS as the collect agreement party because you use the business unit that is closest to the external customer to settle with the external customer (ship-to party CU-DLS Dallas). In this example, the customer in Dallas, who is the ship-to party CU-DLS_Dallas, pays the sales organization in Dallas (STA-DLS, Dallas import station).

Nuremberg export station STA-NUE pays the carrier for the pre-carriage. The Rotterdam gateway CFS-RTM pays the carrier for the main carriage and settles internally with Nuremberg export station STA-NUE to recover the cost of the main carriage. The Nuremberg export station STA-NUE recovers the cost of the main carriage from Dallas import station STA-DLS. Also the Dallas import station STA-DLS settles with the ship-to party in Dallas (CU-DLS Dallas) for the main carriage and the on carriage.

Cargo Movement

Nuremberg export station STA-NUE picks up the cargo from the shipper CU-NUE at Nuremberg. STA-NUE moves the cargo in the pre-carriage stage to Rotterdam export container freight station CFS-RTM. The export gateway at Rotterdam is now responsible for planning and moving the cargo. It moves the cargo in the main stage to Newark import container freight station CFS-NWK. The import gateway at Newark collects the container at the port and brings it to the container freight station in Newark. Dallas import station STA-DLA then moves the cargo in the on-carriage stage to the ship-to party CU-DLS in Dallas. You can use export/import processing to ensure that the export organization and the import organization have the required information and business documents to transport freight across customs borders. For more information, see [Export/Import Processing](#).

Agreements

The following table explains the agreements that are in place:

Agreement Type	Agreement Parties	Reason for Agreement
Forwarding agreement	Nuremberg export station STA-NUE as the organization and Dallas import station STA-DLA Dallas as the business partner	Provide main carriage services
Forwarding agreement	Nuremberg export station STA-NUE as the organization and shipper CU-NUE Nuremberg as the business partner	Move the cargo from Nuremberg to Rotterdam as agreed in the Incoterm FOB
Forwarding agreement	Dallas import station STA-DLS as the organization and ship-to party CU-DLS Dallas as the business partner	Provide freight services from Rotterdam to the ship-to party location in Dallas
Freight agreement	Nuremberg export station STA-NUE as the organization and Carrier 1 as the business partner	Provide pre-carriage services
Freight agreement	Rotterdam export gateway CFS-RTM as the organization and the ocean carrier Carrier 2 as the business partner	Provide ocean freight services
Freight agreement	Dallas import station STA-DLS as the organization and Carrier 3 as the business partner	Provide on-carriage services
Internal agreement	The Rotterdam gateway CFS-RTM as the organization and Nuremberg export station STA-NUE as the business partner	Provide main carriage and other consolidation services

Forwarding Settlements

The forwarding settlement document is a business document that contains the freight costs, and is a request to settle those costs with the customers. The customers can be external customers or internal customers who are the other organizations participating in the overall transportation. Transportation Management (TM) sends the forwarding settlement document to Sales and Distribution (SD) to create a billing document to be sent to the customer.

Nuremberg export station STA-NEU sends a billing document to the shipper CU-NEU Nuremberg for USD 500. The shipper pays for the pre-carriage cost.

Dallas import station STA-DLS sends a billing document to the ship-to party CU-DLS Dallas for USD 2800. The ship-to party pays for the main carriage and on-carriage costs.

Nuremberg export station STA-NEU also sends a billing document to the internal organization Dallas import station STA-DLS for USD 2200. The Dallas import station pays for the main-carriage cost.

Freight Settlements

The freight settlement document is a business document that contains the freight costs. You can use the freight settlement document to record the expected costs that you want to settle with your logistics service providers or carriers. Carrier 1 invoices Nuremberg export station STA-NEU for USD 400, the cost of providing the pre-carriage transportation service. In TM, STA-NEU creates a freight settlement document for the pre-carriage and posts these costs in materials management (MM). Nuremberg export station then uses this information in MM as a basis to verify the invoice it receives from Carrier 1. If the details match, it pays Carrier 1 from MM.

Rotterdam export gateway CFS-RTM and Dallas import station STA-DLS Dallas follow the same process. They pay carriers 2 and 3 USD 2000 and USD 400 respectively.

Internal Settlements

Rotterdam export gateway CFS-RTM creates an internal settlement on Nuremberg export station STA-NEU to recover the costs it incurred in providing the transportation services for the main carriage. CFS-RTM receives an invoice from its carrier for USD 2000, but it can also charge STA-NEU USD 2200 based on the internal rates between the organizations. As a result CFS-RTM can make USD 200 profit from the transportation.

Your company can also base the internal settlement document on the actual costs from the carrier. In this case CFS-RTM does not make a profit and bills STA-NEU for USD 2000.

Consolidated Shipment

The graphic illustrates a single shipment scenario. You can also manage a consolidated shipment scenario where Rotterdam export gateway CFS-RTM consolidates multiple forwarding orders (movements of cargo for an external customer) from different organizations in your company. CFS-RTM moves the cargo for the forwarding orders to Newark import container freight station CFS-NWK. CFS-NWK deconsolidates the cargo. The individual import freight stations then deliver the cargo to the different ship-to parties.

Rotterdam export gateway CFS-RTM sends an internal settlement document for the main-carriage cost to each of the internal export organizations for the forwarding orders. It can base the settlements on the internal rates agreed between the organizations or the cost of executing the transportation services.

Settlement Based on Agreed Rates

Rotterdam export gateway CFS-RTM can use a standard agreed rate as a basis for its freight costs. It can use the agreed rate to settle the cost it incurs in servicing each forwarding order that it consolidates and ships. For more information, see [Standard Charge Calculation](#).

Settlement Based on Freight Costs

If CFS-RTM bases the settlements on the costs, it must determine the share of the overall execution cost for which each organization is liable. CFS-RTM uses cost distribution to share the costs to the organizations. Using cost distribution, CFS-RTM can determine the share of the overall freight booking cost that applies to each of the forwarding orders, by distributing the freight booking cost at the forwarding order level. It can then use the distributed costs for each of the forwarding orders to create the settlement.

For more information about distributing freight costs, see [Cost Distribution Management](#). For more information about transferring distributed costs from the freight side to the forwarding side, and using the costs as a basis for settling with a customer or an internal organization, see [Cost Pull](#).

More Information

Subject	More Information
Overview of how the forwarding settlement process works	Forwarding Settlement
Overview of how the freight settlement process works	Freight Settlement
Overview of internal settlement management	Internal Settlement Management

Forwarding Settlement

Use

You can trigger the creation of an invoice to give to a customer for transportation service charges. You can also trigger the creation of a credit memo to give a credit to a customer. The Sales and Distribution (SD) system creates an invoice and a credit memo.

Integration

Forwarding Settlement is a component of Transportation Management (TM) and is integrated with the following TM components:

Component	Integration Type
Forwarding Order Management	A forwarding order provides the basis for a forwarding settlement document. A forwarding settlement document provides the basis for credit memos for forwarding orders.
Charge Management and Service Product Catalogs	As a prerequisite for creating a forwarding settlement document, you must create the relevant charge management master data.

Forwarding Settlement is also integrated with SD, as the creation of the invoice and credit that are relevant for accounting takes place in SD.

Constraints

This component triggers the invoice and credit creation and the actual invoices and credit memos are created in the SD.

Forwarding Settlement Document

Definition

A document that is posted to Sales and Distribution (SD) to request the creation of an invoice to be sent to a customer.

Use

You can use the forwarding settlement document (FWSD) to enter the data for transportation invoicing for your business partners on the sold-to party side, and forward it to SD for invoicing. Transportation Management (TM) calculates the transportation charges on the basis of a forwarding order (FWO). When you post the FWSD, the system creates an invoice in SD.

You can create an individual FWSD for one FWO or a collective FWSD for multiple FWOs.

Structure

The FWSD data is taken from the FWOs.

The FWSD consists of the following tab pages:

- **General Data**

The system displays general data such as the following:

- Invoice date
- Sales organization

- Payment terms
- Total amount
- Document currency

- **Business Partner**

The system displays the sold-to party, bill-to party and payer by default. You can use business partner determination to specify how the system assigns business partners to forwarding settlement documents if they are not defined in the forwarding orders.

- **Charges**

The system displays the transportation charges for each FWSD item. The FWSD item could be either an FWO or individual items of the FWO, depending on the calculation level settings in the calculation profile. For more information about the calculation profile, see Customizing for [Transportation Management](#) under [Basic Functions > Charge Calculation > Basic Settings > Define Calculation Profiles](#). In the case of collective FWSDs, the system shows the total amount for all the FWOs assigned to the FWSD.

For execution-based settlement scenarios such as buyer's consolidation, trailer-unit-based settlement or route-based settlements, the settlement items are the container item of the booking, the trailer unit and the freight order respectively.

- **Orders**

The system displays the list of FWOs or FWO items that are being settled with this FWSD, depending on the calculation level settings in the calculation profile. The FWSD item could be either an FWO or individual items of the FWO.

For FWSDs that you create for FWOs that are part of a buyer's consolidation, the system displays the freight booking document that has the container item of the buyer's consolidation. In the **Details** area of the screen, the system displays the container item details and the FWOs that are part of the container.

For FWSDs that you create for FWOs that have a trailer-unit-based settlement basis, the system displays the relevant trailer unit. In the **Details** area of the screen, the system displays the FWOs in the trailer unit for a bill-to party. For more information, see [Trailer-Unit-Based Settlement](#).

For FWSDs that you create for FWOs that have a route-based settlement basis, the system displays the relevant freight order document. In the **Details** area of the screen, the system displays the forwarding orders associated with the freight order. For more information, see [Route-Based Settlement](#).

- **Document Flow**

The system displays the predecessor and successor business documents as a document flow, including the corresponding follow-on billing documents created in SD on posting the FWSD. You can access the business documents from the **Document Flow** tab of the document.

- **Application Log**

The system displays any error or warning messages that were encountered when the FWSD was posted to system Internal SD. This information can help you to correct the errors and post the FWSD successfully.

The life cycle status indicates the current status of the FWSD in the business process. Important life cycle statuses are listed in the following table:

Life Cycle Status	Description
In Process	The FWSD has some incomplete or missing information.
Ready for Posting	The FWSD is ready for you to post to SD.
Posted	SD successfully created the billing document.

Life Cycle Status	Description
Posting Started	You posted the FWSD to SD to create a billing document and the posting process is not yet finished.
Cancellation Requested	You requested the cancellation of the FWSD from SD and the cancellation is not yet finished.
Canceled	You canceled the FWSD. You also canceled the billing document in SD.
Canceled by Credit Memo	You created and posted a credit memo for the full amount of the FWSD.

The confirmation status is used primarily to track the integration of FWSD with SD, and indicates the result of the last integration interaction between TM and SD. Important confirmation statuses are listed in the following table:

Confirmation Status	Description
Not Yet Posted	The FWSD is not yet posted to SD to create the billing document.
Posting Successful	SD successfully created the billing document.
Posting Failed	SD failed to create the billing document. You must manually check why the billing document was not created.
Cancellation Successful	SD successfully canceled the billing document.
Cancellation Failed	SD failed to cancel the billing document.
Posting Started	You transferred the FWSD to SD and the status in SD is not yet known.

Integration

The FWSD is integrated with a number of other components in TM, for example:

- [Forwarding Order Management](#)

You can start the generation of an FWSD from an FWO.

- [Charge Management and Service Product Catalogs](#)

The system calculates the transportation charges based on the setup of master data objects in the Charge Management and Service Product Catalogs component, for example calculation sheets, rate tables, and scales.

- [Agreement Management](#)

The system can use the forwarding agreement to include master data objects for transportation charges in the order process, such as calculation sheets or rate tables.

- FWSD also integrates with SD to create the invoice that you send to the customer and also to post to financial accounting.

More Information

[Forwarding Settlement](#)

[Charge Calculation](#)

Execution-Based Settlement

Features

As a logistics service provider (LSP) or carrier, you need flexibility in how you settle for the transportation activities you carry out on behalf of your customer. Transportation Management (TM) enables you to settle on the basis of individual forwarding orders as standard. You can also settle on the basis of an execution criteria, such as buyer's consolidation, route, or trailer unit.

Buyer's Consolidation

In a buyer's consolidation scenario, you are an LSP and to keep the transportation costs low for your customer, you consolidate and load the goods from different suppliers with whom your customer has an established business relationship, into one container before the main carriage. This means you pass the full container load (FCL) rate for the main carriage and on-carriage to the customer.

For more information, see [Buyer's Consolidation Settlement](#).

Execution-Based Criteria

In road transportation scenarios, you may also need to settle with your customer based on an execution unit such as a trailer unit, or on execution attributes such as the type of vehicle used to provide the service, or on the distance a truck traveled to fulfill the service.

For example, in a route-based settlement business scenario, your customer needs you to pickup goods from multiple locations such as warehouses and distribution centers and deliver the goods to a series of customer locations or retail outlets. Your customer wants to settle with you on the basis of an agreed rate for the full journey your truck makes as it performs the pickup and delivery activities, and not on the basis of the individual forwarding orders that you execute during the journey. You need to use an execution-based settlement criteria to enable this type of settlement.

Settlement Basis in TM

You can specify a settlement basis for route-based and trailer-unit-based scenarios in the **Settlement Basis** field of an agreement item.

By default, TM uses the information in an individual forwarding order to create forwarding settlement documents. For example, you create a particular forwarding settlement document for a particular forwarding order. Using the settlement basis gives you the flexibility to use different execution criteria to settle with your customers. You can settle based on how you executed the order and not by individual forwarding orders.

The table explains some of the main differences between execution-based settlement and settlement based on individual forwarding orders:

	Execution-Based Settlement	Forwarding-Order-Based Settlement
Settlement item	Execution document such as trailer unit or container	Forwarding order
Resolution base grouping rule	In execution document	In forwarding order

	Execution-Based Settlement	Forwarding-Order-Based Settlement
	Across forwarding orders in execution document for execution stages	
Data source (stage and logistics quantities)	Execution document such as freight booking item or trailer unit	As specified in settlement profile

In execution-based settlement, the system uses the stages from the freight order or trailer unit and not the constituent forwarding orders to calculate the settlement amount. You can settle on the basis of route-based or trailer-unit-based settlement criteria. For more information, see [Route-Based Settlement](#) and [Trailer-Unit-Based Settlement](#).

When you specify a resolution base grouping rule in the basic data of a calculation sheet, the grouping rule applies to items in a forwarding order. However, in the execution-based settlement scenario the item is a forwarding order. Therefore, the grouping logic applies across forwarding orders that are part of the execution document. For more information on the resolution base grouping rule, see [Calculation Sheet](#).

Constraints

You have specified the settings required for forwarding settlements. For more information, see [Forwarding Settlement](#).

You have specified the appropriate settlement basis in the forwarding agreement item.

If FWO is settled on any of the execution-based settlements, the charges for the FWO for the execution-based stages are not calculated in the individual FWO. The charges are only calculated in the forwarding settlement document. In the charge items for the execution-based stages, the execution-based charge item is enabled in the charges tab.

Example

You have 7 different forwarding orders that you deliver to a customer location on one trailer unit.

If you do not specify **Trailer Unit** in the **Settlement Basis** field, by default the system rates and settles each of the 7 forwarding orders individually. If you create a collective settlement for the 7 forwarding orders, the system still bases the settlement on each individual forwarding order. It performs the following tasks:

- Rates each of the forwarding orders individually
- Creates one forwarding settlement document with each of the 7 forwarding orders specified as a line item in the settlement

In contrast, if you specify **Trailer Unit** in the **Settlement Basis** field of the relevant forwarding agreement, you execute the 7 forwarding orders in one trailer unit and you settle with your customer based on a rate for the trailer unit. This rate covers all 7 forwarding orders in the trailer unit.

Buyer's Consolidation Settlement

Use

In a buyer's consolidation scenario, you are a logistics service provider (LSP) or carrier and your customer regularly buys goods from a number of suppliers in a particular country or region, and needs to transport them to a single location. Your customer needs to keep transportation costs as low as possible and therefore requests you to do a buyer's consolidation. To enable this, you can consolidate and load the goods from the different suppliers into one container before the main carriage and deliver the

consolidated container to the customer's location. This means you do not charge a less than container load (LCL) rate, and you pass the full container load (FCL) rate for the main carriage and on-carriage to the customer.

Prerequisites

You specify the following settings to enable the system to settle for the charges in a buyer's consolidation:

- **BCO** (buyer's consolidation) in the **Buyer's or Shipper's Consolidation** field of the forwarding order on the export side.
- An agreement item for buyer's consolidated transportation for main carriage and on-carriage. The agreement item includes the following settings:
 - A separate calculation sheet for the buyer's consolidation
 - **BCO** (buyer's consolidation) in the **Buyer's or Shipper's Consolidation** field

Features

Settlement as a Variant of an Execution-Based Settlement

On the export side, the system bases the settlement charges for main carriage on the information from the freight booking container item. The freight booking is the execution document and therefore the system settles for a buyer's consolidation as a variant of an execution-based settlement.

Settlement Fallback

The system tries to find an agreement line item with a **BCO** (buyer's consolidation) setting. If it does not find one, it tries to find an agreement line with a shipping type of **FCL** (full container load). For more information, see [Buyer's Consolidation](#).

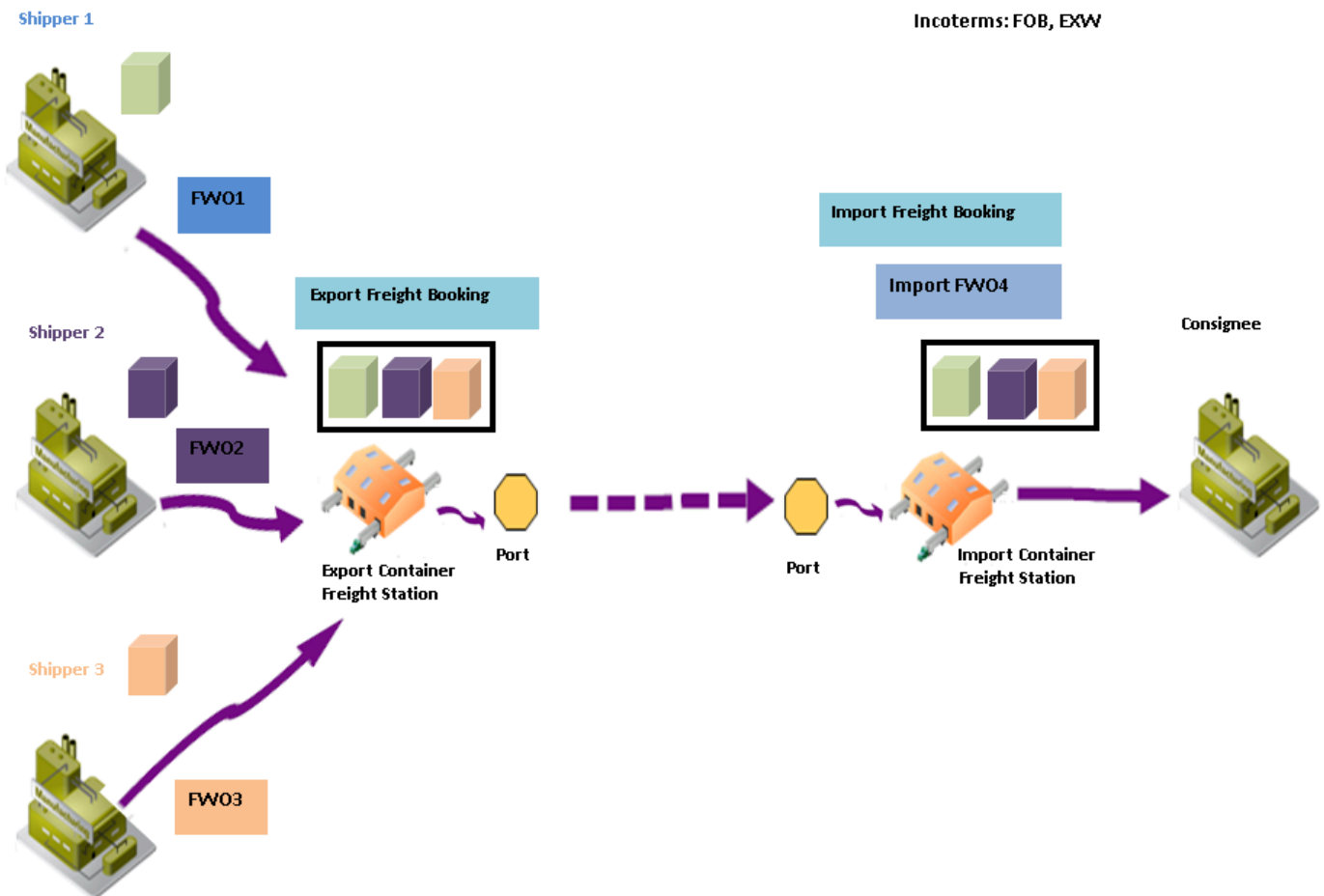
Settlement for Container Items

The system determines if the **Buyer's or Shipper's Consolidation** field in the freight booking for the container items has the **BCO** (buyer's consolidation) setting. If the setting exists, the system then creates one consolidated settlement for all the forwarding orders in the container item for the main carriage.

Example

In a buyer's consolidation with an Incoterm of **ExWorks (EXW)**, the ship-to party pays for all the transportation costs, including pre-carriage, main carriage, and on-carriage.

The concept is illustrated in the following figure, and is followed by an explanation:



For an Incoterm of **EXW**, on the export side there are multiple forwarding orders with one for each pickup location. The export business unit creates a settlement document that bills the import business unit, for each of the following activities:

- Pickup for forwarding order 1
- Pickup for forwarding order 2
- Pickup for forwarding order 3
- Main carriage

The system determines that the container item in the export freight booking has a buyer's or shipper's consolidation specification of **BCO** for all three forwarding orders. The system uses the container information from the export freight booking to create a single forwarding settlement document for the main carriage.

On the import side, the system automatically creates a copy of the forwarding order. This forwarding order contains a reference to the container item for the buyer's consolidation. The container item contains all the cargo information for the three forwarding orders, including the pickup locations. The import business unit creates a settlement document on the ship-to party, for each of the following activities:

- Pickup for forwarding orders, using the cargo items in the container
- Main carriage, using the cargo information for the container item
- Delivery to the ship-to party, using the cargo information for the container item

More Information

Trailer-Unit-Based Settlement

Features

You load a trailer with cargo items such as pallets and packages. The pallets and packages are from different forwarding orders from your customer. According to the agreement, you want to settle for delivering the trailer unit and not for delivering individual forwarding orders. You agree a set of rates that pass on to the customer the benefit of consolidating orders that have a common destination into a single trailer unit.

For example, you agree a rate of USD 4,000 to deliver a trailer unit to a location. You use a single trailer unit to deliver 2 forwarding orders (FWO1 and FWO2). FWO1 weighs 14,000 kg and FWO2 weighs 10,000 kg. You settle with the customer at the level of trailer unit. The following table describes how the system calculates the settlement amount:

Charge Type	Calculation (USD)	Charge (USD)
Trailer-unit charge	1 x 4,000	4,000
Settlement amount	Not applicable	4,000

In contrast, for example, you agree with your customer that you do not settle on the basis of an execution criteria such as trailer unit, and instead settle on the basis of each individual forwarding order (the default setting in the **Settlement Basis** field). The following table describes how the system calculates the settlement amount:

Charge Type	Calculation (USD)	Charge (USD)
Haulage FWO1	14,000 Kg X 0.2 cents / kg	2,800
Haulage FWO2	10,000 Kg X 0.2 cents / kg	2,000
Settlement amount for FWO1 and FWO2	Not applicable	4,800

More Information

Route-Based Settlement

Features

You can settle forwarding orders in a route-based settlement if you have a transportation journey with one or more pickups and deliveries. A route-based settlement can accommodate the following business situations:

- Multiple pickup and single delivery

Multiple pickups from plants or warehouses with a single delivery to an end location, such as a warehouse.

- Single pickup and multiple delivery

Pickup from a single location such as a distribution center and multiple deliveries to a series of locations such as retail outlets or customer delivery locations.

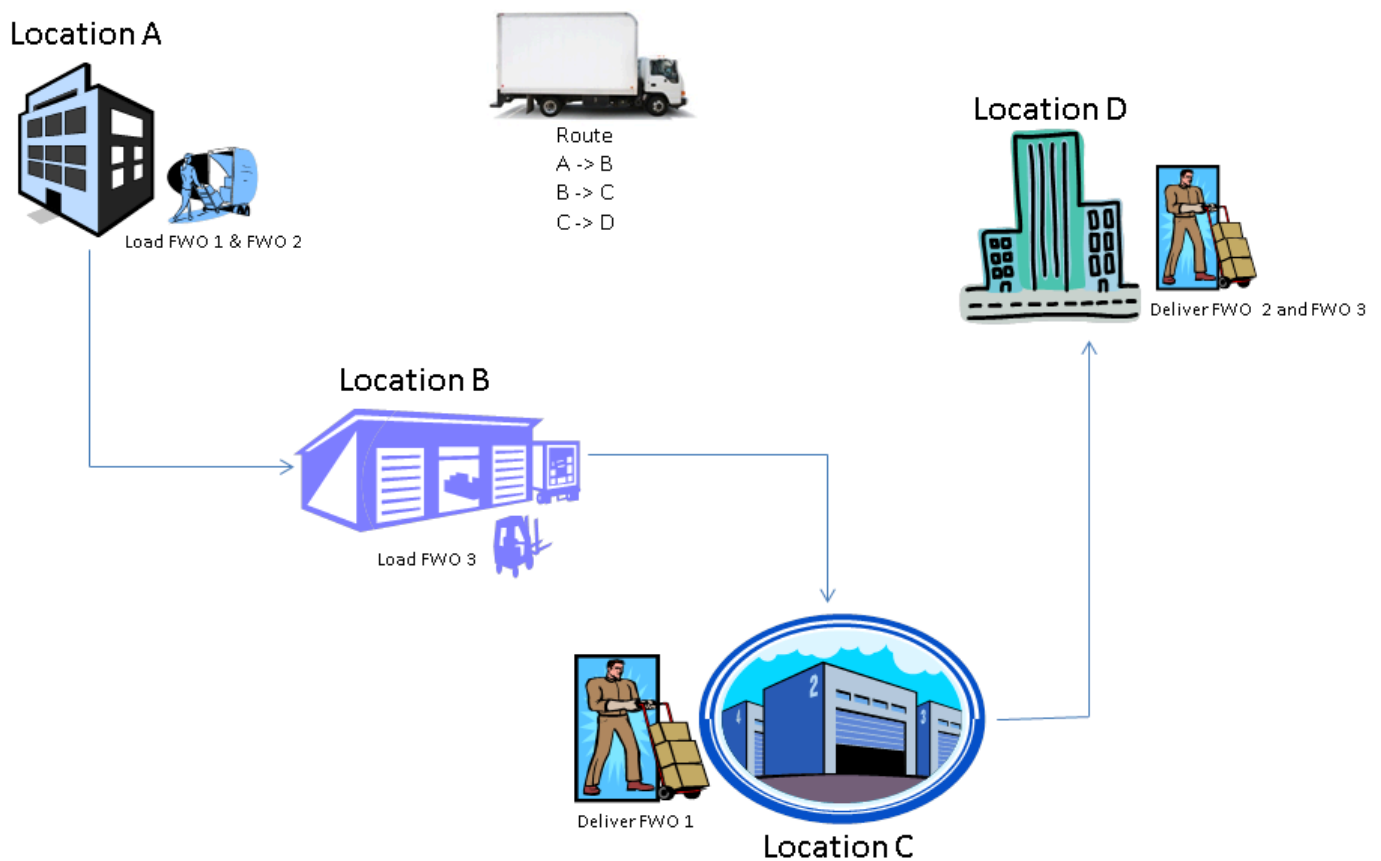
- Multiple pickup and multiple delivery

Multiple pickups from locations such as warehouses and distribution centers followed by deliveries to customer locations or retail outlets.

- Predefined start and end

Start from a specified location, pick up or delivery at scheduled locations, and return to the start location, or finish at a specified end location.

The figure below illustrates a multiple pickup and multiple delivery scenario and is followed by an explanation:



The following table describes the transportation requirements for the 3 forwarding orders:

Forwarding Order	Pickup Location	Delivery Location
FWO1	A	C
FWO2	A	D
FWO3	B	D

The following table describes the planning output. In planning, you consolidate the forwarding orders into freight order FO1. You use a 20-tonne truck.

From Location	To Location	Activity
Depot	A	Pickup FWO1 and FWO2 at location A
A	B	Pickup FWO3 at location B
B	C	Deliver FWO1 at location C
C	D	Deliver FWO2 and FWO3 at location D

You agree to a settlement basis of route-based with your customer. The system charges for the 3 forwarding orders as follows:

Charge Type	Calculation	Charge (USD)
Base charge (as distance traveled times the rate per km)	250 km X USD 1.20	300
Flat charge per truck (based on type of truck)	USD 150 for a 20-tonne truck	150
Stop charge (number of stops times the charge per stop)	2 stops X USD 50 per stop	100
Total freight cost	Not applicable	550

The system uses the information contained in freight order FO1, which executes the 3 forwarding orders, to create the forwarding settlement. On the user interface, the system displays the amount of USD 550 in line 1 of the settlement document for freight order FO1. It displays FWO1, FWO2, and FWO3 in the details screen with the full details for the journey.

More Information

[Trailer-Unit-Based Settlement](#)

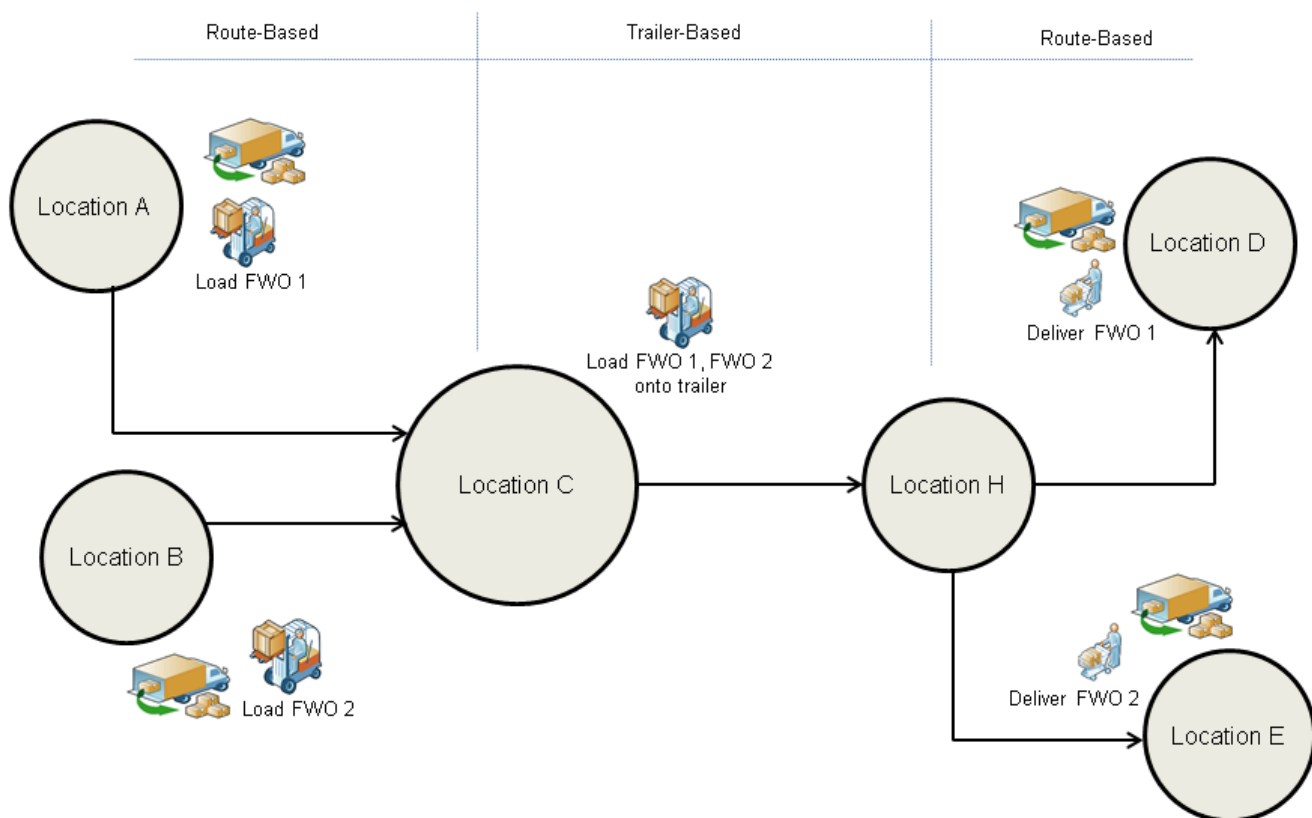
[Execution-Based Settlement](#)

Mixed Execution-Based Settlement

Use

Business Requirement

The business requirement is illustrated in the following figure, and is followed by an explanation:



You are a carrier and your customer needs you to perform the following transportation tasks:

1. Pick up different cargo (different forwarding orders) from multiple locations

For example, you pick up cargo from customer suppliers who are based in different parts of your region.
2. Consolidate the cargo on a trailer to move it to a location as instructed by your customer

For example, you put the cargo on a trailer in a distribution center that your customer owns in a different region.
3. Deliver the cargo from the distribution center to other locations as instructed by your customer

For example, you deliver the cargo to different plants or retail outlets in the region.
4. Settle with the customer using a route-based settlement for the pick-up (pre-carriage) and delivery (on-carriage) stages.

Use a trailer-unit-based settlement to settle for the main carriage.

The following table describes the transportation requirements for the forwarding orders:

Forwarding Order	Source Location	Destination Location
FWO1	A	D
FWO2	B	E

You plan the following business documents in your system for FWO1 and FWO2:

- Freight order 1 to pick up FWO1 and bring it to location C

- Freight order 2 to pick up FWO2 and bring it to location C
- Freight order 3 to pick up FWO1 and FWO2, now consolidated on a single trailer, and bring it from C to H
- Freight order 4 to deliver FWO1 to location D
- Freight order 5 to deliver FWO2 to location E

You use a small truck to pick up and deliver the forwarding orders.

You plan to include the following charge types in the settlement:

- Base freight

Calculated as distance traveled times the rate per km

- Flat charge per truck

Calculated based on the type of truck you use to execute the forwarding orders. This charge is not applicable for the main carriage where the goods travel on a trailer.

- Base charge for trailer

Calculated as distance traveled times the rate per km for the type of trailer you use to execute the main carriage

System Settings

You must specify the settings in the following table in the relevant forwarding agreement for your customer:

Line Item	Stage Category	Settlement Basis
1	Pre-carriage	Route
2	Main carriage	Trailer Unit
3	On-carriage	Route

Trailer Document for Main Carriage

You plan to move FWO1 and FWO2 on a single trailer between locations C and H in the main carriage. For more information on how to plan a trailer, see [Use of Trailers](#).

Pre-Carriage and On-Carriage

You create the freight orders and plan for the pre-carriage and on-carriage stages of FWO1 and FWO2. For example, in the **Actual Route** screen area of the forwarding order, you can select the relevant stage and choose **Freight Document Create**. The system adds the forwarding order to a freight order.

Settlement Documents

When you create a forwarding settlement document for FWO1 the system creates 3 settlement documents for each of the following documents:

- Trailer document, for the main carriage of both FWO1 and FWO2
- Pre-carriage of FWO1
- On-carriage of FWO1

When you create a settlement document for FWO2, the system creates 2 settlement documents as follows (it has already created a settlement document for the main carriage, as the main carriage is part of the trailer-unit based settlement contained in FWO1):

- Pre-carriage of FWO2
- On-carriage of FWO2

To create a single settlement document for your customer for all the transportation activities, you can use the **Collective** option under **Create Forwarding Settlement Document** when you select 2 or more forwarding orders on the forwarding order overview screen.

Charges

The system charges for the 2 forwarding orders as follows:

- Distance and type of truck for A to C and for B to C, pre-carriage
- Distance traveled by trailer for C to H, main carriage
- Distance and type of truck for H to D and for H to E, on-carriage

The following table explains how the system applies the charges:

Forwarding Order Stage	Charges	Calculation	Amount (USD)
Main carriage	Base freight	400 km X USD 1.20	480
	Base charge for trailer	400 km X USD 0.20	80
Pre-carriage for FWO1	Base freight	50 km X USD 1.40	70
	Flat charge for type of truck	USD 200	200
Pre-carriage for FWO2	Base freight	80 km X USD 1.40	112
	Flat charge for type of truck	USD 200	200
On-carriage for FWO1	Base freight	30 km X USD 1.40	42
	Flat charge for type of truck	USD 200	200
On-carriage for FWO2	Base freight	20 km X USD 1.40	28
	Flat charge for type of truck	USD 200	200

More Information

[Execution-Based Settlement](#)

Forwarding Settlement

Use

In Transportation Management (TM), you can perform forwarding settlement with your ordering parties by creating a forwarding settlement document (FWSD) and sending it to Sales and Distribution (SD) for creating the billing document (invoice) and posting to financials.

Prerequisites

- Agreements

You have the option of specifying the settlement basis at the agreement item level, where the system uses execution data as the basis to perform invoicing. By default, the system uses the settlement basis at forwarding order level.

- Types of FWSD and number range intervals

In Customizing, you have defined types of FWSD as well as default types and number range intervals. For more information, see Customizing for [Transportation Management](#) under [►►Settlement ► Forwarding Settlement ►](#).

- Forwarding settlement profile

You have created a forwarding settlement profile in Customizing for [Transportation Management](#) under [►►Settlement ► Define Settlement Profile ►](#).

You have assigned the forwarding settlement profile to the charges profile that you assign to the sales organization that you use in your forwarding order.

You can assign a forwarding settlement profile to a charge profile in Customizing for [Transportation Management](#) under [►►Basic Functions ► Charge Calculation ► Basic Settings ► Define Charges Profiles ► Set. Prof. \(Forwarding Settlement Profile\) ►](#). You can assign a charge profile to a sales organization in Customizing for [Transportation Management](#) under [►►Master Data ► Organizational Management ► Organizational Model ► Change Organizational Model \(transaction PPOME\) ►](#). Select the relevant sales organization [►►Org. Data ► Detail Settings ► Charges Profile ID ►](#).

When the system creates a forwarding order, it checks for a forwarding settlement profile in the charges profile of the sales organization.

If you do not specify a forwarding settlement in the charges profile, the system displays an error message when you create a forwarding order.

- Calculation profile

You have specified a calculation profile in Customizing for [Transportation Management](#) under [►►Basic Functions ► Charge Calculation ► Basic Settings ► Define Calculation Profiles ►](#). For more information about specifying calculation profiles, see [Calculation Profiles](#).

You can assign a calculation profile to a charge profile in Customizing for [Transportation Management](#) under [►►Basic Functions ► Charge Calculation ► Basic Settings ► Define Charges Profiles ► Calculation Profile ►](#).

You can assign a charge profile to a sales organization in Customizing for [Transportation Management](#) under [►►Master Data ► Organizational Management ► Organizational Model ► Change Organizational Model \(transaction PPOME\). ►](#). Select the relevant sales organization [►►Org. Data ► Detail Settings ► Charges Profile ID ►](#).

When the system creates a forwarding order, it checks for a calculation profile in the charges profile of the sales organization.

If you do not specify a calculation profile in the charges profile, the system displays an error message when you create a forwarding order.

- Process controller

In Customizing, you can set up the process controller for settlement documents. For example, if you want to use a strategy to create or preprocess FWSDs differently than the standard way, you can define your own strategy using the process controller and assign it to a specific service. For more information, see Customizing for [Transportation Management](#) under [►►Settlement ► Configure Process Controller for Settlement ►](#).

- Define default agreement partner functions for stages

You have specified the agreement partner function for a stage type and Incoterm combination. When you specify an Incoterm in a forwarding order (FWO), the system uses the agreement partner function to identify the default agreement party for a stage. For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) [Define Default Agreement Partner Functions for Stages](#).

- Master data for Charge Management

You have set up the master data in the Charge Management component. The system automatically calculates the transportation charges based on this master data. For more information, see [Charge Calculation](#).

- Business partner determination

You have the option of specifying a business partner determination profile, to enable the system to automatically assign business partners to the settlement document. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) [Business Partners](#) [Define Partner Determination Profiles](#).

i Note

The system includes the bill-to party, sold-to party, and payer in the settlement document by default. The system does not transfer business partners other than the sold-to party, the bill-to party and the payer to SD billing documents.

- Forwarding order

The system contains one or more forwarding orders (FWOs).

The system determines the logical system behind the business system group linked to the sales organization data of the forwarding settlement document. The business system group indicates the ERP system from which the sales organization is replicated in the organization model.

If this logical system is different to the current client logical system, the settlement document is posted to the external SD using enterprise services. If the logical system is the same as the current client logical system, the settlement document is posted to internal SD.

For more information on the organizational model see Customizing for [Transportation Management](#) under [Master Data](#) [Organizational Management](#) [Organizational Model](#) [Create and Merge Organizational Hierarchy](#).

- Integration settings for posting to SD

For integration to SD on an internal system, you have specified the relevant settings in Customizing for [Transportation Management](#) under [Settlement](#) [Forwarding Settlement](#) [Settings for Posting Forwarding Settlements](#).

For integration to SD on an external SAP S/4 HANA or SAP ERP system, you have specified the relevant settings in Customizing under [Integration with Other SAP Components](#) [Transportation Management](#) [Invoice Integration](#) [Billing](#).

- Required execution status for settlement creation

As standard, you can create a forwarding settlement document at the following execution statuses in a forwarding order:

- [Execution Not Started](#)
- [In Execution](#)
- [Executed](#)

You can specify the execution status as follows:

- In the relevant forwarding order type in Customizing for [Transportation Management](#) under [Forwarding Order Management](#) [Forwarding Order](#) [Define Forwarding Order Types](#) [Settlement Creation Status](#)
- In the relevant settlement profile type in Customizing for [Transportation Management](#) under [Settlement](#) [Define Settlement Profile](#) [Status for Settlement Creation](#)

If you don't specify an execution status, the system does not take the execution status into account when it creates a settlement document.

For example, you can use the **Executed** execution status setting of the forwarding order to settle the forwarding order.

You can also use the execution status as specified in a settlement rule as a basis for creating a forwarding settlement document.

By default, the system includes the header-level charges with the stage-level charges in the first settlement document it creates for a forwarding order.

Process

1. You create the FWSD.

You can create an individual FWSD for one FWO or a collective FWSD for multiple FWOs. You can review the FWOs from your personal worklist or in the **Forwarding Order Management Overview** or **Forwarding Settlement Overview**. The system creates the FWSD based on the data in the FWOs and calculates the transportation charges.

You can also create settlement documents for the following FWOs:

- FWO with Incoterms (see [Incoterms in Forwarding Orders](#))
- FWO with freight terms (see [Freight Terms in Forwarding Orders](#))

The system uses certain criteria to divide settlements when you choose to create a collective settlement document. This can result in the system creating more than one settlement. For more information, see [Split Criteria in Settlement Document Creation](#).

You can also create settlements for groups of charge types, charge categories, and charge subcategories from a forwarding order. You can apply different settlement creation criteria to the different groups. You can define the factory calendar for the billing schedule you want the system to use for a group of charges in a settlement group. This gives you control over when you create settlements for different charges. For more information, see [Flexible Creation of Forwarding Settlements](#).

i Note

You can create multiple forwarding settlement documents using the **Creation of Forwarding Settlement Documents** function. For more information, see [Creation and Transfer of Forwarding Settlement Documents](#).

2. You review the FWSD details.

You can preview the invoice as it would appear in SD. The preview contains information from the FWSD and tax information determined from SD.

You can use the preview for FWSDs that are at all life cycle statuses except the following:

- **Posting Started**
- **Posted**
- **Cancellation Requested**
- **Canceled**

The consistency status of the document must also be **Consistent**. When you preview an invoice, the system does not permanently store any data.

i Note

You can overwrite the address of the business partner that the system has automatically determined. For example, if you need to send an invoice to a bill-to party at another address instead of the regular address, you can change the address

of the bill-to party on the **Business Partner** tab page of the FWO or FWSD. When you post the settlement, the SD billing document has the deviating address of the bill-to party in the SD billing document. For more information, see [Creation of a Forwarding Order](#).

3. You transfer the data to SD.

You choose **Save and Post** to send the FWSD to create the billing document in SD. If you have SD on an external SAP S/4 HANA or SAP ERP system, the system uses enterprise services to post between the systems.

4. SD confirms the billing document creation.

If SD created the billing document, you receive a confirmation from SD. The confirmation is displayed with the appropriate status of the FWSD. The life cycle status changes to **Posted**.

If SD did not create the billing document, the confirmation status shows that posting failed. If posting is done to an internal SD system, you must manually check in the **Application Log** tab of the settlement document for detailed messages about why the invoice was not created. If you have SD on external SAP S/4 HANA or SAP ERP system, you must check the error logs for the enterprise service message, in transaction SXI_MONITOR.

i Note

You can request that the FWSD is canceled at any time, even if its life cycle status is **Posted**. After you request cancellation of the FWSD, SD cancels the billing document. If the billing document is canceled, the FWSD has the life cycle status **Canceled**. If the billing document cannot be canceled, the FWSD has the confirmation status **Cancellation Failed**.

You can create an FWSD for a canceled FWO to recover any costs you incurred in processing the order before cancellation. You must manually add the charges to the FWSD that is associated with the cancellation.

If you trigger the creation of an FWSD for an FWO that is a part of a buyer's consolidation, the system takes the following steps:

- Determines all the FWOs that are in the container item of the buyer's consolidation
- Creates one FWSD for the all the FWOs in the container item instead of individual FWSDs for each of the FWOs

If you trigger the creation of an FWSD for an FWO that has an execution-based settlement basis, the system creates one FWSD for all the FWOs that are part of the relevant execution document for one customer. It bases the settlement on the execution document instead of the individual FWOs.

For a settlement basis of trailer unit, the execution document is the trailer unit. For a settlement basis of route-based, the execution document is a freight order.

Tracking Manual Changes

If you make a manual change to an individual forwarding settlement document or internal settlement document, the system flags the settlement document as manually changed.

The system regards changes to the following attributes in the forwarding settlement document as manual changes:

- Payment terms
- Notes and attachments
- Charges
- Invoice date
- Service date

- Payer
- Deviating address for a bill-to business partner

You can view the manual change status on the personal object worklist for forwarding settlement documents.

Creation Method

The system displays the approach you take when you create a forwarding settlement document or an internal settlement document. The system displays the creation method on the POWL screen of the settlement document.

The system can display the following information:

- **Manual**

You have not previously created a settlement document for the forwarding order. You can manually create a forwarding settlement document as follows:

- Choose ► **Charges/Settlement** ► **Create Forwarding Settlement Document** ►
- Select one or more forwarding orders on the forwarding orders POWL screen, and choose the appropriate option under **Create Forwarding Settlement Document**

- **Background Processing**

You use the **Creation and Transfer of Forwarding Settlement Documents** background processing function to create a settlement document. You have not previously created a settlement document for the forwarding order and you have not changed the forwarding order after previously creating a settlement document.

- **Change and Background Processing**

You have changed the associated forwarding order, enabled the relevant automatic change processing features, and run the background processing function to create the forwarding settlement document.

Follow-Up Forwarding Settlement Documents

You can create a second or subsequent forwarding settlement document after you have already created a settlement document. When you create a follow-up forwarding settlement document, the system uses the following approach to propose the invoiced amount:

1. Copy the charges from the forwarding order

Depending on the calculation option you specify in the relevant settlement profile, you can copy all the charge lines from the order to the settlement as they are, or you can copy the fixed charges and recalculate the other charges.

For more information, see Customizing for **Transportation Management** under ► **Settlement** ► **Define Settlement Profile** ►

2. Calculate the invoiced amount

The system takes the following steps:

- For each charge line, subtract the invoiced amounts that are already included in the previously created but not canceled forwarding settlement documents
- Propose the result of the subtraction as the **Invoiced Amount** in the follow-up forwarding settlement document.

This can result in the following invoiced amounts in the follow-up settlement document:

- An overall negative amount for the settlement document, where you have over-invoiced your customer

- A zero amount where you have completely invoiced your customer

To prevent you from transferring the follow-up settlement document to Sales and Distribution until the zero or negative amount has been removed, the system assigns a consistency status of **Inconsistent** to the follow-up settlement document.

You can perform the following tasks:

- Run the background processing function to create and post settlement documents (program /SCMTMS/CFIR_CREATE_BATCH) if you have enabled the change management process to correct the settlement documents automatically.
- Manually cancel the existing settlement document and replace it with a new settlement document that contains the changed amount. The relevant forwarding order must not have a life cycle status of **Completed**.
- Manually create a credit memo for the over-invoiced amount

More Information

Subject	See
Create a buyer's consolidation	Buyer's Consolidation
Create execution-based FWSDs	Execution-Based Settlement
Determine business partners automatically	Business Partner Determination
Create, edit, and display a trailer unit	Creation and Editing of Transportation Units
Change management process for settlement documents	Automatic Change Management Process for Settlements
Determine a rate for a collective settlement document	Collective Rate Determination

Shipper's Consolidation Settlement

Use

In a shipper's consolidation scenario, you are a logistics service provider (LSP) or carrier and you have a single customer who needs you to ship goods from a single location and deliver them to a number of different locations in a particular country or region.

Your customer needs to consolidate the goods over the pre-carriage and main carriage to optimize the freight cost. As an LSP, you need to settle for the freight cost, including the main carriage cost, with your customer. You transport the goods in one container for the pre-carriage and main carriage, and only separate the goods from the container for on-carriage delivery to the customer's location. You charge your customer a full container load (FCL) charge for the pre-carriage and main carriage.

Prerequisites

You must specify the following settings to enable the system to settle for charges in a shipper's consolidation:

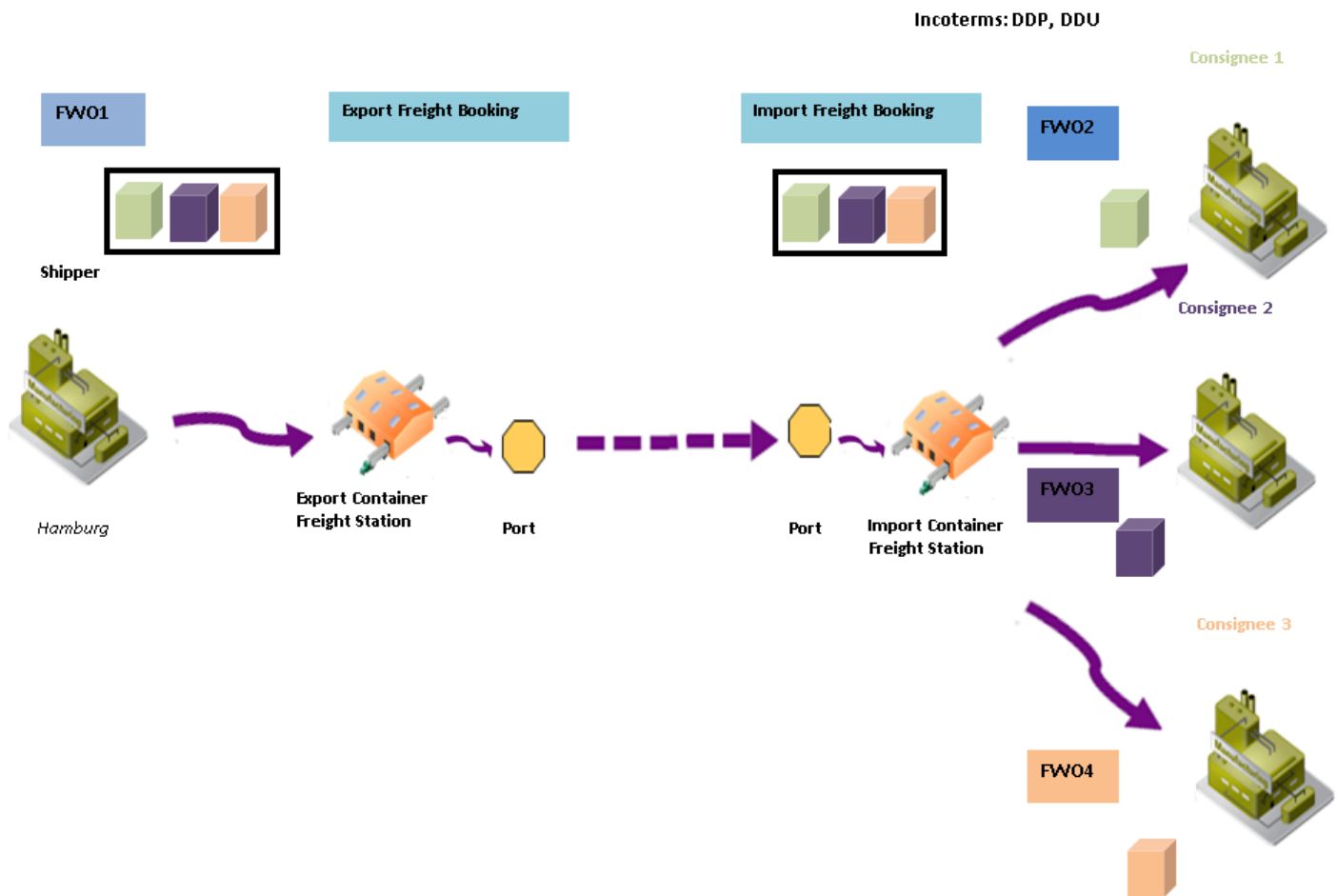
- **SCO** (shipper's consolidation) in the **Buyer's or Shipper's Consolidation** field of the forwarding order on the export side
- An agreement item for shipper's consolidated transportation for pre-carriage and main carriage. The agreement item includes the following settings:
 - A calculation sheet for the shipper's consolidation

- **SCO** (shipper's consolidation) in the **Buyer's or Shipper's Consolidation** field

Example

In a shipper's consolidation with an Incoterm of **Delivered Duty Paid (DDP)**, the seller (the shipper, your customer) pays for all the transportation costs, including pre-carriage, main carriage, and on-carriage. To enable this, the system makes the shipper, ship-to party, source location, and destination location information available for each item in the forwarding order.

The concept is illustrated in the following figure and is followed by an explanation:



For an Incoterm of DDP, on the export side, you create one forwarding order with one container item. In the container item, you enter the cargo you collected from the single location that you deliver to different ship-to parties. The system uses the container as the basis for creating the forwarding settlement. The export business unit creates a settlement document on the shipper, for each of the following activities:

- Pre-carriage
- Main carriage
- On-carriage 1 in forwarding order 1 for item 1 in the container
- On-carriage 2 in forwarding order 1 for item 2 in the container
- On-carriage 3 in forwarding order 1 for item 3 in the container

In the settlement document, the system includes each of the on-carriage stages for the individual cargo items from the forwarding order.

On the import side, the system automatically creates a copy of the forwarding order for each of the on-carriages to the delivery locations. The import business unit creates a forwarding settlement document on the export business unit, for each of the on-carriages as follows:

- On-carriage 1 in forwarding order 2
- On-carriage 2 in forwarding order 3
- On-carriage 3 in forwarding order 4

The system includes each of the individual on-carriage forwarding orders in the settlement document.

More Information

[Shipper's Consolidation \(SCO\) in Import/Export](#)

Flexible Creation of Forwarding Settlements

Use

You can create a forwarding settlement document for a group of charges in a forwarding order. This means that you can create more than one forwarding settlement document for a forwarding order, where each settlement document is for a different group of charges in the order. This enables you to create a settlement document for a group of charges that you want to settle at the same time in your transportation process.

To enable this, you specify settlement groups and settlement rules in Customizing for **Transportation Management** under **Settlement** ► **Define Settlement Groups and Settlement Rules** ►.

You assign charge types, charge categories, and charge subcategories to a settlement group. You assign a settlement group to a settlement rule. You can also specify the following additional criteria in a settlement rule:

- Minimum execution status

You can specify the minimum execution status required for a forwarding order before the system creates a forwarding settlement document.

- Service date rule

You can specify the service date rule that the system uses to determine the service date of the forwarding settlement document.

- Billing schedule

You can specify a factory calendar for the billing schedule that determines the settlement date for a settlement group.

When you create a forwarding settlement document, the system checks if a charge is contained in one of the charge types, charge subcategories, or charge categories that you have assigned to a group. If it finds a charge, it creates the forwarding settlement document according to the following criteria:

- Settlement group

The system creates a separate settlement for each settlement group.

- Settlement rule

The system creates a separate settlement for a group of charges when a forwarding order is in a particular execution status or when you reach a particular date.

Specifying settlement groups and settlement rules is optional. If you do not specify settlement groups, the system includes all the charge types from a forwarding order in one forwarding settlement document.

For more information, see Customizing for [Transportation Management](#) under [►►Settlement ► Define Settlement Groups and Settlement Rules ►](#).

Prerequisites

General Prerequisites

- You have specified settlement groups, settlement rules, and assigned the necessary settlement groups to settlement rules in Customizing for [Transportation Management](#) under [►►Settlement ► Define Settlement Groups and Settlement Rules ►](#).
- You have assigned a settlement rule to a settlement profile. For more information, see Customizing for [Transportation Management](#) under [►►Settlement ► Define Settlement Profile ►](#).

Settlement Group Prerequisites

- If you want to assign charge types to settlement groups, you have specified charge types in Customizing for [Transportation Management](#) under [►►Basic Functions ► Charge Calculation ► Basic Settings ► Define Charge Types ►](#).
- If you want to assign charge subcategories to settlement groups, you have specified charge subcategories in Customizing for [Transportation Management](#) under [►►Basic Functions ► Charge Calculation ► Basic Settings ► Define Charge Subcategories ►](#).
- If you want to assign charge categories to settlement groups, you have specified charge categories in Customizing for [Transportation Management](#) under [►►Basic Functions ► Charge Calculation ► Basic Settings ► Define Charge Categories ►](#).

Settlement Rule Prerequisites

- If you want to use a service date rule, you have specified the necessary settings in Customizing for [Transportation Management](#) under [►►Settlement ► Define Service Date Rules and Rule Prioritization ►](#).
- If you want to specify a billing schedule, you have specified a factory calendar for a billing schedule in Customizing for [Transportation Management](#) under [►►Master Data ► Calendar ► Maintain Factory Calendar ►](#).

Activities

Determine the following information:

- The set of charges you include in forwarding settlement documents
- The set of charges you create settlements for after a forwarding order reaches a particular execution status
- The set of charges you create settlements for at regular periods of time

Example

Business Requirements

You need to settle the following charges:

- Base freight
- Fuel surcharge
- Demurrage
- Other value added services (VAS)

You need to settle base freight and fuel surcharges together. You want to settle demurrage charges in a separate settlement so that payment for base freight and fuel surcharge is not held up by disputes on the demurrage charges. You need to settle for the base freight and other VAS charges every week, and for the demurrage charges every month.

You have the following business rules to determine the service date:

- Service date for demurrage charges is the system date
- Service date for base freight, fuel surcharges, and other VAS is the pick-up date at the first location

You need to create settlement documents for basic freight charges (including base freight and fuel surcharges) and other VAS charges when the forwarding order is in execution. You need to create settlement documents for demurrage charges when the forwarding order has completed execution.

Customizing Settings

You specify the following settlement groups in Customizing for **Transportation Management** under **►►Settlement ► Define Settlement Groups and Settlement Rules ►**:

- **Basic freight charges**
You assign the base freight and fuel surcharge charges to this group.
- **Demurrage**
You assign the demurrage charge to this group.
- **Other VAS**
You assign the charge types for VAS to this group.

You specify the settlement rule **Standard Process Customer 1** that contains the following settings:

- Line item 1 of the **Assign Settlement Groups** table:
 - Settlement group: **Basic Freight Charges**
 - Execution status: **In Execution**
 - Service date rule: a rule that contains a date type of **Actual Start Date of Complete Carriage** at priority 1
 - Billing schedule: a weekly factory calendar
- Line item 2 of the **Assign Settlement Groups** table:
 - Settlement group: **Demurrage**
 - Execution status: **Executed**
 - Service date rule: leave empty

The system uses the service date rule from the settlement profile. You must use a profile that contains an empty service date rule (if the [Service Date Rule](#) field is empty, the system uses the system date).

- Billing schedule: a monthly factory calendar
- Line item 3 of the [Assign Settlement Groups](#) table:
 - Settlement group: Other VAS
 - Execution status: In Execution
 - Service date rule: a rule that contains a date type of [Actual Start Date of Complete Carriage](#) at priority 1
 - Billing schedule: a weekly factory calendar

Freight Settlement

Use

You use this component to trigger the accrual of the freight costs and to prepare for the verification of an invoice, received from a service provider or carrier for transportation charges, against a freight document, such as freight orders, freight bookings, or service orders.

In Transportation Management (TM), you can perform freight settlement with your service providers or carriers by creating a freight settlement document and posting it to Materials Management (MM). The transportation charges are calculated in TM on the basis of a freight order, freight booking, service order, or consignment order. TM relies on the integration to MM for the follow-on processes in financials. The freight settlement document forms the basis of representing the freight document cost from TM to MM. The information in the settlement document is used to create a purchase order and a service entry sheet (SES). The posting of the SES confirmation forms the basis for invoice verification or the evaluated receipt settlement (ERS) process in MM.

i Note

Settlement and posting of freight costs for service orders are available only when you use Freight Settlement Integration using MM External Services Management.

Prerequisites

- Types of settlement document and number range intervals

In Customizing, you've defined freight settlement document types as well as default types and number range intervals. For more information, see Customizing for [Transportation Management](#) under ► [Settlement](#) ► [Freight Settlement](#) ►.

- Freight settlement profile

You've created a freight settlement profile in Customizing for [Transportation Management](#) under ► [Settlement](#) ► [Define Settlement Profile](#) ►.

You've assigned the freight settlement profile as follows:

- To the business partner and purchasing organization that you use in your freight order

You use the app [Manage Business Partner Master Data](#) to work with the business partner. Select the relevant business partner ► [Display in BP Role: Carrier](#) ► [Vendor Org. Data](#) ►. Choose the relevant purchasing organization ► [Additional Settings](#) ► [FSD Profile ID](#) ►.

For more information, see [Business Partner](#).

- To the charges profile that you assign to the purchasing organization that you use in your freight order

You can assign a freight settlement profile to a charge profile in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charges Profiles > Fre. Set. Prof. (Freight Settlement Profile)**.

You can assign a charge profile to a purchasing organization in Customizing for **Transportation Management** under **Master Data > Organizational Management > Organizational Model > Change Organizational Model** (transaction PPOME). Select the relevant purchasing organization **Org. Data > Detail Settings > Charges Profile ID**.

When the system creates a freight order, it checks for a freight settlement profile as follows:

1. Business partner of the purchasing organization
2. Charges profile of the purchasing organization

If you don't specify a freight settlement profile in the business partner or in the charges profile, the system displays an error message when you create a freight order.

- Calculation profile

You've specified a calculation profile in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Calculation Profiles**. For more information about specifying a calculation profile, see [Calculation Profiles](#).

You've assigned the calculation profile as follows:

- To the business partner of the purchasing organization that you use in your freight order

See **SAP Menu > Logistics > Transportation Management > Master Data > BP - Maintain Business Partner** (transaction BP). Select the relevant business partner **Display in BP Role: Carrier > Vendor Org. Data**. Choose the relevant purchasing organization **Additional Settings > Calculation Profile ID**.

You can use the **Manage Business Partner Master Data** app to maintain the business partner. For more information, see [Business Partner](#).

- To the charges profile that you assign to the purchasing organization that you use in your freight orders

You can assign a calculation profile to a charges profile in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charges Profiles > Purch. Calc. Profile (Purchasing Calculation Profile)**.

You can assign a charge profile to a purchasing organization in Customizing for **Transportation Management** under **Master Data > Organizational Management > Organizational Model > Change Organizational Model** (transaction PPOME). Select the relevant purchasing organization **Org. Data > Detail Settings > Charges Profile ID**.

When the system creates a freight order, it checks for a calculation profile in the following objects:

1. Business partner of the purchasing organization
2. Charges profile of the purchasing organization

If you don't specify a calculation profile in the business partner or in the charges profile, the system displays an error message when you create a freight order.

- Process controller

In Customizing, you can set up the process controller for use with settlement documents. For example, if you want to use a strategy to create or preprocess settlement documents differently than the standard way, you can define your own strategy

using the process controller and assign it to a specific service. For more information, see Customizing for [Transportation Management](#) under [► Settlement ► Configure Process Controller for Settlement ►](#).

- Master data for charge management

You've set up the master data in the Charge Management component. The system automatically calculates the transportation charges based on this master data. For more information, see [Charge Calculation](#).

- Business partner determination

You have the option of specifying a business partner determination profile, to enable the system to automatically assign business partners to a settlement document. For more information, see [Business Partner Determination](#).

i Note

The system includes the invoicing party, carrier, and payee in the settlement document by default. The system doesn't transfer business partners other than the carrier, invoicing party, and payee to MM.

- Business document

The system contains one or more freight orders, service orders, or freight bookings.

Integration

Freight Settlement is a subcomponent of Transportation Management (TM) and is integrated with the following TM subcomponents:

Subcomponent	Integration Type
Freight Order Management	Freight orders, freight bookings, and service orders provide the basis for the following business documents: • Freight settlement documents • Credit memos for freight orders
Charge Management and Service Product Catalogs	As a prerequisite for creating a freight settlement document, you must create the relevant charge management master data.

This component prepares the freight settlement document for invoice verification and relies on the integration with Materials Management (MM) using External Services Management or Lean Services for posting accruals and verifying invoices. For more information, see:

- [Freight Settlement Integration using MM External Services Management](#).
- [Freight Settlement Integration using Lean Services](#).

Freight Settlement Integration using MM External Services Management

In Transportation Management (TM), you can perform freight settlement with your service providers or carriers by creating a freight settlement document and posting it to Materials Management (MM) using external services management.

Prerequisites

- You've enabled the checkbox **TM Settlement Integration based on MM External Services Mgmt** in the Customizing of **Transportation Management** under **Basic Functions > General Settings > Define General Settings for TM**. This setting ensures that you've enabled external service-based integration to Materials Management (MM).
- To integrate to an internal MM system, you've configured the following Customizing settings:
 - **Assign Purchasing Information for Posting**
 - **Assign Service Master Record and Account Assignment Category**
 - Tax determination settings under **Tax**
 - Mapping for cost distribution in a shipper scenario, when you've enabled cost distribution in the freight settlement document

For more information, see Customizing for **Transportation Management** under **Settlement > Freight Settlement > Integration for Settlement Posting > External Services Management**.

To integrate to an external MM system on SAP S/4HANA or SAP ERP, you've made the relevant settings in the Customizing activities under **Integration with Other SAP Components > Transportation Management > Invoice Integration > Invoicing** in the external SAP ERP or SAP S/4HANA system.

The system determines the logical system behind the business system group that you've linked to your purchase organization data for the freight settlement document. The business system group represents the SAP ERP or another SAP S/4HANA system from which you replicate the purchasing organization in the organization model.

If the logical system is different to the current client logical system, the system uses enterprise services to post the settlement document to an external MM system. If the logical system is the same as the current client logical system, the system posts the settlement document directly to an internal MM system. For more information about organizational models, see [Organizational Management](#).

- **Company code and Plant**

For more information, see [Company Code and Plant in Posting Freight Settlements](#).

Process

1. You create the freight settlement document.

Note the following system features:

- You can create an individual settlement document for a freight order, service order, or freight booking, or a collective settlement document for multiple freight orders, service orders, or freight bookings. You can access freight orders, service orders, or freight bookings from your personal worklist, in the relevant document overview, or in the **Freight Settlement Overview**. The system automatically creates the settlement document on the basis of the data in the freight orders, service orders, or freight bookings and calculates the transportation charges.

You can create multiple freight settlement documents using the **Mass Creation of Freight Settlement Documents** function. You can run transaction /SCMTMS/SFIR_CREATE (report /SCMTMS/SFIR_CREATE_BATCH). For more information, see [Creation and Transfer of Freight Settlement Documents](#).

- You can create settlement documents for individual stages in freight orders that contain non-linear stages. This means you can create a settlement document for each pairing of a source location and destination location. For more information, see [Freight Orders with Non-Linear Stages](#).
- You can create separate freight settlement documents for each of the invoicing carriers in a rail freight order, where you have a different invoicing carrier for each stage in the order. This feature gives you the flexibility to work with rule 11 business situations in North America.

In rule 11 situations, the header carrier receives the order from the customer and can execute one or more of the stages. The header carrier is the carrier that you specify in the ► **General Data** ► **Transportation** screen area of the freight order. The invoicing carrier is the carrier who bills the customer for the shipment. This is the carrier that you specify at stage level in the rail freight order.

For more information about rail freight orders, see [Rail Transportation](#).

- You can create a settlement document for each additional agreement party if you have an existing freight order or freight booking that contains one or more additional agreement parties. For more information, see [Additional Agreement Parties](#).
- The system uses certain criteria to split settlements when you choose to create a collective settlement document. This can result in the system creating more than one settlement. For more information, see [Split Criteria in Settlement Document Creation](#).
- The system settles a parcel freight order in the same way as it settles a standard freight order. For more information about parcel freight orders, see [Parcel Shipment](#).
- When you create a freight settlement document, by default the system uses today's date as the settlement date. You have the following options to manually override the system default:
 - Implement **BADl: Determination of Settlement Date for Freight Settlement Documents**. The system uses the settings that you specify in this BADl to determine the settlement creation date. For more information, see Customizing for **Transportation Management** under ► **Business Add-Ins (BADIs) for Transportation Management** ► **Settlement** ► **Freight Settlement** .
 - Enter a settlement date in the **Mass Creation of Freight Settlement Documents** background processing function. The system uses the date that you specify in this background function as the settlement date if you haven't specified a rule in the BADl.

2. You review the settlement document details for consistency.

You can review the settlement document details to check their accuracy and you can update some of the data as needed. After you save any changes you make, you can check the consistency of the settlement document. If the automatic consistency check is successful, the system changes the consistency status of the settlement document to **Consistent** and the lifecycle status of the settlement document to **Ready for Accruals**.

3. You post the settlement document to MM to post accruals.

Note the following system features:

- TM sends the settlement document data to MM when you choose **Post**. TM transfers the settlement information, including the source location and destination location information for each stage in the settlement, to MM. The location information enables MM to determine different tax rates for different locations, countries/regions in the purchase order item or, if enabled, in the service item.
- When the SES is accepted and posted, accruals are booked to accounting. The SES serves as the basis for invoice verification in MM.

The settlement date from the freight settlement document is used as the **Doc. Date** (Document Date) in the header of the purchase order.

The SES dates on the header of the **Accept. Data** (Acceptance Data) tab page are taken from the following places:

- Posting date from the settlement date in the freight settlement document
- Document date as the system date

Posting Periods

You can post the freight settlement document only if the posting periods in MM and Financial Accounting (FI) are both open for the posting date of the SES. If the posting period is closed, you have the following options:

- Open the closed posting period

You must open the posting period for both MM and FI.

In FI, open the posting period in Customizing under ► **Financial Accounting** ► **Financial Accounting Global Settings** ► **Ledgers** ► **Fiscal Year and Posting Periods** ► **Posting Periods** ► **Open and Close Posting Periods** ►.

In MM, open the posting period in the SAP Menu under ► **Logistics** ► **Materials Management** ► **Material Master** ► **Other** ► **MMPV – Close Period** ► or **MMRV - Allow Posting to Previous Period**.

- Change the settlement date of the freight settlement document to match the open posting period and post the settlement document again.
- In TM, you can have multiple charge types in a charge item. As standard, MM creates a purchase order item for a charge item, and applies tax to the purchase order item. If you need the flexibility to apply tax to individual charge types in a charge item, you must specify the settings in the following Customizing activities in Customizing for **Transportation Management** under ► **Settlement** ► **Freight Settlement** ► **Integration for Settlement Posting** ► **Tax** ►:

- **Define Tax at Charge Type Level for Countries/Regions**

You can define the countries/regions in which you apply tax at charge type level.

- **Assign Tax Indicators for Charges**

You can define tax indicators for the charge categories, charge subcategories, and charge types that you use in a country/region. You can use charge categories and charge subcategories to specify tax settings for a group of charge types in one step. This avoids the need to specify tax settings for a large number of individual charge types.

For more information, see Customizing for **Transportation Management** under ► **Settlement** ► **Freight Settlement** ► **Integration for Settlement Posting** ► **External Services Management** ► **Tax** ►. Also see SAP Note [2622403](#) 📄.

4. When the settlement document is posted successfully, its lifecycle status is set to **Posted**. If issues are encountered during posting, the lifecycle status is set either to **In Process** or **Posting Started**, and the confirmation status has the status of the corresponding step in which the issue was encountered. For example, the confirmation status **Accruals Failed** means that the entire FSD posting encountered an issue, or the status **SES Posting Failed** indicates that the purchase order was created successfully but the SES creation and posting failed.

If you encounter issues when you post to internal MM, the system logs the issues. You can review the issues on the **Application Log** tab of the settlement document.

5. Invoice verification and Evaluated Receipt Settlement (ERS)

- MM bases the expected invoice on the SES confirmation. When the invoice is posted against the freight document, the invoice values are updated to the freight settlement document and the lifecycle status of the settlement document is set to **Invoice Verified**.
- The automatically generated invoice is based on the data from the settlement document, if you perform ERS.

i Note

You can edit the settlement document until the point at which it's posted to MM.

You can't cancel the settlement document when the invoice is posted against the settlement document in MM and the settlement document has a lifecycle status of **Invoice Verified**.

When you successfully cancel the settlement document, the posting date of the canceled SES is determined as follows:

- a. Original posting date of the SES, if both the MM posting period and FI posting period are open on the original posting date.
- b. Cancellation date of the freight settlement document if both the MM posting period and FI posting period are open on the cancellation date.

More Information

Subject	More Information
Enable the system to determine the business partners that you use in different business document types.	Business Partner Determination
Change management process for freight settlements	Automatic Change Management for Freight Settlements
Verify Invoices for TM documents	Integration: Invoice Processing (MM-IV/SD-BIL) with TM
Perform self-billing for suppliers for TM documents	Evaluated Receipt Settlement (ERS) (LO-INT-TM)
Make certain changes to your freight costs in the freight order, freight booking, or service order after you post accruals	Changes After Posting
Allow for unplanned changes in your freight settlement process	Reversing a Freight Settlement Document
Work with collective freight settlement documents, to combine the settlements for freight documents into a single freight settlement document	Split Criteria in Settlement Document Creation
Determine a rate for a collective settlement document	Collective Rate Determination

Freight Settlement Document

A freight settlement document is a document that is used for posting the freight cost accruals, and that forms the basis of invoice verification in Materials Management (MM) for freight documents such as freight orders, freight bookings, and service orders.

Use

The transportation charges are calculated in Transportation Management (TM) on the basis of a freight order, service order, or freight booking. TM relies on the integration to MM for the follow-on processes in financials. The freight settlement document forms the basis of representing the freight document cost from TM to MM. On posting the freight settlement document, a purchase order for services and a service entry sheet (SES) is created with the values from TM charge lines copied to these business documents. The SES confirmation also is done simultaneously, which creates the necessary financial entries for the value of the freight cost.

When the invoice is received, logistics invoice verification in MM checks it against the data replicated in MM from the freight settlement document. If you perform evaluated receipt settlement, the automatically generated invoice is also based on the data replicated in MM from the freight settlement document.

You can create an individual freight settlement document for one freight order, service order, or freight booking. You can create a collective freight settlement document for multiple freight orders or freight bookings. You can also create multiple freight settlement documents at the same time.

Structure

The freight settlement document data is taken from the business documents (for example, freight orders). However, you can still change some of the data.

The freight settlement document contains the following tab pages:

- **General Data**

The system displays general data, such as the following:

- Settlement document type
- Settlement date
- Procuring company code
- Purchasing organization and purchasing group
- Payment terms
- Net amount
- Document currency
- Verified invoice amount

You can change some of the data, such as the settlement date and payment terms.

- **Business Partner**

The system displays the carrier, invoicing party, and payee by default. You can use business partner determination to specify how the system assigns business partners to freight settlement documents.

- **Charges**

The system displays the transportation charges for each freight settlement document item. The freight settlement document item could be either referring to a business document (that is, freight order, service order, or freight booking), or to individual items of the business document, depending on the calculation level settings in the calculation profile. For more information about the calculation profile, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [Charge Calculation](#) > [Basic Settings](#) > [Define Calculation Profiles](#) . In the case of collective freight settlement documents, the system shows the total amount for all the freight orders, service orders, or freight bookings assigned to the freight settlement document.

- **Orders**

The system displays the list of business documents or business document items that are being settled with this freight settlement document, depending on the calculation level settings in the calculation profile. The list of all the freight orders, service orders, or freight bookings assigned to the freight settlement document is shown.

- **Document Flow**

The system displays the predecessor and successor business documents as a document flow, including the corresponding MM documents, such as the purchase order, SES, and invoice. You can access the business documents from the [Document Flow](#) tab of the document.

- **Application Log**

The system displays any error or warning messages that were encountered when the freight settlement document was posted to an internal MM system. This information can help you to correct the errors and post the freight settlement document successfully.

The life cycle status indicates the current status of the freight settlement document in the business process. Important life cycle statuses are listed in the following table:

Life Cycle Status	Description
In Process	The freight settlement document is not yet ready to be posted to MM.
Ready for Accruals	The freight settlement document is ready for you to post accruals in MM.
Posting Started	You have initiated posting of the freight settlement document and the freight settlement document posting process is not yet complete.
Posted	The freight settlement document has been posted successfully and the purchase order (PO) and service entry sheet (SES) have been created in MM.
Invoice Verified	At least one invoice has been posted against the freight settlement document in MM.
Cancellation Requested	You requested the cancellation of the freight settlement document but the cancellation process is not yet complete.
Canceled	You canceled the freight settlement document.
Reversal Requested	You have requested the reversal of the freight settlement document but the process of posting the reversal is not yet complete.
Reversed	The freight settlement document document has been successfully reversed in MM.
Credit Memo Created	A credit memo has been posted against a reversed freight settlement document in MM.

The confirmation status is used primarily to track the posting status of the freight settlement document to MM, and indicates the result of the last posting status. Important confirmation statuses are listed in the following table:

Confirmation Status	Description
Not Posted	The freight settlement document is not yet posted to MM.
Not Confirmed	The freight settlement document posting has been initiated but is not yet complete.
Confirmed	The freight settlement document has been posted successfully to MM.
Posting Failed	The posting of freight settlement document is not successful in MM.
Cancellation Posting Failed	The cancellation posting of freight settlement document is not successful in MM.
No Confirmation	You posted the freight settlement document to MM but and the status in MM is not yet known.
Reversal Failed	The reversal posting of the freight settlement document failed.
Purchase Order Creation Failed	The creation of the purchase order failed when the freight settlement document was posted. For information about the error, see the Application Log tab in the freight settlement document.
SES Creation Failed	The creation of the SES failed when the freight settlement document was posted. For information about the error, see the Application Log tab in the freight settlement document.
SES Cancellation Failed	The cancellation of the SES failed when the freight settlement document was canceled. For information about the error, see the Application Log tab in the freight settlement document.

Confirmation Status	Description
Purchase Order Cancellation Failed	The cancellation of the purchase order failed when the freight settlement document was canceled. For information about the error, see the Application Log tab in the freight settlement document.
Cost distribution document creation failed	The creation of the settlement management documents with the cost distribution data failed when the freight settlement document was posted. For information about the error, see the Application Log tab in the freight settlement document.
Cost distribution document cancellation failed	During the cancellation of the freight settlement document, the cancellation posting of settlement management documents with the cost distribution data failed. For information about the error, see the Application Log tab in the freight settlement document.
SES Reversal failed	During the reversal posting of the freight settlement document, the creation of the SES failed. For information about the error, see the Application Log tab in the freight settlement document.
Cost distribution document reversal failed	During the reversal of freight settlement document, the reversal posting of settlement management documents with the cost distribution data failed. For information about the error, see the Application Log tab in the freight settlement document.

Integration

The freight settlement document is integrated with a number of other components in TM, for example:

- [Freight Order Management](#)

You can start the generation of a freight settlement document from a freight order, service order, or freight booking.

- [Charge Management and Service Product Catalogs](#)

The system calculates the transportation charges based on the setup of master data objects in the Charge Management and Service Product Catalogs component, for example calculation sheets.

- [Agreement Management](#)

The system can use the freight agreement to include master data objects for transportation charges in the order process, such as calculation sheets or rate tables.

The freight settlement document also integrates with MM to create the purchase order (PO) and service entry sheet (SES) to post accruals, and to be the basis for invoice verification.

More Information

[Freight Settlement](#)

[Charge Calculation](#)

[Business Partner Determination](#)

Reversing a Freight Settlement Document

In your business you are a requester of transportation services, such as a shipper, and you have a close working relationship with a provider of transportation services, such as a carrier. You have enabled a self-billing (evaluated receipts settlement) process with a

particular service provider. You automatically pay the service provider according to the charge details in your freight order, freight booking, or service order.

Occasionally you can have the following business scenario:

1. Your carrier started to execute your transportation request.
2. You accrued the freight cost in Materials Management (MM).
3. You created a self-billing invoice to update the open item in your accounts payable.

Then your carrier encounters some unplanned issues. For example, the carrier does not deliver a particular transportation service due to execution issues such as a vehicle breakdown. Also, you may encounter incorrect pricing information in Transportation Management (TM)..

To resolve the issues, you need to reverse the freight settlement document in TM. When you transfer the reversal to MM, you need to reverse the associated purchase order, service entry sheet (SES), and accruals posting in MM. You need MM to then run the evaluated receipts settlement for the particular freight order. This posts a credit memo that reverses the self-billing invoice that was created earlier. You can only reverse the freight settlement document when it has the life-cycle status **Invoice Verified**.

The following table describes the steps you must take to reverse the freight settlement document:

Module	Action	Result	Life-Cycle Status of FSD
TM	Choose Reverse Document in the freight settlement document.	TM initiates the reversal process. The system creates an SES with a returns flag, for the complete value of the freight settlement document.	Reversal Requested
MM	System creates an SES in which it automatically enables the returns indicator.	System creates the reversal posting for the accruals in MM.	Reversed
MM	You run transaction MRRL (Evaluated Receipt Settlement (ERS) with Logistics Invoice Verification). Under the Transportation Management screen area, you enter the relevant TM freight order number.	This creates a credit memo to reverse the previously posted invoice.	Credit Memo Created

TM updates the invoicing status of the relevant freight order to **Not Invoiced**, and you can create a new freight settlement document for the freight order.

Related Information

[Evaluated Receipt Settlement \(ERS\)](#)

Company Code and Plant in Posting Freight Settlements

The company code in the freight settlement process is used to book the freight accruals to financial accounting (FI) under the company code. The plant is required in the posting process during the creation of the service purchase order as part of the posting step for a freight settlement document.

Procuring Company Code

The procuring company code is responsible for the procurement of a transportation service from the carrier or transportation service provider of a freight document. The procuring company code is available in the freight document under ►►[General Data](#)►►[Organizational Data](#)►.

If a unique company code has been assigned to a purchasing organization, the system uses this company code automatically as the procuring company code of the freight document. If the system cannot determine a unique company code, the field remains empty. Procuring company code determination is enabled only when you have specified in Customizing for the freight order type, freight booking type, or consignment order type that the freight document is relevant for subcontracting and have specified in Customizing for the freight order type, freight booking type, or service order type in the [Subcontract. Checks](#) field that the system is to perform basic additional checks.

If the procuring company code is defined in the freight document, it is transferred to the freight settlement document.

When the freight settlement document is posted, the system determines the company code and plant as follows during the creation of the service purchase order and the service entry sheet (SES):

- **Company Code**

The system determines the company code as follows:

- Procuring company code available in the freight settlement document header
- Procuring company code of the plant that is specified in Customizing for [Transportation Management](#) under ►►[Settlement](#)►►[Freight Settlement](#)►►[Integration for Settlement Posting](#)►►[Assign Purchasing Information for Posting](#)► to integrate to an internal Materials Management (MM) system.

To integrate to an external MM system on SAP S/4 HANA or SAP ERP, see the Customizing activities under ►►[Integration with Other SAP Components](#)►►[Transportation Management](#)►►[Invoice Integration](#)►►[Invoicing](#)►►[Mapping of Organizational Units](#)►.

- **Plant**

The system determines the plant to be populated in the purchasing order item as follows:

- The plant specified at freight settlement document type Customization to determine the plant and procuring company code. For more information see, Customizing for [Transportation Management](#) under ►►[Settlement](#)►►[Freight Settlement](#)►►[Integration for Settlement Posting](#)►►[Assign Purchasing Information for Posting](#)►►[Assign Purchasing Information for Posting](#)► to integrate to an internal MM system.

To integrate to an external MM system on SAP S/4 HANA or SAP ERP, see the Customizing activities under ►►[Integration with Other SAP Components](#)►►[Transportation Management](#)►►[Invoice Integration](#)►►[Invoicing](#)►►[Mapping of Organizational Units](#)►.

- The plant from the underlying transactions (orders and deliveries) if they are all unique from the linked freight units in the freight document.
- Use the plant defined as the default plant for the company code in Customizing for [Transportation Management](#) under ►►[Settlement](#)►►[Freight Settlement](#)►►[Integration for Settlement Posting](#)►►[Assign Default Plant for Company Code](#)► to integrate to an internal MM system.

To integrate to an external MM system on SAP S/4 HANA or SAP ERP, see the Customizing activities under ►►[Integration with Other SAP Components](#)►►[Transportation Management](#)►►[Invoice Integration](#)►►[Invoicing](#)►►[Mapping of Organizational Units](#)►.

i Note

If you have enabled in Customizing for the freight order type, freight booking type, or consignment order type that the freight document is relevant for subcontracting and in the **Subcontract. Checks** field that the system is to perform **Basic Additional Checks**, the company code is mandatory in the freight document and the same is taken over into the freight settlement document. In this case, SAP recommends that you don't specify the plant in the Customizing activity **Assign Purchasing Information for Posting**. This allows the plant to be determined (as described above) during the freight settlement document posting. This allows the company code and plant to be determined more flexibly and not to be tied to the freight settlement document type.

Related Information

[Company Codes](#)

Changes After Posting

Changes After You Post Accruals

You can include certain changes in the freight costs of a freight order, freight booking, or service order after you post accruals.

You can add or change charge amounts directly in the existing freight order, freight booking, or service order. You can transfer these changes in the freight costs to Materials Management (MM).

If automatic change management is enabled, you can use the background function **Creation and Transfer of Freight Settlement Documents** (program /SCMTMS/SFIR_CREATE_BATCH) to create, change, and post freight settlement documents to MM (see [Creation and Transfer of Freight Settlement Documents using Integration with MM External Services Management](#)). The freight settlement document contains the changes from the freight order. For more information, see [Automatic Change Management for Freight Settlements](#).

i Note

If automatic change management isn't enabled, the following conditions apply:

- If the new or changed amount results in the invoicing status of the freight order changing to **Partially Invoiced**, you can create a new freight settlement document to post the changed amounts to MM.
- If the new or changed amount results in the invoicing status of the freight order changing to **Over Invoiced**, you can create a credit memo in TM to send the changed amounts to MM.
- If you don't want to create a new settlement document or credit memo, you can cancel the existing settlement document provided the life-cycle status of the settlement document isn't **Invoice Verified**. You can then create a new settlement document that contains the complete amount including the changes.

When you post the credit memo or the new settlement document to MM, a new purchase order and service entry sheet (SES) are created.

Follow-Up Freight Settlement Document

If automatic change management is not enabled, you can create a second or subsequent freight settlement document after you have already created a settlement document. You can transfer these changes in the freight costs to Materials Management (MM) as follows::

1. Copy the charges from the freight order

Depending on the calculation option you specify in the relevant settlement profile, you can copy all the charge lines from the order to the settlement as they are, or you can copy the fixed charges and recalculate the other charges. For more

information, see Customizing for [Transportation Management](#) under [► Settlement ► Define Settlement Profile ►](#).

2. Calculate the invoiced amount

The system takes the following steps:

- For each charge line, subtract the invoiced amounts that are already included in the previously created but not canceled freight settlement documents
- Propose the result of the subtraction as the **Invoiced Amount** in the follow-up freight settlement document

This can result in the following invoiced amounts in the follow-up settlement document:

- An overall negative amount for the settlement document, where you have over-accrued costs for your carrier or logistics service provider (LSP)
- A zero amount where you have completely accrued the costs for your carrier or LSP.

To prevent you from posting the settlement document to MM until the zero or negative amount has been removed, the system assigns a consistency status of **Inconsistent** to the settlement document.

You can run the background processing function to create and transfer settlement documents (program /SCMTMS/SFIR_CREATE_BATCH) if you have enabled the change management process to correct the settlement documents automatically

If you have not enabled the automatic change management process, you can perform the following tasks:

- Manually cancel the existing settlement document and replace it with a new settlement document that contains the changed amount. The relevant freight order must not have a life cycle status of **Completed**.
- Manually create a credit memo for the over-accrued amount.

Related Information

[Automatic Change Management for Freight Settlements](#)

Dispute Management for Freight Settlements

Use

As a requester of transportation services, such as a shipper, you can have one of the following freight settlement processes in place with a provider of transportation services, such as a carrier:

- Self-billing

Your service provider does not submit an invoice. Instead, you settle automatically based on the information in your freight order. Self-billing is also named evaluated receipt settlement (ERS).

- Submission of an invoice by the service provider

You pay the service provider the amount contained in the invoice.

In both processes, you can have differences and potential disputes between what you expect to pay against what the service provider expects to be paid. To keep good business relationships, you need to work closely with your service provider to resolve these differences in a collaborative way. You need to get clarification on what the differences are, and converge on an agreement with the service provider on how to resolve the differences.

In Transportation Management (TM), you can use the invoice submission settlement process with enterprise services. You are a shipper who requests transportation services from your service provider. You can enable your service provider to use enterprise services to submit invoices for the freight orders that the service provider executes. The service provider creates an invoice document in their external system and uses enterprise services to send you the invoice. You use the external invoice to create a carrier invoice in the TM module of your S/4 HANA system.

Your service provider may have included additional unplanned freight costs or may have different charges to what you expect to pay in your freight order. Your TM system automatically evaluates the external invoice. It automatically creates a dispute case when your service provider submits an invoice that contains different values to what you expect. The system captures these differences in a dispute case. The system captures the individual differences in charges as dispute items.

On the dispute case screens, you as a shipper can approve or reject a dispute case. Open the [Dispute Cases Worklist](#) app from the launchpad. Choose [Freight Settlement](#) > [Dispute Case](#) > [Overview Freight Settlement Dispute Cases](#).

If you approve a dispute case, the system updates the freight order with information for the charge type, such as rate and quantity information. However, the system does not update logistics information, such as gross weight, gross volume, and total distance of the freight order.

When you resolve disputes early, you have the following business advantages:

- Inclusion of changes and unplanned costs in one invoice
- Accuracy in payment to the service provider

More Information

Subject	More Information
Structure of an individual dispute case, including working with a dispute case	Freight Settlement Dispute Case
Structure of an individual carrier invoice, including working with a carrier invoice	Carrier Invoice
Necessary Customizing settings required before you can use dispute management	Prerequisites for Dispute Management
Individual aspects of dispute management	Features of Dispute Management
Details of how freight cost accruals are updated in Materials Management (MM)	Automatic Change Management for Freight Settlements
Evaluated Receipt Settlement process	Evaluated Receipt Settlement (ERS) (LO-INT-TM)
Supplier invoice web services (B2B)	Supplier Invoice - Create (B2B, Inbound, Asynchronous) .

Dispute Management for Invoice Submission

Your logistics service provider may have included additional unplanned freight costs or have differences in the charges to what you expect in your freight order. Your system automatically evaluates the external invoice. It creates a dispute case when your service provider submits an invoice that contains different values to what you expect. Your system captures these differences in a dispute case. It captures the individual charge element differences as dispute items.

Your service provider creates an invoice document in their external system and uses enterprise services to send this invoice to you. You create a carrier invoice in the TM module of your S/4 HANA system.

The following are the typical steps that can happen in an invoice submission process:

- Transportation Management (TM) creates a freight settlement document for a freight order and posts the freight settlement document to Materials Management (MM). For example, you have specified an event in your organization where the system triggers the creation and transfer of a freight order when a freight order gets into execution.
- You can use the background processing function [Creation and Transfer of Freight Settlement Documents](#) to create and transfer freight settlement documents. For more information, see [Creation and Transfer of Freight Settlement Documents](#).
- On posting, purchase order and service entry sheet (SES) entries are created for the freight settlement document.
- Your service provider uses enterprise services to submit an invoice in line with the terms you have agreed with your service provider.

Transportation Management creates a carrier invoice in the shipper's system. The carrier invoice captures the invoice details submitted by your service provider. TM creates a dispute case if your service provider submits an invoice in which there are charges that are different from the charges in the original freight order. The system links the dispute case to the carrier invoice.

The system does not create a dispute case if your service provider does not submit any changes against the charge amounts or logistics data in the carrier invoice.

- You work closely with your service provider and either accept or reject the dispute case for the carrier invoice. When the carrier invoice is successfully posted to Materials Management (MM), the life cycle status of the carrier invoice is set to **Accepted**.
 - You reject the carrier invoice when you do not agree with the service provider on the carrier invoice. If the carrier invoice has an associated open dispute case, the system closes the dispute case when you reject the carrier invoice. You can add a note to the carrier invoice to record the reason for your rejection.

The life cycle status of the carrier invoice is set to **Rejected**.

- You accept the dispute case for the carrier invoice.

TM updates the freight order with the new or changed charges that are contained in the carrier invoice. You can use the background processing function [Creation and Transfer of Freight Settlement Documents](#) to update the new or changed charges for the freight orders to MM.

- You use the background processing function [Posting of Carrier Invoices](#) to transfer the carrier invoices that are ready for posting to Materials Management (MM). For more information, see [Posting of Carrier Invoices](#). The system creates an incoming invoice in MM. The system posts the carrier invoices to MM for posting only when the following conditions are satisfied:
 - No dispute case exists or the dispute case has a life cycle status of **Resolved**.
 - The carrier invoice has a life cycle status of other than **Rejected**.
 - All the freight orders associated with the carrier invoice have the invoicing status **Completely Invoiced**.
 - All the freight settlement documents associated with the freight orders have the life cycle status **Accruals Posted**.
 - The carrier invoice has a block status of **Not Blocked**.
 - The carrier invoice has a consistency status of **Consistent**.

More Information

Subject	More Information
Structure of an individual carrier invoice, including working with a carrier invoice	Carrier Invoice
Necessary Customizing settings required before you can use dispute management	Prerequisites for Dispute Management
Individual aspects of dispute management	Features of Dispute Management
Details of how freight cost accruals are updated in Materials Management (MM)	Automatic Change Management for Freight Settlements

Freight Settlement Dispute Case

Use

A freight settlement dispute case is an individual business document that captures differences in logistics item quantities or charge amounts for a freight order or a carrier invoice. As a requester of transportation services, such as a shipper, you own the information in the freight order.

In the invoice submission process, the process of settlement between you and your service provider is based on an invoice that your service provider submits to you for a freight order. For example, your service provider can submit an invoice that contains changes to logistics details such as gross weight or gross volume, or changes to charge details such as rate or an additional charge line for an unplanned charge. In these situations, the system captures the differences in a dispute case and links the dispute case to the invoice your service provider submits.

If the dispute case fails the tolerance limits you specify in Customizing, you must manually review the dispute case on the **Freight Settlement Dispute Cases** user interface (UI).

Structure

General Data

The **General Data** tab page contains high-level information about the dispute case and associated freight order or carrier invoice, for example:

- Carrier name, carrier invoice reference, and freight order reference
- Source location and destination location
- Logistics information, such as gross weight, gross volume, and total distance at freight order level
- Status information
- Amount information, at freight order level

Logistics Details

On the **Logistics Details** screen area, the system displays logistics dispute information for a dispute item submitted by the service provider. For example, your carrier claims the total distance is 510 km, and not 500 km as you expect. The system displays 500 km in the **Order Quantity** field, and 510 km in the **Dispute Quantity** field.

The system also displays the dispute resolution status of each logistics item in a dispute case.

i Note

For an invoice submitted through an enterprise service, these details are not captured because the logistical details for the order are not part of the enterprise service.

Charge Details

On the [Charge Details](#) screen area, the system displays the charge dispute information for a dispute item submitted by the service provider. For example, your carrier adds an unplanned cost for 4 hours detention at the rate of USD 50 per hour, totaling USD 200. The system displays the following information on the [Charge Details](#) screen area:

- [Dispute Item Description: New Charge Line](#)
- [Charge Description: Detention](#)
- [Currency: USD](#)
- [Order Amount: 0](#)
- [Dispute Amount: 200](#)

The system also displays the dispute resolution status of each charge item in a dispute case.

Dispute Charge Amounts

The system displays the proposed charge amount from the service provider for the total freight order or carrier invoice on the [Dispute Charge Amounts](#) screen area. This is what your service provider expects to be paid in total and is how the freight order or carrier invoice would look if you accept the proposal from your service provider.

In an invoice submission process, your service provider adds the unplanned cost for detention of USD 200 to the carrier invoice. The system displays USD 5,200 in the [Dispute Charge Amounts](#) screen area of the carrier invoice.

Proposed Charge Amounts

You accept or reject the dispute. The system displays the effect your decision has on the total charges in the freight order.

For example, the total order amount is USD 5,000. This is what you expect to pay your carrier. Your carrier submits a dispute case for an unplanned cost for detention of USD 200. You reject the dispute item. The system displays USD 5,000 in the [Dispute Charge Amounts](#) screen area.

Taking Action on a Dispute Case

For dispute cases linked to the carrier invoice created using an enterprise service, you as a requester of the transportation service can perform the following tasks:

- Choose [Approve All](#) to accept all the dispute items. This means that you accept the carrier invoice sent by the carrier.
- Choose [Reject Invoice](#) to reject the invoice from the carrier. This also closes the dispute case. The dispute case has the life-cycle status [Rejected](#).

In both the [Logistics Details](#) and [Charge Details](#) screen areas, you as a requester of transportation services can perform the following tasks:

- Choose the [Approve](#) button to accept a dispute item.
- Choose the [Reject](#) button to reject a dispute item.

For information about dispute case statuses, see [Statuses in Dispute Management](#). You can also save your work. This enables you to save a draft while you finalize your decisions. When you save and confirm a dispute case, you finalize your action on the dispute case.

You can also use the **Publish Freight Settlement Dispute Cases** background processing function (program /SCMTMS/SET_CARRIER_RELEVANCE) to automatically save and confirm the dispute cases that pass the tolerance check criteria. With this function, the system uses the charge details from a dispute case that is resolved to update the changed charge information in the freight order. For more information, see [Publication of Freight Settlement Dispute Cases](#). The system publishes your decision for your service provider.

Notes

On this tab page, you can add any relevant notes, enabling you to share information all through the process of dispute resolution. You can use notes to collaborate with your colleagues internally.

Attachments

On this tab page, you can upload relevant files, enabling you to share files all through the process of dispute resolution. You can use attachments to share information with your colleagues internally.

Tolerance Check Log

On this tab page, the system displays the following information:

- Success messages for the charge type dispute items that are inside your tolerance limits and that the system passed automatically
- Failure messages for the dispute items that are outside your tolerance limits and that the system failed automatically

The system displays the dispute cases that failed the automatic tolerance check on the **Freight Settlement Dispute Cases - All Freight Settlement Dispute Cases** personal object worklist (POWL) screen. You must manually review these dispute cases.

You can specify tolerance limits in Customizing for **Transportation Management** under **► Settlement ► Freight Settlement ► Freight Settlement Dispute Management ► Define Dispute Tolerance Rules ►**.

Work Items

The system displays information on the work items created for the dispute case, including the approval level, the status of the work item, the approver, and the time key events in the lifecycle of the work item take place. For more information, see [Workflow for Dispute Management](#).

Integration

- Freight settlement dispute cases are integrated with freight orders. When you accept a change in a dispute case, the system uses the dispute resolution details from a dispute case to update the changed charge information in the freight order. For more information, see [Freight Order Management](#).

For the freight orders for which you have resolved disputes, you use the background processing function **Creation and Transfer of Freight Settlement Documents** to update the changed charges to materials management (MM).

The system posts the freight settlement documents, which are created from freight orders, to Materials Management. It does not create a freight settlement document from a dispute case. For more information about integration, see [Integration: Invoice Processing \(MM-IV/SD-BIL\) with TM](#).

- You can use an enterprise service to create a freight settlement dispute case from outside Transportation Management (TM) in a self-billing scenario. You can send the details of the dispute case outside TM as an enterprise service message. For

more information, see <Provide link to the enterprise services documentation of the dispute case>.

More Information

Subject	More Information
Introduction to dispute management	Dispute Management for Freight Settlements
Necessary Customizing settings required before you can use dispute management	Prerequisites for Dispute Management
Individual aspects of dispute management	Features of Dispute Management
Details of how freight cost accruals are updated in MM	Automatic Change Management for Freight Settlements
Exchange the charge information that's under dispute between a carrier and a shipper	Transportation Dispute Case Notification

Carrier Invoice

Use

In an invoice submission process, the process of settlement between you and your service provider is based on a carrier invoice that your service provider submits to you against one or more freight orders. Your service provider has executed the transportation requirements in your freight order before sending the invoice.. You have a trusted relationship with your service provider, and your service provider sends you a carrier invoice.

Prerequisites

You must specify a carrier invoice type in Customizing for **Transportation Management** under ► **Settlement** ► **Freight Settlement** ► **Define Carrier Invoice Types** ►.

You must specify a carrier invoice type as the default type.

You must specify the charge types for tax by enabling the **Tax** setting in the **Define Charge Types** Customizing activity under ► **Transportation Management** ► **Basic Functions** ► **Charge Calculation** ► **Basic Settings** ►.

You have specified the following settings in Customizing for **Transportation Management** under ► **Integration** ► **Enterprise Services** ► **Carrier Invoice Integration** ►:

- Map the condition type received in the interface to the TM charge type, in the Customizing activity **Assign Carrier Condition Type to TM Charge**.
- Assign the TM charge type for an invoice that TM processes, under the Customizing activity **Assign Carrier Tax Types to TM Tax Charge Type**.

To post a carrier invoice, you must assign the tax code to the TM charge type in the **Assign Tax Code to Charge Types** Customizing activity under ► **Transportation Management** ► **Integration** ► **Enterprise Services** ► **Carrier Invoice Integration** ►. The system uses this setting when it posts the carrier invoice to an internal materials management (MM) system.

To post the carrier invoice to an external MM system running on SAP S/4 HANA or SAP ERP, you must specify the mapping settings in Customizing under ► **Integration with Other SAP Components** ► **Transportation Management** ► **Invoice Integration** ► **Tax** ► **Map SAP TM Tax Setting** ►.

Integration

The carrier invoice is integrated with the following objects and systems:

- Freight settlement dispute case

Your service provider can submit an invoice for one or more freight orders.

Transportation Management (TM) captures the invoice details submitted by your service provider. If there are differences between the submitted invoice and the expected values for the freight order, TM creates a dispute case and links it to the carrier invoice. If your service provider submits an invoice that matches the expected value for the freight order, TM does not create a dispute case.

- Posting of the carrier invoice to MM

You can use the [Posting of Carrier Invoices](#) background function to post the carrier invoices that are ready for posting to MM. MM creates incoming invoices. For more information, see [Posting of Carrier Invoices](#).

You can also post one or more carrier invoice directly as follows:

- Under [Freight Settlement](#), choose [Worklist](#) > [Overview Freight Settlement Documents](#) > [Carrier Invoices](#) >.
- Under [Freight Settlement](#), choose [Carrier Invoice](#) > [Display Carrier Invoice](#) >.

You can also automate the carrier invoice posting directly by enabling [Post Invoice Immediately](#) for the carrier invoice type in Customizing for [Transportation Management](#) under [Settlement](#) > [Freight Settlement](#) > [Define Carrier Invoice Types](#) >.

The system determines the logical system behind the business system group that you have linked to your purchase organization data for the freight order, freight booking, or service order that is a part of the carrier invoice. The business system group represents the SAP ERP system or SAP S/4 HANA system from which you replicate the purchasing organization in the organization model.

If the logical system is different to the current client logical system, the system uses enterprise services to post the carrier invoice to an external MM system. If the logical system is the same as the current client logical system, the system posts the carrier invoice to an internal MM system.

For more information on organizational model see [Create and Merge Organizational Hierarchy](#) under [Transportation Management](#) > [Master Data](#) > [Organizational Management](#) > [Organizational Model](#) >.

You use the InvoiceRequest_In interface to create a carrier invoice in TM. For more information see [Carrier Invoice - Create \(B2B\)](#).

Features

Possible Actions for Carrier Invoice

The following table explains possible actions that are available to you:

Action	Action Description
Reject Invoice	You can reject a carrier invoice at any time, provided the carrier invoice is not yet posted to MM. When rejecting a carrier invoice, you can capture notes giving your reasons for rejection. Your service provider must create a new carrier invoice for the freight order. When you reject a carrier invoice that has an open dispute case, the system automatically closes the dispute case. This saves you from taking extra steps to cancel the dispute case manually.

Action	Action Description
Post	You can post the carrier invoice to MM provided the carrier invoice satisfies all of the following conditions: • No dispute case exists or the dispute case has a life cycle status of Resolved • Has a life cycle status of other than Rejected • All associated freight orders have the invoicing status Completely Invoiced • All associated freight settlement documents have the status Accruals Posted • Has a block status of Not Blocked
Release	Releases the carrier invoice in the life cycle status Draft. When there are carrier invoices with charges linked to the resolution base of the freight order, the carrier invoice is created in life cycle status Draft. When the resolution base information is assigned to the repeating charges in the carrier invoice, this action can be used to release the invoice for further processing.
Set to Draft	Recall a released invoice in case you want to change the resolution base information for the charge lines. This action cancels the relevant dispute case for the associated carrier invoice.

Life Cycle Status

The life cycle status indicates the resolution status of the overall carrier invoice. The following table explains what the key life cycle statuses mean:

Status	Status Description
New	Your service provider has not yet submitted the carrier invoice.
Draft	When there are carrier invoices with charges linked to the resolution base of the freight order, the carrier invoice is created in life cycle status Draft .
In Process	The carrier invoice satisfies one or more of the following conditions: • All the charge lines where you expect resolution base information have been resolved. • Submitted by your service provider but not yet accepted and posted to MM • Has disputes that are not yet resolved • Not yet posted to MM • Transferred, but failed to post to MM
Ready for posting	The carrier invoice satisfies all of the following conditions: • No dispute case exists or the dispute case has a life cycle status of Resolved. • Has a life cycle status of other than Rejected • All associated freight orders have the invoicing status Completely Invoiced. • All associated freight settlement documents have the status Accruals Posted. • Has a block status of Not Blocked • Has a consistency status of Consistent
Posting started	You have transferred the carrier invoice for posting to MM.

Status	Status Description
Carrier invoice posted	The carrier invoice has been successfully posted as an invoice document in MM.
Rejected	You rejected the carrier invoice for business reasons. For example, the tax amounts in the invoice were incorrect

Dispute Case Status in a Carrier Invoice

In an invoice submission process, your service provider submits a carrier invoice against one or more freight orders. You service provider can also submit an invoice that has a different charge amount to what you expect as a shipper. TM subsequently creates a dispute case so that you can come to a consensus on the disputed items with your service provider. The dispute case status reflects the processing status of the dispute on the carrier invoice. The following table explains what the individual dispute case statuses mean:

Status	Status Description
No Dispute	The default status of a carrier invoice, where there are no dispute cases on the charges or logistics information in the freight orders covered by the carrier invoice.
Awaiting Your Approval	You as the service requester must act to approve, make a proposal, or reject the dispute created by your service provider.
Resolved	Both you as the service requester and your service provider have agreed on common ground in the dispute and you both have agreed on a decision. You have accepted the dispute case.
Rejected by You by Carrier Invoice	You as a service requester reject a carrier invoice from your service provider. This closes the associated dispute case.

Block Status

The block status prevents the carrier invoice from being posted to MM. For example, your service provider can use an invoice number reference from their own transportation system, such as a forwarding order number, a waybill number, or an invoice number. When your service provider submits a carrier invoice, TM checks to ensure that your service provider has not used the same invoice number reference ID to submit an invoice previously in the same day. If your service provider has used the same invoice number reference previously in the same day, as standard TM blocks the carrier invoice and displays the reason [Invoice number reference already used](#) on the carrier invoice screen. This protects you from creating duplicate invoices when you submit invoices electronically, for example, when you use machine-to-machine (M2M) messages or B-to-B (B2B) messages.

You can specify your own additional blocking checks. For example, you can block a carrier invoice if the freight orders in the carrier invoice have a delivery date that is later than the current date. As the shipper, this enables you to pay for a transportation only when all of the parts of the transportation are completed. For more information, see Customizing for [Transportation Management](#) under [Business Add-ins \(BAdIs\) for Transportation Management](#) [Settlement](#) [Freight Settlement](#) [Freight Settlement Dispute Management](#) [BAdI: Additional Checks for Blocking Carrier Invoices](#).

You cannot further process a carrier invoice when it is blocked. To unblock the carrier invoice, choose [Unblock](#) on the carrier invoice screen. For more information on block statuses, see Customizing for [Transportation Management](#) under [Basic Functions](#) [General Settings](#) [Define Block Reason Codes](#).

Confirmation Status

You can use the confirmation status to track the posting of a carrier invoice to MM. The status indicates the result of the last integration interaction between TM and MM. The following table explains what the main confirmation statuses mean:

Status	Status Description
Not Posted	TM has not posted the carrier invoice to MM.
Not Confirmed	TM has initiated the posting of the carrier invoice to MM, but the posting process is not yet complete.
Confirmed	Carrier invoice was posted successfully and an invoice was created in MM.
Invoice Posting Failed	Posting of carrier invoice could not be completed due to errors.

Tax Handling

You must specify the charge types that represent tax by selecting the **Tax** checkbox in Customizing under ► **Transportation Management** ► **Basic Functions** ► **Charge Calculation** ► **Basic Settings** ► **Define Charge Types** ►.

When you post the carrier invoice to an internal MM system, you must assign a tax code to a TM charge type in the **Assign Tax Code to Charge Types** Customizing activity under ► **Transportation Management** ► **Integration** ► **Enterprise Services** ► **Carrier Invoice Integration** ►.

When you post the carrier invoice to an external MM system running on SAP S/4 HANA or SAP ERP, you must specify the mapping settings in the **Map SAP TM Tax Settings** Customizing activity under ► **Integration with Other SAP Components** ► **Transportation Management** ► **Invoice Integration** ► **Tax** ►.

TM does not verify the tax amounts in the carrier invoice. Instead, MM verifies the tax amounts when you post the invoice in MM.

If there is a discrepancy between what your carrier has submitted and what MM expects to pay, MM does not post the carrier invoice. You must manually reject these carrier invoices in TM. You can find the carrier invoices that contain a discrepancy by filtering the following statuses:

- Life cycle status **In Process**
- Confirmation status **Invoice Posting Failed**

More Information

Subject	More Information
Background processing function for posting carrier invoices to MM	Posting of Carrier Invoices
Individual aspects of dispute management	Features of Dispute Management
Process a carrier invoice using Web Services	Carrier Invoice - Create (B2B)

Processing of a Carrier Invoice Using a B2B Interface

Use

You have a trusted relationship with your service provider, and your service provider sends a carrier invoice to you electronically using the B2B message interface for one or more freight orders. Transportation Management (TM) captures the invoice details submitted by your service provider. If there are differences between the submitted invoice and the expected values for the freight

order, TM creates a dispute case and links it to the carrier invoice. If your service provider submits an invoice that matches the expected value for the freight order, TM doesn't create a dispute case.

The [InvoiceRequest_In](#) interface is used to process a carrier invoice that refers to a document such as a freight order, a service order, or a freight booking in TM. For more information on the B2B interface, see [Carrier Invoice - Create \(B2B\)](#).

Process Flow

1. You as a shipper use a freight order, freight booking, or service order to request a transportation service from your carrier.
2. Your carrier provides the transportation services.
3. The carrier creates the invoice document in their external system and sends you the invoice through Web Services.
4. Your system creates a carrier invoice in the TM module.
5. Your carrier may include an additional unplanned freight cost. The carrier may send an invoice that differs in a charge amount to what you expect in your freight order.

If your carrier submits an invoice that contains differences in values to what you expect, your system automatically evaluates the incoming invoice and creates a dispute case. The system captures the individual charge element differences as dispute items.

You resolve the dispute. For more information, see [Dispute Management for Invoice Submission](#).

6. If the values in the invoice your carrier sends match the values you expect in your freight order, the system doesn't create a dispute case.

You resolve the disputes by accepting the carrier values if they are acceptable to you and are for the correct reasons. For example, the carrier invoice contains waiting as an additional charge. The system creates a dispute for this invoice. As part of the dispute resolution you find that there was a waiting time for the carrier and hence the carrier's claim for the waiting charge is correct. You accept this carrier invoice including the waiting charges.

Alternatively, the carrier invoice doesn't contain any disputes. You post the carrier invoice to material management (MM). For more information see [Carrier Invoice](#).

Restrictions

You as a shipper have the following restrictions in processing the carrier invoice and credit memo in TM. In all cases, the system doesn't process the carrier invoice or credit memo and creates an error against the B2B message. You must work with your carrier to resolve these errors. Your carrier must then resend a new error-free B2B message.

Complete Cancellation

TM only supports a credit memo that cancels the complete invoice that the carrier previously sent.

A credit memo that contains partial amounts from the previously sent invoice, or with different charge types than in the previously sent invoice, causes an error. The system doesn't process the credit memo.

For example, your carrier sends you an invoice with the reference INV984500 for USD 1000. This comprises of a base freight amount of USD 975 and a loading charge of USD 25. You accept the invoice. Later, the carrier realizes that the invoice is incorrect. The loading charge isn't applicable for the freight order that is part of the invoice INV984500.

If your carrier uses the B2B message to send you the credit memo for the credit on the loading charge of USD 25, the system doesn't process the credit memo and creates an error. Instead, your carrier must send a complete credit memo for all the previously sent invoice. Your carrier must then send you a corrected invoice. In this example, your carrier must send a credit memo

with a canceled invoice ID of INV984500 for the whole amount of USD 1000. The carrier must then send second corrected invoice for USD 975 for the base freight charge of USD 975.

Repeating of Condition Types

The system doesn't process invoices for a freight order, freight booking, or service order in which the TM charge calculation contains a repeating charge line that is not due to the calculation resolution base. Instead, the system creates an error. The interface doesn't support an invoice with a repeating condition type for an item in an invoice.

For example, you have a freight order for a less-than-truck-load (LTL) shipment that consists of two packages. You agree with your carrier that the carrier charges you a freight charge for each package and a handling charge for loading and unloading services. You have an agreement with the carrier in which the system calculates charges for the freight order with header-level calculation and has freight charges for each package and a handling charge for the loading and unloading. You expect the carrier to send you an invoice as follows:

Charge Type	Amount (USD)	Explanation
Handling charge	5	Due to loading
Handling charge	5	Due to unloading
Freight charge	25	For package 1
Freight charge	40	For package 2
Total	75	

The freight charge is repeated for this freight order, and the invoice does not cause an error due to this charge as the system handles such invoices for charges which repeat due to the resolution base. However, the system cannot process the handling charges in the invoice above that are repeating and are not with a resolution base. Instead, the system creates an error in inbound processing.

Through-Rate Calculation

The system creates an error and doesn't process an invoice for a TM freight order or freight booking in which the TM charge calculation is **Through Rate**. For more information, see [Through Rates](#).

For example, you agree stage-level calculation with your carrier. You have a freight order with following transportation stages:

Stage ID	Stage
10	Stage A to stage B
20	Stage B to stage C

Your carrier gives you a through rate for a freight order from stage A to stage C. The system doesn't process the carrier invoice and creates an error if your carrier submits the freight order through the B2B interface.

Partial Charge Line Resulting in a Dispute Case

The system creates a dispute case when your carrier submits an invoice that contains a partial charge line.

For example, your carrier sends you an invoice with a loading charge of USD 25. You expect the loading charge to be USD 50. The system captures this deviation as a dispute case. It displays the information on the **Dispute Charge Amounts** screen area of the dispute case.

Different Currencies

The system processes an invoice sent by your carrier in the same currency as that which you expect for the freight document. If the carrier sends you an invoice with a different currency than you expect, the system creates an error and doesn't process the invoice.

For example, you expect your carrier to send you an invoice in the currency USD. If the carrier sends the invoice through the B2B interface in the currency EUR, the system doesn't process the invoice document and creates an error.

Related Information

[Carrier Invoice](#)

[Carrier Invoice - Create \(B2B\)](#)

[Working with Resolution-Based Charges](#)

Working with Resolution-Based Charges

You have a freight order for a less-than-truck-load (LTL) shipment that consists of two packages. You agree with your carrier that the carrier charges you a freight charge for each package and a loading and handling charge for the transport. You have an agreement with your carrier in which the system calculates charges for the freight order with header-level calculation and has freight charges for each package and handling charge for the loading and unloading. You expect the carrier to send you an invoice as follows:

Charge Type	Amount (USD)	Explanation
Handling charge	5	Due to loading and unloading
Freight charge	25	For package 1
Freight charge	40	For package 2
Total	70	

The InvoiceRequest_In interface is used to process a carrier invoice, and a carrier credit memo that refers to such a freight order cannot carry the reference of the packages for each charge. When these carrier invoices or credit memos are processed, the system handles these invoices in the following way:

- Creates a carrier invoice that contains charge lines with a resolution base in the life cycle status **Draft**.
- The carrier invoice or carrier credit memo can only be processed when the logistical references for the charge lines for the resolution base are assigned. The assignment can be done by the system automatically or by the user manually.

To enable the automatic determination of the logistical references, the **Auto Prop Res Base Charge Line** is enabled in the settlement profile.

To manually assign the logistical reference, you can navigate to the carrier invoice or carrier credit memo by using the worklist **Freight Settlement Worklist** **All Carrier Invoices in Draft Status** or **All Carrier Credit Memos in Draft Status**. In the carrier invoice or carrier credit memo, you can assign the logistical reference by going to the screen area **Charges with Logistical Refs**. Alternatively, you can trigger the automatic determination of the logistical references by using the action **Propose Logistical References**.

- When the logistical references are assigned to the charge lines, the carrier invoice can be released for further processing.
- Once the carrier invoice is released, the logistical reference assigned to the charge lines in the carrier invoice can be changed by setting the carrier invoice life cycle status to **Draft** using the action **Set to Draft**. Any associated open dispute

case with the carrier invoices are cancelled on executing this action.

The processing of a carrier credit memo with charge lines that contain a resolution base is done in a similar way.

i Note

The following criteria are fulfilled for a carrier invoice that is set to **Draft** status by the user.

- Only a carrier invoice and a carrier credit memo with charge lines expecting logistical references due to the resolution base.
- Only a carrier invoice and carrier credit memo that are not posted.
- Only a carrier invoice with a dispute case, where the life cycle status of the dispute case is not in status **Resolved**.

A carrier credit memo cannot be set to **Draft** after it has been released.

Related Information

[Determine Logistical References Automatically](#)

Determine Logistical References Automatically

The automatic match algorithm is used to automatically propose the logistical references for charge lines with missing logistical references in the carrier invoice and carrier credit memo. The charges in the freight document that are modelled using a resolution base expect the logistical reference to which the charges are referenced. The logistical reference determination in the carrier invoice and carrier credit memo can be enabled as follows:

- Enable **Auto Prop Res Base Charge Line** in the settlement profile.
- In the carrier invoice or carrier credit memo, you can trigger the automatic determination of the logistical references by using the action **Propose Logistical References**.

The following steps show how the feature works:

1. In the carrier invoice, each charge line that expects a logistical reference is compared with respective charges in the freight document based on matching a charge line with the same charge type with a rate amount, rate currency, price per unit, UoM, final amount, final amount currency, and quantity. If a match is found, then the logistical reference is assigned from the charge line of the freight document to the charge line in the carrier invoice.

For the carrier credit memo, the charge lines in the carrier credit memo are compared with the charge lines in the corresponding carrier invoice referenced by the credit memo in the field **Cancelled Supplier Invoice ID**.

2. In the carrier invoice, if there is no complete match found, then in the carrier invoice, each charge line that expects a logistical reference is compared with respective charges in the freight document based on matching with the same charge type with a final amount, final amount currency, and quantity.
3. If the above exact matches are not feasible due to invoices with a variance or missing information from the carrier such as quantity, the system tries to do a near match. In the carrier invoice, each charge line that expects a logistical reference is compared with respective charges in the freight document based on the matching with the same charge type. If there are more than one charge lines with the same charge type in the freight document, the invoiced amount and currency of the charge line in the carrier invoice is compared with the final amount and currency of the charge line in the freight document and the logistical reference of the freight document charge line which has the least difference with the carrier invoice charge line is assigned to the charge line of the carrier invoice.

The following tables contain an example that explains how the feature works:

You have freight order 810002, that contains the following cargo items:

Cargo Item	Cargo Item Details	Quantity	UoM	Gross Weight	UoM
Package	Pallet 1	50	PC	10	KG
	Product 1	50	PC	10	KG
	Product 2	5	PC	20	KG
	Product 3	10	PC	25	KG

Master data (calculation sheet):

Charge Type	Resolution Base	Rate Table/Rate Value	Currency
Base charge	Product	RT_BASE_WT	USD
Loading charge		10	USD
Packaging charge	Package	RT_PACK_PTYPE	USD

Freight order 810002 has the following charges:

Line Number	Charge Type	Rate (USD)	Price per Unit	Quantity	Logistical Reference	Final Amount (USD)
10	Base charge	10	/1 KG	10 KG	Product 1	100
20	Base charge	10	/1 KG	20 KG	Product 2	200
30	Base charge	10	/1 KG	25 KG`	Product 3	250
40	Loading charge	10				10
50	Packaging charge	50	/1 KG	1 KG	Pallet 1	50

The carrier invoice for freight order 810002 is received with the following charges through the enterprise service. As the [InvoiceRequest_In](#) interface does not carry the logistical references for the charge lines, the logistical references will be empty in the carrier invoice:

Carrier Invoice # CAR10209

Line	Charge Type	Rate (USD)	Price per Unit	Quantity	Logistical Reference	Invoice Amount (USD)
1	Base charge	10				100
2	Base charge	10				210
3	Base charge	10				255
4	Loading charge	10				10

Line	Charge Type	Rate (USD)	Price per Unit	Quantity	Logistical Reference	Invoice Amount (USD)
5	Packaging charge	50				60

On executing the automatic logistical reference proposal, the system identifies the three charge lines for the base charge and one charge line for the packaging charge that require the logistical references to be populated in the carrier invoice. The system runs as follows on the charge line to propose the logistical reference:

1. For the first base charge line in the carrier invoice for 100 USD, the system tries to find the matching charge line in the freight order 810002, based on the charge type, rate, price per unit, quantity, and invoice amount. As in this case there is no such match, the charge line is processed for a near match in the next steps.

For the second and third base charge lines, as there is no exact match based on the charge type, rate, price per unit, quantity, and invoice amount, the charge lines are processed for a near match in the next steps.

2. For the packaging charge, the same matching logic is run and also the system doesn't find a matching value. The charge is processed for a near match in the next steps.
3. The system tries to find a match by comparing the three base charge lines based on the charge type individually from the carrier invoice to the charge lines in the freight order. In this case for the first base charge line in the carrier invoice, the system finds the match to the charge type and invoice amount to the charge type and final amount in the freight order referring to the product one, the system proposes the product one as the logistical reference for the carrier invoice base charge line in the carrier invoice. For the second and third base charge lines, the amounts do not match with the freight order charges. For the packaging charge also, there is no match to the packaging charge line in the freight order based on the charge type and amount comparison. These charge lines will be processed further in the nearest match step.
4. The second base freight charge line of USD 210 in the carrier invoice is compared with the base charges in the freight order, base charge 200 (line 20) and base charge 250 (line 30). For the carrier invoice for the second base charge line, the system determines the nearest match to find the charge line with the least deviation. In this case, the system finds that the charge line in the freight order for product two is with the least amount of deviation of 10 USD when compared to the base charge for product three in the freight order, which has a deviation of 40 USD.

The third base freight charge line of USD 255 is compared with the only remaining base freight line and the logistical reference of product three is assigned to the charge line in the carrier invoice. For the packaging charge, as there is only one packaging charge in the freight order, the logistical reference of the charge line is assigned to the charge line in the carrier invoice, in this case pallet one.
5. The system does not run the automatic logistical reference process on the loading charge as the loading charge does not have a logistical reference in the freight order.

After the system runs the process, it updates the carrier invoice as follows:

Carrier invoice CAR10209

Line	Charge Type	Rate (USD)	Logistical Reference	Invoice Amount (USD)
1	Base charge	10	Product 1	100
2	Base charge	10	Product 2	210
3	Base charge	10	Product 3	225
4	Loading charge	10		10
5	Packaging charge	50	Pallet 1	60

For the carrier credit memo, the cancelled supplier invoice ID is used as a reference to trace the credit memo to the supplier invoice it is cancelling. The missing logistical references are established by the matching logic of the charge line information between the carrier credit memo and the carrier invoice, based on exact charge line attributes such as charge type, rate, quantity, and credit amount, or based on the charge type and credit amount.

For example, you have carrier credit memo CMCAR9002 with cancelled supplier invoice ID CAR10209:

Line	Charge Type`	Rate (USD)	Credit Amount (USD)
1	Base charge	10	100
2	Base charge	10	210
3	Base charge	10	255
4	Loading charge	10	10
5	Packaging charge	50	60

On executing the automatic logistical reference proposal process on the carrier credit memo, the system identifies the three charge lines for the base charge and one charge line for the packaging charge. These require the logistical references to be populated in the carrier credit memo does the following:

The system runs the following steps on the charge line to propose the logistical reference:

1. For the first base charge line in the carrier credit memo, the base charge for USD 100, the system tries to find the matching charge line in carrier invoice CAR10209, based on the charge type, rate, price per unit, quantity, and invoice amount. As the exact matches are found for the first base charge line, the logistical reference to product one from the carrier invoice is copied over. The system does the same matching for the other two base charge lines and packaging charge, resulting in product two, product three, and pallet one being assigned as the logistical references.
2. The system does not run this process on the loading charge as the loading charge does not have any logistical reference in the carrier invoice.

After running the automatic logistical reference proposal, the system makes the following updates, for carrier credit memo CMCAR9002 and cancelled supplier invoice CAR10209:

Line	Charge Type	Rate (USD)	Logistical Reference	Credit Amount (USD)
1	Base charge	10	Product 1	100
2	Base charge	10	Product 2	210
3	Base charge	10	Product 3	225
4	Loading charge	10		10
5	Packaging charge	50	Pallet 1	60

i Note

1. You cannot add a logistical reference to a carrier invoice that does not exist in the freight document charge.
2. If there are additional charge lines that cannot be assigned a logistical reference in the carrier invoice, the carrier invoice remains in the draft status and cannot be processed.

Prerequisites for Dispute Management

Settlement Profile Settings

In Customizing for **Transportation Management** under **► Settlement ► Define Settlement Profile ►**, you can specify the freight settlement profile settings. You have the option of specifying the following settlement profile settings:

- Approval workflow to enable you to use an approval workflow for all freight settlement dispute cases that fall outside the specified tolerances
- Tolerance groups to enable you to specify lower and upper limits on disputed charge type amounts and total freight order amounts within which the system automatically approves dispute cases
- Dispute item profile to specify the individual logistics item or charge type against which a service provider can create a dispute case

Dispute Management Settings

In Customizing for **Transportation Management** under **► Settlement ► Freight Settlement ► Freight Settlement Dispute Management ►** you must specify the following settings:

- Dispute item types

In the **Define Dispute Item Types** Customizing activity, you must specify the dispute item types that a service provider can use to create dispute items. The service provider can only enter dispute amounts and create dispute cases for the enabled fields.

You add a dispute item type to a dispute item profile, so that the system can subsequently use the item type when the service provider creates a dispute case for the freight order.

- Dispute item profiles

In the **Define Dispute Item Profiles** Customizing activity, you must group the dispute item types against which you want to allow a service provider to create dispute items.

i Note

For a carrier invoice created from an enterprise service, it is not mandatory to define the dispute item profile.

We deliver the item profile **DEFAULT**. The following table contains the item types we deliver as standard in the **DEFAULT** item profile:

Item Type	Function	Use Example
CHARGELINE . DISPUTE	Propose a rate change on an existing charge line.	Create a dispute case against the base freight rate.
MANUAL . CHARGE . DISPUTE	Propose a rate change for charge lines you have manually added from an agreement.	Loading charges
NEW . CHARGE . LINE . DISPUTE	Add a new charge type.	Detention or demurrage
ROOT . DISTANCE	Propose a change for the total distance of a freight order.	Not applicable

Item Type	Function	Use Example
ROOT . GROSS . VOLUME	Propose a change on total gross volume of a freight order.	Not applicable
ROOT . GROSS . WEIGHT	Propose a change on total gross weight of a freight order.	Not applicable

For example, you do not want your carrier to propose a new charge line in a dispute case. You use the DEFAULT item profile and exclude the item type NEW . CHARGE . LINE . DISPUTE. Also, in a full truck load scenario, you do not want your carrier to create a dispute case for the total distance. You use the DEFAULT item profile and exclude the item type ROOT . DISTANCE.

- Tolerance rules

In the [Define Dispute Tolerance Rules](#) Customizing activity, you must specify tolerance limits for charge types that enable you to reduce administration overhead and automatically approve dispute cases. You can specify more than one tolerance in a tolerance group.

You can use the tolerance settings to enable automatic approval for changes to the following charge amounts:

- Charge line rate amount
- Charge line final amount
- Overall amount in a freight order

For example, you can specify a tolerance for the total amount in a freight order for a dispute case in a self-billing process, or a tolerance for the net amount in a carrier invoice submission process.

When the system creates a dispute case, it performs a tolerance check. It checks each charge type amount and the overall amount in a dispute case against the tolerance deviations contained in the relevant dispute case. If the charge type amount or overall amount falls within the tolerance limits, the system automatically passes the dispute case. If the amount falls outside the limits, the system adds the dispute case to the work list on the [Freight Settlement Dispute Case](#) screen for manual attention.

If you do not specify tolerance rules, you must manually review each dispute case on the [Freight Settlement Dispute Case](#) screen.

- Dispute case reason codes and types

In the [Define Dispute Case Reason Codes and Types](#) Customizing activity, you must specify the reason code that you assign to a freight settlement dispute case or a freight settlement dispute case line item to specify the reason why you created the dispute case or to record the reason why you approved, accepted with a proposal, or rejected the dispute case or line item.

You must also specify the freight settlement dispute case types that the system uses when it creates a freight settlement dispute case.

- Printouts, email, and enterprise service

If you want to exchange dispute case information with your carrier in a printout, email, or enterprise service, you must assign the action profiles in the [Output Profile](#) and [Additional Output Profile](#) fields in Customizing for [Transportation Management](#) under [► Settlement ► Freight Settlement ► Freight Settlement Dispute Case ► Define Dispute Case Reason Codes and Types ►](#).

For more information, see [Output Management](#).

To send an email to your carrier, you must enter an email address in the [Address Independent Communication](#) section of the business partner master data for the carrier.

- Approval levels and workflow

If you want to specify an approval workflow, you must specify the appropriate settings in the settlement profile in Customizing for **Transportation Management** under ► **Settlement** ► **Define Settlement Profile** ►. The system uses an approval workflow for dispute cases created against freight orders for a service provider.

If you want to specify the approval levels a dispute case must pass through, and the roles that the system applies to each approval level, you must specify the appropriate settings in the **Define Approval Levels and Roles for Workflow** Customizing activity.

You can specify approval levels for level-1 users and level-2 users. Level-1 users and level-2 users can approve, accept and propose, or reject a change from a service provider. For more information, see [Workflow for Dispute Management](#).

If you want to add or remove workflow approvers based on one or more of the details in a dispute case, you must specify **BAdI: User Determination for Dispute Case Workflow** in Customizing for **Transportation Management** under ► **Business Add-Ins (BAdIs) for Transportation Management** ► **Settlement** ►. The approver you specify in this BAdI overrides the approver you specify in Customizing.

- You can use the Business Configuration Set (BC Set) **Customizing Workflow Agents Approval Levels and Roles for Workflow** (/SCMTMS/INV_DISP_WFL_AGENTS_C) for the **Define Approval Levels and Roles for Workflow** Customizing activity.

More Information

Subject	More Information
Structure of an individual dispute case, including working with a dispute case	Freight Settlement Dispute Case
Individual aspects of dispute management	Features of Dispute Management
Details of how freight cost accruals are updated in Materials Management (MM)	Automatic Change Management for Freight Settlements
Working with the master data for the vendor	Manage Supplier Master Data

Features of Dispute Management

Your service provider, such as your carrier or additional agreement party for providing other services, submits invoices for the freight orders. The carrier invoice may be different from what you expected to pay the carrier. This results in the system creating a dispute case.

Tolerance Check

When the service provider submits invoices with disputes, Transportation Management (TM) applies the tolerance settings you specified in the **Define Dispute Tolerance Rules** Customizing activity. If the dispute items are within the tolerance limits, the system automatically accepts the dispute case. If the dispute case is outside the limits, it fails the tolerance check and the dispute case must be reviewed by a user on the ► **Freight Settlement** ► **Dispute Case** ► **Overview Freight Settlement Dispute Cases** ► screen.

The system stores the results of the tolerance checks in the **Tolerance Check Log** tab page of the individual dispute case. You have a record of the automatic acceptance and automatic failure results.

Dispute Resolution Cycle

A dispute is resolved when both you and your service provider arrive at a common agreement on the charge or logistics item under dispute. To arrive at common ground, you either reject or accept the dispute.

You can have only one dispute case for one service provider at a time for a freight order. However, in freight orders with a carrier and an additional agreement party, you can have two open disputes, one for the carrier and one for the additional agreement party. Note that the system does not allow more than one dispute case for a freight order for the same service provider to be open at the same time.

Workflow and Work Items

You can use a workflow process to structure the approval process for your dispute cases. You can ensure that one or more users in your organization approve a dispute case before you accept or reject a proposed change to a freight order from a service provider.

If you enable a workflow process to approve dispute cases, the system creates work items for your level-1 and level-2 users. You can review all the work items that are open on the [Freight Settlement Dispute Case Work Items](#) screen. You can review the status of the work items for an individual dispute case on the [Dispute Case > Work Items](#) screen. You can navigate to the dispute case screen from the work item. For more information, see [Workflow for Dispute Management](#).

Dispute Cases Statuses

The following table contains the statuses that can apply to a dispute case:

Status	Answers the Question
Life cycle status	Is the overall dispute case open or resolved?
Action status	Who needs to act on the overall dispute case?
Workflow status	What is the status of an individual work item?
Dispute item status in the Logistics Details tab page and the Charge Details tab page	What is the approval status of an individual dispute item?
Dispute status for total amount	What is the overall approval status for the total amount of the freight order?

For more information, see [Statuses in Dispute Management](#).

Dispute Case Status in Freight Order

You can review the dispute case status in a freight order on the [Statuses > Subcontracting](#) screen. The status indicates the resolution status of a dispute case in a freight order, and is explained in detail in the following table:

Status	Status Description
No Dispute	The default status of the freight order when there are no disputes created against the freight order.
Awaiting Your Approval	You as the service requester must act to approve, make a proposal, or reject the dispute created by your service provider. Note that if the freight order is with a carrier and an additional agreement party, this status applies to the open dispute cases from your service providers. If there are any open dispute cases, you must act to approve, make a proposal, or reject the dispute.
Awaiting Carrier Approval	Your service provider must act on a dispute case. If the freight order is with a carrier and an additional agreement party, your service providers must act on their respective dispute case. This means you as the service requester have taken one of the following actions:

Status	Status Description
	- Rejected the claim from your service provider. If your service provider agrees with your decision, they must accept the dispute. - Proposed a solution that is acceptable to you against the claim from the service provider. If your service provider agrees with your decision, they must accept the dispute.
Resolved	Both you as the service requester and your service provider have agreed on common ground in the dispute and you both have agreed on a decision. If the freight order is with a carrier and an additional agreement party, all the open dispute cases have been agreed on common grounds by the service requester and the respective service provider.
In Dispute	This status only applies to the Dispute Case Status on the Status tab of a freight document such as a freight order or a freight booking. In a freight order with a carrier and an additional agreement party, there are unresolved dispute cases from both the carrier and additional agreement party. The dispute cases for each of the service providers may have different dispute case statuses. For example, one dispute case may have the status Awaiting Your Approval and the other dispute case may have the status Awaiting Carrier Approval. In such scenarios, you must navigate to the dispute case document for the freight order to know the actual status of the dispute case.

Dispute Management with Carriers and Additional Agreement Parties

If you specify one or more carriers or additional agreement parties in your freight order, you can have multiple disputes in a single freight order. However you can have just one dispute for each party.

The service provider such as, a carrier or an additional agreement party, can submit an invoice to the shipper.

In the freight carrier invoice scenario, the service provider can submit a freight carrier invoice with the disputed items. The TM system creates a dispute on a carrier invoice if there is a mismatch of values between the expected invoice value and the values submitted by the carrier. The dispute gets resolved when you as the shipper approve the proposal submitted by the carrier. When the dispute is resolved the system copies the charges approved in the dispute based on the freight carrier invoice and updates the freight order.

For more information, see [Additional Agreement Parties](#).

Dispute Resolution Statuses

In dispute management scenario with multiple service providers such as, carriers or additional agreement parties, you have multiple service providers who may create multiple dispute cases. The status of one dispute case may differ from the other. Hence, the system displays the dispute resolution status for every charge item in a freight order. Based on the dispute statuses of the charge items, the system displays a freight order dispute resolution status in the **Dispute Case Status** field on the **Statuses** tab.

The following table explains how the system computes the freight order dispute resolution status:

Dispute Resolution Status of Carrier 1	Dispute Resolution Status of Carrier 2	Dispute Resolution Status of Freight Order
No Dispute	No Dispute	No Dispute
No Dispute	Awaiting Your approval	Awaiting Your Approval
No Dispute	Dispute Resolved	Dispute Resolved

Dispute Resolution Status of Carrier 1	Dispute Resolution Status of Carrier 2	Dispute Resolution Status of Freight Order
Awaiting Your approval	No Dispute	Awaiting Your Approval
Awaiting Your approval	Awaiting Your approval	Awaiting Your Approval
Awaiting Your approval	Dispute Resolved	Awaiting Your Approval
Dispute Resolved	No Dispute	Dispute Resolved
Dispute Resolved	Awaiting Your approval	Awaiting Your Approval
Dispute Resolved	Dispute Resolved	Dispute Resolved

For more information on the dispute statuses, see [Carrier Invoice](#).

Edit Dispute Case on Behalf of Carrier

As a shipper, you can act on behalf of a carrier to capture inputs on dispute items from the carrier. You can capture offline inputs such as an email or phone conversation from your carrier and add the inputs to a dispute case on behalf of the carrier. To add this information, the dispute case must be in the dispute resolution status of **Awaiting Carrier Approval**.

More Information

Subject	More Information
Structure of an individual dispute case, including working with a dispute case	Freight Settlement Dispute Case
Necessary Customizing settings required before you can use dispute management	Prerequisites for Dispute Management
Introduction to dispute management	Dispute Management for Freight Settlements

Statuses in Dispute Management

Life Cycle Status

The life cycle status indicates the resolution status of the overall dispute case. The following table explains what the individual status values mean:

Status	Status Description
In Dispute	Indicates that the transportation service requester and the transportation service provider have not yet agreed a resolution to the dispute case. Both parties disagree on the logistics information or charge information in the freight order.
Cancelled	Your service provider has submitted a carrier invoice that has an associated dispute case. You have rejected the carrier invoice. In the background, the system cancels the dispute case and changes the dispute case status of the carrier invoice to Resolved. This saves you from taking extra steps to cancel the open dispute case manually. Your service provider has submitted a credit memo to cancel an existing carrier invoice. The carrier invoice had an associated open dispute case that is cancelled by the system.

Status	Status Description
Dispute Resolved	Indicates that the transportation service requester and the transportation service provider have agreed a resolution to the dispute case. This means that you as a service requester have taken one of the following actions: • Accepted the claim from the service provider • Rejected the claim of the service provider and the service provider accepted this decision • Proposed a solution that is acceptable to you against the claim from the service provider, and the service provider accepts your proposal
Dispute Rejected	Indicates that your transportation service provider has agreed to withdraw a dispute case.

Action Status

The action status indicates the party that owns the next action in the dispute resolution process. The following table explains what the individual status values mean:

Status	Status Description
Awaiting Your Approval	You as the service requester must act to approve, make a proposal, or reject the dispute created by your service provider.
Completed	Both you as the service requester and your service provider have agreed on a decision for the dispute case.

Workflow Status

The workflow status indicates the status of the workflow for a dispute case. The following table explains what the individual status values mean:

Status	Status Description
Workflow Not Enabled	You have not enabled a workflow to resolve disputes with your service provider in the Customizing settings for the relevant settlement profile (see Prerequisites for Dispute Management). The dispute case does not go through an automatic and structured approval process.
Automatically Approved	The system has used the tolerance settings you specified in Customizing to approve the dispute case automatically in the background (see Prerequisites for Dispute Management). You do not need to take any action.
Awaiting Approval	You are waiting for one or more of the approvers in the workflow to take action on the dispute. The approvers in the workflow must accept, accept with a proposal, or reject the dispute case (see Workflow for Dispute Management).
Completed	The workflow has no more work items and the dispute case has an action status of Completed .

Dispute Item Status

The dispute item status indicates the resolution status of a dispute item for an individual logistics item or an individual charge type. The following table explains what the individual status values mean:

Status	Status Description
New	You as the service requester must act on the dispute item. You must accept, accept with a proposal, or reject the dispute item.
Approved by You	You as the service requester have approved the dispute item.
Accepted by You with Proposal	You as the service requester make a proposal against the dispute submitted by your service provider, and send the proposal to your service provider for action.
Rejected by You	You as the service requester reject the dispute submitted by your service provider.
Approved by Tolerance Check	The system has used the tolerance settings you specified in Customizing to approve the dispute item automatically in the background (see Prerequisites for Dispute Management). You do not need to take any action.

Workflow for Dispute Management

As standard, we deliver a 2-level approval process for your dispute case approval workflow. In this process, when your service provider submits a carrier invoice that has a dispute, the system sends the dispute case to a level-1 approver for approval. When the level-1 approver takes action, the system sends the dispute case to a level-2 approver for approval.

Both the level-1 approver and level-2 approver accept or reject the dispute case. When the level-2 approver takes action, the system updates the life-cycle status of the dispute case. If the level-2 approver rejects the dispute case, the carrier invoice is rejected. If the level-2 approver accepts the dispute case, the carrier invoice is accepted.

You must enable the relevant settlement profile for the approval workflow. For more information, see [Prerequisites for Dispute Management](#).

Features

Workflow Configuration

We have implemented the 2-level approval workflow WS61200016 as standard. You can review the configuration of the standard workflow in transaction SWDD ([Workflow Builder](#)).

We have added an action to the standard Post Processing Framework (PPF) profile for dispute cases in freight settlement. We have used the settings in the following table to specify the standard workflow in the PPF profile:

Object Type	Object Name
Application	/SCMTMS/TRANSPORTATION
Action profile	/SCMTMS/TOR_INV_PREP (Post Freight Settlement Document to MM)

Object Type	Object Name
Action definition	/SCMTMS/FDI_WFL (Workflow for Dispute Invoice)
Processing type	Workflow (Default)
Workflow template	WS61200016

You can review these settings in Customizing for [Cross Application Components](#) under [Processes and Tools for Enterprise Applications](#) [Reusable Objects and Functions for BOPF Environment](#) [PPF Adapter for Output Management](#) [Maintain PPF Settings](#).

If you enable the approval workflow in a settlement profile, the system triggers the workflow when a dispute case is created in either the self-billing or invoice submission processes. For more information, see Customizing for [Transportation Management](#) under [Settlement](#) [Define Settlement Profile](#).

You can implement your own approval workflow. To do this you must create a workflow configuration to suit your business needs in transaction SWDD, and specify the workflow in the [Workflow](#) processing type, under the /SCMTMS/FDI_WFL action definition.

User and Role Specification

The system uses the role and the user assigned to the role to determine the approver for a work item. You specify the user in the purchasing organization of the freight order. For more information, see [Assigning Employees to Organizational Units](#). You specify the role in Customizing for [Transportation Management](#) under [Settlement](#) [Define Settlement Profile](#) [Define Approval Levels and Roles for Workflow](#). For more information, see [Prerequisites for Dispute Management](#) under [Dispute Management Settings](#) [Approval levels and workflow](#).

The system uses the approval level of the work item and the role assigned to the user to assign a work item to an approver.

For example, you assign employees John Brown and Josephine Green to a purchasing organization in your company. You assign a role of settlement specialist for John Brown and a role of transportation manager for Josephine Green in transaction PFCG ([Role Maintenance](#)). In the [Define Approval Levels and Roles for Workflow](#) activity, you specify approval level 1 for the settlement specialist role and approval level 2 for the transportation manager role. You enable the settlement profile for the approval workflow.

When the system creates a dispute case for freight orders in the purchasing organization, it first sends a work item for the dispute case to John Brown. When he finishes with the work item, the system then sends the work item to Josephine Green. If you assign more than one user to the same role, the system sends the work item to all the users assigned to the role. However, the system moves the workflow on to the next stage if just one of the users takes action on the work item.

More Information

Subject	More Information
Resolution of errors in workflow	Workflow Error Resolution for Dispute Management
Necessary Customizing settings required before you can use dispute management	Prerequisites for Dispute Management
Individual aspects of dispute management	Features of Dispute Management
Information on statuses in dispute management	Statuses in Dispute Management
Information on specifying roles	SAP Help Portal at http://help.sap.com/netweaver . Choose the appropriate release and in the application help, choose SAP NetWeaver Library: Function-Oriented View Application Server Application Server ABAP Other Services Services for

Subject	More Information
	Application Developers > SAP Business Workflow > Role Documentation

Workflow Error Resolution for Dispute Management

You can use transaction SWI2_DIAG to check if the system does not process a work item due to an error in your system settings or a technical error. You must run the transaction to check for errors. The system provides information such as the cause of the error, the workflow step at which the error happened, and the ID of the work item. When you resolve an error, you must choose the **Restart Workflow** button on the **Diagnosis of Workflow with Errors** screen to process the work item again. The system continues with the workflow from the point where the error occurs.

Resolution of Specific Errors

The following list contains details on the errors you may encounter, and the steps you must take to resolve the errors:

- Users not assigned to your purchasing organization

You have not assigned users to the purchasing organization to which the freight order belongs. The service provider has created the dispute case against the freight order. The system cannot choose a user and therefore cannot create a work item. In transaction PPOME (**Organization and Staffing Change**) -> **Staff Assignments**, you must ensure your level-1 approver and level-2 approver are assigned to the purchase organization that you use in the freight order.

- Incorrect roles assigned to approval levels

You have assigned a level-1 approver and a level-2 approver to the correct purchasing organization. You have specified the role that the system uses for level-1 approval and level-2 approval in Customizing for **Transportation Management** under **Settlement** > **Freight Settlement** > **Freight Settlement Dispute Management** > **Define Approval Levels and Roles for Workflow**. However, although you have specified the level-1 approver and level-2 approver in the purchasing organization, you have not assigned the users to a role that you have used in the Customizing activity.

For example, the following table contains the settings you specify in transaction PPOME for the users in your purchasing organization:

User	Role	Role Description
Josephine Green	/SCMTMS/TRANSPORTATION_MANAGER	Transportation manager
John Brown	/SCMTMS/SETTLEMENT_SPECIALIST	Settlement specialist

The following table contains the settings you specify in the **Define Approval Levels and Roles for Workflow** Customizing activity:

Approval Level	Role	Role Description
1	/SCMTMS/TRANSPORTATION_PLANNER	Transportation planner
2	/SCMTMS/PURCHASING_CLERK	Purchasing clerk

You have not specified a role for a transportation manager and settlement specialist in the **Define Approval Levels and Roles for Workflow Customizing** activity. The system does not create a work item.

- Approval level number or role not specified in the [Define Approval Levels and Roles for Workflow](#) Customizing activity

You must specify an approval level for level 1 and level 2 of the workflow in Customizing for [Transportation Management](#) under [Settlement](#) [Freight Settlement](#) [Freight Settlement Dispute Management](#) [Define Approval Levels and Roles for Workflow](#). You must also specify a role for each approval level in the [Define Approval Levels and Roles for Workflow](#) Customizing activity. This ensures that the system selects a role for an approval level. The system automatically sets the approval level when you use the Business Configuration Set (BC Set) [Customizing Workflow Agents Approval Levels and Roles for Workflow](#) for the [Define Approval Levels and Roles for Workflow](#) Customizing activity.

- Incorrect logic in [BAdI: User Determination for Dispute Case Workflow](#)

You may have implemented incorrect logic in the [BAdI: User Determination for Dispute Case Workflow](#) in Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) [Settlement](#) [Freight Settlement](#) [Freight Settlement Dispute Management](#). The system uses your standard settings to determine the approvers for your workflow. However, you can use this BAdI to overwrite the standard settings, to add or remove workflow approvers according to your own logic. For example, the system determines John Brown as the level-1 user. You implement this BAdI to override this standard determination. However, due to an error in the BAdI logic, the system does not identify an approver.

i Note

If you specify levels and roles in the [Define Approval Levels and Roles for Workflow](#) Customizing activity but do not enable the approval workflow in the relevant settlement profile, the system takes the following action:

- Displays a workflow status of [Workflow Not Enabled](#)
- Does not use the level and roles settings as you have not enabled the workflow

More Information

Subject	More Information
Workflow process for dispute management	Workflow for Dispute Management
Prerequisites you need for dispute management	Prerequisites for Dispute Management
Key aspects of dispute management	Features of Dispute Management
Explanation of dispute management statuses	Statuses in Dispute Management
Standard SAP workflow process	SAP Help Portal at http://help.sap.com/netweaver . Choose the appropriate release and in the application help, choose SAP NetWeaver Library: Function-Oriented View Application Server Application Server ABAP Other Services Services for Application Developers SAP Business Workflow SAP Business Workflow: Reference Documentation .
Resolution of errors in standard SAP workflow process	SAP Help Portal at http://help.sap.com/netweaver . Choose the appropriate release and in the application help, choose SAP NetWeaver Library: Function-Oriented View Application Server Application Server ABAP Other Services Services for Application Developers SAP Business Workflow Role Documentation Roles in Workflow System Administrator Business Workflow Administration Workflow System Administration Workflow Runtime Administration Diagnosing Workflows with Errors .

Dispute Case Resolution and Accruals Update

A dispute is resolved when both you and your service provider arrive at a common agreement on the charge or logistics item under dispute. The system updates the freight order with the changed charge information.

You use the background function [Creation of Freight Settlement Documents](#) to create, change, and post the freight settlement documents to Materials Management (MM) (see [Creation and Transfer of Freight Settlement Documents](#)). The freight settlement document contains the changes from the freight order.

Change Process for Freight Settlement Documents

The system uses the change process for freight settlement documents that you specify in the relevant settlement profile in Customizing to update MM. For more information, see Customizing for [Transportation Management](#) under [Settlement](#) [Define Settlement Profile](#).

For example, you have a freight order with a transportation charge of USD 2000. Your carrier creates a dispute case for the transportation rate, and you resolve the dispute by changing the transportation charge to USD 2300.

Transportation Management (TM) uses the change process for freight settlement documents and the life cycle status of the freight settlement document to post the freight settlements to MM. Depending on the update strategy you select, TM updates the accruals to MM as follows:

- [Create New FSD for Delta Amount](#)

Life Cycle Status of FSD	System Action	Example
New/In Process/Ready for Accruals	Include the new charges in the new FSD when the invoicing status of the freight document is Partially Invoiced . If the invoicing status of the freight document is Over Invoiced , the system cancels the existing FSD and recreates the new FSD for the new values.	A new FSD for the value of USD 300 is created if the invoicing status of the freight document is Partially Invoiced . If the invoicing status of the freight document is Over Invoiced , the system cancels the existing FSD for USD 2000 and recreates the new FSD for a new value of, for example, USD 1800.
Accruals Posted/Invoice Verified	◦ Create a new freight settlement document for a positive delta amount if the invoicing status of the freight order is Partially Invoiced. A positive amount is where you owe your carrier an extra amount as a result of the dispute resolution. ◦ If the invoicing status of the freight document is Over Invoiced, the system executes the change process for FSD as Reverse and Repost with New FSD.	◦ TM creates a new freight settlement document for the delta amount of USD 300. On posting this FSD in MM, a new purchase order and SES entry for USD 300 is created. ◦ If the transportation charge were updated to USD 1800 after you resolved the dispute case, TM would create a credit memo for the entire previous invoice amount for USD 2000. It would then create a new FSD for the revised amount of USD 1800.
Canceled	Create a new freight settlement document with the updated amount.	TM creates a new freight settlement document for USD 2300.

- [Reverse and Repost with New FSD](#)

Life Cycle Status of FSD	System Action	Example
New/In Process/Ready for Accruals	Cancels the existing FSD and recreates the new FSD for the new values.	TM changes the charges in the freight settlement document from USD 2000 to USD 2300. All FSDs that were created earlier are canceled and a new FSD for the revised value USD 2300 created.
Accruals Posted/Invoice Verified	Creates a credit memo for the FSD and reverses the FSD amount. A new FSD is created for the revised value of the freight document charges.	A credit memo for USD 2000 is created and linked to the FSD, and a new FSD for the revised value of USD 2300 is created.
Canceled	Create a new freight settlement document with the updated amount.	TM creates a new freight settlement document for USD 2300.

i Note

- You can run the **Creation of Freight Settlement Documents** background function to post the freight settlement documents to Materials Management (MM) (see [Creation and Transfer of Freight Settlement Documents](#)).
- Regardless of the settings for the **Change Process for FSD** in the settlement profile, if a freight settlement document has the life cycle status **Posting Started**, **Cancellation Requested**, or **Reversal Requested** when you run the background function, the system reports an error. You must wait until the life cycle status of the settlement document is **Accruals Posted** or **In Process** before you take further action.

Background Function to Publish Freight Settlement Dispute Cases

You can use a background function to publish freight settlement dispute cases in batches in the background. When you run the background function, you as a requester of transportation services confirm the decisions you make to accept the relevant dispute cases. The system updates the resolution status of the dispute case to **Dispute Resolved**. Also the system uses the charge details from a dispute case that is resolved to update the changed charge information in the freight order. For more information, see [Publication of Freight Settlement Dispute Cases](#).

Background Function to Create and Transfer Freight Settlement Documents

You can include, exclude, or include only the freight orders for which all dispute cases are resolved when you create and transfer freight settlement documents in the background. For more information, see [Creation and Transfer of Freight Settlement Documents](#).

More Information

Subject	More Information
Structure of an individual dispute case, including working with a dispute case	Freight Settlement Dispute Case
Necessary Customizing settings required before you can use dispute management	Prerequisites for Dispute Management
Individual aspects of dispute management	Features of Dispute Management
Integration with MM	Integration: Invoice Processing (MM-IV/SD-BIL) with TM

Charge Correction Advice Management

Use

The charge correction advice process enables you as a logistics service provider (LSP) to create a charge correction advice document (CCAD) to correct errors in invoices that you have received from a carrier for delivering air freight transportation services.

The process enables you to simulate in the Transportation Management (TM) system the invoicing process that takes place externally between you and a carrier. The CCAD enables you to post corrections to the freight settlements that you receive from your airline carrier.

You can settle an invoice that contains charges for shipments, and corrections for errors from a previous invoice. For example, you receive invoices from your carrier at the end of each month. After you settle an invoice for the month of July, the carrier contacts you saying there was an error on a particular invoice. You paid for the transportation of a particular item that weighed 50 kg. The actual weight of the item was found to be 75 kg. You agree with the carrier to pay the extra amount due, for the extra weight. At the end of August, the carrier sends you a revised invoice that contains the corrected amount due for the extra 25 kg from the invoice in July.

Process

1. The carrier provides the air freight service for an air freight booking. You create a freight settlement document (FSD). This is the planned invoice from the carrier, and simulates the invoice that the carrier creates in an external system for the service.
2. You post the settlement document to Materials Management (MM).

The life cycle status of the FSD changes to **Posted**. In MM, the system creates a purchase order (PO) and a service entry sheet (SES) for the settlement document. The amount of the PO and the total amount in document currency in the settlement document match.

3. In Logistics Invoice Verification, you verify the invoice from the carrier and pay the settlement amount.
4. After a short period of time, the carrier contacts you with a correction. The correction could be due to different reasons, such as an error in the weights, or incorrect rates.
5. You navigate to the relevant air freight booking and make the correction. For example, you change the weight of the item from 100 kg to 120 kg. You create a CCAD for this from the air freight booking.

The CCAD contains the amount that has already been settled, the new delta amount to be corrected, and the final correct settlement amount for the freight booking.

i Note

The CCAD is a new category of freight settlement document type. You can only create a CCAD for an air freight booking that has an associated FSD that is at a life cycle status of **Posted**.

6. You post the CCAD to MM. In MM, a new PO is created with the following details:
 - o PO lines for the amounts that were previously invoiced, and SES lines with the amounts that were previously invoiced. Both documents have the **Returns Indicator** enabled.
 - o PO lines for the new amounts after correction and SES lines for the new amounts after correction that are newly determined in the CCAD

Posting the SES with the returns indicator set reverses the previously invoiced amount. MM also posts a new SES entry for the fully adjusted correct amount. The life cycle status of the CCAD after successful posting in TM changes to **Posted**.

Example

The following table illustrates how the CCAD process works for an air freight booking:

Steps in TM	Amount in FSD/CCAD (USD)	Steps in MM	Comment
Total freight cost in air freight booking in TM	2400	None	None
Generate the FSD	2400	PO created, for example, PO1 with line 10. SES created, for example, SES 1, linked to PO1/10 with amount USD 2400	Regular FSD process
Update the booking to reflect the correction, for example, weight correction. New total freight cost in air freight booking	3000	None	None
Generate a new CCAD	600	- No change to PO1, created in step 1 - PO2 has the following items: - Item 10 for USD 3000 reflecting the new changed amount - Item 20 for USD 2400 with returns indicator selected - SES 2 created for PO2/10, for USD 3000 - SES 3 returned for PO2/20, for USD 2400	MM posts a return for the old settlement document amount of USD 2400. It also posts the new amount of USD 3000.

The following table illustrates how the CCAD process works for an air freight booking, where the process results in negative amounts:

Steps in TM	Amount in FSD / CCAD (USD)	Steps in MM	Comment
Total freight cost in air freight booking	2400	None	None
Generate the FSD	2400	PO created, for example, PO1 with line 10. SES created, for example, SES 1, linked to PO1/10 with amount USD 2400.	Regular FSD process
Update the booking to reflect the correction, for example, weight correction. New total freight cost in air freight booking	2000	None	None
Generate a new CCAD	-400	- No change to PO1, created in step 1 - PO2 has the following items: - Item 10 for USD 2000 reflecting the new changed amount	MM posts a return for the old settlement document amount of USD 2400. It also posts the new amount of 2000.

Steps in TM	Amount in FSD / CCAD (USD)	Steps in MM	Comment
		◦ Item 20 for USD 2400 with returns indicator selected • SES 2 created for PO2/10 for USD 2000 • SES 3 returned for PO2/20, for USD 2400	

Related Information

[Freight Settlement Integration using MM External Services Management](#)

[Management of Air Freight Bookings for Airlines](#)

Freight Settlement Integration using Lean Services

In Transportation Management (TM), you can perform freight settlement with your service providers or carriers by creating a freight settlement document (FSD) and posting it to Materials Management (MM) using Lean Services.

Prerequisites

- You've disabled the checkbox **TM Settlement Integration based on MM External Services Mgmt** in the Customizing of **Transportation Management** under **Basic Functions > General Settings > Define General Settings for TM**. This ensures that you've enabled lean service-based integration with Materials Management (MM).
- You can also enable lean service-based integration only for selected freight documents, when the rest of the system is using external service-based integration. To do so, you've to specify combinations of company codes and carriers in the Customizing activity **Enable Lean Services based Settlement Integration**. You can do that from the Customizing of **Transportation Management** under **Settlement > Freight Settlement > Integration for Settlement Posting > Service Procurement - Lean Services**. The system enables lean service-based integration only for those freight documents where it finds the combination of these company codes and carriers.
- You've integrated TM that enables you to post supplier invoices that refer to freight orders. These freight orders are based on purchase orders with lean service items. You can do that from the customizing activity **Activate Integration of Transportation Management (TM)** in Customizing of **Materials Management** under **Logistics Invoice Verification > Incoming Invoice**.
- To integrate to an internal MM system, you've configured the following Customizing settings:
 - **Assign Purchasing Info for Posting Lean Service Integration**
 - **Assign Val. Class, Acc. Assignment Category, and Conditions to Charge Types**
 - **Mapping for Cost Distribution** when you've enabled cost distribution in the freight settlement document

For more information, see Customizing for **Transportation Management** under **Settlement > Freight Settlement > Integration for Settlement Posting > Service Procurement - Lean Services**.

The system determines the logical system from the freight units that are assigned to the freight document. The freight documents are created from the requirement documents, such as sales orders, purchase orders, and so on. For more information about organizational models, see [Organizational Management](#).

- **Company Code and Plant**

For more information, see [Company Code and Plant in Posting Freight Settlements using Lean Services](#)

- **Freight cost confirmation and execution status**

For more information, see [Freight Cost Confirmation and Posting](#).

Process

In lean service, the freight settlement document is created and posted as a part of the posting step from the related freight documents. For more information, see [Monitor Freight Documents for Accrual](#).

1. You create and post the freight settlement document.

- o The system creates and posts the individual or collective settlement document for a freight order, freight booking, or consignment order. You can see the freight documents from the **Monitor Freight Documents for Accrual** app. The system automatically creates and posts the settlement document based on the data in the freight orders or freight bookings. For more information, see [Monitor Freight Documents for Accrual](#).

You can create and post multiple FSDs using the **Mass Creation of Freight Settlement Documents** function. You can run transaction /SCMTMS/SFIR_CREATE (report /SCMTMS/SFIR_CREATE_BATCH). For more information, see [Creation and Transfer of Freight Settlement Documents using Integration with Lean Services](#).

- o You can create separate freight settlement documents for each of the invoicing carriers in a rail freight order, where you have a different invoicing carrier for each stage in the order. This feature gives you the flexibility to work with rule 11 business situations in North America.

In rule 11 situations, the header carrier receives the order from the customer and can execute one or more of the stages. The header carrier is the carrier that you specify in the ► **General Data** ► **Transportation** ► screen area of the freight order. The invoicing carrier is the carrier who bills the customer for the shipment and you specify this carrier at stage level in the rail freight order.

For more information about rail freight orders, see [Rail Transportation](#).

- o You can create a settlement document for each additional agreement party if you have an existing freight order or freight booking that contains one or more additional agreement parties. For more information, see [Additional Agreement Parties](#).
- o The system settles a parcel freight order in the same way as it settles a standard freight order. For more information about parcel freight orders, see [Parcel Shipment](#).
- o When you create and post a freight settlement document, by default the system uses the date from the relevant **Event for Freight Cost Confirmation** as the settlement date. For more information, see [Freight Cost Confirmation and Posting](#)
- o You can have multiple charge types in a charge item. As standard in lean services, MM creates a purchase order item for every independent charge type and applies tax to the purchase order(PO) item. Dependent charge types are represented as condition types for its parent service in the PO item.
- o TM transfers the settlement information, including the source location and destination location information for every independent charge types to MM. The location information enables MM to determine different tax codes/rates for different locations in the purchase order item.

For more information, see [Setting Up Tax Rates for Purchasing Transportation Services](#).

- o When the SES is accepted and posted, accruals are booked to accounting. The SES serves as the basis for invoice verification in MM.

The settlement date from the freight settlement document is used as the **Doc. Date** (Document Date) in the header of the purchase order.

The SES dates on the header of the **Accept. Data** (Acceptance Data) tab page are taken from the following places:

- Posting date from the settlement date in the freight settlement document
- Document date as the system date

Posting Periods

You can post the freight settlement document only if the posting periods in MM and Financial Accounting (FI) are both open for the posting date of the SES. If the posting period is closed, you have the following options:

- Open the closed posting period

You must open the posting period for both MM and FI.

In FI, open the posting period in Customizing under **Financial Accounting > Financial Accounting Global Settings > Ledgers > Fiscal Year and Posting Periods > Posting Periods > Open and Close Posting Periods**.

In MM, open the posting period in the SAP Menu under **Logistics > Materials Management > Material Master > Other > MMPV – Close Period** or **MMRV - Allow Posting to Previous Period**.

- Change the **Posting Date** in the relevant freight document for a **Business Partner** under **Charges > Posting Date** tab to match the open posting period and post the freight document again.

2. When the settlement document is posted successfully, its lifecycle status is set to **Posted**. If issues are encountered during posting, the lifecycle status is set to **Posting Failed** and the confirmation status has the status of the corresponding step in which the issue was encountered. For example, the confirmation status **Purchase Order Creation Failed** indicates that the purchase order creation failed.

If you encounter issues when you post to internal MM, the system logs the issues. You can review the issues on the **Application Log** tab of the freight settlement document.

3. Invoice verification and Evaluated Receipt Settlement (ERS)

- MM bases the expected invoice on the SES confirmation. When the invoice is posted against the freight document, the lifecycle status of the settlement document is set to **Invoice Verified**.
- The automatically generated invoice is based on the data from the settlement document, if you perform ERS.

i Note

You can't cancel the settlement document when the invoice is posted against the settlement document in MM and the settlement document has a lifecycle status of **Invoice Verified**.

When you successfully cancel the settlement document, the posting date of the canceled SES is determined based on the event for freight cost confirmation. Once the posting date is determined based on the event for freight cost confirmation, the same date is also copied over to the cancellation posting date. This ensures that while canceling the accruals, the cancellation of posting is done in the same financial period where the posting was done earlier.

More Information

Subject	More Information
Enable the system to determine the business partners that you use in different business document types.	Business Partner Determination
Make certain changes to your freight costs in the freight order or freight booking after you post accruals	Flexibility in Freight Charge Changes

Subject	More Information
Allow for unplanned changes in your freight settlement process	Reversing a Freight Settlement Document
Combination of life cycle and confirmation status	Life Cycle Status of Freight Settlement Documents
Invoicing of freight documents	Invoicing Freight Documents

Life Cycle Status of Freight Settlement Documents

The life cycle status shows the current status of the freight settlement document in the business process. The confirmation status indicates the result of the last posting step of the freight settlement document to Materials Management (MM).

The following table lists the combination of important life cycle and confirmation statuses:

Life Cycle Status	Confirmation Status	Description
Ready for Accruals	Not Posted	Freight settlement document created in blocked status but posting isn't yet initiated
Posted	Not Confirmed	Freight settlement document initiated for posting but MM documents aren't yet created
Posted	Confirmed	Freight settlement document posted successfully
Canceled	Not Confirmed	Freight settlement document initiated for cancellation but MM documents aren't yet canceled
Canceled	Confirmed	Freight settlement document canceled successfully
Posting failed	Purchase order creation failed	The creation of the purchase order failed when the freight settlement document was posted. For information about the error, see the Application Log tab in the freight settlement document.
Posting failed	SES Creation Failed	The creation of the service entry sheet failed when the freight settlement document was posted. For information about the error, see the Application Log tab in the freight settlement document.
Posting Failed	Cost distribution document creation failed	The creation of the settlement management documents with the cost distribution data failed when the freight settlement document was posted. For information about the error, see the Application Log tab in the freight settlement document.
Cancellation posting failed	Purchase Order Cancellation Failed	The cancellation of the purchase order failed when the freight settlement document was canceled. For information

Life Cycle Status	Confirmation Status	Description
		about the error, see the Application Log tab in the freight settlement document.
Cancellation posting failed	SES Cancellation Failed	The cancellation of the SES failed when the freight settlement document was canceled. For information about the error, see the Application Log tab in the freight settlement document.
Cancellation posting failed	Cost distribution document cancellation failed	During the cancellation of the freight settlement document, the cancellation posting of settlement management documents with the cost distribution data failed. For information about the error, see the Application Log tab in the freight settlement document.

Company Code and Plant in Posting Freight Settlements using Lean Services

The company code in the freight settlement process is used to book the freight accruals to financial accounting (FI) under the company code. The plant is required in the posting process during the creation of the service purchase order as part of the posting step for a freight settlement document.

Prerequisites

Procuring company code determination is enabled only when you've specified the following in Customizing for the freight document type:

- The document type is [Relevant for Subcontracting](#).
- [Subcontract. Checks](#) field has the value [Basic Additional Checks](#).

For more information, see the [Basic Settings](#) section in the following Customizing:

- [Freight Order Management](#) > [Freight Order](#) > [Define Freight Order Types](#) >
- [Freight Order Management](#) > [Freight Booking](#) > [Define Freight Booking Types](#) >
- [Freight Order Management](#) > [Consignment Order](#) > [Define Consignment Order Types](#) >

Procuring Company Code

The procuring company code is responsible for the procurement of a transportation service from the carrier or transportation service provider of a freight document. The procuring company code is available in the freight document under [General Data](#) > [Organizational Data](#) >.

The procuring company code is mandatory in the freight document and as is taken over in the freight settlement document. If the procuring company code is defined in the freight document, it's transferred to the freight settlement document.

Plant

The system determines the plant to be populated in the purchasing order item as follows:

- The plant from the underlying transactions (orders and deliveries) if they're all unique from the linked freight units in the freight document.
- Use the plant defined as the default plant for the company code in Customizing for [Transportation Management](#) under [Settlement](#) > [Freight Settlement](#) > [Integration for Settlement Posting](#) > [Assign Default Plant for Company Code](#) to integrate to an internal MM system.

Freight Cost Confirmation and Posting

Freight cost confirmation is a prerequisite step to trigger the posting of the freight cost to financial for a freight document. Freight costs can be confirmed based on the execution status of the freight document or automatically using business rules.

Prerequisites

The freight document has reached the following state:

- Freight orders: [Checkout](#) of last or only stop belonging to your organization.
In cases where the source and destination stop don't belong to your organization, then the [Departure](#) at the source stop is used.
- Ocean freight bookings: [Shipped on Board](#)
- Air freight bookings: [Uplift Confirmed](#)

Freight Cost Confirmation Based on Execution Status

To confirm the freight costs, the system considers the execution events for the freight documents. Execution status is used as one of the prerequisites to create freight settlement document. You can configure the event relevant for the freight cost confirmation in the [Execution Settings](#) section in the Customizing for [Transportation Management](#) under:

- [Freight Order Management](#) > [Freight Order](#) > [Define Freight Order Types](#)
- [Freight Order Management](#) > [Freight Booking](#) > [Define Freight Booking Types](#)
- [Freight Order Management](#) > [Consignment Order](#) > [Define Consignment Order Types](#)

Following are the relevant events and their effects on freight cost confirmation:

- **Not Relevant:** If you select this value in the associated freight order, freight booking, or consignment order type, the system doesn't change the invoicing status to [Ready for Confirmation](#). The system can't confirm the freight costs and perform the follow-on processes unless the invoicing status is [Ready for Confirmation](#).

i Note

For freight documents using lean service-based integration, you can't use **Not Relevant** as the value for the Customizing setting [Event for Freight Cost Confirm](#), because the invoicing status [Ready for Confirmation](#) is a mandatory requirement for the follow-on processes.

- **Checkout, Departure:** On the execution of these events, the system sets the status of the freight document as [Ready for Confirmation](#) and you can start the follow-on processes.
- **Freight Booking: Start of Main Carriage:** This is relevant only for freight bookings and on execution of this event the system sets the status of the freight booking as [Ready for Confirmation](#) and you can start the follow-on processes.

Automatic Freight Cost Confirmation

You can automate the confirmation and posting of freight costs during freight document processing by creating a condition with the condition type /SCMTMS/TOR_FRC_CONF (Automatic Freight Cost Confirmation).

When creating the condition, ensure you specify the following details:

- Carrier/service provider
- Procuring company code
- Purchasing organization
- Enable automatic confirmation

When a freight document matches the condition and has reached the relevant state, the system automatically confirms and posts the freight costs. For more information, see [Condition](#).

Related Information

[Invoicing Status of Freight Documents using Lean Services](#)

Collective Settlement

The collective settlement process enables you to group multiple freight documents into a single freight settlement document (FSD) and then post to Materials Management (MM) for further processing in financials.

You use the [Monitor Freight Document for Accrual](#) app to post your freight documents to MM for further processing in financials. You can post the freight documents either individually or collectively. When you post collectively, then the system doesn't redetermine the charges in the settlement document and copies all the charges from the freight documents to the FSD. This copying over of charges ensures that there are no mismatches in charges between the freight documents and the FSD.

Prerequisites

To do collective posting of FSDs that use Lean Service-based integration, ensure that the [Calculation Option](#) in the relevant settlement profile is [Copy all Charges](#). You can do that from the Customizing for [Transportation Management](#) under [Settlement](#) ► [Define Settlement Profile](#) ►.

Features

Split Criteria to Create Collective Settlements

For more information on split criteria, see [Split Criteria in Settlement Document Creation](#).

Invoice date determination for collective settlements

In general, the invoicing date of the FSD is determined from the posting date of the charge item for each business partner that is involved in the freight document. The posting date of the charge item is based on the dates on which the events for the freight cost confirmation are reported for the freight documents. For more information on events, see [Freight Cost Confirmation and Posting](#).

But in the collective settlement process, you combine a given set of freight documents, and it's possible that all these freight documents don't have the same posting date. In this case, the system uses the current system date as the invoice date.

Cancellation Posting Date

In general, during the posting cancellation process, the posting date of the charge item is copied to the cancellation posting date at the header-level of the FSD. This date is then used to cancel the MM document. You can also cancel the FSD in a different posting period by manually changing the posting date in the freight document and then initiating the cancellation.

For the collective settlement process, when you cancel the posting, the system picks the latest posting date from the charge items of all the freight documents under that collective FSD. This latest date is used as the cancellation posting date and it can be the same as the invoice date or different if you've manually changed the posting date in any of the freight documents.

❖ Example

Scenario 1

We have a collective FSD, **FSD1** with three freight orders:

Freight Order	Posting Date
FO1	15.08.2025
FO2	18.08.2025
FO3	21.08.2025

In this scenario, the dates are:

- Invoicing date for **FSD1**: **25.08.2025** (assuming this date is the system date)
- Cancellation posting date: **21.08.2025**

Scenario 2

Now, if there's a change in the posting date of the order, **FO3**, before you've initiated the cancellation of **FSD1**:

Freight Order	Posting Date	New Posting Date
FO1	15.08.2025	no change in posting date
FO2	18.08.2025	no change in posting date
FO3	21.08.2025	30.08.2025

In this scenario, the dates are:

- Invoicing date for **FSD1**: **25.08.2025** (assuming this date is the system date)
- Cancellation posting date: **30.08.2025**

Scenario 3

You want to cancel collective **FSD1** after the posting period (in this case after the month of August) for the following freight orders in collective **FSD1**:

Freight Order	Posting Date
FO1	15.08.2025
FO2	18.08.2025

Freight Order	Posting Date
F03	21.08.2025

The cancellation of **FSD1** fails and the system changes the lifecycle status to **Cancellation Posting Failed** as the posting period is already over. In this case, you have to manually change the posting date of one of the freight order in **FSD1** from the **Monitor Freight Settlement for Accruals** app. This updates the cancellation posting date as per the latest date in the freight order list:

Freight Order	Posting Date	New Posting Date
F01	15.08.2025	no change in posting date
F02	18.08.2025	no change in posting date
F03	21.08.2025	30.09.2025

Now, the dates are:

- Invoicing date for **FSD1**: **25.08.2025** (assuming this is the system date)
- Cancellation posting date: **30.09.2025**

Creation and posting of collective settlements

In collective settlement, multiple freight documents are combined into an FSD and these freight documents form settlement items in the FSD. When the FSD is posted to MM for further processing, purchase order (POs), service entry sheets (SES), and agency billing documents (ABDs) are created. Each PO item corresponds to the independent charge lines from freight documents. Follow-on documents for collective FSD use the same APIs that are used to create follow-on documents for individual settlements.

i Note

A collective FSD can't have more than 9999 charge lines for all the freight documents combined. If any FSD has more than 9999 charge lines, then the system labels those FSDs as inconsistent and they aren't processed further. You've to regroup the freight documents again for collective settlement.

Block handling for charge mismatches

As the calculation option in Customizing is **Copy All**, all the charges from the freight documents are copied over to the FSD so that there are no mismatches during the settlement process. The subsequent posting creates the collective settlement and posts to MM seamlessly.

Handling changes in charges for collective settlement after it's posted

There can be new charge lines that are added or a charge line is changed in a freight document after the collective freight settlement is posted for processing. In this case, a new FSD is created for the changed freight document which can be posted individually or can also be part of the collective settlement, which is created later. The corresponding follow-on documents (PO, SES, and ABD) are also created.

♣ Example

Addition of a new charge line:

There are 2 freight orders, **F01** and **F02** with the following details:

Freight Order	Charges	Rate (in USD)	Final Amount (in USD)	Invoicing Status	Follow-on Documents
FO1	BP Charge Item		220	Posted	
	Base Freight	200		Posted	FSD1, PO1, SES1, ABD1
	Loading Charges	20		Posted	FSD1, PO1, SES1, ABD1
FO2	BP Charge Item		220	Posted	
	Base Freight	200		Posted	FSD1, PO1, SES1, ABD1
	Loading Charges	20		Posted	FSD1, PO1, SES1, ABD1

The collective settlement **FSD1** is:

Collective FSD1			
	Invoice Amount		440 USD
	Freight Documents	FO1	220 USD
		FO2	220 USD

If you add a new charge line **Unloading Charge** to **FO1** and post **FO1**, the system creates a new FSD, **FSD2** with the delta charge, and new follow-on documents (PO, SES, and ABD).

Freight Order	Charges	Rate (in USD)	Final Amount (in USD)	Invoicing Status	Follow-on Documents
FO1	BP Charge Item		230	Posted	
	Base Freight	200		Posted	FSD1, PO1, SES1, ABD1
	Loading Charges	20		Posted	FSD1, PO1, SES1, ABD1
	Unloading Charge	10			FSD2, PO2, SES2, ABD2
FO2	BP Charge Item		220	Posted	
	Base Freight	200		Posted	FSD1, PO1, SES1, ABD1
	Loading Charges	20		Posted	FSD1, PO1, SES1, ABD1

After you confirm the new **Unloading Charge** for **FO1**, the system starts the accrual postings for **FO1** and creates a new FSD for the delta charge for **FO1**. You can post this new FSD individually, or can combine it with the collective settlement and create new follow-on documents (PO, SES, and ABD).

Collective FSD1			
	Invoice Amount		440 USD
	Freight Documents	FO1	220 USD
		FO2	220 USD

FSD2			
	Invoice Amount		10 USD
	Freight Documents	FO1	10 USD

❖ Example

Updating an existing charge line:

There are 2 freight orders, **FO1** and **FO2** with the following details:

Freight Order	Charges	Rate (in USD)	Final Amount (in USD)	Invoicing Status	Follow-on Documents
FO1	BP Charge Item		220	Posted	
	Base Freight	200		Posted	FSD1, PO1, SES1, ABD1
	Loading Charges	20		Posted	FSD1, PO1, SES1, ABD1
FO2	BP Charge Item		220	Posted	
	Base Freight	200		Posted	FSD1, PO1, SES1, ABD1
	Loading Charges	20		Posted	FSD1, PO1, SES1, ABD1

The collective settlement **FSD1** is:

Collective FSD			Invoice Amount
FSD1			440 USD
	Freight Documents	FO1	220 USD
		FO2	220 USD

Now for **FO1**, the **Loading Charge** is changed to 30 USD, then the invoicing status of **FO1** changes to **Ready for Confirmation**. When you save **FO1** the lifecycle status of the FSD is set to **Canceled** and confirmation status to is set to **Not Confirmed**. This status change also triggers the cancellation of the follow-on documents (PO, SES, and ABD) associated with the FSD. Once all the accruals are canceled successfully, the FSD confirmation status changes to **Confirmed**. Also, the invoicing status of **FO2** changes to **Ready for Posting**:

The collective **FSD1** is:

Collective FSD			Invoice Amount	Lifecycle Status
FSD1			440 USD	Canceled
	Freight Documents	FO1	220 USD	
		FO2	220 USD	

Related Information

[Monitor Freight Documents for Accrual](#)

Invoicing Freight Documents

You can post an invoice for a freight document and carrier or logistics service provider combination at the level of a charge element. You can review the amounts and quantities associated with a freight charge element during invoice processing.

Prerequisite

You've integrated TM that enables you to post supplier invoices that refer to freight orders. These freight orders are based on purchase orders with lean service items. You can do that from the customizing activity [Activate Integration of Transportation Management \(TM\)](#) in Customizing of [Materials Management](#) under [Logistics Invoice Verification](#) > [Incoming Invoice](#).

Key Features

- You can create an invoice receipt item for an independent charge. Wherever applicable, the system merges or includes a dependent charge in an invoice receipt item.
- When you post an invoice for any charge, the system changes the posting status of the freight charge to **Invoice Verified**. The system includes the freight document in the document flow.

- **Variances in Freight Invoices**

For a price variance, the system automatically distributes the price difference at the level of logistics document and delivery item in the freight document. The system creates a freight cost allocation document for a price variance in a freight invoice. It then automatically posts the price difference to an expense or material account depending on the business scenario. You also get a price variance when you enter an unplanned freight cost for an invoice. For more information see, [Manage Variance Posting Errors](#).

- **Unplanned Freight Cost**

In the [Create Supplier Invoice - Advanced](#) app (App ID: MIRO) you use the fields **Freight Order** and **Charge Type** within the G/L Account section to support the posting of supplier invoices with unplanned freight costs.

If you want to create an invoice that refers to a freight order with unplanned freight costs against a charge type which isn't part of your freight order, the unplanned freight cost is posted as an unplanned G/L account item for which the transportation management fields **Charge Type** and **Freight Order** are filled accordingly.

In addition, with the [Supplier Invoice - Create \(B2B, Inbound, Asynchronous\)](#) service, also known as **InvoiceRequest_In**, you can now post supplier invoices with unplanned freight costs. For more information, see [InvoiceRequest_In - Item](#).

The system doesn't update the freight document with the unplanned freight cost.

Invoice Processing

You can process an invoice for a freight document as follows:

- [Create Supplier Invoice - Advanced](#)

You can use this app to create and verify an invoice from your carrier or service provider.

- [Supplier Invoice - Create \(B2B, Inbound, Asynchronous\)](#)

Your carrier or service provider can use an electronic business-to-business (B2B) message to submit an invoice to you, the shipper.

- For all transactions related to invoicing, see [Invoicing Freight Orders](#).

i Note

As a first step, the MM (Materials Management) invoice is created and posted, and then the Freight Carrier Invoice (FCI) in Transportation Management (TM) is generated from it.

Related Information

[Invoicing Status of Freight Documents using Lean Services](#)

Manage Variance Posting Errors

App ID: F7090

With this app, you can monitor the errors that arise when you post a price variance for a carrier invoice. You can review the reason for why a variance posting failed and you can also repost the variance after you correct the errors. The app displays the variance posting errors that the system detects when it posts a carrier invoice for which it couldn't create a freight cost allocation document.

After you correct the errors and successfully repost the variance, the system removes the relevant carrier invoice from the app.

Key Features

You can use this app to do the following:

- Monitor carrier invoices for errors when you post a variance
- View the reason for why a variance posting failed
- Repost the failed variance posting for the carrier invoice
- Update the posting date when you repost the carrier invoice for failed variance postings

Additional Details

You can update the posting date when you repost the variance from the app. When you repost the variance, the system creates a freight cost allocation document. It distributes the variance amounts to the relevant delivery item or predecessor order item associated with the freight document. It then posts the variance amount to material valuation or to an expense account (general ledger) based on the business scenario.

Supported Device Types

- Desktop
- Tablet

Reversing a Freight Settlement Document

In your business, you are a requester of transportation services, such as a shipper, and you have a close working relationship with a provider of transportation services, such as a carrier. You've enabled a self-billing (evaluated receipts settlement) process with a particular service provider. You automatically pay the service provider according to the charge details in your freight order, freight booking, or service order.

Occasionally you can have the following business scenario:

1. Your carrier started to execute your transportation request.
2. You accrued the freight cost in Materials Management (MM).
3. You created a self-billing invoice to update the open item in your accounts payable.

Then your carrier encounters some unplanned issues. For example, the carrier doesn't deliver a particular transportation service due to execution issues such as a vehicle breakdown. Also, you can encounter incorrect pricing information in Transportation Management (TM).

To resolve the issues, you need to reverse the freight settlement document in TM. When you transfer the reversal to MM, you need to reverse the associated purchase order, service entry sheet (SES), and accruals posting in MM. You need MM to then run the evaluated receipts settlement for the particular freight order. This posts a credit memo that reverses the self-billing invoice that was created earlier. You can only reverse the freight settlement document when it has the life-cycle status **Invoice Verified**.

The following table describes the steps that you must take to reverse the freight settlement document:

Module	Action	Result	Life Cycle Status of FSD
TM	Choose Reverse Document in the freight settlement document.	TM initiates the reversal process.	Reversed
MM	The approval of a service entry sheet is revoked and the status of the service entry sheet is reset to In Process .	System revokes the SES.	Reversed
MM	You run transaction MRRL (Evaluated Receipt Settlement (ERS) with Logistics Invoice Verification). Under the Transportation Management screen area, you enter the relevant TM freight order number by selecting this from the search help. For more information, see Invoicing Freight Documents .	This creates a credit memo to reverse the previously posted invoice.	Credit Memo Created

TM updates the invoicing status of the relevant freight order to **Ready for Posting**, and you can create a new freight settlement document for the freight order.

Related Information

[Evaluated Receipt Settlement \(ERS\)](#)

[Invoicing Freight Orders](#)

Setting Up Tax Rates for Purchasing Transportation Services

You use Transportation Management (TM) in SAP S/4HANA, private cloud edition, and you post freight cost accruals to Financials. When you post freight cost accruals, the system needs extra information to determine the tax code. You must specify extra conditions to meet your business requirements. The system determines tax in the service purchase order item when it creates the item during the posting step of freight cost accruals.

If you use the evaluated receipt settlement (ERS) process with your carriers and suppliers, to perform any follow-on steps you must determine taxes when you post freight cost accruals.

To enable tax determination in the posting step of freight cost accruals, the system determines the value of the field KOMP-TAXIL (Tax Indicator: Import) as described in the following table:

TAXIL (Tax Indicator: Import)	Description
3	Transportation service for export of goods
4	Transportation service for export of goods in EU (European Union)
5	Transportation service for domestic movement of goods
6	Transportation service for import of goods
7	Transportation service for import of goods in EU

The system uses the following criteria to specify the value of the TAXIL field:

- Country/region of departure and country/region of arrival of the freight document or stage
- Country/region of the freight procuring company

The following table gives you an example of the settings that can apply:

Country/region Code of Freight Procuring Company	Source Location Country/Region of Freight Document/Stage	Destination Location Country/Region of Freight Document/Stage	TAXIL Indicator	TAXIL Indicator Description
DE (Germany)	DE (Germany)	U.S. (United States)	3	Transportation service for export of goods
DE (Germany)	DE (Germany)	FR (France)	4	Transportation service for export of goods in EU
DE (Germany)	DE (Germany)	DE (Germany)	5	Transportation service for domestic movement of goods
DE (Germany)	U.S. (United States)	DE (Germany)	6	Transportation service for import of goods

Country/region Code of Freight Procuring Company	Source Location Country/Region of Freight Document/Stage	Destination Location Country/Region of Freight Document/Stage	TAXIL Indicator	TAXIL Indicator Description
DE (Germany)	FR (France)	DE (Germany)	7	Transportation service for import of goods in EU

The system also takes the tax indicator KOMP-TAXIR (Tax Indicator: Region) into account when it determines the tax code. The system determines the TAXIR (Tax Indicator: Region) as described in the following table:

TAXIR (Tax Indicator: Import)	Description
0	Same region
1	Different region

The system compares the regions of the source and destination locations and determines the TAXIR (Tax Indicator: Region) as described in the following table:

Source Location Region	Destination Location Region	TAXIR (Tax Indicator: Region)	TAXIR Description
HH (Hamburg)	HH (Hamburg)	0	Same region
HH (Hamburg)	BW (Baden-Wuerttemberg)	1	Different region
HH (Hamburg)	CA (California)	1	Different region
IN (Indiana)	HH (Hamburg)	1	Different region

You can also configure tax determination for source and destination country/region and postal code by following the configuration steps in [Pricing and Conditions](#).

The fields that can be used to determine the country/region and postal code are described in the following table:

Field Name	Description
KOMK-SLAND1_A	Tax country/region departure
KOMK- SLAND1_Z	Tax country/region destination
KOMK- REGIOA	Departure region (federal state, county, province, duchy)
KOMK- REGIOZ	Destination region (federal state, county, province, duchy)
KOMK- PSTLZA	Departure postal code
KOMK- PSTLZZ	Target postal code

For more information about specifying the tax determination settings, see [Set Tax Rates - Purchasing](#) .

Monitor Freight Documents for Accrual

App ID: F4329

With this app, you can monitor your freight documents to post accruals. You can confirm a freight document to post the freight cost to Financial Accounting for accruals. You can monitor your freight documents in case there are errors during the posting process. To post the freight costs to Financial Accounting, you must confirm that the charges for a freight document are a correct reflection of the planned costs to be paid to the carrier. When you confirm a freight document for posting, the system posts the freight costs to Financial Accounting. You can also identify freight documents for which you have successfully posted the freight costs.

Key Features

You can use this app to perform the following tasks:

- Confirm freight costs for posting to Financial Accounting for accruals
- View freight documents that are successfully posted to Financial Accounting
- Cancel postings for freight documents that are successfully posted to Financial Accounting
- Monitor freight documents that have errors when you post them to Financial Accounting
- Display the errors that the system encountered when it posted your freight documents
- Post or cancel posting for freight documents for which the system encountered errors during posting

Additional Details

You can check out road and rail freight orders from the [Create Road Freight Order](#) or [Create Rail Freight Order](#) app. When you check out the last or only stop of the freight order and the stop belongs to your organization, the system gives the freight order a posting status of **Ready for Confirmation**. In cases where the source and destination stop don't belong to your organization, the posting status of the freight order is set to **Ready for Confirmation** after departure at the source stop. For more information, see [Freight Cost Confirmation and Posting](#).

You can set an ocean freight booking to **Shipped on Board**. The system gives the freight booking a posting status of **Ready for Confirmation**. For more information, see [Freight Booking](#). You can set an air freight booking to **Uplift Confirmed**. The system gives the air freight booking a posting status of **Ready for Confirmation**. For more information, see [Management of Air Freight Bookings for Airlines](#).

You can view these freight documents in this [Monitor Freight Documents for Accrual](#) app, in the **For Confirmation** view.

When your freight documents are in the invoicing status **Ready for Confirmation** and you've verified that the planned costs are correct, you can confirm the freight cost for posting to Financial Accounting.

The system sets the invoicing status to **Posted** when the posting is successful or to **Posting Failed** if it encounters errors during posting. In the **Errors** view, you can show the list of freight documents that have an invoicing status of **Posting Failed**. In the **Posted** view, you can see the list of freight documents that have an invoicing status of **Posted**.

You can see the freight documents with the invoicing status **Ready for Posting** under the **For Posting** view.

You can **Post** freight documents to financials in any of the following ways:

- Individual: You select only one document and post it individually to the financials.
- Collective: You can select multiple freight documents and post them.

i Note

You can do collective settlement only for those freight documents whose **Calculation Option** in the **Settlement Profile** is **Copy All Charges**. You can see the **Calculation Option** in the Customizing for Transportation Management, under ►

- Using Profile Settings: This posts the freight documents to the financials based on the value defined in the [Calculation Option](#) in the [Settlement Profile](#) that's relevant to the freight document.

Supported Device Types

- Desktop
- Tablet

Monitor Freight Settlements for Accrual

App ID: F8720

With this app, you can monitor the freight settlement postings for your freight documents.

Key Features

This app provides the following key features:

- Unblock or cancel blocked freight settlement documents

You can see a list of freight documents under the tab [Ready for Accruals](#) that are blocked. You can navigate to these documents and see the reason for their block and act upon it. You can then either [Unblock](#) the freight document and then [Post](#) the freight settlement to Materials Management (MM) for further processing or [Cancel](#) the blocked freight settlement document.

- View posting failed freight settlement documents

You can see a list of freight documents for which posting of accruals has failed under the [Posting Failed](#) tab and navigate to these documents to see the specific errors that occurred during the accrual process. Posting typically fails when the creation of a follow-on document encounters an error.

Once you resolve the errors, you can [Reprocess](#) the document.

- View posted freight settlement documents

You can see a list of freight documents that are successfully posted to MM, under the [Posted](#) tab. You navigate to these documents and then to the [Document Flow](#) tab to see any follow-on documents that were created. If necessary, you can also [Cancel](#) the accruals.

- View cancellation failed freight settlement documents

You can see a list of freight documents for which cancellation failed, under the [Cancellation Failed](#) tab. You can view the reason for the failure under [Application Log](#). Once you've resolved the error, you can [Reprocess](#) the freight settlement document. Cancellation typically fails when an error prevents the cancellation of a related follow-on document.

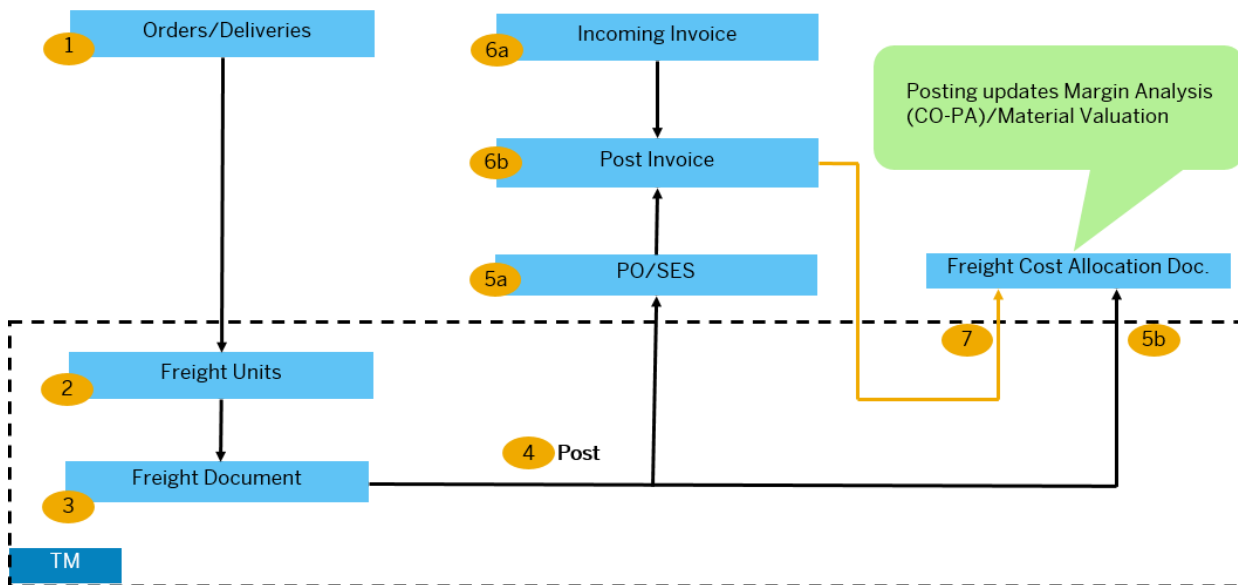
Supported Device Types

- Desktop
- Tablet
- Smartphone

Posting of Allocated Freight Costs

You can allocate freight costs at the level of document and delivery item in a freight document. The system posts the costs to a material valuation in an inbound process, and to an expense account (general ledger) in an outbound process.

The following graphic gives a high-level overview of the settlement process for shippers, and is followed by an explanation:



- In step 1, you create an order in the **Sales** and **Sourcing and Procurement** components.
- In 2, you create a freight unit (FU) out of the order.
- In 3, you plan the FUs in a freight document.
- In 4, you post the freight document.
- In 5a, the system creates a service purchase order (PO) and service entry sheet (SES) in **Sourcing and Procurement**.
- In 5b, the system also posts the freight cost to material valuation for an inbound process or expense for an outbound process.
- In 6a, you receive the invoice from the carrier.
- In 6b, you post the invoice against the accrued freight cost, from step 5a.
- 7 applies if you have a price difference in an invoice variance. The system posts the variance amount to a material valuation for an inbound process or to an expense for an outbound process.

For example, you have a net amount in document currency of USD 1845 for a freight document consisting of two deliveries, with a cost distribution of USD 800 for product 1 for delivery 1, and USD 1045 for product 2 for delivery 2. When you post the freight document, the system takes the following steps:

- Creates a service purchase order for USD 1845 and creates an SES for USD 1845.
- Creates a freight cost distribution document with each product as an item.

Each item has the amount of the distributed cost, USD 800 for product 1, and USD 1045 for product 2.

You can post the distributed cost to material valuation or general ledger. Depending on whether the process is inbound or outbound, the freight cost allocation document posts the freight cost to the respective cost objects.

For example, in an outbound process you have a freight document with cargo items that are linked to the deliveries in a sales order. The system posts the freight cost to the general ledger. When you post the freight document with a freight cost distribution, the

system passes on the distributed costs to the freight cost allocation document. The system takes the following steps:

- Copies the references from the freight document cost distribution and gets the linked underlying delivery items.
- Copies the product and quantities in the delivery item to the items in the freight cost distribution document for product 1 and product 2.
- Copies the account assignments for the respective item in the freight cost distribution document from the underlying sales order item to which the delivery item is linked.
- Copies the allocated amounts to the items in the freight cost allocation document (USD 800 for product 1 and USD 1045 for product 2).

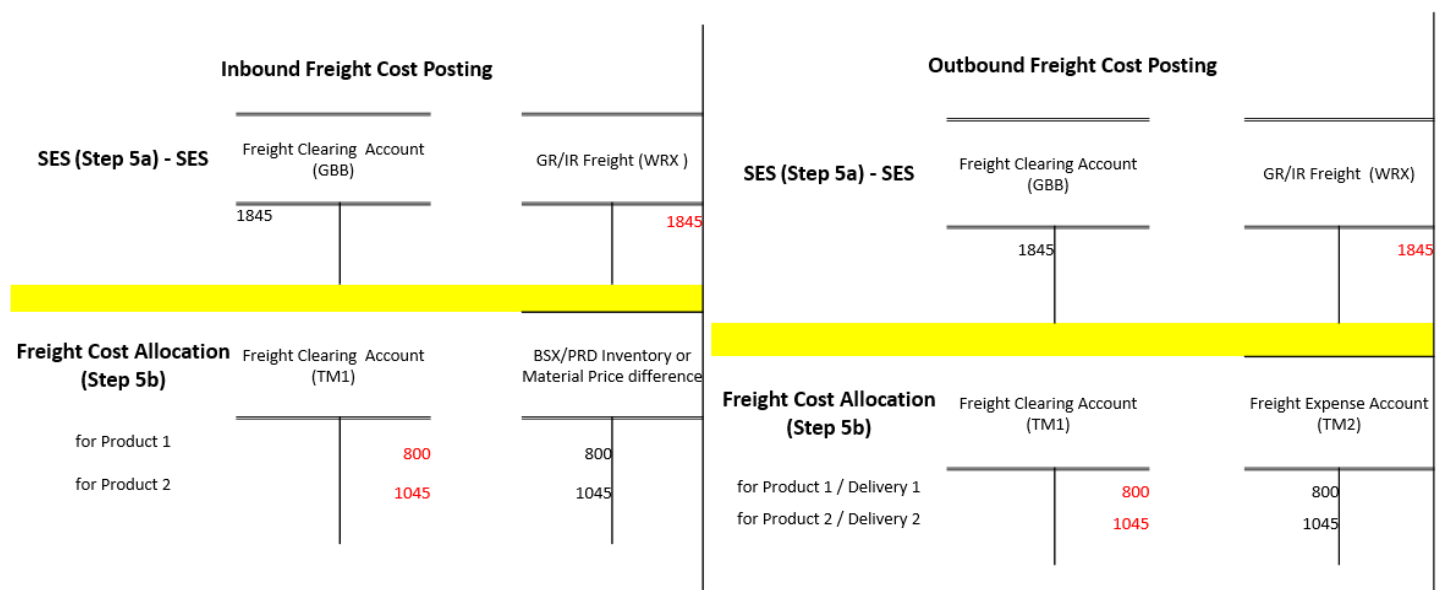
Examples of an account assignment can be cost objects including a cost center, profit center, Controlling – Profitability Analysis (CO-PA) segment, or an internal order.

Additional Details

You can review the relevant freight cost allocation document in the [Manage Freight Cost Allocation Documents](#) app. You can view the freight cost allocation documents by entering the freight order or freight booking in the [Monitor Freight Cost Allocation Document Items](#) app. The app provides list reporting at the level of freight cost allocation document item. The system automatically releases the freight cost allocation document to accounting.

In special cases, you can release the freight cost allocation document in the [Manage Freight Cost Allocation Documents](#) app or the [Schedule Document Release](#) app.

The following graphic displays a posting pattern for a freight document and is followed by an explanation:



In step 5a, the system uses the SES to post the total freight cost to an intermediate freight clearing account. In step 5b, the system uses the freight cost distribution document to allocate the freight cost to **Sourcing and Procurement** for an inbound process, or to a general ledger expense account for an outbound process. The system uses the material information and the account assignments from the underlying sales orders or purchase orders.

In step 5a, you post freight costs in an inbound and outbound process. The system uses the account posting key GBB as the offset entry for inventory posting. This points to a freight clearing account or a similar clearing account.

In step 5b, you post freight costs in an inbound process. You post the freight cost to a material valuation. The freight clearing account key comes from the pricing procedure (also known as a calculation schema) in the freight cost allocation document. The material and its attributes determine the inventory posting account. The system posts the costs to BSX inventory posting or PRD

cost price differences. The system bases the postings on the typical material posting that you set up, depending on the price control of the material and the material stock position.

Also in step 5b, you post the freight costs in an outbound process. You post the freight cost to an expense account. The system uses the pricing procedure from the freight cost allocation document to determine the account keys for freight clearing and freight expense.

Prerequisites

- Distribution profile

The system uses criteria such as gross weight, net weight, gross volume, and distance times weight to perform freight cost distribution. To configure a distribution profile, see the Customizing Activity [Define Charges Profiles](#) under Customizing for [Transportation Management](#) [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#).

- G/L account settings

You must change the [Manage G/L Account Master Data](#) accounting settings.

For more information, see the setup instruction guide for any of the Transportation Management (TM) scope items under the best practices for the solution scenario at https://me.sap.com/processnavigator/SolS/EARL_SolS-055. Navigate to [Supply Chain](#) [Delivery and Transportation](#) [Freight Settlement \(6W4\)](#).

Example of Freight Cost Distribution with a Business Use Case

The following graphic contains an example of a freight document with a distribution level of [Order/Delivery Item](#). It's followed by an explanation.

Item Hierarchy of Freight Order or Freight Booking

Item hierarchy	Item #	Gross Weight (KG)	Tare Weight (KG)
Container 1	10	6,500	1,000
Package 1	20	2,250	0
Product 1	30	2,250	0
Package 2	40	3, 250	500
Product 2	50	2,750	0

The above table contains the item hierarchy of a freight order or freight booking. In the document, you have two products. You packed each product in a separate package with appropriate labels for each package. You placed both packages in the same container.

Charges for the Freight Order or Freight Booking

Charges	Logistical Reference	Amount (USD)
Base Freight	Root	1000
Container Charges	Container	500
Packaging	Package (For Package 1)	50
Packaging	Package (For Package 2)	100
Labeling	Product (For Product 1)	75
Labeling	Product (For Product 2)	120
Total Charges		1,845

The above table contains the charges for the freight document. The system bases some charges on the packaging or product. It also bases other charges on the root or the container, as you can see in the **Logistical Reference** column. The system distributes some charges by distribution rule. It distributes other charges directly. You can see this distribution in the following graphic:

Distribution by Gross Weight

Distribution Item/Charge Type	Value (USD)	Calculation/Remarks
Product 1 (2,250 KG)	800	
Base Freight	450	$1000 * 2,250 / (2,250 + 2750)$
Container Charges	225	$500 * 2,250 / (2,250 + 2750)$
Packaging	50	Assigned directly (package 1 has only product 1)
Labeling	75	Assigned directly as its for product 1
Product 2 (2,750 KG)	1,045	
Base Freight	550	$1000 * 2,750 / (2,250 + 2750)$
Container Charges	275	$500 * 2,750 / (2,250 + 2750)$
Packaging	100	Assigned directly (package 2 has only product 2)
Labeling	120	Assigned directly as its for product 2

The system assigns the costs for packaging and labeling directly. The distribution rule has no influence when the costs are allocated directly.

Flexibility in Freight Charge Changes

After you've posted accruals for a freight document, you can do the following changes to the freight costs:

- Add or update charge amounts directly in the existing freight documents and also transfer these changes in the freight costs to Materials Management (MM).
- Add an independent freight charge for a freight document and carrier or service provider combination, after you've verified the existing charges in an invoice. You can subsequently confirm the newly added freight charge for posting accruals.

i Note

- You can't change or delete an existing charge line with the invoicing status **Invoice Verified**.
- You can't add a dependent charge that refers to an independent charge line which has the invoicing status **Invoice Verified**.

For a freight document and carrier or service provider combination where some charges are invoice verified, you have the following conditions:

- Where some charges are invoice verified and other charges are posted, you can't change an invoice verified charge line. Also, you can't include dependent charges that reference charges that are already invoice verified.
- Where some charges are invoice verified and you have a dependent charge that references a charge range, you cannot include an independent charge in the range of dependent charges.

For freight settlement integration using lean services, the system identifies the change process strategy based on the nature of change to the charge line.

i Note

The change process strategy for that you've defined in the Customizing activity for **Transportation Management** under **Settlement** ► **Define Settlement Profile** ► isn't considered for the freight settlements using integration with lean services.

The system uses change management where the previous freight settlement document (FSD) along with the purchase order (PO) and service entry sheet (SES) is canceled and a new freight settlement document with the updated amount is created and posted.

You can use the background function **Creation and Transfer of Freight Settlement Documents** (program /SCMTMS/SFIR_CREATE_BATCH) to create and post freight settlement documents to MM. For more information see, [Creation and Transfer of Freight Settlement Documents using Integration with Lean Services](#). The freight settlement document contains the changes from the freight document.

Changes to Invoice Verified Freight Documents

When there are changes to the charges for the business partners in the freight document which is already invoice verified, then the system handles the changes in the following ways:

- When you add a new charge line, a new FSD and follow-on documents (Purchase Order (PO) and Service Entry Sheet (SES)) are created only for the new delta charge line without any changes to the existing follow-on documents.
- Any change to the already invoiced charge lines isn't allowed.

The following table has some scenarios where there can be changes to **Invoice Verified** freight documents:

Scenario	Description
Agreement Redetermination	Once the invoice is verified for the business partners in the freight document, you as a shipper along with your carrier notice that there's a Toll Charge in the route and you decide to include it in the freight agreement. This additional charge in the freight document is posted and invoiced in separate follow-on PO and SES documents. In case there are changes in the already invoiced charge line due to agreement re-determination, then charge line isn't updated in the freight document charges.

Scenario	Description
Change in Logistical Data	There can also be instances where there are changes to the logistical data. Example, the weight of the cargo during the planning to the actual weight can differ due to the packaging activities which occurs during the warehouse activity. In such cases the information can be updated later in the process after the charges in the freight document are already posted. The change in logistical data doesn't have any impact on the Invoice Verified charge lines.
Unplanned Charges	Unplanned charges are additional services that are provided or received as part of the execution step. Example, there could be additional toll charges since the route used can be tolled roads or can have fumigation charges incurred during execution. In such cases the information can be updated later in the process after the charges in the freight document are already posted or the invoice verified.

Changes to Posted Freight Documents

You can modify a charge line which has the invoicing status **Posted**. For posted charge lines in a freight document if the system has already created the follow-on documents (like FSD, PO, or SES) the system cancels the existing follow-on documents and creates new follow-on documents with updated amounts.

When you modify a **Posted** charge line, the system updates the status of the modified charge line to **Ready for Confirmation** and the same is rolled-up to charge item and charge header level. Once you've confirmed freight documents with the status **Ready for Confirmation**, the status changes set to **Ready for Posting**.

The following table shows the scenarios where the invoicing status is set to **Ready for Confirmation** after the FSD is posted and the level at which the status change happens:

Charge Level	Scenario	Impact
Charge Element	Any manual or automatic change that changes the amount of a charge element	Rolled-up to charge item level and then charge header level, based on the following priority: 1. Ready for Confirmation 2. Posting Failed 3. Ready for Posting 4. Posted 5. Invoice Verified 6. Not Ready for Posting
	A new charge line is added under the charge item manually or comes in from an agreement	
	Logistical data change resulting in rate change	
	Change in resolution base of a charge line	

Examples

The following examples show various scenarios and the follow-on documents that are created:

❖ Example

Change to charge amount in **Posted** charge line

The table shows the setup of the charge lines in a freight order:

Line No.	Charge Type	Rate (in USD)	Final Amount(in USD)	Invoicing Status	Follow-on Documents
Charge Header				Posted	
BP Charge Item			220	Posted	
10	Base Freight	200		Posted	FSD1, PO1, SES1
20	Loading Charge	20		Posted	FSD1, PO1, SES1

Due to unforeseen circumstances, the loading charge is changed to 25 USD. As part of accrual posting, the system cancels the FSD1, PO1, SES1 is canceled and creates FSD2, PO2, SES2. Also, for the freight cost allocation posting, the system creates the reversal freight cost allocation document to reverse the amount 220 USD in the already posted freight cost allocation document and creates a new freight cost allocation posting document for the updated amount 225USD.

Line No.	Charge Type	Rate (in USD)	Final Amount(in USD)	Invoicing Status	Follow-on Documents
Charge Header				Ready for Confirmation	
BP Charge Item			225	Ready for Confirmation	
10	Base Freight	200		Posted	FSD2, PO2, SES2
20	Loading Charge	25		Ready for Confirmation	FSD2, PO2, SES2

❖ Example

Addition of a new charge line to **Posted** freight documents:

For new charge line additions, the system creates FSD and subsequent follow-on documents only for the delta amount (newly added amount) in the freight document without canceling the existing postings/ follow-on documents (FSD, PO, and SES). The system determines if a delta settlement needs to be created based on the nature of the change:

- For accrual postings, the system creates new FSD, PO, and SES documents for the newly added (delta) charge in the freight document without canceling the existing FSD, PO, and SES.
- For freight cost allocation postings, the system creates new freight cost allocation document for the newly added (delta) charge in the freight document. The system doesn't change the already created freight cost allocation document and also doesn't create any reversal freight cost allocation document.

The table shows the setup of the charge lines in a freight order:

Line No.	Charge Type	Rate (in USD)	Final Amount(in USD)	Invoicing Status	Follow-on Documents
Charge Header				Posted	
BP Charge Item			220	Posted	
10	Base Freight	200		Posted	FSD1, PO1, SES1

Line No.	Charge Type	Rate (in USD)	Final Amount(in USD)	Invoicing Status	Follow-on Documents
20	Loading Charge	20		Posted	FSD1, PO1, SES1

Due to changes in the route, there's **Toll Charge** incurred and you update it in the freight document. As part of accrual posting, FSD2, PO2, SES2 is created only for the delta charge **Toll Charge** without canceling the existing documents.

Line No.	Charge Type	Rate (in USD)	Final Amount(in USD)	Invoicing Status	Follow-on Documents
Charge Header				Ready for Confirmation	
BP Charge Item			230	Ready for Confirmation	
10	Base Freight	200		Posted	FSD1, PO1, SES1
20	Loading Charge	20		Posted	FSD1, PO1, SES1
30	Toll Charges	10		Ready for Confirmation	FSD2, PO2, SES2

Also, for the freight cost allocation posting, the system creates a new freight cost allocation posting document for the delta amount of 10 USD.

Related Information

[Invoicing Status of Freight Documents using Lean Services](#)

Blocking and Unblocking of Freight Documents

Generally, the freight settlement document (FSD) is created and posted when you confirm and post the freight document. But there can be scenarios where charges are different for the freight document and the FSD. This difference can be due to redetermination of a different freight agreement or redetermination of the rates in the same freight agreement during the creation of FSD. The redetermination of freight agreements or rates can result in charges mismatch between the freight document and the FSD.

The system can redetermine a freight agreement while FSD creation, if you've selected **Copy Fixed Charges and Redetermine Others** as **Calculation Option** in the Customizing for **Transportation Management** under **Settlement > Define Settlement Profile > Settlement Profile Settings**.

Whenever the system creates and posts an FSD from a freight document it checks the following:

- If there are no charges mismatches during FSD creation, the FSD is created and posted. The invoicing status of the freight document is set to **Posted** or **Posting Failed** based on whether the FSD was posted successfully or not.
- When there are mismatched charges between the freight document and FSD during FSD creation, the system checks whether those mismatched charges can be overruled or not.

Blocking during Mismatched Charges

Charges mismatch can be categorized into two types:

- Mismatches which are acceptable and can be overridden

In this case, you can review the mismatch that occurred in the freight document and FSD charges. If these mismatches are acceptable, for example if you and your carrier have mutually agreed to a rate change in master data, you can override the block manually in FSD and continue the accrual process.

Alternately, you can recalculate the charges in the freight document, create the FSD, and continue the accrual process.

- Mismatches which are inconsistent and can't be overridden

In these cases, mismatches occur when master data is deleted, for example charge lines or agreements. When you start the accrual posting from the freight document, then the system detects these changes and does the following:

- Stops the FSD creation.
- Sets an invoicing block on the freight document to stop any further settlement process until the freight document charges are corrected.
- The invoicing status of the freight document is set to **Posting Failed**.

Examples

Scenario: No mismatch in charges

If there's no mismatch in charges, then the freight document and the FSD aren't blocked.

Scenario: Mismatch in charges due to definition in agreement (order relevant charges)

The following tables show the setup of a freight order **FO1** and the corresponding freight settlement document **FSD1**:

Freight Order	Charge Type	Rate (in USD)	Final Amount (in USD)
FO1	FB00	100	150
	LOAD (Order Relevant)	50	

Freight Settlement Document	Charge Type	Rate (in USD)	Final Amount (in USD)
FSD1	FB00	100	100

In this case, both the **FO1** and **FSD1** aren't blocked as the mismatch is only for order relevant charges.

Scenario: Mismatch in charges due to definition in agreement (invoice relevant charges)

The following tables show the setup of a freight order **FO1** and the corresponding freight settlement document **FSD1**:

Freight Order	Charge Type	Rate (in USD)	Final Amount (in USD)
FO1	FB00	100	100

Freight Settlement Document	Charge Type	Rate (in USD)	Final Amount (in USD)
FSD1	FB00	100	150

Freight Settlement Document	Charge Type	Rate (in USD)	Final Amount (in USD)
	LOAD (Invoice Relevant)	50	

In this case, both the **FO1** and **FSD1** aren't blocked as the mismatch is only for invoice relevant charges.

Scenario: Amount Mismatch

The following tables show the setup of a freight order **FO1** and the corresponding freight settlement document **FSD1**:

Freight Order	Charge Type	Rate (in USD)	Final Amount (in USD)
FO1	FB00	100	150
	LOAD	50	

Freight Settlement Document	Charge Type	Rate (in USD)	Final Amount (in USD)
FSD1	FB00	120	170
	LOAD	50	

In this case, both the freight order and FSD have the status **Blocked** as there's an amount mismatch. To override, you can:

- Manually override the block in FSD and then post to accruals. You can use the **Monitor Freight Settlements for Accrual** app to unblock the FSD. For more information, see [Monitor Freight Settlements for Accrual](#).
- Recalculate the charges in the freight order which updates the charges in the freight order and removes the invoicing block. The system then creates and posts the new FSD.

Scenario: Missing charges due to change in master data

The following tables show the setup of a freight order **FO1** and the corresponding freight settlement document **FSD1**:

Freight Order	Charge Type	Rate (in USD)	Final Amount (in USD)
FO1	FB00	100	150
	LOAD	50	

Freight Settlement Document	Charge Type	Rate (in USD)	Final Amount (in USD)
FSD1	FB00	100	100

In this case, there are changes in the master data and the charges aren't updated in the freight order. To override the block, you've to recalculate the charges in **FO1** which updates the charges in **FO1** and the system removes the invoicing block and creates a new FSD and posts it.

Scenario: Charges mismatch due to agreement redetermination

The following tables show the setup of a freight order **FO1** with the agreement **AGR1** and the corresponding freight settlement document **FSD1** with the agreement **AGR2**.

Freight Order	Charge Type (in charge items of AGR1)	Rate (in USD)	Final Amount (in USD)
FO1	FB00	100	150
	LOAD	50	

Freight Settlement Document	Charge Type (in charge items of AGR2)	Rate (in USD)	Final Amount (in USD)
FSD1	BASE	100	100
	DOCUMENTATION	50	

In this case, due to redetermination of a new freight agreement during FSD creation, charges aren't updated in the freight order. To override the block, you've to recalculate the charges in **FO1** which updates charges in **FO1** and the system removes the invoicing block and creates a new FSD and posts it.

i Note

If you've selected **Copy Fixed Charges and Redetermine Others** as **Calculation Option** in the Customizing, then for charge calculation in the follow-on documents, the system picks only that freight agreement and calculation sheet that has been determined and used for the charge calculation in the freight document, and the freight agreement and calculation sheet is still valid.

Handling Calculated Charges in Lean Services

In charge calculation, the final amount for a freight document is derived from the calculation parameter which is $(\text{Rate} \times \text{Amount} * \text{Quantity}) / \text{Price Unit}$. For accrual postings using lean services, the calculation parameters from the charge element are mapped to the corresponding purchase order (PO) items.

There can be scenarios where after you've calculated the charges for a freight document and the final amount of a charge element can be different from the calculation parameters. If these calculation parameters are mapped to the PO, then the final amount in PO is different from the final amount that is calculated in the freight document resulting in inconsistency.

The scenarios where there can be inconsistency are:

- Charge Calculation Methods
 - Clipping/Clipping_V2
 - External calculation methods, such as **SMC3**
 - Air freight: standard rating
 - Air freight: break-weight rating
- Multiple calculation rules in rate tables
- Amount based calculation bases, such as **GOODS_VALUE**

For more information on calculation methods, see [Calculation Methods](#) and on calculation rules, see [Rate Table and Local Rate Table](#).

Deficit Weight Rate (DWR) Calculation Method

When you use the calculation method **DWR** for a charge line, an additional charge line for the deficit amount with the same charge type and same rate amount is created after charge calculation. The deficit weight charge line is a generated charge line which isn't part of any agreement or is planned between the involved parties. Hence, the deficit weight charge line isn't sent to the PO for further processing. Rather, the quantities of the deficit weight charge line are merged with the quantity of the parent charge line(s) and then sent to PO for further processing.

The resolution base of the deficit charge line and the parent charge line can be **ROOT** or different resolution base, such as **PRODUCT** or **PACKAGE**.

When both the deficit weight charge line and the parent charge line have the resolution base as **ROOT**, the system adds two charge lines (deficit weight charge line and parent charge line) with the same charge type and same resolution base. But when the freight document is posted then only the parent charge line is sent to the PO and the quantity of the deficit weight charge line is merged with the parent charge line.

When the deficit weight charge line and parent charge line have a resolution base apart from **ROOT**, the deficit weight charge line isn't sent to the PO but the quantity of the deficit charge line is merged with the quantity of the parent charge line, and the parent charge line is sent to PO. The weighted average of the quantity of the deficit weight charge line (with respect to the quantity of each parent charge line) is calculated and added to the quantity of the corresponding parent charge line. The other parameters like rate amount, calculation rule, and price unit isn't changed for any of the charge lines.

❖ Example

The following table shows a freight order, **FO1** with two charge lines with the same resolution base as **ROOT**. One is a parent line with charge type **SD02** and the other is deficit weight charge line **SD02**(DWR). When the freight document is posted, the **SD02** (DWR line) isn't sent to PO. The main **SD02** line is sent where the quantities are combined (1800 KG + 200 KG = 2000 KG).

Charge Line	Rate Amount (in EUR)	Quantity (in KG)	Merged Quantity in KG (to be sent to PO)	Amount in EUR (to be calculated in the PO item)
SD02	25	1800	1800+200=2000	2000*25=50000
SD02 (DWR)	25	200	Not sent to PO	

❖ Example

The following table shows a freight order **FO2** with three charge lines, two with the same charge types **SF02** and the one deficit weight charge line **SF02** (DWR). All three have the same resolution base as **PRODUCT**.

Charge Line	Rate Amount (in EUR)	Quantity (in KG)	Weighted Quantity from DWR (in KG)	Merged Quantity (in KG, to be sent to PO)	Amount in EUR (to be calculated in the PO item)
SF02 (Product 10)	2	30	$10 * (30 / (30 + 50)) = 3.75$	$30 + 3.75 = 33.75$	$33.75 * 2 = 67.5$
SF02 (Product 20)	2	50	$10 * (50 / (30 + 50)) = 6.25$	$50 + 6.25 = 56.25$	$56.25 * 2 = 112.5$
SF02 (DWR)	2	10	Not sent to PO	Not Sent to PO	
Total Amount					$(67.5 + 112.5) = 180$

Here the weighted average for the deficit weight charge line is derived from the other two charge lines and the total amount is calculated based on the weighted average.

The weighted average values are added to the quantities of the corresponding parent charge line and that parent charge line is sent to PO.

Adjusted Charge Type (ADJ)

In freight tendering, you as a shipper can tender a road freight order or ocean freight booking to one or more carriers. Tendering can be based on freight Request for Quotation (RFQ) (only for road freight orders) or direct tendering. In some cases there can be a difference between the amounts calculated by the shipper and proposed by the carrier. In those cases, **ADJ** charge type is used as the adjusted amount. The following steps explain the tendering process and the usage of **ADJ** charge type:

- You as a shipper, request for a quotation from the carriers for the transportation services. You can also set a price limit for the carrier to consider your request.
- The carrier responds to your quotation based on the price limits.
- You can reject or accept the quotation or have further negotiation with the carrier.

When you accept (award) the quotation from the carrier, the system adjusts the charges in the freight order or freight booking with the amount based on the response received from the carrier.

- Assuming there's already an agreement between you and the carrier, and the amounts are calculated based on that agreement, the system adjusts the difference in the charges grid in the form of a new charge line with charge type **Adjusted/Quoted Amount (ADJ)**. **ADJ** is a delta amount between charges from the shipper and amount the proposed by the carrier.

Based on the difference in the amount provided by the carrier and the one calculated from the agreement, the amount for **ADJ** line can be negative or positive. **ADJ** charge line is created automatically to handle the adjustment in charges so it doesn't depict any transportation charges and isn't passed on to PO. Instead, the **ADJ** charge amount is distributed to the other charge lines under the same charge item based on weighted average.

i Note

- You can't edit the amount of an **ADJ** charge type manually.
- As the **ADJ** lines aren't part of accruals posting process, its posting status is always blank.

Example

❖ Example

The following example shows the distribution of **ADJ** charge amount based on weighted average when the adjusted charge (**ADJ**) is positive:

The following are the amounts for various charge lines in freight order **FO3** calculated from a freight agreement.

Charge Type	Rate Amount (in USD)	Amount (in USD)
FB00	200	200
BASE_3PL	20	570
Total Amount		770

The total amount is 770 USD. There's a carrier who has proposed 1100 USD, and the shipper has agreed to award the carrier. In this case, the adjusted (**ADJ**) charge amount is $(1100 - 770) \text{ USD} = 330 \text{ USD}$, which is automatically calculated by the system.

Now, the charges in **FO3** look like :

Charge Type	Rate Amount (in USD)	Amount (in USD)
FB00	200	200
BASE_3PL	20	570
ADJ		330
Total Amount		1100

To fit this adjusted amount (**ADJ** charge line), the system calculates the weighted average and then calculates the charges based on the weighted average and sends to PO as shown in the following table:

Charge Type	Rate Amount (in USD)	Amount(in USD)	Weighted Average/Adjusted Value	Amount to be sent to PO (Amount+Adjusted Value) (in USD)
FB00	200	200	$(200 / (200 + 570)) * 330 = 85.71$	285.71
BASE_3PL	20	570	$(570 / (200 + 570)) * 330 = 244.29$	814.29
Total amount				1100

The adjusted charge line (**ADJ**) isn't sent to the PO, but after the adjusted amount is calculated, the charge lines with charge type **FB00** and **BASE_3PL** are sent to the PO.

Example

♣ Example

The following example shows the distribution of **ADJ** charge amount based on weighted average, when the adjusted charge (**ADJ**) amount is negative:

The following are the amounts for various charge lines in freight order **FO4** calculated from a freight agreement.

Charge Type	Rate Amount (in USD)	Amount (in USD)
FB00	450	450
BASE_3PL	550	550
Total Amount		1000

The total amount is 1000 USD. There's a carrier who has proposed 900 USD, and the shipper has agreed to award the carrier. In this case, the adjusted (**ADJ**) charge amount is $(900-1000)$ USD = -100 USD, which is automatically calculated by the system. Now, the charges in **FO4** look like :

Charge Type	Rate Amount (in USD)	Amount (in USD)
FB00	450	450
BASE_3PL	500	550
ADJ		-100

Charge Type	Rate Amount (in USD)	Amount (in USD)
Total Amount		900

To fit this adjusted amount (**ADJ**) charge line), the system calculates the weighted average and then calculates the charges based on the weighted average and sends to PO as shown in the following table:

Charge Type	Rate Amount (in USD)	Amount(in USD)	Weighted Average/Adjusted Value	Amount to be sent to PO (Amount+Adjusted Value) (in USD)
FB00	450	450	$(450 / (450 + 550)) * (-100) = -45$	405
BASE_3PL	550	550	$(550 / (450 + 550)) * (-100) = -55$	495
Total amount				900

The adjusted charge line (**ADJ**) isn't sent to the PO, but after the adjusted amount is calculated, the charge lines with charge type **FB00** and **BASE_3PL** are sent to the PO.

Related Information

[Calculation Methods](#)

[Charges Overview](#)

Posting of Allocated Freight Costs in Advanced and Multistage Intercompany Business Processes

You can use a freight order or freight booking to transport cargo in a business process for advanced or multistage intercompany sales or advanced or multistage intercompany stock transfer. The difference between the advanced intercompany and multistage intercompany scenario is that, apart from the selling and delivering company, additional companies, known as intermediate companies, can be involved in the process.

The responsibility for the freight cost depends on the Incoterms that are in place between the relevant companies.

- In an advanced intercompany sales process, the responsibility is with the delivering and selling company.
- In an advanced intercompany stock transfer process, the responsibility is with the delivering and receiving company.
- In a multistage intercompany sales or stock transfer process, the responsibility is with the delivering and/or receiving, selling company as well as intermediate companies.

The system uses the responsibility for the logistics stage of the freight unit to determine the paying company code, based on the incoterms of the stock transfer order or the intercompany sales order. The procuring company code of the freight order or freight booking can be that of the delivering, selling, receiving company, or intermediate companies.

The system posts the allocated freight costs to expenses or material valuation in financial accounting based on the answers to the following questions:

- Which company code is responsible for the freight cost?

- Is the freight cost part of an advanced/multistage intercompany sales or advanced/multistage intercompany stock transfer process?
- Is this company the delivering, receiving, or selling company code?

The system posts the freight cost allocation to expenses or to a material account if the material is relevant for valuation in financial accounting.

Advanced Intercompany Sales

For more information about this process, see [Advanced Intercompany Sales](#).

The system uses the Incoterms agreed between the delivering company and selling company to determine the responsibility for the freight costs. It posts the distributed costs as follows:

- In the delivering company, as an expense for the intercompany sales order item (see step 4 in [Advanced Intercompany Sales](#)) and the outbound delivery item (see step 5).
- In the selling company, as an expense for the standard sales order item (see step 2) and the outbound delivery item (see step 5).

Advanced Intercompany Stock Transfer

For more information about this process, see [Advanced Intercompany Stock Transfer](#).

The system uses the Incoterms agreed between the delivering company and receiving company to determine the posting of the freight cost. It determines the posting of the freight cost in the delivering company as an expense because the freight movement is outbound for the intercompany sales order item (see step 4 in [Advanced Intercompany Stock Transfer](#)) and the outbound delivery item (see step 5).

If the material being transported is relevant for valuation in the delivering company, and you have enabled group valuation in universal parallel accounting, the system also posts the freight cost in the group valuation ledger of the delivering company code that is considered for group valuation. The receiving company adjusts the material valuation. For more information about a group valuation ledger, see [3191636](#) .

Multistage Intercompany Sales

For more information about this process, see [Multistage Intercompany Sales](#).

The system uses the Incoterms agreed between the delivering company, intermediate company, and selling company to determine the responsibility for the freight costs. It posts the distributed costs as follows:

- In the delivering company, and the (relevant/subsequent) intermediate company, as an expense for the intercompany sales order item (see step 4 in [Multistage Intercompany Sales](#)) and the outbound delivery item (see step 5).
- In the selling company, as an expense for the standard sales order item (see step 2) and the outbound delivery item (see step 5).

Multistage Intercompany Stock Transfer

The system uses the Incoterms agreed between the delivering company and receiving company to determine the posting of the freight cost. It determines the posting of the freight cost in the delivering company as an expense because the freight movement is outbound for the intercompany sales order item (see step 4 in [Multistage Intercompany Stock Transfer](#)) and the outbound delivery item (see step 5).

For more information about this process, see and [Multistage Intercompany Stock Transfer](#).

If the material being transported is relevant for valuation in the delivering company, and you've enabled group valuation in universal parallel accounting, the system also posts the freight cost in the group valuation ledger of the delivering company code that is considered for group valuation. The receiving company adjusts the material valuation. For more information about a group valuation ledger, see [3191636](#) .

Prerequisites

To enable freight cost posting with freight cost allocation for an advanced/multistage intercompany sales and an advanced/multistage intercompany stock transfer process, you must specify the following settings:

- Freight cost allocation creation criteria of always by settlement documents in Customizing for **Logistics – General** under **Settlement Management** > **Basic Settings** > **Specify Extended Scope of Settlement Management** .
- Criteria for creating settlement management documents in Customizing for **Transportation Management** under **Settlement** > **Freight Settlement** > **Integration for Settlement Posting** > **Mapping for Cost Distribution** > **Posting Using Settlement Management** > **Basic Settings for Creation of Settlement Management Documents** . You must specify the following settings:
 - **Document Category**: always specify **1 Settlement**
 - For the settlement process type, choose the delivered standard process type **2TME – Freight Cost Allocation (Expense)** and a use case type of **31 Freight Cost Distribution on G/L Account**. Also choose a settlement process type of **2TMM – Freight Cost Allocation (ML-Account)** and a use case type of **30 Freight Cost Distribution on Material Account**.

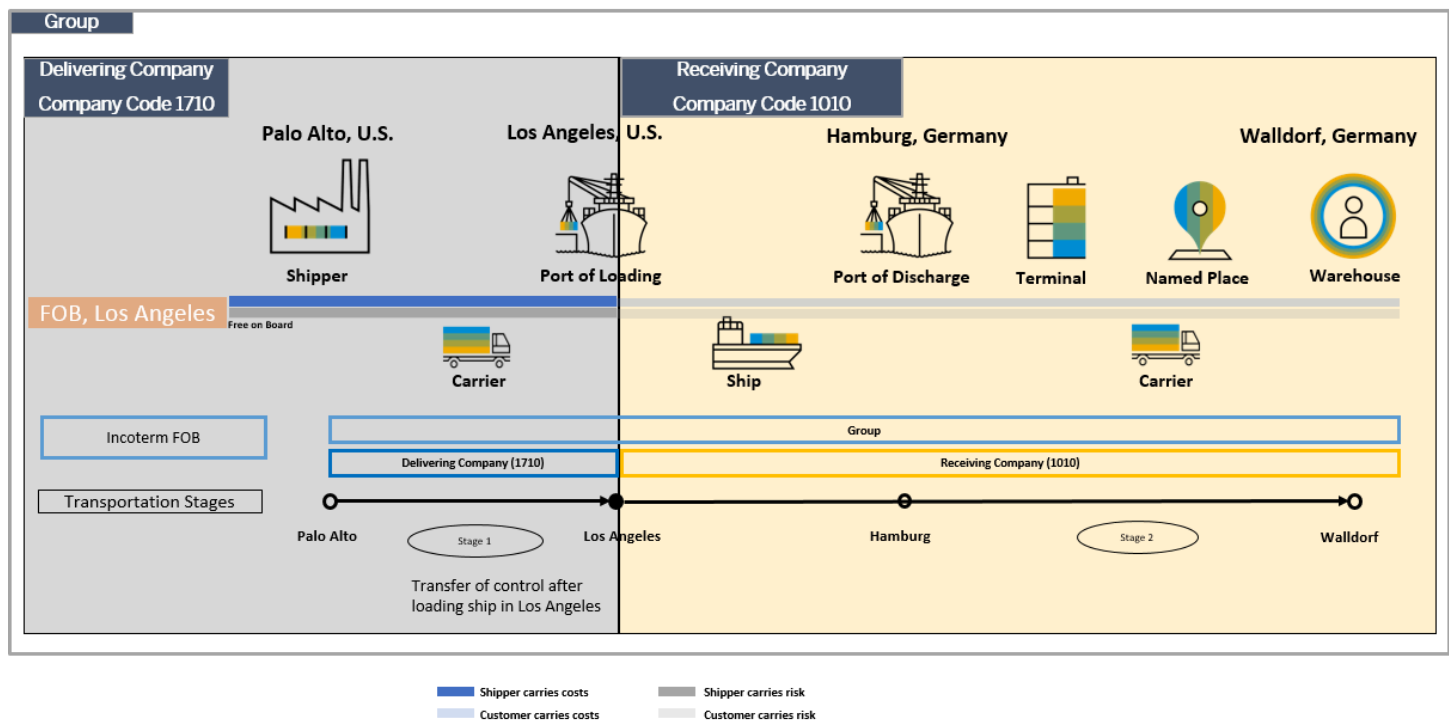
i Note

If you already use freight cost allocation features and you decide to also use these intercompany features, you must change the settings you have already made to the settings we recommend here.

Examples

Advanced Intercompany Stock Transfer Process

The following graphic outlines an example of an advanced intercompany stock transfer order, and is followed by an explanation:



In an advanced intercompany stock transfer process, you can transfer stock between different legal entities in a corporate group. You can transfer stock from one plant to another if the plants belong to different company codes.

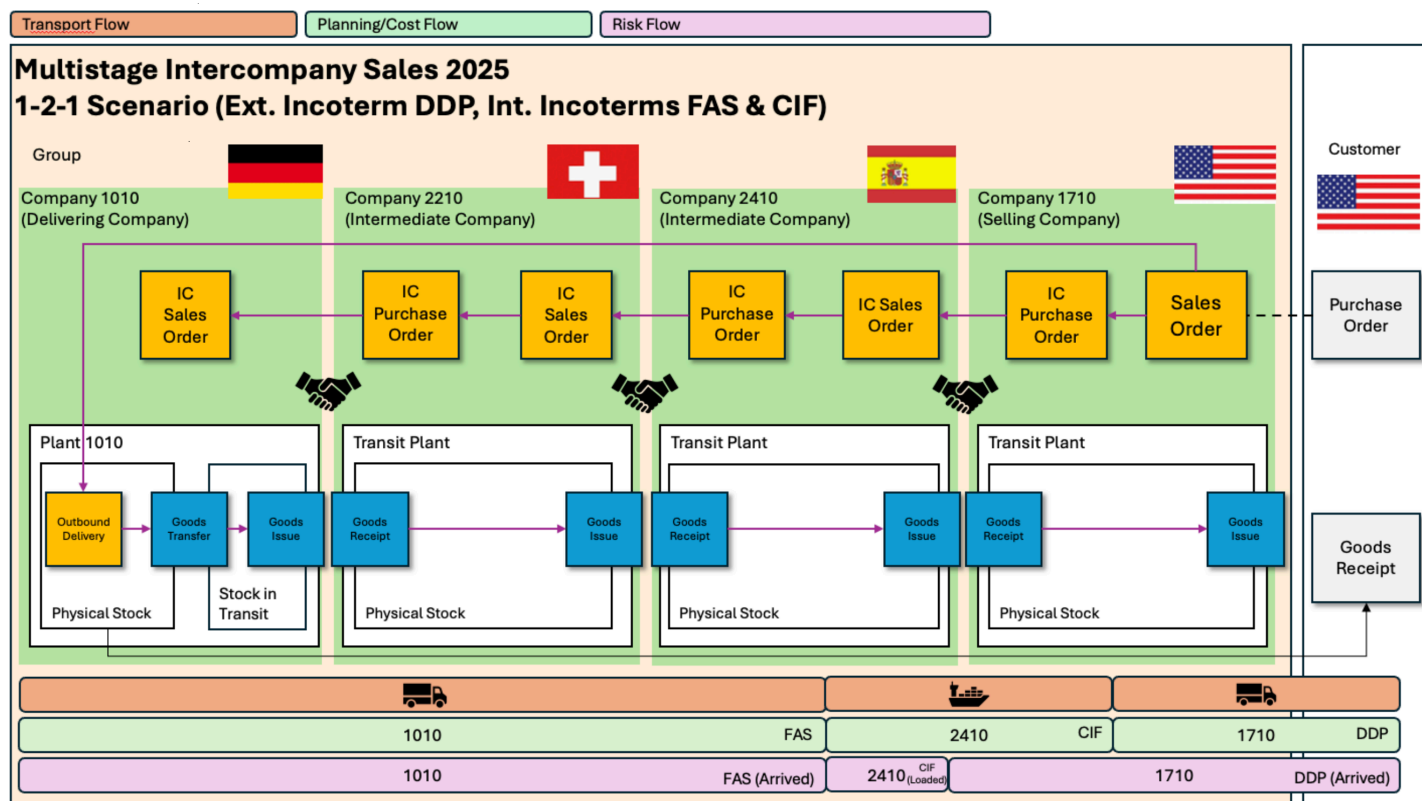
For example, a receiving company, 1010, located in Germany, needs a product that is manufactured by another legal entity in the United States (U.S.). A purchasing organization in the receiving company creates an intercompany purchase order in the receiving company, where the supplier represents a plant that is part of a different company code, in this case, in the U.S.

The contractual agreement with the internal supplier for the internal business is stored in the intercompany purchase order in the receiving company. This includes the Incoterms that define the terms of sale and the passing of risk for the importation and exportation of the goods.

In our example, the Incoterm in the intercompany purchase order is **Free on Board (FOB)**. This means that the delivering company is responsible for organizing and paying the transportation of the goods from its plant to the port of departure, and for loading the goods on the ship. The receiving company is responsible for organizing the transport and paying from the port of departure to the final destination at its receiving location.

For an example of an advanced intercompany sales process, see [Transportation Management in Advanced Intercompany Business Processes](#)

Multistage Intercompany Sales with two Intermediate Company (1-2-1)



A sales office (selling company 1710) receives a purchase order from a customer and creates a sales order. During sourcing, it's determined that the material is shipped from a plant with company code (1010) which is different from the company code of the selling company (1710). Two intermediate companies 2410 and 2210 are also involved, apart from the delivering company 1010. Therefore, two intercompany (IC) purchase orders and three intercompany (IC) sales orders are created.

The sales order contains Incoterm Delivery Duty Paid (DDP). This means that the group is responsible for organizing the whole transportation route. In the sales order, Incoterm location 1 (in this case, the customer's address in the USA) is relevant for stage building. The group carries the planning and cost responsibility for transporting the goods to the customer's address in the USA.

Based on the Incoterms in the documents, TM creates a single freight unit with the following stages:

Stage	Executing Company Code	Document
Stage 1: From the delivering company (1010) in Germany to the port of loading in Spain (intermediate company 2410)	Comp Code 1010	Freight Order 1 (FO1)
Stage 2: From intermediate company (2410) in Spain to port of discharge in the USA (1710)	Comp Code 2410	Freight Booking 1 (FB1)
Stage 3: From the port of discharge in the USA to the customer in USA (Incoterm location 1 in the sales order), organized by selling company 1710	Comp Code 1710	Freight Oder 2 (FO2)

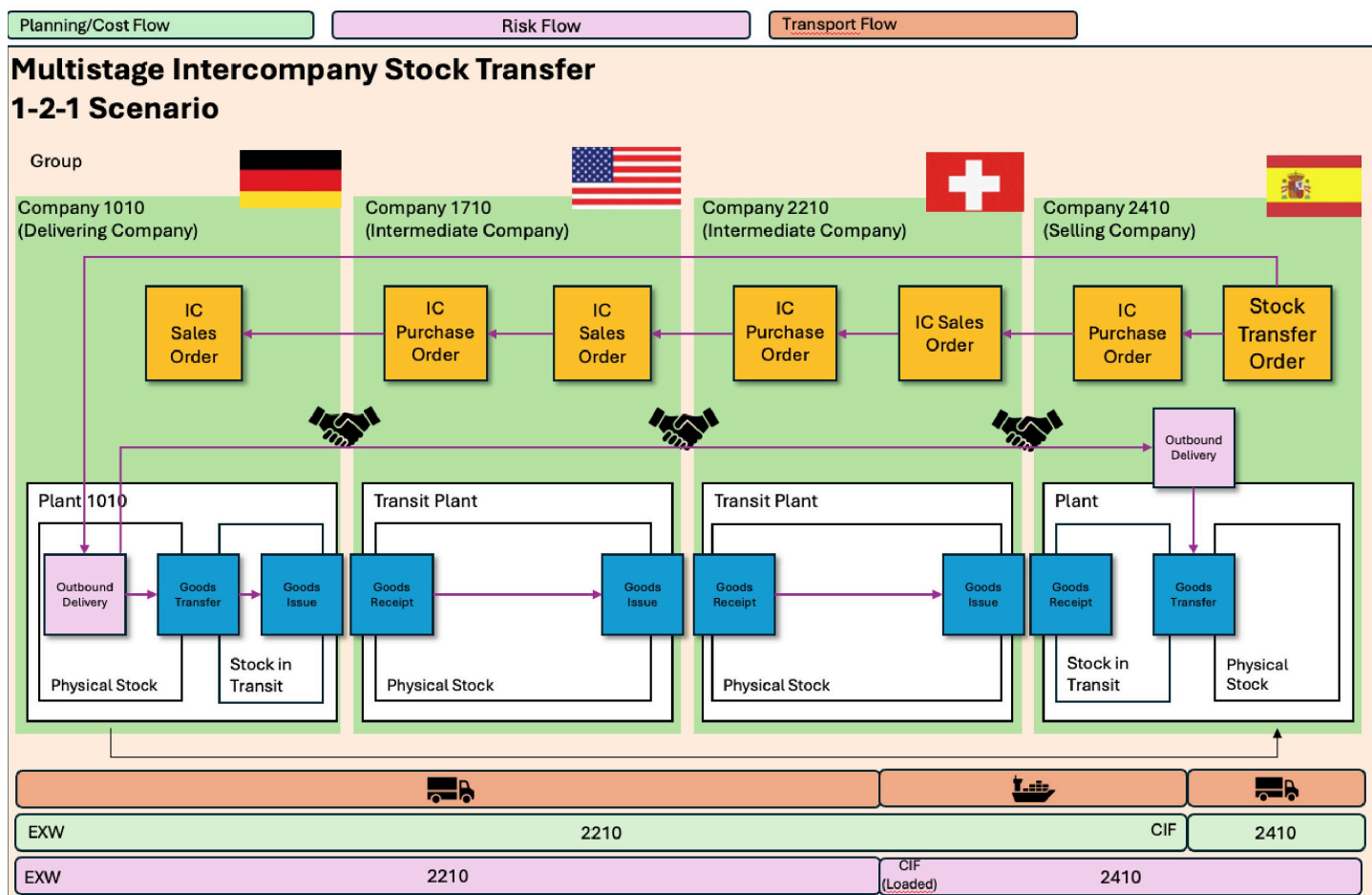
As per the defined incoterms, the company 2210 isn't involved in the transportation.

Based on your settings, the following are the company code and plant determination for freight settlement document (FSD) posting:

	Company Code	Plant
FSD1 (FO1)	1010	1010
FSD2 (FB1)	2410	2430
FSD3 (FO2)	1710	1730

For an example of an multistage intercompany sales process, see [Transportation Management in Multistage Intercompany Business Processes](#).

Multistage Intercompany Stock transport with two Intermediate Company (1-2-1)



A selling company 2410, in Spain, receives a stock transfer order. The delivering company 1010, in Germany, buys the goods from the intermediate companies (1710 and 2210). The intermediate company sends goods to the delivering company, which then ships them to the receiving company. In a multistage stock transfer process, there's no external customer involved. Only the transfer of stock between different legal entities of a group is covered.

Based on the Incoterms in the documents, TM creates a single freight unit with the following stages:

Stage	Executing Company Code	Document
Stage 1: From the delivering company (1010) in Germany to the port of discharge in Spain (intermediate company 2410)	Comp Code 1010	Freight Booking 1 (FB1)
Stage 2: From port of discharge to the final delivery point of the stock	Comp Code 2410	Freight Order 1 (FO1)

As per the incoterms defined, the intermediate companies 2210 and 1710 aren't involved in the transportation.

Based on your settings, the following are the company code and plant determination for freight settlement document (FSD) posting:

	Company Code	Plant
FSD1 (FB1)	1010	1010
FSD2 (F01)	2410	2430

Viewing Freight Cost Allocation Documents

You can use the following apps to view freight cost allocation documents:

- **Manage Freight Cost Allocation Documents**

With this app, you can easily manage freight cost allocation documents. You can search for and navigate to specific freight cost allocation documents. The app provides a detailed view of a freight cost allocation document including information about the amounts, administrative data, and document items.

- **Monitor Freight Cost Allocation Document Items**

With this app, you can get an overview of items in a freight cost allocation document. You can quickly search for and analyze specific freight cost allocation document items. In particular, you can search these allocation documents by freight order or freight booking.

- **Schedule Document Release - Freight Cost Allocation**

With this app, you can manually schedule a job to release freight cost allocation documents to Finance. Normally, when you create a freight cost allocation document, the system automatically releases the document to Finance. However, if the system can't automatically release the document, you must manually schedule a job to release it. The system then releases the document at the appointed time.

Alternatively, you can use transaction code WAB_RELEASE or program RWLF2056 to schedule the periodic release of freight cost allocation documents in a background processing program.

For more information, see [Freight Cost Allocation](#).

Related Information

[Transportation Management in Advanced Intercompany Business Processes](#)

[Advanced Intercompany Sales](#)

[Advanced Intercompany Stock Transfer](#)

[Transportation Management in Multistage Intercompany Business Processes](#)

[Multistage Intercompany Sales](#)

[Multistage Intercompany Stock Transfer](#)

Credit Memos

Use

You can create a credit memo for the following business reasons:

- Damaged goods
- Delays in transports
- Service level agreements not met
- Overcharges for transportation service

The type of credit memo you create depends on your role in the transportation process. You can create the following credit memo types:

- A credit memo for a forwarding order to reflect the credit you provide as a logistics service provider (LSP) to your customer. Here you create a credit memo with reference to a forwarding settlement document. For more information, see [Credit Memos for Forwarding Orders](#).
- A credit memo for a freight order to reflect the credit your carrier provides to you. Here you can create a credit memo with reference to a freight settlement document. For more information, see [Credit Memos for Freight Orders](#).

You can post a credit memo to financial accounting for further processing. You can create more than one credit memo for a particular forwarding settlement document or freight settlement document. However, at any given time, there can be only one credit memo for a particular forwarding settlement document or freight settlement document that hasn't yet been posted successfully.

i Note

You can't use credit memos for freight settlements using lean-service based integration.

Prerequisites

In Customizing, you have to set up the process controller for use with settlement documents. For example, if you want to use a strategy to create or preprocess forwarding settlement documents differently than the standard way, you can define your own strategy using the process controller and assign it to a specific service. For more information, see Customizing for [Transportation Management](#) under [► Settlement ► Configure Process Controller for Settlement ►](#).

Features

Charges

The system displays the transportation charges that have been invoiced in the [Already Invoiced Amount](#) field. The system displays the difference between the amount that has been invoiced and the amounts credited in all credit memos entered against the settlement in the [Credit Remaining Amount](#) field.

You can enter a credit amount against each of the charge types up to the amount in the [Credit Remaining Amount](#) field. The system controls the credit amount that you can provide because you can't exceed the invoiced amount.

Related Information

[Forwarding Settlement Document](#)

[Freight Settlement Document](#)

Credit Memos for Forwarding Orders

Use

You can use a credit memo to give a credit to a customer in a forwarding settlement where you have over-invoiced for transportation services. You use a forwarding settlement document as a reference when you create a forwarding credit memo.

Prerequisites

- Types of credit memo and credit memo reason codes

In Customizing, you have defined a credit memo type that the system uses to automatically enter certain settings when you create a credit memo. You have also defined the reasons for why the credit memo is necessary. For more information, see Customizing for [Transportation Management](#) under [►►Settlement ► Forwarding Settlement ► Define Credit Memo Reason Codes and Types for Forwarding SDs ►](#).

- Influence invoicing status

You can specify the [Influence Invoicing Status](#) setting in the relevant default credit memo type. When you create a credit memo, either manually or using the background processing function, the credit memo has a role in determining the invoicing status of the forwarding order.

- Automatically enter credit amount

When you specify that the system automatically enters the credit amount, the system automatically copies the credit remaining amount to the credit amount in a credit memo. This means that you do not have to manually enter the credit amount in a credit memo. For more information, see Customizing for [Transportation Management](#) under [►►Forwarding Settlement ► Define Credit Memo Reason Codes and Types for Forwarding SDs ► Automatically Propose Credit Amount ►](#).

- Integration settings for posting to Sales and Distribution (SD)

You have made the relevant settings in Customizing for [Transportation Management](#) under [►►Settings for Posting Forwarding Settlements ► Settlement ► Forwarding Settlement ►](#).

The system determines the logical system behind the business system group that you have linked to your sales organization data for the credit memo. The business system group represents the SAP ERP system from which you replicate the sales organization in the organization model.

If the logical system is different to the current client logical system, the system uses enterprise services to post the credit memo to an external SD system. If the logical system is same as the current client logical system, the system posts the credit memo to an internal SD system.

For integration to SD on an external system (SAP S/4HANA or SAP ERP), you have made the relevant settings in Customizing under [►►Integration with Other SAP Components ► Transportation Management ► Invoice Integration ► Billing ►](#) in the external system.

- Forwarding settlement documents

The system contains a forwarding settlement document that is associated with one or more forwarding orders and has the life cycle status [Posted](#).

Features

Credit Memo Preview

You can preview a credit memo as it would appear in Sales and Distribution (SD). The preview contains information from the credit memo and tax information from SD.

You can use the preview for credit memo documents that are at all life cycle statuses except the following:

- [Credit Memo Posting Started](#)

- [Credit Memo Posted](#)
- [Cancellation Requested](#)
- [Canceled](#)

The credit memo must also be consistent. In a preview, the system does not permanently store any data.

Activities

You can create a credit memo from a forwarding settlement document. You can access the credit memos from the forwarding settlement documents overview screen. The system creates the credit memo on the basis of the data in the associated forwarding settlement documents and displays the amount that has already been invoiced.

The forwarding settlement document you reference can be one of the following types:

- Individual forwarding settlement document that references just one forwarding order
- Collective forwarding settlement document that references more than one forwarding order

Related Information

[Credit Memos](#)

[Automatic Change Management Process for Settlements](#)

Credit Memos for Freight Orders

Use

You can use a credit memo to get a credit from a carrier in a freight settlement process where you have overpaid or over accrued in financials for a transportation service. You use a freight settlement document as a reference when you create a freight credit memo.

i Note

For air freight bookings, you use a charge correction advice document (CCAD) to process changes in charges, including issuing a credit. Alternatively, you could use the credit memo for handling credit from your carrier.

Prerequisites

- Types of credit memo and credit memo reason codes

In Customizing, you have defined a credit memo type that the system uses to automatically enter certain settings when you create a credit memo. You have also defined the reasons for why the credit memo is necessary. For more information, see Customizing for [Transportation Management](#) under [► Settlement ► Freight Settlement ► Define Credit Memo Reason Codes and Types for Freight SDs ►](#).

- Influence invoicing status

You can specify the [Influence Invoicing Status](#) setting in the relevant default credit memo type. When you create a credit memo, either manually or by using the background processing function, the credit memo has a role in determining the invoicing status of the freight orders, freight bookings, or service orders. For more information, see Customizing for

[Transportation Management](#) under [Freight Settlement](#) > [Define Credit Memo Reason Codes and Types for Freight SDs](#) > [Define Credit Memo Types](#) > [Influence Invoicing Status](#) .

- Automatically enter credit amount

When you specify that the system automatically enters the credit amount, the system automatically copies the remaining credit amount to the credit amount in a credit memo. This means that you do not have to manually enter the credit amount in a credit memo. For more information, see Customizing for [Transportation Management](#) under [Freight Settlement](#) > [Define Credit Memo Reason Codes and Types for Freight SDs](#) > [Define Credit Memo Types](#) > [Automatically Propose Credit Amount](#) .

i Note

The system includes the invoicing party, carrier, and payee in the credit memo by default. The system does not transfer business partners other than the invoicing party and the payee to materials management (MM).

- Business document

The system contains one or more freight orders, service orders, or freight bookings.

The system determines the logical system behind the business system group that you have linked to your purchase organization data for the credit memo. The business system group represents the SAP ERP system from which you replicate the purchasing organization in the organization model.

If the logical system is different to the current client logical system, the system uses enterprise services to post the credit memo to an external MM system. If the logical system is same as the current client logical system, the system posts the credit memo to an internal MM system.

For more information on organizational model see [Create and Merge Organizational Hierarchy](#) under [Transportation Management](#) > [Master Data](#) > [Organizational Management](#) > [Organizational Model](#) .

- Integration settings for posting freight settlement documents

For integration to an internal MM system, you have configured the following settings in Customizing:

- [Assign Purchasing Information for Posting](#)
- [Assign Service Master Record and Account Assignment Category](#)
- Tax determination setting under [Tax](#)
- Mapping for cost distribution in a shipper scenario when you have enabled cost distribution in the freight settlement document

For more information, see Customizing for [Transportation Management](#) under [Settlement](#) > [Freight Settlement](#) > [Integration for Settlement Posting](#).

For integration to MM on an external SAP S/4 HANA or SAP ERP system, you have made the relevant settings in Customizing under [Integration with Other SAP Components](#) > [Transportation Management](#) > [Invoice Integration](#) > [Invoicing](#) in external the SAP S/4 HANA or SAP ERP system.

Activities

You can create a credit memo from a freight settlement document on the overview screen for the freight settlement documents.

When you create a credit memo from a freight settlement document, the system includes the information from the freight settlement document in the credit memo. The system takes the amount that is already invoiced from the freight settlement document.

When you use the automatic change process, the system uses the existing freight settlement document to create a credit memo. This enables the system process a large-scale change to the freight document charges, for example, a change in the agreement or in the calculation sheet, or the addition or deletion of a charge line. For more information, see [Automatic Change Management for Freight Settlements](#).

Example

The following table illustrates how the freight credit memo process works for an ocean freight booking:

Steps in TM	Amount in FSD / Freight Credit Memo (USD)	Steps in MM	Comments
Total freight cost in ocean freight booking	2400	None	None
Create the freight settlement document and post it to MM.	2400	Purchase order (PO) created, for example, PO1 with line 10. Service entry sheet (SES) created, for example, SES 1. Linked to PO1/10 with amount USD 2400.	Regular freight settlement document process and posting of the FSD
Update the booking to reflect the correction, for example, weight correction. New total freight cost in freight booking.	2000	None	None
Create the freight credit memo and post it to MM to reverse the old FSD amount. Create a new FSD for the updated freight cost..	Credit of 2400 using the Credit Memo FSD for 2000 for the new, updated freight cost	When the credit memo is posted, a purchase order (PO) created, for example, PO2 with line 10. A service entry sheet (SES) is created, for example, SES 2. This is linked to PO1/10 with the amount of USD 2400. However, in this case the PO item and the SES are both marked as returns items, which reverses the previous amounts posted by the FSD. When the new FSD is posted, a new purchase order is created, for example, PO3 with line 10. A service entry sheet (SES) created, for example, SES 3. This is linked to PO1/10 with the amount of USD 2000.	PO based on returns item created on posting of the credit memo. Regular freight settlement document process and posting of the new FSD created for the revised freight cost.

Related Information

[Credit Memos](#)

[Automatic Change Management for Freight Settlements](#)

Internal Settlement

Use

You can send an internal settlement document from an internal organization in your company to another internal organization to recover costs incurred in delivering transportation services for a forwarding order (see [Internal Settlement Management](#)).

You can also create internal settlement documents to settle the cost of providing internal resources for freight orders (see [Internal Settlement for Resources](#)).

In the standard internal settlement process, the internal settlement is between the purchasing organization of the freight order and the sales organization of the forwarding order. In the internal settlement process for resources, the settlement is between the organizations that own the resources that are used to execute the order and the purchasing organization of the freight order.

Prerequisites

- You have specified the purchasing organization of the freight order or freight booking that is executing the stage as a forwarding house. The forwarding house acts as an organization in the internal agreement. To work with internal settlements, you must model the purchasing organization as a forwarding house.
- You assign a business partner that represents your forwarding house or sales organization to the organization in organization management. For the internal settlement process, the business partner assigned to the forwarding house or the sales organization of the forwarding order is treated as a business partner of the purchasing organization. The purchasing organization is modeled as a forwarding house in the freight order or freight booking for the internal charge calculation in the forwarding order and internal settlement.
- You have specified a forwarding settlement document type for internal settlement documents in Customizing for [Transportation Management](#) under [► Settlement ► Forwarding Settlement ► Define Forwarding Settlement Document Types ►](#).
- You have specified the general prerequisites for calculating charges. For more information, see [Charge Calculation](#).
- You have specified the relevant settlement profiles and calculation profiles. The process for specifying a settlement profile and a calculation profile for an internal settlement document is similar to the process for specifying a forwarding settlement profile and a calculation profile for a forwarding settlement document. For more information, see the prerequisites for settlement profiles and calculation profiles in [Forwarding Settlement](#).
- By default, TM creates an intercompany settlement document when the sales organization of the forwarding order and the purchase organization (which is modeled as a forwarding house in organization management) of the freight order or freight booking belong to different company codes. These internal settlement documents for intercompany settlement are posted to SD to create an invoice and for posting to financial accounting.

The system determines the logical system behind the business system group that you have linked to your sales organization data for the internal settlement document. You model the sales organization as a forwarding house. The business system group represents the SAP ERP system from which you replicate the sales organization in the organization model.

If the logical system is different to the current client logical system, the system uses enterprise services to post the settlement document to an external SD system or to financial accounting in your SAP S/4 HANA or SAP ERP system. If the logical system is same as the current client logical system, the system posts the settlement document to an internal SD or financial accounting system.

For more information on the organizational model, see Customizing for [Transportation Management](#) under [► Master Data ► Organizational Management ► Organizational Mode ► Create and Merge Organizational Hierarchy ►](#).

- To integrate to an internal SD system, you have specified the relevant settings in Customizing for [Transportation Management](#) under [► ► Settlement ► Forwarding Settlement ► Settings for Posting Forwarding Settlements ►](#).
- To integrate to an external SD on an external SAP S/4 HANA or SAP ERP system, you have specified the relevant settings in Customizing under [► Integration with Other SAP Components ► Transportation Management ► Invoice Integration ► Billing ►](#) in the external SAP S/4 HANA or SAP ERP system. To post the SD billing document to your internal company as an incoming vendor invoice in financial accounting, please see the SAP note [2157070](#) .

- To use an intracompany settlement when the sales organization of the forwarding order and the purchase organization (which is modeled as a forwarding house in organization management) of the freight order or freight booking belong to the same company organization, you have selected the **Intracompany Settlement** checkbox in the **Org. Data** tab page of the company organization. You can access the structure using transaction PPOCE.

These internal settlement documents for intracompany settlement are posted to financial accounting to create the financial accounting documents.

- To integrate to an internal financial accounting module, you have specified the relevant settings in Customizing under ► **Integration with Other SAP Components** ► **Transportation Management** ► **Settlement** ► **Forwarding Settlement** ► **Settings for Posting Intracompany Settlements** ►.
- To integrate to financial accounting on an external SAP S/4 HANA, you have specified the relevant settings in Customizing under ► **Integration with Other SAP Components** ► **Transportation Management** ► **Invoice Integration** ► **Billing** ► **Settings for Posting Intracompany Settlements** ► in the external SAP S/4 HANA.
- To integrate to financial accounting on an external SAP ERP, you have specified the relevant settings in Customizing for **Transportation Management** under ► **Invoice Integration** ► **Billing** ► **Intracompany Settlement** ► in the external SAP ERP.

You have assigned a sales organization unit (cost receiver organization) in Transportation Management (TM) to an internal order or cost center in Controlling (CO) for cost distribution. The sales organization is the receiver of the cost for an internal settlement.

You have assigned a purchasing organization unit (cost sender organization) in CO to an internal order or cost center account for cost distribution. The purchasing organization is the sender of the cost for an internal settlement.

Features

Intercompany Settlements

When you post the internal settlement document to Sales and Distribution (SD), the system creates an SD billing document.

Intracompany Settlements

When you post the internal settlement document, the system reposts the primary costs to an internal order or cost center in financial accounting to create the financial accounting documents. The TM system performs the following tasks:

- Displays the accounting document in the document flow of the internal settlement document
- Uses an internal settlement document to move the cost from the purchasing organization of the execution document such as freight order or freight booking, to the sales organization of the forwarding order that is part of the execution documents.

Related Information

[Integration: Invoice Processing \(MM-IV/SD-BIL\) with TM](#)

Internal Settlement Management

Use

You can send an internal settlement document from an internal organization in your company to another internal organization to recover costs incurred in delivering transportation services for a forwarding order.

If the organizations involved belong to the same company code, the settlement is an intracompany settlement. If the organizations belong to different company codes, the settlement is an intercompany settlement.

Prerequisites

- You have specified an internal settlement rule in Customizing for **Transportation Management** under **Forwarding Order Management** > **Define Stage Type Sequence for Movement Types**. A stage in a forwarding order must reach the execution status you specify here before you can create and edit internal settlements.
- For internal settlements (both intracompany and intercompany settlements), you must have an internal agreement in place between the purchasing organization (forwarding house) of the freight order or freight booking that is executing the stage, and the sales organization in the forwarding order.
- You have specified all the relevant internal settlement prerequisites (see [Internal Settlement](#)).
- You have the option of specifying your own logic to set or unset the **Ready for Invoicing** checkbox for a header-level internal charge item, with the **BAdI: Readiness for Invoicing of Header-Level Charges**.

For example, you can specify that the header-level charges for a purchasing organization in the freight order or freight booking should be set to **Ready for Invoicing** only when the stage with the executing organization has a handling execution status of **Loaded**.

For more information, see Customizing for **Transportation Management** under **Business Add-Ins (BADIs) for Transportation Management** > **Settlement** > **Forwarding Settlement** > **BAdI: Readiness for Invoicing of Header-Level Charges**.

Activities

Determining the Appropriate Agreement

You perform planning on a forwarding order to finalize its execution. The forwarding order contains a breakdown of the stages you require to fulfill the order. Based on this breakdown, the system calculates the charges for the order. When it calculates charges it determines the appropriate internal agreement to use to perform the calculation.

Manually Creating an Internal Settlement Document

You can manually create an internal settlement document in the **Forwarding Orders for Internal Settlement** screen by selecting the appropriate freight order or freight booking.

The system checks the following information in the forwarding order:

- Purchasing organization of the freight order or freight booking executing the stage
- Sales organization of the overall forwarding order

Depending on the internal organizations that are involved, the system creates internal settlements that are either intercompany or intracompany settlements.

Creating Multiple Internal Settlement Documents

You can also create multiple internal settlement documents using the **Creation of Forwarding Settlement Documents** report, and selecting a category of **Internal Settlement Document**.

Creating an Internal Settlement Document for Header-Level Charges

You can have charges in a forwarding order that apply to an entire transportation, even though you incur the charges during a stage of the transportation. For example, you can incur a charge for export documentation during the pre-carriage stage of a transportation. This charge applies to the entire transportation, and you may want to settle this charge separately from the other pre-carriage charges.

For internal settlements, you can settle header-level charges separately from your stage-level charges. You can also create internal settlement documents for the header-level charges only.

The internal charges in the forwarding order represent the cost on the sales organization of the forwarding order, from the purchasing organizations that execute the stages of the forwarding order. You can identify the purchasing organization in the forwarding order as follows:

1. Choose the **Actual Route** tab page
2. In the **Document** column, choose the executing document
3. Under **General Data** **Organizational Data**, see the **Purchasing Organization**

To enable you to create separate internal settlements for header-level charges, you have the following features:

- **Ready for Settlement** checkbox in the forwarding order

For each header-level internal charge item, the system automatically specifies the **Ready for Settlement** checkbox when the forwarding order goes into an execution status of **In Execution**.

The system only includes a header-level charge item in an internal settlement when the **Ready for Settlement** checkbox is selected.

- **BAdI: Readiness for Invoicing of Header-Level Charges**

You can specify your own logic to set or unset the **Ready for Invoicing** checkbox for a header-level internal charge item. For more information, see the prerequisites section.

- **Creation and Transfer of Forwarding Settlement Documents** background processing function (program /SCMTMS/CFIR_CREATE_BATCH)

You can create internal settlement documents for header-level charges separately from stage-level charges. You can also create internal settlements only for header-level charges. For more information, see [Creation and Transfer of Forwarding Settlement Documents](#).

Working with Forwarding Orders for Internal Settlement

You can display all the forwarding orders, and the freight orders or freight bookings that contain forwarding orders, that must be settled internally on the **Forwarding Orders for Internal Settlement** personal object worklist screen.

The individual line items can contain the following information:

- Charge item in a forwarding order that covers the stage that the freight order or freight booking executed
- Charge item that applies to the entire forwarding order, from the purchasing organization that is linked to the freight order or freight booking.

The system does not display a link to the execution document for these charge items.

Specifying a Forwarding House

You must specify the purchasing organization of the freight order or freight booking that is executing the stage as a forwarding house. The forwarding house acts as the organization in the internal agreement. The business partner of the sales organization in

the forwarding order acts as the business partner in the agreement, and is treated as a customer of the forwarding house.

Example

A sales organization receives a forwarding order from a customer. The sales organization delegates the execution of the main carriage to an internal organization. This internal execution organization uses an internal settlement document to bill the sales organization that received the forwarding order, to recover the costs of executing the main carriage.

In Customizing, you have specified an internal settlement rule of **In Execution** in the **Define Stage Type Sequence for Movement Types** Customizing activity. When the execution status of the freight booking that executes the main carriage goes to **In Execution**, the main carriage of the forwarding order is ready for internal settlement. You can create the internal settlement document in the following ways:

- Manually from the **Forwarding Orders for Internal Settlement** screen
- Automatically in the background, using the **Creation and Transfer of Forwarding Settlement Document** background processing program

If the internal organization that executed the main carriage belongs to the same company code as the sales organization that received the forwarding order, the system creates an intracompany settlement. If the organizations belong to different company codes, the system creates intercompany settlements.

More Information

Subject	See
Using internal settlement documents to settle the costs of providing internal resources	Internal Settlement for Resources
Role and structure of an agreement	Agreement
Using forwarding settlements	Forwarding Settlement
Using cost distribution for logistics service providers (LSPs)	Cost Distribution for LSPs
Creating internal settlement documents in the background	Creation and Transfer of Forwarding Settlement Documents

Internal Settlement for Resources

Use

You can create internal settlement documents to recover costs for the internal resources you use in executing orders, for example, trailers and trucks.

i Note

The features described in this document are not yet released.

Prerequisites

- You have enabled a freight order type for internal settlements if you use your own resources to execute freight orders. You have also assigned the internal settlement document type that you use to create internal settlements for your own

resources to the relevant freight order type. For more information, see Customizing for [Transportation Management](#) under [►Freight Order Management►Freight Order►Define Freight Order Types►](#).

- You have defined a resource in which the organizational unit to which the resource belongs is defined as a sales organization, sales group, sales office, or a forwarding house. You do this in the [/SCMTMS/RES01 – Resource](#) transaction under [►SAP Menu►Logistics►Transportation Management►Master Data►Resource►](#) or in the [Define Resources](#) app.
- For freight orders in which you have used your own resources, you must have an internal agreement in place between the organizational unit assigned to the resource (forwarding house, sales organization, sales group, or sales office) and the business partner assigned to the purchasing organization or purchasing group of the freight order that is using the resource.
- You have specified [Resource](#) in the [Resource](#) field on the [General Data](#) tab page of the relevant internal agreement.
- You have put in place all the relevant prerequisites for managing internal settlement documents (see [Internal Settlement Management](#)).

TM creates an intercompany settlement document when the sales organization of the resource and the purchase organization of the freight order belong to different company codes. You post these internal settlement documents for intercompany settlement to Sales and Distribution (SD) to create the invoice and for financial accounting.

TM creates an intracompany settlement when the sales organization of the resource and the purchase organization of the freight order belong to the same company code. You must select the [Intracompany Settlement](#) checkbox on the [Org. Data](#) tab page of the company organization. You can access the structure using transaction PPOCE. You post these internal settlement documents for intracompany settlement to financial accounting to create the financial accounting documents.

The system determines the logical system behind the business system group that you have linked to your sales organization data for the internal settlement document. You model the sales organization as a forwarding house. The business system group represents the SAP ERP system from which you replicate the sales organization in the organization model.

If the logical system is different to the current client logical system, the system uses enterprise services to post the settlement document to SD or financial accounting in an external SAP S/4 HANA or SAP ERP system. If the logical system is same as the current client logical system, the system posts the settlement document to an internal SD of financial accounting system.

For more information on organizational models, see Customizing for [Transportation Management](#) under [►Master Data►Organizational Management►Organizational Model►Create and Merge Organizational Hierarchy►](#).

- To integrate to an internal SD module, you have specified the relevant settings in Customizing for [Transportation Management](#) under [►Settlement►Forwarding Settlement►Settings for Posting Forwarding Settlements►](#).
- To integrate to SD on an external SAP S/4 HANA or SAP ERP system, you have specified the relevant settings in Customizing under [►Integration with Other SAP Components►Transportation Management►Invoice Integration►Billing►](#) in the external SAP S/4 HANA or SAP ERP system.
- To integrate to an internal financial accounting model, you have made the relevant settings in Customizing for [Transportation Management](#) under [►Settlement►Forwarding Settlement►](#).
- To integrate to financial accounting on an external SAP S/4 HANA system, you have specified the relevant settings in Customizing under [►Integration with Other SAP Components►Transportation Management►Invoice Integration►Billing►Settings for Posting Intracompany Settlements►](#) in the external SAP S/4 HANA system.
- To integrate to financial accounting on an external SAP ERP system, you have made the relevant settings in Customizing under [►Integration with Other SAP Components►Transportation Management►Invoice Integration►Billing►Intracompany Settlement►](#) in the external SAP ERP system.

You have assigned a sales organization unit (cost sender organization) in TM to an internal order or cost center in Controlling (CO) for cost distribution. The sales organization of the resource is the sender of the cost for an internal settlement.

You have assigned a purchasing organization unit (cost receiver organization) in CO to an internal order or cost center account for cost distribution. The purchasing organization is the receiver of the cost for an internal settlement.

Features

You can use an internal settlement for a resource when you are a logistics service provider and you do not subcontract the movement of goods to a carrier. Instead, you move the goods using your own resources, such as trucks and trailers.

Different organizations in your company can provide the resources. The organizations that provide the resources need to get paid. You can use the internal settlement for resources feature to enable this payment.

For example, you have a trailer that organization 1 provides, and a truck and driver that organization 2 provides. Organization 3 is the purchasing organization that manages the freight order. Organizations 1 and 2 create internal settlements on organization 3 to recover the cost of providing their resources. Usually, you base an internal settlement document on a forwarding order. However, for the freight orders that you execute using your own resources, you base the internal settlement on a freight order.

Activities

You perform planning on a forwarding order to finalize its execution. You create freight orders to execute the forwarding orders using your own resources, such as trailers or trucks. You create internal settlement documents to settle the costs of using your own resources.

You can manually create an internal settlement document for a resource in a freight order by choosing **Follow Up > Create Internal Settlement Document**. You can also create multiple internal settlement documents using the **Mass Creation of Freight Settlement Documents** report, and selecting a create option of **Internal Settlement Document**.

The system checks the following information in the freight order:

- Organizational unit that is assigned to the resource
- Purchasing organization of the overall freight order

Depending on the internal organizations that are involved, the system creates an internal settlement that is either an intercompany settlement or an intracompany settlement. If the organizational unit of the resource and the purchase organization of the freight order belong to the same company code, the system creates an intracompany settlement. If the organizations belong to different company codes, the system creates an intercompany settlement.

The system displays a list of the resources that you include in the particular internal settlement document.

❖ Example

A sales organization receives a forwarding order from a customer. The sales organization delegates the execution of the main carriage to an internal purchasing organization. The purchasing organization uses internal resources, such as trailers and trucks, that are assigned to other internal organizations to plan and execute the order. The forwarding order is executed using the internal resources. The organizations that own the trucks and trailers bill the purchasing organization of the freight order to recover the costs of providing the trucks and trailers.

More Information

Topic	Link
Overview of internal settlement document management	Internal Settlement Management

Topic	Link
Role and structure of agreements	Agreement
Role and structure of trailer units	Trailer Unit
Creating, editing, and displaying trailer units	Creation and Editing of Transportation Units

Cost Distribution Management

Use

Freight costs are an important part of financial reporting in your businesses. You need the ability to allocate freight costs at charge element level across a number of charge sources to manage the following business processes:

- Freight costs
- Material valuations
- Profitability of customers, products, and lines of business
- Customer billing for logistics service providers (LSPs)

As an LSP, you can use cost distribution to meet the following business needs:

- Manage and assess the cost of executing freight orders, including an analysis of the following elements:
 - Your own company resources
 - Subcontracted costs such as the cost of resources and base freight
- Create customer settlements based on distributed internal and subcontracted costs
- Use distributed costs from a freight order for the following activities:
 - Customer billing for LSPs
 - Profitability analysis of forwarding orders

As a shipper, freight costs cover the following stock movements:

Inbound

- Purchases
- Sales returns
- Intercompany or intracompany stock transfers

Outbound

- Customer sales
- Purchase returns
- Intercompany or intracompany stock transfers

As an LSP, you can use the freight cost from external sources, such as a freight order, freight booking, or service order, and internal costs, such as the cost of providing internal resources such as trucks and trailers, as a basis for creating settlement documents. You can base settlement documents on costs as follows:

- Use the distributed cost as a basis of settling with your customers. For more information, see [Cost Pull](#).
- Use a mixture of distributed cost, such as the cost of providing internal resources, and rate tables to settle with your customers. For more information about rate tables, see [Standard Charge Calculation](#).
- Use the distributed cost, or a mixture of the distributed cost and rate tables, to create internal settlements between your internal companies. For example, you can create an internal settlement based on the distributed cost, between the purchase organization in the freight order or freight booking that provided the freight service, and the sales organization in the forwarding order. For more information, see [Internal Settlement Management](#).
- Use the distributed cost for a profitability analysis of your forwarding orders. For more information, see [Profitability Analysis](#).

The system uses the freight charges in the freight order, freight booking, service order, and freight settlement document as the basis for distributing freight costs from external sources, such as carriers or LSPs.

The system uses the internal freight charges that you incur when you use internal resources as a basis for distributing internal costs. The system calculates internal freight charges in the freight order.

You can distribute costs in a freight document where you have a logistics item that comes from a sales order, purchase order, delivery item, or a forwarding order.

Prerequisites

- For both shippers and LSPs, you have specified the cost distribution settings in Customizing for **Transportation Management** under **Basic Functions > Cost Distribution**.
- For both shippers and LSPs, you have used the following settings to assign the distribution profile to the relevant organization:
 1. Assigned the distribution profile to the relevant charges profile in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charges Profiles**.
 2. Assigned the charges profile to the relevant purchasing organization in Customizing for **Transportation Management** under **Master Data > Organizational Management > Organizational Model > Change Organizational Model**. You assign the charges profile on the **Org. Data** (Organization Data) tab page of the relevant purchasing organization.
- For both shippers and LSPs who want to use the freight cost in a freight document for internal settlement, you must enable freight document for cost distribution.

For more information, see the following Customizing activities:

- For freight orders, see Customizing for **Transportation Management** under **Freight Order Management > Freight Order > Define Freight Order Types**.
- For freight bookings, see Customizing for **Transportation Management** under **Freight Order Management > Freight Bookings > Define Freight Booking Types**.
- For service orders, see Customizing for **Transportation Management** under **Freight Order Management > Service Order > Define Service Order Types**.
- For consignment orders, see Customizing for **Transportation Management** under **Freight Order Management > Freight Order > Define Consignment Order Types**.

i **Note**

You can perform cost distribution at either consignment or freight order levels, not both simultaneously. You must choose to activate cost distribution for one and assure the other is disabled in Customizing.

- For shippers who want to transfer the freight cost at order/delivery item level, you must enable the freight settlement document for cost distribution. For more information, see Customizing for [Transportation Management](#) under [► Settlement ► Freight Settlement ► Define Freight Settlement Document Types ►](#).

i **Note**

You cannot enable internal cost distribution for a freight booking. You can use an internal resource such as a truck or a trailer to execute a freight order. You can distribute costs for an internal resource in a freight order.

You cannot distribute costs in a consistent way where you have a freight order, freight booking, service order, or freight settlement document that contain a mix of forwarding order items and order/delivery items. In these cases, you can only distribute costs to one distribution level, the forwarding order level or the order/delivery item level.

Features

Method, Rule, and Level

You can view the results of a cost distribution on the user interface (UI). You can only change the distribution settings in Customizing. The following are the main cost distribution features:

- Distribution method

The system distributes costs in a direct or hierarchical distribution. In a direct distribution, the system distributes costs according to the attributes you specify in the distribution rule and distribution level without reference to where in the item hierarchy you package a product. In a hierarchical distribution, the system uses a step-by-step approach to take each level of the packaging hierarchy into account in the distribution.
- Distribution rule

The system uses criteria such as gross weight, net weight, gross volume, and distance times gross weight to perform a cost distribution.
- Distribution level

The system uses a target level of forwarding orders or order/delivery items in a distribution.

Service Orders

You can distribute costs in a service order to allocate the cost of the service order to the underlying distribution item.

More Information

Topic	Link
Use cost distribution for shippers, including individual prerequisites	Cost Distribution for Shippers
Use cost distribution for LSPs, including individual prerequisites	Cost Distribution for LSPs
Use cost distribution in service orders for both shippers and LSPs, including prerequisites and business case examples	Cost Distribution in Service Orders for Shippers and LSPs

Topic	Link
Use of cost distribution in a group logistics scenario	Group Logistics Company Process
Transfer distributed costs from the freight side to the forwarding side, and use the costs as a basis for settling with a customer or an internal organization.	Cost Pull
Configure agency business documents	SAP Note 2535942 

Cost Distribution for Shippers

Use

As a shipper, you create documents such as purchase orders, sales orders, and deliveries. Transportation Management (TM) creates a freight unit (see [Integration of Orders and Deliveries in Transportation Planning](#)). You use these freight units to plan the execution by freight orders or freight bookings. You create a freight settlement document for the freight order to post the transportation costs to Materials Management (MM). This enables you to post accruals and to match invoices from the transportation service provider.

You must perform cost distribution at the level of order/delivery item in the freight settlement document.

For example, you have a net amount in document currency of USD 1000 for a freight order consisting of two deliveries, with a cost distribution of USD 700 for product 1 for delivery 1, and USD 300 for product 2 for delivery 2. When you post the settlement document, the system takes the following steps:

- Creates a purchase order for USD 1000 and creates a service entry sheet for USD 1000
- Creates a settlement management document with each product as an item. Each item has the amount of the distributed cost, USD 700 for product 1 and USD 300 for product 2.

You specify the type of posting that you want to do with the distributed cost. For example, you can post the distributed cost to material valuation or general ledger. Depending on the settings you've specified, the settlement management document posts to these or other cost objects. Settlement management uses the underlying sales order and delivery information from the freight settlement document cost distribution information to determine from the source order or delivery document the particular cost object against which it posts the distributed costs.

For example, you have a freight order with cargo items that are linked to the deliveries in a sales order. You specify that you want to post the freight cost to the general ledger if the underlying freight cost distribution item has a reference to a delivery item. When you post the freight settlement document with cost distribution for this freight order, the distributed cost is passed on to settlement management. Settlement management takes the following steps:

- Creates a settlement management document with a type of **Settlement Process** and a document category of **Settlement**
- Copies the references from the freight settlement document cost distribution information and gets the linked underlying delivery items
- Copies the product and quantities in the delivery item to the settlement process document item (product 1 and product 2)
- Copies the account assignments for the respective settlement process document item from the underlying sales order item to which the delivery item is linked
- Copies the freight cost distribution amounts for the settlement process document item (USD 700 for product 1 and USD 300 for product 2)

Examples of account assignments can be cost objects including a cost center, profit center, CO-PA segment, or an internal order.

Integration

The system uses the freight settlement document to enable you to integrate the freight cost back to the underlying order/delivery items.

Prerequisites

To integrate to an internal settlement management system to post the freight cost distribution, you've configured the settings in the following Customizing activities:

- Enable cost distribution for the freight settlement document type in Customizing for [Transportation Management](#) under [► Settlement](#) [► Freight Settlement](#) [► Define Freight Settlement Document Types](#) [►](#) and [Define Credit Memo Reason Codes and Types for Freight SDs](#)

For freight settlement integration using Materials Management (MM) External Services Management, you've specified the settings in the following Customizing activities under [► Transportation Management](#) [► Settlement](#) [► Freight Settlement](#) [► Integration for Settlement Posting](#) [► External Services Management](#) [►](#):

- [Assign Purchasing Information for Posting](#)
- [Assign Service Master Record and Account Assignment Category](#)
- Tax determination settings under [Tax](#)
- [Mapping for Cost Distribution](#)

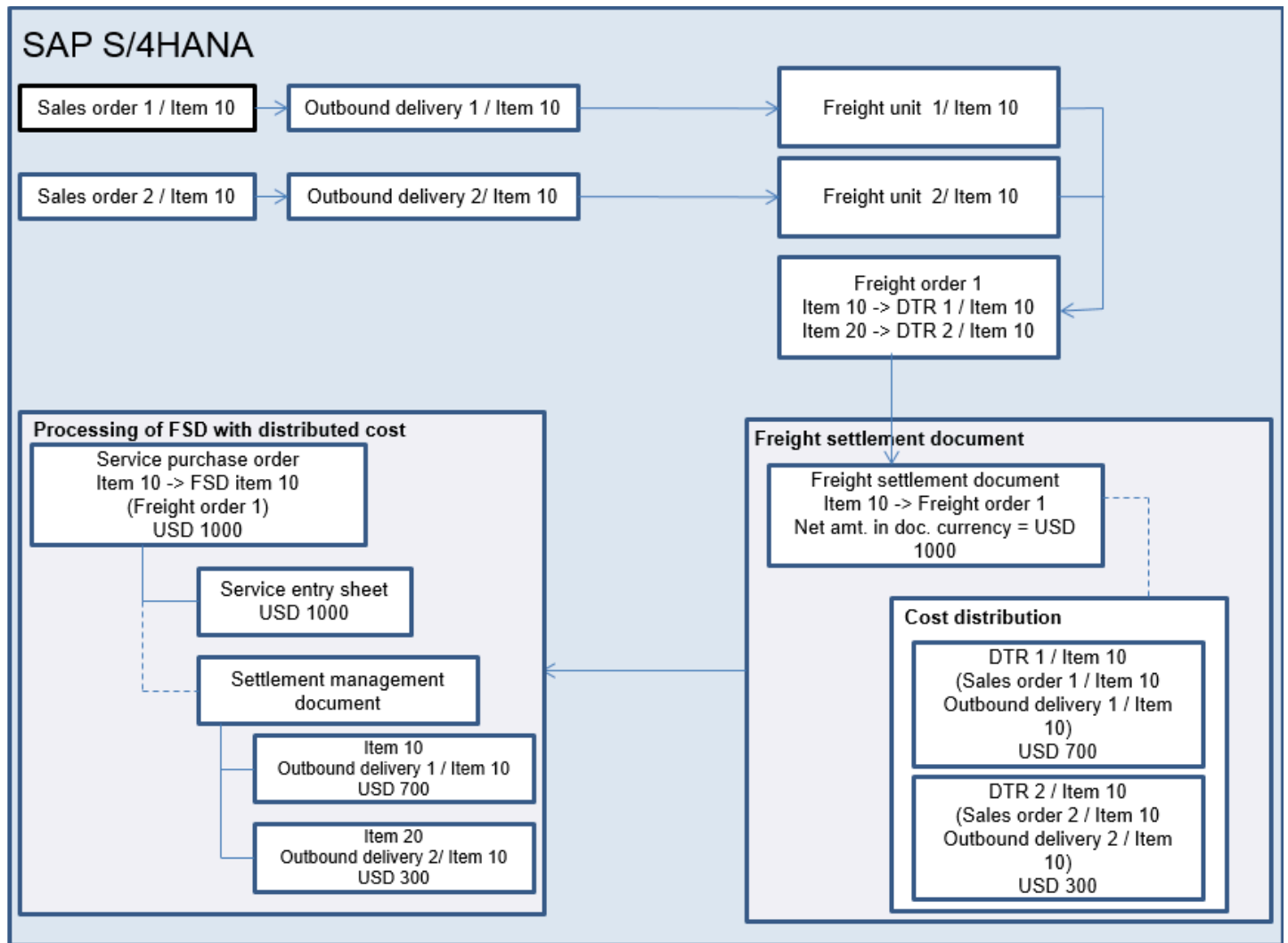
For freight settlement integration using Lean Services, you've specified the settings in the following Customizing activities under [► Transportation Management](#) [► Settlement](#) [► Freight Settlement](#) [► Integration for Settlement Posting](#) [► Service Procurement - Lean Services](#) [►](#):

- [Assign Purchasing Information for Posting Lean service Integration](#)
- [Assign Val. Class, Acc. Assignment Category, and Conditions to Charge Types](#)
- [Mapping for Cost Distribution](#) when you've enabled cost distribution in the freight settlement document

To integrate to settlement management on an external SAP S/4HANA or SAP ERP system, you've specified the relevant settings in the Customizing activities under [► Integration with Other SAP Components](#) [► Transportation Management](#) [► Invoice Integration](#) [► Mapping for Cost Distribution](#) [►](#).

Example

The following figure illustrates the process in which deliveries of two sales orders are planned in the freight order in Transportation Management. The freight costs are then posted for freight orders with freight cost distribution.



Processing Distributed Costs

You include two freight units for outbound deliveries in a freight order. The total amount in document currency of the freight order is USD 1000. In the associated freight settlement document, the system distributes the freight cost to the underlying delivery items by gross weight.

When you post the freight settlement document, the system takes the following steps:

1. Creates a service purchase order and includes the items that correspond to the total amount in document currency for the freight settlement document.
2. Posts a service entry sheet (SES) to confirm the services in the service purchase order
3. Creates a settlement management document with two line items. Each line item corresponds to the individual line items in the freight order.

The system copies the account assignment details from the relevant sales order line item into the settlement management document. In this example, the system copies the account assignments from item 10 of sales order 1 to item 10 of the settlement management document and item 10 of sales order 2 to the item 20 of the settlement management document.

More Information

Subject	More Information
Work with invoices for transportation management	SAP S/4HANA > Enterprise Business Applications > Supply Chain > Logistics - General (LO) > Integration with SAP Transportation Management > Integration: Invoice Processing (MM-IV/SD-BIL) with TM
Work with freight cost allocation documents	Freight Cost Allocation
Use SAP best practices for Transportation Management in SAP S/4HANA 1709 on premise, to configure settlement management documents	SAP Note 2535942 

Billing the Freight Cost in TM to a Customer in SD

When you sell your products to a customer you may also take the responsibility of delivering these products to that customer. For this you incur costs for transportation, for example, through a carrier. You can pass on these costs to your customer.

You use integration of sales orders and deliveries to facilitate transportation planning and execution in Transportation Management (TM). The TM component creates or updates freight units directly for sales orders or deliveries. You plan transportation for these freight units by creating freight orders that you then subcontract to the carrier. The freight order determines the freight costs to be paid to the carrier. You configure TM to distribute the freight costs in the freight order according to delivery items. You can then add the distributed costs for each delivery item to the final amount in the billing document that you send to your customer.

This involves integration of the following processes:

- Freight orders in TM
- Pricing in SD
- Billing in SD

Example

As the seller of a product, you have a freight order that combines two deliveries for your customer. The freight order contains the following information:

Item Hierarchy	Item	Gross Weight (KG)	Tare Weight (KG)	Delivery Reference
Product 1	20	2,250	0	Outbound delivery 81000100/10
Product 2	50	2,750	0	Outbound delivery 81000900/10

You have the following charges in your freight order:

Charge	Resolution Base	Amount (USD)
Base freight	Root	1,000
Fuel surcharge	Root	250

Charge	Resolution Base	Amount (USD)
Packaging	Product (for product 1)	50
Packaging	Product (for product 2)	100
Total Charges		1,400

You can distribute costs in the freight order to the level of each delivery item, based on the gross weight of each delivery item, as follows:

Distribution Item/Charge Type	Amount (USD)	Details
Outbound delivery 81000100/10 (2,250 KG)	612.50	
Base freight	450	$1000 * 2,250 / (2,250 + 2750)$
Fuel surcharge	112.50	$250 * 2,250 / (2,250 + 2750)$
Packaging	50	Assigned directly for product 1
Outbound Delivery 81000900/10 (2,750 KG)	787.50	
Base freight	550	$1000 * 2,750 / (2,250 + 2750)$
Fuel surcharge	137.50	$250 * 2,750 / (2,250 + 2750)$
Packaging	100	Assigned directly for product 2

If you enable the SD billing process for your delivery items, the system takes the distributed cost from TM into account when it creates the billing document for the delivery items in SD. You create a billing document for the outbound deliveries 81000100/10 and 81000900/10 in SD. The system uses the freight cost of 612.50 USD and 787.50 USD for each of the delivery items respectively in the SD billing document.

i Note

- You must integrate SD with an internal TM component to be able to include the freight cost in the billing document for all deliveries of products.
- You can only include the freight cost in a billing document when you have one freight order document for the delivery and the freight order has only one currency in which you calculate the cost.

Related Information

[Billing of Freight Costs - Integration with Transportation Management](#)

[Cost Distribution for Shippers](#)

Cost Distribution for LSPs

Use

As a logistics service provider (LSP), the customer contacts you for a transportation and you create a forwarding order in Transportation Management (TM). An organization in your company works with customers to handle order placement and a different organization works with carriers to handle order execution. You use a freight order to execute the transportation service. You use the standard charge calculation and freight settlement processes to settle costs.

You can use cost distribution for external or internal costs in the following settlements:

- Forwarding settlements to settle with your external customers
- Internal settlements to settle with your internal organizations

When the system calculates internal and external charges in a forwarding order, and the internal and external charge calculation is based on cost pull, it automatically pulls the distributed costs from the associated freight documents, such as freight orders, freight bookings, or service orders. It also uses the distributed cost to calculate the planned revenue in a profitability analysis.

Integration

Financial Accounting

The system uses the internal agreement and internal settlement document to integrate with financial accounting. For more information, see [Internal Settlement](#).

Analysis of Profitability in Forwarding Order

You can use cost distribution as a prerequisite for analyzing profitability in a forwarding order. For more information, see [Profitability Analysis](#).

Charge Management

You can use the distributed freight cost to settle with your external customers and to settle with your internal organizations. To enable the system to use the distributed freight cost in a freight order or freight booking, you must select a calculation method type of **Internal Charge Calculation** and the appropriate cost pull strategy in the relevant calculation sheet. You use the calculation sheet in the agreement between your sales organization and your customer for a forwarding settlement, or between your sales organization and purchase organization for an internal settlement.

When you calculate charges in the relevant forwarding order, the system checks the freight order that executed the forwarding order. It uses the result of the cost distribution you performed on the forwarding order that was executed in the freight order as a basis for the forwarding settlement document or internal settlement document. For more information, see [Cost Pull](#).

Prerequisites

You have performed the following tasks:

- Enabled the freight order or freight booking for cost distribution in Customizing for **Transportation Management** under **Freight Order Management > Freight Order > Define Freight Order Types and Freight Order Management > Freight Booking > Define Freight Booking Types**.

This enables you to use the cost distribution information with a cost-pull strategy when you calculate charges.

- Calculated charges in the freight document
- Specified an internal agreement between the purchase organization of the freight document and the business partner of the sales organizations in the forwarding orders. For more information, see [Internal Settlement Management](#).
- Specified a forwarding agreement between the sales organization in a forwarding order and the agreement party in the same forwarding order. For more information, see [Forwarding Settlement](#).

- Created a forwarding agreement or an internal agreement that contains a calculation sheet in which you have assigned a calculation method type of **Internal Charge Calculation** to a charge type.

This enables you use cost distribution information when you calculate charges.

The system uses the cost distribution for external and internal costs as a basis for the calculation. For example, you can use this calculation method to calculate the following costs and settlements:

- Between the purchase organization in the freight document and the sales organization in the forwarding order
- Between the sales organization and the agreement party in the forwarding order

Alternatively, you can use calculation sheets with charge types that are linked to rate tables. You can use the rate tables as the basis for creating settlements. In this case, you do not need cost distribution in the freight order, freight booking, or service order to calculate internal settlements.

i Note

When you use a calculation sheet in which you have assigned a calculation method type of **Internal Charge Calculation** to a charge type, the system uses the cost distribution amounts from both your external costs and internal costs. For more information on the internal charge calculation methods see [Cost Pull](#).

You have specified all the prerequisites for internal settlement (see [Internal Settlement Management](#)).

You have specified all the prerequisites for forwarding settlement (see [Forwarding Settlement](#)).

Example

You are an LSP and you have 2 customers who need to transport cargo.

Organizations

The following table explains the role of the relevant organizations in your company:

Organization	Role
SalesOrg_1	Sells services to the customer and the customer places forwarding orders with this organization
PurchaseOrg_1	Buys transportation services from carriers to execute freight orders

Agreements

The following tables give you an explanation of the external and internal agreements that you create to transport the cargo:

External agreements

Agreement	Agreement Type	Organization	Business Partner	Reason
Cust_Agmt_1	Forwarding agreement	SalesOrg_1	Customer_1	Pay for transportation services in FWO_1
Cust_Agmt_2	Forwarding agreement	SalesOrg_1	Customer_2	Pay for transportation services in FWO_2

Agreement	Agreement Type	Organization	Business Partner	Reason
Int_Agmt_1	Internal agreement	PurchaseOrg_1	BP_of_SalesOrg_1 (carrier)	Pay haulage cost

Internal agreement

Agreement	Agreement Type	Organization	Business Partner	Reason
Int_Agmt_2	Internal agreement	PurchaseOrg_1	SalesOrg_1	Pay haulage cost
Int_Agmt_Res_3	Internal agreement for resource	ResOrg_1 (forwarding house)	BP_of_PurchaseOrg_1	Pay for usage of the trailer resource

Orders

You create two forwarding orders in TM, with one forwarding order for each customer. The carrier transports the cargo (the carrier is the business partner of SalesOrg_1). In the freight order you have external costs from the carrier.

Cost Distribution

You distribute costs by gross weight. You perform cost distribution at the level of **Forwarding Order** in the freight order. You also have internal costs against the purchase organization of the freight order for using a trailer resource from another internal organization. The following tables describe how the system distributes costs in the freight order or freight booking:

External cost for the carrier, viewed as a distributed cost for the two forwarding orders that are part of the freight order.

	Base Freight (USD)	Haulage (USD)	Total (USD)
Net amount in document currency for the freight order	700	300	1000
Distributed cost for forwarding order FWO_1	500	200	700
Distributed cost for forwarding order FWO_2	200	100	300

Internal cost for the trailer

	Cost of Trailer (USD)	Use of Trailer over Distance (USD)	Total (USD)
Internal cost for provision of Trailer	200	100	300
Distributed cost for forwarding order FWO_1	140	70	210
Distributed cost for forwarding order FWO_2	60	30	90

Internal Settlement with Distributed Costs

FWO_1 and FWO_2 belong to SalesOrg_1. PurchaseOrg_1 performs the following tasks:

- Executes the freight order
- Pays the execution costs
- Incurs the cost of using the trailer from the internal organization to which the trailer belongs

PurchaseOrg_1 uses the external costs it incurred for both forwarding orders to settle internally with SalesOrg_1. The following table describes how PurchaseOrg_1 settles with SalesOrg_1:

	External Cost (USD)	Internal Cost (USD)	Markup Charge (USD)	Total (USD)
FWO_1	700	210	100	1010
FWO_2	300	90	50	440

More Information

Subject	See
Create and change agreements	Agreement
Create and change calculation sheets	Calculation Sheet and Local Calculation Sheet Maintenance
Calculate transportation charges according to a specific logic	Calculation Methods
Manage internal settlements	Internal Settlement Management
Manage cost distribution	Cost Distribution Management
Analyze profit	Profitability Analysis
Transfer distributed costs from the freight side to the forwarding side, and use the costs as a basis for settling with a customer or an internal organization	Cost Pull
Create an invoice for TM	Integration: Invoice Processing (MM-IV/SD-BIL) with TM

Cost Distribution in Service Orders for Shippers and LSPs

Use

A service order is linked to a freight document or a freight document item. The system distributes the service order cost to the eventual consumer of the service. The system uses the underlying freight document or freight document item to establish the consumer of the service. This can be a forwarding order or a delivery or a delivery item that is contained in the freight document.

The system uses the distribution level and the distribution rule that you specify to determine the distribution items.

Prerequisites

You have specified the following settings in Customizing for **Transportation Management**:

1. Distribution item and distribution rule in the relevant distribution profile under **SAP Menu > Basic Functions > Cost Distribution > Define Cost Distribution Profiles**
2. Distribution profile under **Basic Functions > Charge Calculation > Basic Settings > Define Charges Profile**

3. Charges profile in the relevant purchasing organization under ►► **Master Data** ► **Organizational Management** ► **Organizational Model** ► **Change Organizational Model** ► (transaction PPOME)

You include the purchasing organization in a freight document. You link a service order to the freight document.

You have enabled the relevant service order type for cost distribution in Customizing for **Transportation Management** under ►► **Freight Order Management** ► **Service Order** ► **Define Service Order Types** ►. For more information, see [Creation and Editing of Service Orders](#).

Features

Changing a Logistics Item

If you make a change to a logistics item in a freight document or freight document line item, and you have linked a service order or service order item to the logistics item, the system automatically triggers cost distribution in the linked service order.

You can change a logistics item as follows:

- Change the quantity of a logistics item where the logistics item is specified in the distribution rule

For example, you as an LSP create a service order for an item in a freight document so that a third party can deliver a service. You specify a distribution rule of **Gross Weight** for the cost distribution.

You then change the gross weight of the item in the freight document. When you save the change, the system automatically distributes the costs again in the service order.

- Add, delete, or reposition a logistics item

For example, you as an LSP remove a product from a package in a freight document. You have created a service order for the freight document. When you save the change to the freight document, the system automatically adjusts the cost distribution in the service order.

Changing a Freight Document

You have a freight document with an associated service order. You change the freight document, but the change does not directly impact the associated service order. However, because of the change, the system distributes the costs again in the freight document. In this case, the system also automatically distributes costs again in the service order.

More Information

Topic	Link
Example 1 for a shipper	Shipper Example: One Service Order for an Entire Freight Booking
Example 2 for a shipper	Shipper Example: One SO for a Service Item in a Freight Booking
Example 1 for an LSP	LSP Example: One Service Order for an Entire Freight Booking
Example 2 for an LSP	LSP Example: One SO for a Service Item in a Freight Booking

Shipper Example: One Service Order for an Entire Freight Booking

The following table gives an example of a freight booking:

Item Hierarchy Level 1	Sub-Level 2	Sub-Level 3	Item Type Description	Original Order/Delivery Reference	Quantity	Gross Weight (KG)
Ocean freight booking FB71000						7000
20 ft dry cont. 20x8			Container		1	7000
	Freight Unit 70				1	3000
		Product 10	Product	Delivery 950250/item 10		800
		Product 20	Product	Delivery 950250/item 20		2200
	Freight Unit 71				1	4000
		Product 30	Product	Delivery 950290/item 10		2400
		Product 40	Product	Delivery 950290/item 20		1600

You create one service order to cover the provision of a service for the entire booking (for example, the provision of fumigation and cleaning).

The following table gives an example of the information on the [Service Orders](#) tab page of your freight booking:

Service Provider	Service Order
SERVICE_PROVIDER_1	SO71000

The following table gives an example of the information on the [General Data](#) > [Services](#) screen of the service order SO71000:

Item Hierarchy Level 1	Sub-Level 2	Original Document	Item	Item Type	Service Type	Service Type (Description)
Booking		71000				
	Service		9090	Service	FUM	Fumigation
	Service		9091	Service	CLN	Cleaning

i Note

The system does not display any other information about the ocean freight booking on the service order screens.

The following table gives the freight cost in service order SO71000:

Freight Cost in Service Order SO71000	Charge Type	Amount (USD)
Fumigation charge	SRV_FUM	200

Freight Cost in Service Order SO71000	Charge Type	Amount (USD)
Cleaning charge	SRV_CLN	50
Total		250

When the system distributes costs, it applies the service order costs to the order/delivery items in the ocean freight booking that used the service. In this example, it applies the service order costs to delivery 950250, items 10 and 20, and delivery 950290, items 10 and 20.

The following table gives an example of the cost distribution information for the service order SO71000, where the distribution rule is **Gross Weight**:

Distribution Hierarchy Level 1	Sub-Level 2	Charge Type	Distribution Percentage	Distribution Amount (USD)	Quantity (KG)	Logistical Reference
Freight document SO71000 – delivery 950250/item 10			11.0	28.57		Delivery 950250/item 10
	Fumigation charge	SRV_FUM	11.0	22.86	3000	
	Cleaning charge	SRV_CLN	11.0	5.71	3000	
Freight document SO71000 – delivery 950250/item 20			31.0	78.57		Delivery 950250/item 20
	Fumigation charge	SRV_FUM	31.0	62.86	3000	
	Cleaning charge	SRV_CLN	31.0	15.71	3000	
Freight document SO71000 – delivery 950290/item 10			34.0	85.71		Delivery 950290/item 10
	Fumigation charge	SRV_FUM	34.0	68.57	4000	
	Cleaning charge	SRV_CLN	34.0	17.14	4000	
Freight document SO71000 – delivery 950290/item 20			23.0	57.14		Delivery 950290/item 20
	Fumigation charge	SRV_FUM	23.0	45.71	3000	

Distribution Hierarchy Level 1	Sub-Level 2	Charge Type	Distribution Percentage	Distribution Amount (USD)	Quantity (KG)	Logistical Reference
	Cleaning charge	SRV_CLN	23.0	11.43	3000	

Shipper Example: One SO for a Service Item in a Freight Booking

The following table gives an example of a freight booking:

Item Hierarchy Level 1	Sub-Level 2	Sub-Level 3	Item Type Description	Original Order/Delivery Reference	Quantity	Gross Weight (KG)
Ocean freight booking FB77000						7000
20 ft dry cont. 20x8			Container		1	7000
	Freight unit 70				1	3000
		Product 10	Product	Delivery 950250/item 10		800
		Product 20	Product	Delivery 950250/item 20		2200
	Freight unit 71				1	4000
		Product 30	Product	Delivery 950290/item 10		2400
		Product 40	Product	Delivery 950290/item 20		1600

You purchase a packaging and labeling service for delivery 950250, item 10. You create a service order for delivery 950250 item 10. Delivery 950250 item 10 is in the container.

You can review the service order information on the [Items](#) tab page of the ocean freight booking FB77000. You select the relevant item and choose the [Service Orders](#) tab page.

The following table gives an example of the information on the [Service Orders](#) tab page of your freight booking:

Service Provider	Service Type	Service Order
SERVICE_PROVIDER_1	Packaging and labeling	S077000

The following table gives an example of the freight cost in service order S077000:

Freight Cost in Service Order S077000	Charge Type	Amount (USD)
Packaging and labeling charges	SRV_PAKLBL	70

Freight Cost in Service Order S077000	Charge Type	Amount (USD)
Total		70

The following table gives an example of the information on the [General Data](#) > [Services](#) screen of the service order S077000:

Item Hierarchy Level 1	Sub-Level 2	Original Document	Item	Item Type	Service Type	Service Type (Description)
Booking		77000	90010			
	Service		9090	Service	PKL	Packaging and labeling

When the system distributes costs, it applies the service order costs to delivery 950250 item 10, as this is the only delivery item that is associated with the service order.

The following table gives an example of the cost distribution information for service order S077000, where the distribution rule is [Gross Weight](#):

Distribution Hierarchy Level 1	Sub-Level 2	Charge Type	Distribution Percentage	Distribution Amount (USD)	Quantity (KG)	Logistical Reference
Freight document FB77000 - delivery 950250/item 10			100	900		Delivery 950250/item 10
	Packaging and labeling	PACK	100	70	800	

LSP Example: One Service Order for an Entire Freight Booking

The following table gives an example of a freight booking:

Item Hierarchy Level 1	Sub-Level 2	Sub-Level 3	Item Type Description	Original Order Reference	Quantity	Gross Weight (KG)
Ocean freight booking FB71000						7000
20 ft dry cont. 20x8			Container		1	7000
				Forwarding order 250	1	3000
	Freight unit 70				1	3000
		Product 10	Product			800

Item Hierarchy Level 1	Sub-Level 2	Sub-Level 3	Item Type Description	Original Order Reference	Quantity	Gross Weight (KG)
		Product 20	Product			2200
				Forwarding order 350	1	4000
	Freight unit 71				1	4000
		Product 30	Product			2400
		Product 40	Product			1600

You create one service order to cover the provision of a service for the entire booking (for example, the provision of fumigation and cleaning).

The following table gives an example of the information on the [Service Orders](#) tab page of your freight booking:

Service Provider	Service Order
SERVICE_PROVIDER_1	SO71000

The following table gives an example of the information on the [General Data](#) > [Services](#) screen of the service order SO71000:

Item Hierarchy Level 1	Sub-Level 2	Original Document	Item	Item Type	Service Type	Service Type (Description)
Booking		71000				
	Service		9090	Service	FUM	Fumigation
	Service		9091	Service	CLN	Cleaning

i Note

The system does not display any other information about the ocean freight booking on the service order screens.

The following table gives the freight cost in service order SO71000:

Freight Cost in Service Order SO71000	Charge Type	Amount (USD)
Fumigation charge	SRV_FUM	200
Cleaning charge	SRV_CLN	50
Total		250

When the system distributes costs, it applies the service order costs to the forwarding orders in the ocean freight booking that used the service. In this example, it applies the service order costs to forwarding orders 250 and 350.

The following table gives an example of the cost distribution information for the service order SO71000, where the distribution rule is [Gross Weight](#):

Distribution Hierarchy Level 1	Sub-Level 2	Charge Type	Distribution Percentage	Distribution Amount (USD)	Quantity (KG)	Logistical Reference
Freight document SO71000 – forwarding order 250			43.0	107.14		250
	Fumigation charge	SRV_FUM	43.0	85.71	3000	
	Cleaning charge	SRV_CLN	43.0	21.43	3000	
Freight document SO71000 – forwarding order 350			57.0	142.86	4000	350
	Fumigation charge	SRV_FUM	57.0	114.29	4000	
	Cleaning charge	SRV_CLN	57.0	28.57	4000	

LSP Example: One SO for a Service Item in a Freight Booking

The following table gives an example of a freight booking:

Item Hierarchy Level 1	Sub-Level 2	Sub-Level 3	Item Type Description	Original Order Reference	Quantity	Gross Weight (KG)
Ocean freight booking FB77000						7000
20 ft dry cont. 20x8			Container		1	7000
				Forwarding order 750	1	3000
	Freight unit 90				1	3000
		Product 10	Product			800
		Product 20	Product			2200
				Forwarding order 850	1	4000
	Freight unit 91				1	4000
		Product 30	Product			2400
		Product 40	Product			1600

You purchase a packaging and labeling service for product 10. Product 10 belongs to forwarding order 750. You create a service order for product 10.

You can review the service order information on the **Items** tab page of the ocean freight booking FB77000. You select the relevant item and choose the **Service Orders** tab page.

The following table gives an example of the information on the **Service Orders** tab page:

Service Provider	Service Type	Service Order
SERVICE_PROVIDER_1	Packaging and labeling	SO77000

The following table gives an example of the freight cost in service order SO77000:

Freight Cost in Service Order SO77000	Charge Type	Amount (USD)
Packaging and labeling charges	SRV_PAKLBL	70
Total		70

The following table gives an example of the information on the **General Data > Services** screen of the service order SO77000:

Item Hierarchy Level 1	Sub-Level 2	Original Document	Item	Item Type	Service Type	Service Type (Description)
Booking		77000	9010			
	Service		9090	Service	PKL	Packaging and labeling

When the system distributes costs, it applies the service order costs to the forwarding order. In this example, this is forwarding order 750 to which product 10 belongs.

The following table gives an example of the cost distribution information for service order SO77000, where the distribution rule is **Gross Weight**:

Distribution Hierarchy Level 1	Sub-Level 2	Charge Type	Distribution Percentage	Distribution Amount (USD)	Quantity (KG)	Logistical Reference
Freight document FB77000 – forwarding order 750			100	900		750
	Packaging and labeling	PACK	100	70	800	

Group Logistics Scenario

Use

You can use a group logistics company process to consolidate all the transportation requirements from all the companies in your enterprise into one central logistics company. Your core business activity can be something other than providing transportation services. For example, you have a group of companies that manufactures products. You need to transport the products to your customers. You can have one of the following business scenarios:

- One company in your group of companies manufactures products and also plans and organizes transportation services for your other internal manufacturing companies.
- One company in your group of companies does not manufacture product but instead is a dedicated group logistics company. All your manufacturing companies use the group logistics company for transportation services.

You can have a dedicated group logistics company for the following business reasons:

- Lower transportation costs

A single dedicated logistics company can get cheaper transportation rates from a logistics service provider (LSP). For example, each individual company approaches an LSP separately. Each company gets a less than container load (LCL) rate from the LSP. However, if you approach an LSP as one company with one transportation requirement, you can get a full container load (FCL) rate.

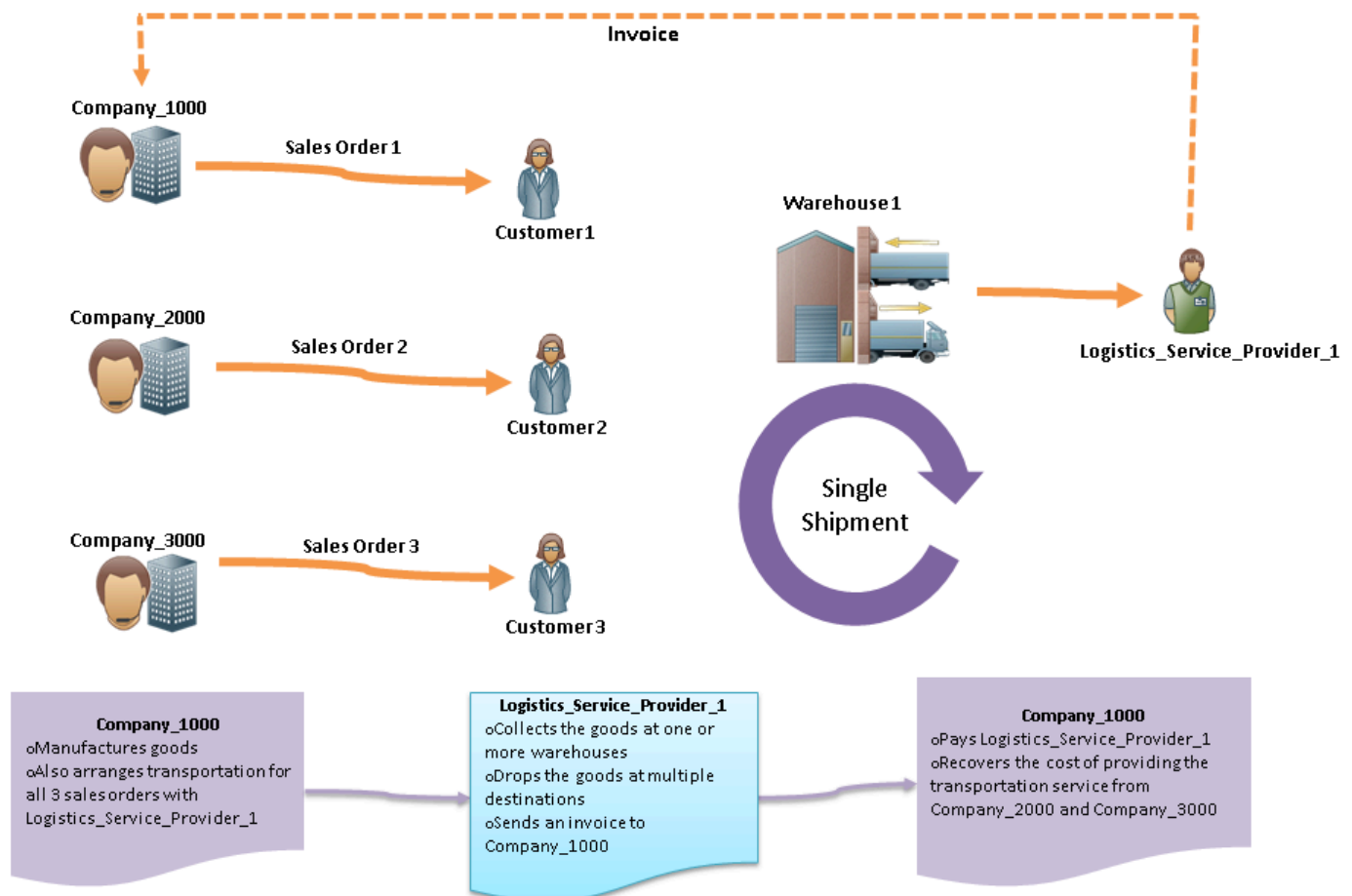
- Standardized logistics process

You have a standard shipment and tracking process across all companies. Resources are not duplicated across companies. For example, your manufacturing companies can concentrate fully on manufacturing.

- Visibility

You can have full visibility to your transportation demands and execution from a centralized operation.

The figure below illustrates a business example of a group logistics scenario, where your companies share a single warehouse, and where an internal company manufactures product and also provides a transportation service to the other manufacturing companies in your group of companies. The graphic is followed by an explanation:



Company_1000, Company_2000, and Company_3000 belong to the same enterprise. Company_1000 manufactures product, but also provides a transportation service for Company_2000 and Company_3000.

Company_2000 and Company_3000 act as internal shipper companies, buying transportation services from Company_1000. Company_1000 acts as a forwarding house. It buys transportation services from Logistics_Service_Provider_1, and sells transportation services to Company_2000 and Company_3000.

Transportation Management (TM) creates an order-based transportation requirement (OTR) or delivery-based transportation requirement (DTR) for each of the sales orders or deliveries that each company has in place with their respective customers in Sales and Distribution on an external S/4 HANA or SAP ERP system. Company_1000 then creates a freight order for the OTRs or DTRs, with itself as the purchasing organization in TM. It sends the freight order to Logistics_Service_Provider_1.

Logistics_Service_Provider_1 collects the cargo at the warehouse and delivers the cargo to multiple destinations. The destination locations are geographically close together. Logistics_Service_Provider_1 sends an invoice to Company_1000 for the freight order. Company_1000 pays Logistics_Service_Provider_1. Company_1000 uses the internal settlement process to recover the cost of providing the transportation service from Company_2000 and Company_3000.

In a group logistics scenario, Company_1000 can create and post internal settlement documents when the life cycle status of the freight order is **In Execution**.

To run a group logistics process, you must integrate orders based on orders created in Sales and Distribution (SD) or Materials Management (MM), and deliveries created in Logistics Execution (LE) from an external S/4 HANA or SAP ERP system with a TM component.

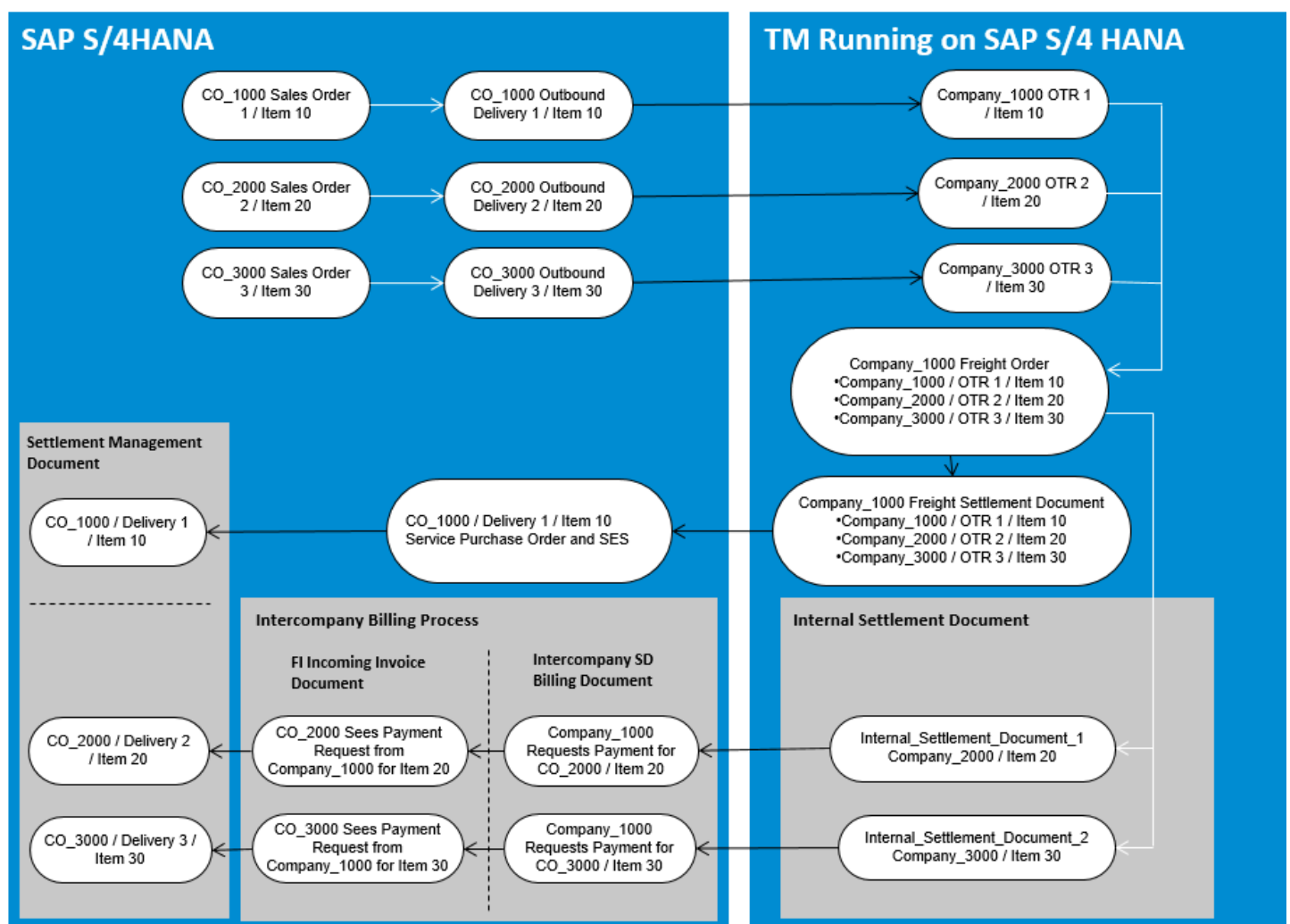
More Information

Subject	See
Detailed process flow for a group logistics company, including cost distribution	Group Logistics Company Process
Comparison between the standard settlement process for shippers with the process for a group logistics company	Comparison: Standard Shipper Process and Group Log. Co. Process

Group Logistics Company Process

Use

The figure below illustrates an example of a process flow for a group logistics company scenario. The graphic is followed by an explanation:



Example of Process Flow for Group Logistics Company Scenario

Agreements

The table gives an example of the agreements that you can put in place:

Agreement Type	Organization	Business Partner
Internal Agreement	Sales organization of Company_1000, the group logistics company. Company_1000 is a forwarding	CO_2000, the company behind sales order 2 and CO_3000, the company behind sales order 3

Agreement Type	Organization	Business Partner
	house with both a purchasing role and a sales role.	
External Agreement	Purchasing organization of Company_1000, the group logistics company. Company_1000 is a forwarding house with both a purchasing role and a sales role.	External logistics service provider (LSP)

The group logistics company (Company_1000) puts an internal agreement in place that covers one of the following charge calculation situations:

- Charge for the actual freight cost for each of the companies, based on the distributed costs in the freight order
- Charge a mark-up in addition to the actual freight cost based on the distributed costs in the freight order
- Use a rate table for a pricing structure that is different to the freight costs

Cost Distribution

As the group logistics company, you have the option of distributing costs in the following business documents:

- Freight order

You can distribute costs in the freight order at order/delivery item level for each of the transportation requirements in the freight order for the following business reasons:

- You want to use the distributed cost as the basis for creating internal settlement documents.
- You want a breakdown of your own transportation costs and the transportation costs of the internal shipper companies. For example, Company_1000 distributes costs in the freight order, and uses the distributed costs to create internal settlement documents for Company_2000 and Company_3000.

- Freight settlement document

As a group logistics company, you can distribute costs in the freight settlement document for the following business reasons:

- Transport your own cargo along with the cargo from the other internal companies
- Pay the external LSP for all the transportation costs
- Post the freight cost to the underlying transportation requirement (that is, the order or delivery item), for example, the sales order item, the delivery item, or the purchase order item

- Internal settlement document

You can distribute costs in an internal settlement document if the shipper companies in your group of companies need to post the freight cost to the underlying transportation requirement, that is, the order or delivery item. The underlying transportation can be a sales order item, a delivery item, or a purchase order item.

For example, Company_2000 and Company_3000 distribute costs in their respective internal settlement documents to post the freight cost to sales order 2 item 20 and sales order 3 item 30 respectively.

Example of Settlement with an External Logistics Service Provider

Company_1000 uses the following process to pay the external LSP for the transportation services:

1. Company_1000 receives the transportation requests from the internal shipper companies as order-based transportation requirements (OTRs) or delivery-based transportation requirements (DTRs) in Transportation Management (TM).
2. Company_1000 uses the OTR or DTR to create a freight order or freight booking. It uses the freight order or freight booking to purchase transportation services from the LSP. Company_1000 can distribute costs in the freight order or freight booking. This enables it to use the distributed cost as the basis for creating internal settlements and to establish what its own share of the freight costs are.
3. Company_1000 creates a freight settlement document. It can distribute costs in the freight settlement document to establish what its own share of the freight costs are. It can also account for its own share of the costs in financial accounting.
 - a. Company_1000 creates a service purchase order and a service entry sheet (SES) in Materials Management (MM) for the entire freight order cost.
 - b. Company_1000 creates a settlement management document to transfer its own share of the costs to its account either as an expense or for material valuation.
4. The LSP (not shown in the graphic) invoices Company_1000. Company_1000 pays the LSP the total amount for the full transportation costs.

Example of Recovering the Cost of Providing Transportation Services

Company_1000 uses the following process to recover the costs of providing the transportation services from Company_2000 and Company_3000:

1. Company_1000 uses the freight order or freight booking to create an internal settlement document for each of Company_2000 and Company_3000.

i Note

The life cycle status of the freight order must be **In Execution** before you can create internal settlement documents for the items in the OTR or DTR. By default, the system uses the billing schedule you specify for the business partner as the invoice date. You can specify the billing schedule under **SAP Menu > Logistics > Transportation Management > Master Data > BP - Maintain Business Partner > Display in BP Role: Customer (FLCU01) > Sales and Distribution > Billing > Invoicing Dates**.

The system uses the billing schedule for Company_2000 and Company_3000 to determine the invoice dates in the internal settlement documents for Company_2000 and Company_3000. Company_2000 and Company_3000 are customers of Company_1000. Company_1000 acts as a forwarding house and sells the transportation service.

2. Company_1000 uses an internal agreement to calculate costs.

You specify Company_1000 as a forwarding house. The agreement is between the purchasing organization of Company_1000 and each of the business partners for Company_2000 and Company_3000.

The agreement can use standard rates, cost pull, or cost pull plus a mark-up as a basis for calculating charges. For more information, see [Charge Calculation](#).

3. Company_1000 uses Company_2000 and Company_3000 as the bill-to parties in the respective internal settlement documents.

Company_1000 has the option of distributing costs at order/delivery item level in both internal settlement documents. This enables Company_2000 and Company_3000 to post the freight cost to the underlying transportation requirement, for example, to the sales order item, delivery item, or purchase order item.

4. Company_1000 posts the internal settlement documents to Sales and Distribution (DS). SD creates two intercompany billing documents for the group logistics company CO_1000. CO_0002 and CO_0003 pay for the freight costs.

5. SD billing uses the output type configuration of the intercompany billing document to create an IDoc. The IDoc posts an incoming invoice for each of the companies CO_2000 and CO_3000, recording the accounts payable to Company_1000. For more information about the settings that you need in place in TM, see [TM Prerequisites for a Group Logistics Company Process](#).
6. If you enable cost distribution in the internal settlement documents, a settlement management document is created when you post the document for each of CO_2000 and CO_3000.

The settlement management document contains the following:

- A line item for the underlying transportation requirement item. This is the order/delivery item on the TM side. For example, this can be a sales order item, a delivery item, or a purchase order item.
- An amount in the line item that matches the distributed amount of the order/delivery item in TM.

i Note

You must have a single SAP S/4HANA client environment to create an intercompany billing document and subsequently to use an IDoc to trigger the corresponding incoming invoice.

Change Process

You can start a change process from TM for an internal settlement document that you have already posted to SD. You can take one of the following steps:

- Cancel the internal settlement document that you have already created
- Change the charges and recalculate in the existing internal settlement document

When you post the new or changed internal settlement document to SD, SD cancels the already existing intercompany billing document. For the new internal settlement document, SD creates a new intercompany billing document. For a changed internal settlement document, SD creates a new intercompany billing document that contains the changes. The system also takes the following actions:

- Uses an IDoc to reverse the accounts payable document for the purchasing company from the SD billing document that was created
- Cancels the settlement management document that contains the distributed freight costs
- Uses an IDoc to create a new Financial Accounting (FI) invoice document that updates accounts payable for the purchasing company with the new or changed values
- Creates a settlement management document that contains the new or changed distributed freight costs

Updates to TM Settlement Documents

The following documents in TM are updated when the settlement documents are posted:

- Freight settlement document with the purchase order ID and SES ID
- Internal settlement document with the Sales and Distribution (SD) billing document ID

You can use the [Freight Cost Distribution Report](#) (transaction WLFLTM) and the [Overview of Freight Cost Distribution Documents](#) function (transaction WLFLTM2) to determine the settlement management documents that are linked to the freight settlement document and the internal settlement document.

More Information

Subject	See
Overview of the business process for a group logistics company	Group Logistics Scenario
Comparison between the standard settlement process for shippers with the process for a group logistics company	Comparison: Standard Shipper Process and Group Logistics Company Process
Prerequisite settings you must specify in TM	TM Prerequisites for a Group Logistics Company Process
Prerequisite settings you must specify in Sales and Distribution	Sales and Distribution (SD) Prerequisites for a Group Logistics Company Process

Partner Function Codes in an OTR or DTR

Use

In a group logistics process, you have the option of specifying an agreement partner function for a stage type and Incoterm combination. You specify an Incoterm in the sales order or delivery. TM includes the Incoterm in the OTR or DTR. TM assigns a party role code of **Supplying Company** or **Receiving Company** to identify the default agreement party for a stage. It assigns the partner function code in the **Business Partner** tab page of the OTR or DTR.

The following table explains how TM assigns the partner function codes **Supplying Company** or **Receiving Company**:

Transportation Requirement	Sales Order or Delivery	Sales Return or Delivery	Purchase Order or Delivery	Purchase Return or Delivery	Stock Transfer Order	Stock Transfer Return or Delivery
OTR header	Company of sales organization as supplying company	Company of sales organization as receiving company	None	None	Company of supplying plant as supplying company	Company of receiving plant as receiving company
OTR item	None	None	Company of receiving plant as receiving company	Company of supplying plant as supplying company	Company of receiving plant as receiving company	Company of supplying plant as supplying company
DTR header	Company of sales organization as supplying company	Company of sales organization as receiving company	Company of receiving plant as receiving company	Company of supplying plant as supplying company	- Company of supplying plant as supplying company - Company of receiving plant as receiving company	- Company of receiving plant as receiving company - Company of supplying plant as supplying company
DTR item	None	None	None	None	None	None

The following table explains how the system uses the **Supplying Company** and **Receiving Company** role codes in charge calculation and in creating settlement documents:

Order or Delivery	Supplying Company Role Code	Receiving Company Role Code	Role Code Used in Calculation and Settlement
Sales order	Company of the sales organization	None	Prepaid party is always the supplying company
Sales returns	None	Company of the sales organization	Collect party is always the receiving company
Purchase order	None	Company of the receiving plant	Collect party is always the receiving company
Purchase return	Company of the supplying plant	None	Prepaid party is always the supplying company
Stock transfer order	Company of the supplying plant	Company of the receiving plant	Role code based on Incoterm (prepaid party as supplying company and collect party as receiving company)
Stock transfer return	Company of the supplying plant	Company of the receiving plant	Role code based on Incoterm (prepaid party as supplying company and collect party as receiving company)

For special cases such as that of stock transfer orders you can allow separate settlements of the different parts of the freight costs to the internal shipper company that acts as a supplying company and to the internal shipper company that acts as a receiving company. To enable this, you can use an Incoterm in combination with the stage type to determine the prepaid and collect parties. For more information, see Customizing for **Transportation Management** under **Forwarding Order Management** > **Define Default Agreement Partner Functions for Stages**.

You can also define your own logic to specify the partner function that the system uses for a stage in Customizing for **Transportation Management** under **Business Add-Ins (BAdIs) for Transportation Management** > **Settlement** > **Forwarding Settlement** > **BAdI: Change of Bill-to Party Types for Stage Items**.

❖ Example

You have a stock transfer order from a supplying company that belongs to Company_1000. You supply the material to a company that belongs to Company_2000, the receiving company. You specify an Incoterm of FOB, with an FOB location of sea port.

Company_1000 pays for all the freight costs from the supplying company to when the cargo is loaded on a ship.

Company_2000 pays for all the freight costs for the main carriage and on-carriage to the delivery location. If you do not specify an Incoterm or stage type in the freight unit that is linked to the OTR or DTR, by default the system uses the collect party as the receiving company in the internal settlement document for the stock transfer order or the stock transfer return.

Comparison: Standard Shipper Process and Group Logistics Company Process

Use

The table explains the differences between the standard shipper process and the process for a group logistics company in Transportation Management (TM):

Item	Standard Shipper Process	Group Logistics Company Process
Business scenario	Supports the following business process: 1. TM consolidates an order-based transportation requirement (OTR) or a delivery-based transportation requirement (DTR) in a freight order or freight booking. The associated freight settlement documents belong to the same company. 2. The same company settles with the external logistics service provider (LSP). 3. The system posts the distributed freight costs to the company that settles with the LSP.	Supports the following business process: 1. The OTR or DTR belong to different companies. TM can consolidate the OTR or DTR in a single freight order or freight booking. 2. Any of the companies can settle the freight costs with the external LSP. This can be a company that has cargo in the freight document or the company that is taking the exclusive role of the group logistics company. 3. The company that pays for the transportation service distributes the freight cost. The paying company also settles the freight cost with the other internal companies that share the transportation service.
Purchasing organization	Option to specify the purchasing organization in the freight order or freight booking in TM as the purchasing organization.	Specify the purchasing organization in the freight order or freight booking in TM as a forwarding house. This enables you to use an internal settlement process for the freight order or freight booking.
Charge calculation	Calculate charges in the freight order or freight booking. Use the rates in the rate table, in line with the freight agreement between the purchasing organization in the freight order or freight booking and the LSP.	In addition to the charge calculation in the freight order or freight booking, calculate internal charges in the OTR or DTR. The internal charges in the OTR or DTR give you, as the group logistics company, the amounts you expect to receive from the internal shipper companies.
Internal settlement document	Not required	You, as a group logistics company, use an internal settlement document to recover the cost of providing the transportation service from the other internal shipper companies. You can create and post an internal settlement document when the life cycle status of the freight order is In Execution .
Cost distribution in the freight order	Not required	Distribute costs in the freight order if you want to use the distributed cost as a basis for internal settlement.
Cost distribution in the freight settlement document	Distribute costs in the freight settlement document to enable you to post freight costs to the underlying transportation requirement, for example, a sales order item, a delivery item, or a purchase order item.	Distribute costs in the freight settlement document if you, as a group logistics company, have the following business requirements: • Transport your own cargo along with the cargo from the other internal shipper companies • Pay the external LSP for all transportation costs • Post the freight cost to an underlying transportation requirement, for example, a

Item	Standard Shipper Process	Group Logistics Company Process
		sales order item, a delivery item, or a purchase order item
Cost distribution in the internal settlement document	Not required	Distribute costs in the internal settlement document if an internal shipper company needs to post the freight cost to an underlying transportation requirement, for example, a sales order item, a delivery item, or a purchase order item.

The table explains the differences in the setup and posting of freight settlement and internal settlement documents between the standard shipper process and the process for a group logistics company:

Item	Standard Shipper Process	Group Logistics Company
Organizations	Standard organizational setup. Purchasing organization is set up with organizational unit function specified as Purchasing .	In TM, if you want to base the internal settlement process on the freight order or freight booking, the purchasing organization in the freight order or freight booking is set up with organizational unit function specified as a forwarding house.
Posting of freight settlement document to Materials Management (MM)	TM posts the freight settlement document, with distributed costs, to MM. Purchase order and service entry sheet (SES): - Creates a service purchase order and includes the items that correspond to the total amount in document currency for the freight settlement document. - Creates and posts an SES to confirm the services in the service purchase order Settlement management document: - Creates a settlement management document with a line item for each line item in the freight order or freight booking - The account assignment details for the line item from the relevant document in the order/delivery item are transferred to the settlement management document items. For example, the assignment details from the sales order item, the purchase order item, or the stock transfer order item are copied to the settlement management items.	TM posts the freight settlement document, with distributed costs, to MM. Purchase order and service entry sheet (SES): - Creates a service purchase order and includes the items that correspond to the total amount in document currency for the freight settlement document - Creates and posts an SES to confirm the services in the service purchase order Settlement management document: - Creates a settlement management document if you, as the group logistics company, transport cargo of your own sales order items, purchase order items, or delivery line items in the freight order or freight booking. You also include cargo from the other internal shipper companies in your orders. - Creates settlement management document items. The freight order items that are linked to own sales order items, purchase order items, or delivery line items for which the group logistics company pays the freight cost.
Posting of internal settlement document to Sales and Distribution (SD)	Not required	Creates an intercompany billing document in Sales and Distribution (SD) billing. This creates the accounts receivable for the group logistics company. The accounts receivable indicates the amounts the group logistics company needs to recover from the other internal shipper companies. SD Billing creates an IDoc as follows:

Item	Standard Shipper Process	Group Logistics Company
		- The SD billing document uses the output type configuration of the intercompany billing document to create an IDoc. - The IDoc posts an incoming invoice for the each of companies involved in the transportation. This records the accounts payable to you, as the group logistics company. For example, the IDoc posts an incoming invoice for Company_2000 and Company_3000, recording the accounts payable to Company_1000. Creates a settlement management document with a line item for each line item in the freight order or freight booking. It copies the account assignment details from the relevant order/delivery item, for example, from the sales order item, purchase order item, or stock transfer order item. The settlement management document establishes links to the sales order item, purchase order item, or delivery line item of the orders that belong to the internal shipper companies.

More Information

Subject	See
Overview of the business process for a group logistics company	Group Logistics Scenario
Detailed process flow for a group logistics company, including cost distribution	Group Logistics Company Process
Prerequisite settings you must specify in TM	TM Prerequisites for a Group Logistics Company Process
Prerequisite settings you must specify for setting up the posting of documents in Sales and Distribution (SD)	Sales and Distribution (SD) Prerequisites for a Group Logistics Company Process

Organizational Prerequisites in TM and Sales and Distribution (SD) for Group Logistics Process

Organizational Settings

You have specified your organizational structure in Transportation Management (TM) in line with the organizational structure in Materials Management (MM) and Sales and Distribution (SD).

For example, the following table contains the organizational structure you can build in SD and MM, along with the corresponding structure in TM:

Enterprise Structure Entity	Enterprise Structure Entity ID	TM Organization Management Entity	TM Organization Management Entity ID
Company	1000	Company organization	CO_1000
Sales organization	S001	Sales organization	SO_S001
Customer	CO_1000	Business partner	CO_1000
Company	2000	Company organization	CO_2000
Sales organization	S002	Sales organization	SO_S002
Customer	CO_2000	Business partner	CO_2000
Company	3000	Company organization	CO_0003
Sales organization	S301	Forwarding house	FH_FH301
Purchasing organization	S301		
Vendor	CO_3000	Business partner	CO_3000

You can specify the organizational master data settings in Customizing for [Enterprise Structure](#) under [Definition](#). You can link the master data objects under [Assignment](#).

You can use the [Create and Merge Organizational Hierarchy](#) Customizing activity to upload your organizational hierarchy. This helps you to copy your organizational settings. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) [Organizational Management](#) [Organizational Model](#).

Alternatively, you can specify the organizational settings in Customizing for [Transportation Management](#) under [Master Data](#) [Organizational Management](#) [Organizational Model](#) [Create Organizational Model](#).

You must specify the following settings under your corporate organizational model:

- All company organizations (company codes) in your corporate organization
- An organization for your group logistics company, with an organizational unit function of [Forwarding House](#).

You specify the group logistics company as a forwarding house because it has both a purchasing role and a selling role. It purchases transportation services from an outside carrier and sells transportation services to the other internal shipper companies in your corporation.

- Enable the group logistics company organization as a group logistics company under [Org. Data](#) [Detail Settings](#) [Group Logistics Company](#).

Organizational Settings for Internal Settlement

To create internal settlement documents in TM and to post the internal settlement documents to SD, you must specify the following settings:

1. Forwarding house in organization management

Define the forwarding house so that the business systems group organizational unit ID of the organization's data in organizational management ID is the same as that used in the ERP enterprise structure for the sales organization and the purchasing organization. The ERP enterprise structure has an identical sales organization ID and purchasing organization ID. For example, you have 0001 for the sales organization and 0001 for the purchasing organization.

2. Business partner with role of Customers

You must create a business partner (BP) for each of the company organizations (company codes) that participate as internal shipper companies in the group logistics process.

3. Assign business partners to company organizations in TM

You must assign the business partner to a company organization (company code) in TM. For more information, see Customizing for [Transportation Management](#) under ► [Master Data](#) ► [Organizational Management](#) ► [Organizational Model](#) ► [Change Organizational Model \(transaction PPOME\)](#) ► Enter the business partner under ► [Org. Data](#) ► [Business Settings](#) ► [Business Partner](#) ► of the relevant company organization.

More Information

Subject	See
How TM assigns a partner function code of Supplying Company or Receiving Company to identify the default agreement party for a stage	Partner Function Codes in an OTR or DTR
Prerequisite settings in Sales and Distribution (SD)	Sales and Distribution (SD) Prerequisites for a Group Logistics Company Process
Explanation of where a group logistics process fits in a business process	Group Logistics Scenario
Detailed explanation of the group logistics process	Group Logistics Company Process

TM Prerequisites for a Group Logistics Company Process

Document Type Settings

- Integration of orders and deliveries

You have specified an order-based transportation requirement (OTR) type and a delivery-based transportation (DTR) type to transfer the transportation requirements.

For example, in this way TM uses the sales orders and deliveries of the internal shipper companies to create OTR and DTR documents for the internal shipper companies in TM.

You must enable the following settings in the OTR or DTR type:

- Internal charge calculation
- Internal settlement
- Internal settlement document type as the default internal settlement document type

You have the option of specifying the following settings in the OTR or DTR type:

- Stage profile, to determine the stage type in a freight unit

For example, you can use the group logistics process in an internal settlement scenario for a stock transfer order. You can settle the different parts of the freight costs to the internal shipper company that acts as a supplying company and to the internal shipper company that acts as a receiving company.

To enable this, you can use an Incoterm in combination with the stage type to determine the prepaid and collect parties. For example, based on an Incoterm of Free on Board (FOB), you can specify that the supplying company pays for the pre-carriage and the receiving company pays for the main carriage and on-carriage.

- Automatic freight unit building

- Freight order type

You have enabled the freight order or freight booking for cost distribution. You use this freight order type when your group logistics company has the following business requirements:

- You want to use the distributed cost as the basis for creating internal settlement documents.
- You want a breakdown of your own transportation costs and the transportation costs of the internal shipper companies. For example, Company_1000 distributes costs in the freight order and uses the distributed costs to create internal settlement documents for Company_2000 and Company_3000.

You can use the distributed costs in a freight order or freight booking to calculate charges for internal settlement. The system uses the distributed costs in the freight order or freight booking as the basis for pulling the costs into the calculation process. For more information, see [Freight Order Management](#).

- Freight settlement document type

You have enabled a freight settlement document type for cost distribution in Customizing for [Transportation Management](#) under [►Settlement►Freight Settlement►Define Freight Settlement Document Types►](#). You use this freight settlement document type when your group logistics company has the following business requirements:

- Transports its own cargo along with the cargo from the other internal shipper companies
- Pays the external logistics service provider (LSP) for all transportation costs
- Posts the freight cost to the underlying transportation requirement, for example, to the sales order item, delivery item, or purchase order item

- Internal settlement document type

- You have specified a forwarding settlement document type for internal settlement documents in Customizing for [Transportation Management](#) under [►Settlement►Forwarding Settlement►Define Forwarding Settlement Document Types►](#).
- You have the option of enabling the internal settlement document type for cost distribution if an internal shipper company needs to post the freight cost to the underlying transportation requirement, for example, to the sales order item, delivery item, or purchase order item.

You use an intercompany settlement in the group logistics process. By default, TM creates an intercompany settlement document for organizations that belong to different company codes.

- Cost distribution settings

- Cost distribution methods

You have specified the relevant cost distribution methods under [►Basic Functions►Cost Distribution►Define Cost Distribution Methods►](#)

- Cost distribution profile

You have specified the relevant settings in a cost distribution profile under [►Basic Functions►Cost Distribution►Define Cost Distribution Profiles►](#)

- Cost distribution profile assignment

You have assigned the cost distribution profile to a charge profile. You have assigned the charge profile to the forwarding house of the group logistics company.

For more information on cost distribution settings in a group logistics process, see [Group Logistics Company Process](#).

Profile Settings

- Forwarding settlement profile and calculation profile on the sales side

You have created a forwarding settlement profile and calculation profile for sales. You have assigned the forwarding settlement profile and the calculation profile to the relevant sales organization. For more information, see [Forwarding Settlement](#).

- Freight settlement profile and calculation profile on the purchasing side

You have created a freight settlement profile and calculation profile for purchasing. You have assigned the freight settlement profile and calculation profile to the relevant forwarding house or purchasing organization. For more information, see [Freight Settlement](#).

Default Agreement Partner Function for a Stage

You have the option of specifying an agreement partner function for a stage type and Incoterm combination. TM uses the agreement partner function to identify the default agreement party for a stage. For more information, see [Partner Function Codes in an OTR or DTR](#).

Calculation Level for a Purchasing Organization

You can plan more than one freight order to deliver the transportation requirements in an OTR or DTR. You can use freight orders that contain different purchasing organizations. However, you must specify the same calculation level for each of the purchasing organizations that you use in each of the freight orders.

More Information

Subject	See
Overview of the business process for a group logistics company	Group Logistics Scenario
Detailed process flow for a group logistics company, including cost distribution	Group Logistics Company Process
Comparison between the standard settlement process for shippers with the process for a group logistics company	Comparison: Standard Shipper Process and Group Logistics Company Process
How TM assigns a partner function code of Supplying Company or Receiving Company to identify the default agreement party for a stage	Partner Function Codes in an OTR or DTR
Organizational settings you must specify for a group logistics process	Organizational Prerequisites in TM and Sales and Distribution (SD) for Group Logistics Process
Prerequisite settings you must specify in Sales and Distribution	Sales and Distribution (SD) Prerequisites for a Group Logistics Company Process

Sales and Distribution (SD) Prerequisites for a Group Logistics Company Process

Integration

- Internal settlement integration

On the external S/4 HANA or SAP ERP, you have enabled integration with SD with customizing activities in [Billing](#) under [Integration with Other SAP Components](#) [Transportation Management](#) [Invoice Integration](#).

- Freight settlement document integration

You have specified the prerequisite settings for posting freight settlement. For more information, see [Freight Settlement](#).

You must specify the relevant settings in Customizing if your group logistics company meets the following business criteria:

- Has its own cargo in a freight order
- Distributes costs at order/delivery item level in the freight order
- Creates a freight settlement document for that freight order

In Customizing in the external S/4 HANA or SAP ERP, you have also specified the following settings:

- Assign Purchasing Information for Posting
Assign Service Master Record and Account Assignment Category
Tax determination setting under Tax
Mapping for Cost Distribution, when cost distribution is enabled in FSD in TM

For more information about the standard process of distributing costs for shippers, see [Cost Distribution for Shippers](#).

Mapping Settings

- Map charge type to condition type

You have specified the charge type that you created in the steps above in a condition type under [Integration with Other SAP Components](#) [Transportation Management](#) [Invoice Integration](#) [Billing](#) [Assignment of Transportation Charge Types](#) [Assign Condition Types](#).

- Map transportation charge to cost distribution

You have specified the relevant settings under [Integration with Other SAP Components](#) [Transportation Management](#) [Invoice Integration](#) [Mapping for Cost Distribution](#) [Assign Transportation Charge Types to Condition Types](#).

You must map these settings if you enable your internal settlement documents for cost distribution in TM.

Settlement Management Document Settings

You have specified the relevant settings in Customizing under [Integration with Other SAP Components](#) [Transportation Management](#) [Invoice Integration](#) [Billing](#) [Mapping for Cost Distribution](#).

Other Settings

For more information about how to enable sales and distribution to post automatically to a vendor account in financial accounting for a group logistics process, see SAP Note [2157070](#).

More Information

Subject	See
Overview of the business process for a group logistics company	Group Logistics Scenario
Detailed process flow for a group logistics company, including cost distribution	Group Logistics Company Process
Comparison between the standard settlement process for shippers with the process for a group logistics company	Comparison: Standard Shipper Process and Group Logistics Company Process
Organizational settings you must specify for a group logistics process	Organizational Prerequisites in TM and Sales and Distribution (SD) for Group Logistics Process
Prerequisite settings you must specify in TM	TM Prerequisites for a Group Logistics Company Process

Automatic Change Management Process for Settlements

Use

As a logistics service provider (LSP), you may need to make changes in your forwarding order process after you have billed your external customer in a forwarding settlement scenario, or your internal customer in an internal settlement scenario. In Transportation Management (TM), you may need to change a forwarding order after you have created a forwarding settlement document or an internal settlement document, and after you have posted the settlement document to Sales and Distribution (SD) to create the billing document.

Alternatively, you are a requester of logistics services, such as a shipper. You have a freight order process in place with an LSP or carrier, to provide transportation services. You may need to make changes in the freight order process after you have created a freight settlement document. In TM, you may need to change a freight order after you have created a freight settlement document and after you have posted the freight settlement document to Materials Management (MM).

Changes can come from the following sources:

- Missing cargo in the transit warehousing process

For example, you are an LSP and in your business process you have integrated TM with Extended Warehouse Management (EWM). You use transit warehousing, where you receive, consolidate, or ship cargo as part of a longer transportation chain. You receive cargo and store it until it is required for transportation to the next stop in the transportation chain.

As part of the transit warehousing process, EWM updates the freight order or freight booking in TM. If there is missing cargo, TM creates a discrepancy. For more information, see [Handling Discrepancies in Transit Warehousing Process](#).

- Direct change in your freight order from your carrier

For example, you are an LSP and you have a forwarding order in process with a requester of transportation services, such as a shipper. You have a freight order in process with a carrier to provide the transportation services. Your carrier reports directly to your office that an item of cargo was missing when the carrier picked up cargo at a source location. You update the freight order that you have in process with the carrier. This in turn updates the forwarding order that you have in process with the shipper.

- Direct change to the charges in your freight order

For example, you are a shipper and you have a freight order in place with a provider of transportation services, such as an LSP or carrier. You realize that your service provider had some additional unplanned charges, for example, demurrage charges because of a delay at a delivery location. You update the freight order with the unplanned charge for demurrage.

Alternatively, you are a shipper and you have a freight agreement with your LSP or carrier and you realize that your rates are revised. You update the rates and then update the freight orders to reflect the revision of the rates by recalculating the charges in the freight order.

The following table describes the role you play in your business, and the documents you work with in TM:

Your Role	Role of Business Partner	Business Document
Provider of logistics services, such as an LSP	Requester of transportation services outside your company, such as a shipper	Forwarding order on an external customer
Provider of logistics services such as an LSP	Requester of transportation services inside your company, such as another business organization	Forwarding order on an internal customer
Requester of transportation services, such as a shipper	Provider of logistics services, such as an LSP or carrier	Freight order on an LSP or carrier
Subcontractor, requesting transportation services from another transportation provider, such as an LSP	Provider of logistics services, such as a carrier	Freight order on a carrier

You can streamline and automate the process of making changes. In a streamlined process, you only need to make a change in one business document, for example, the freight document such as the freight order, freight booking, or service order, and forwarding order. You run the background processing functions [Creation and Transfer of Freight Settlement Documents](#) or [Creation and Transfer of Forwarding Settlement Documents](#). In the background, the system makes the required change in the relevant downstream business documents. The following table describes the downstream documents that the system can update automatically:

Document You Change	Downstream Document in TM	Downstream Document
Forwarding order	- Forwarding settlement document - Internal settlement document - Credit memo	Sales and distribution (SD) billing document
Freight order, freight booking, service order	- Freight settlement document - Charge correction advice document - Credit memo	- Purchase order - Service entry sheet (SES)

Using an automated process also means you have control points at key stages in the change process. These controls ensure you have an efficient change process.

More Information

Subject	More Information
Change management process for forwarding settlement documents	Automatic Change Management for Forwarding Settlements
Change management process for freight settlement documents	Automatic Change Management for Freight Settlements

Subject	More Information
Background processing function to automatically change a forwarding settlement document and an invoice posting	Creation and Transfer of Forwarding Settlement Documents
Standard credit memo process for forwarding orders	Credit Memos for Forwarding Orders
Changing a forwarding order	Editing of a Forwarding Order
How EWM handles discrepancies	Handling Discrepancies in Transit Warehousing Process

Automatic Change Management for Forwarding Settlements

Use

Forwarding and Internal Settlement Change Management Process

The following table describes the steps you, as a logistics service provider (LSP), can take to implement a change in the forwarding order process:

Process Stage	Action
Before you create a forwarding settlement document or an internal settlement document	Make the change directly in the forwarding order.
After you create a forwarding settlement document or an internal settlement document, but before you post the SD billing document.	Cancel the forwarding settlement document or internal settlement document. Change the forwarding order, and create a new settlement document. This enables you to keep a one-to-one relationship between the settlement document in TM and your subsequent billing document in Sales and Distribution (SD). Alternatively, if the forwarding order has an invoicing status of Partially Invoiced, you can leave the existing settlement document in place and create a new settlement document for the delta amount.
After you post a settlement document to create an SD billing document.	Cancel an existing settlement document by creating a credit memo in TM. Post the credit memo to cancel the existing SD billing document. When you create a new settlement document in TM, the new settlement document contains the updated amounts including the changes. You post the settlement document to SD to create an SD billing document for the new amount. This enables you to keep a one-to-one relationship between the forwarding order in TM and your billing document in SD. Alternatively, if the forwarding order has an invoicing status of Partially Invoiced, you can create a new settlement document for the delta amount, and send this to SD to create a new SD billing document for the delta amount.

Change Management Process Settings

You specify the change management strategy for your forwarding orders in the relevant settlement profile under Customizing for **Transportation Management** under **Define Settlement Profile > Settlement Profile Settings > Change Process FWSD**. You can specify a strategy of **Standard Settlement Process, Reverse and Repost**, or **Delta Only**. The system determines the process steps according to the strategy that you choose.

You can control the change process strategy for settled charge lines when there's an increase in the charge lines from the Customizing activity for **Transportation Management** under **Basic Functions > General Settings > General Settings for TM**. For more information, see [Controlled Change Process for Freight and Forwarding Settlements](#).

You can specify the approach the system takes when you change a forwarding order for which a settlement document already exists:

- Reverse and repost a settlement document

When you change a forwarding order, in the background Transportation Management (TM) creates a credit memo that automatically cancels the settlement document in TM and the associated billing document in SD. The system creates a new forwarding settlement document or internal settlement document, and a new billing document for the entire forwarding order.

You must choose this option if you want to ensure that the replacement forwarding settlement document or internal settlement document and replacement billing document are the only documents for the forwarding order that contain the full total amount charge.

For example, in a forwarding settlement scenario, you have the following business documents and charges:

Business Document	Amount (USD)
Forwarding order at first, ID 1234	1000
Forwarding settlement document, ID 810909	1000
Sales and Distribution (SD) billing document, ID 5600121	1000

You then change the charge amount in the forwarding order from USD 1000 to USD 1200. In the reverse and repost strategy, the system creates the following business documents:

- In TM, a credit memo 9560005 for USD 1000 that cancels the forwarding settlement document 810909
- On posting the credit memo 9560005 to SD, an SD credit memo 9951007 for USD 1000

On the overall accounting side, this means that the customer doesn't have to pay any charge, as the amount for the credit memo 9560005 (USD 1000) cancels the amount of the forwarding settlement 810909 (USD 1000).

The system then creates the following documents:

- In TM, the forwarding settlement document 810945 for USD 1200
- On posting the forwarding settlement document to SD, the SD billing document 5600181 for USD 1200

On the overall accounting level, following the original forwarding settlement document, the credit memo, and the second forwarding settlement document, the customer eventually must pay USD 1200.

If you change a forwarding order that is included in a collective forwarding settlement document or internal settlement document, the credit memo that the system creates only cancels the items belonging to the forwarding order from the collective settlement document

For example, you have a collective forwarding settlement document with a lifecycle status of **Invoiced** that includes the following forwarding orders:

Settlement Line Item	Forwarding Order	Amount (USD)
10	1	1000
20	2	2000
30	3	3000

You change to forwarding order 1 from USD 1000 to USD 1200.

The system creates the following documents:

- In TM, a credit memo for USD 1000 for forwarding order 1 only. This is item 10 of the existing settlement document.
- On posting the credit memo to SD, an SD credit memo for USD 1000 for forwarding order 1

On the overall accounting level, the customer doesn't have to pay any charge for this forwarding order at this point in time. The credit memo amount of USD 1000 cancels the forwarding settlement amount of USD 1000 for forwarding order 1.

The system then creates the following documents:

- In TM, a forwarding settlement document for USD 1200 for forwarding order 1
- On posting the forwarding settlement document to SD, an SD billing document for USD 1200

On the overall accounting level, from the original forwarding settlement document, to the credit memo, and on to the second forwarding settlement document, the customer eventually pays USD 1200 for forwarding order 1.

- Delta amount only

When you change a forwarding order, the system doesn't cancel the existing settlement document and the associated billing document. Instead, it creates a new settlement document and a new billing document for the amount that isn't already covered by the existing forwarding settlement document or internal settlement document and billing document.

For example, in a forwarding settlement scenario, you have the following documents and charges:

Business Document	Amount (USD)
Forwarding order 1234 at first	1000
Forwarding settlement document 810909	1000
SD billing document 5600121	1000

You then change the charge amount in the forwarding order from USD 1000 to USD 1200. In the delta only strategy, the system creates the following business documents:

- Forwarding settlement document 810945 for USD 200
- SD billing document 5600181 for USD 200

At the overall accounting level, the customer must pay the amounts in both forwarding settlement documents, a total of USD 1200.

The following table describes the action that the system takes using the different strategies when you change a forwarding order and the forwarding order has an existing settlement document:

Settlement Document Type	Lifecycle Status	Reverse and Repost Invoicing Status of Forwarding Order: Partially invoiced, Over Invoiced, or Not Up-to-Date	Delta Only Invoicing Status of Forwarding Order: Over Invoiced or Not Up-to-Date	Delta Only Invoicing Status of Forwarding Order: Partially invoiced	Follow up Action
Forwarding settlement document or internal settlement document	In Process/Ready for Posting	Cancel the existing settlement document or settlement document item	Cancel the existing settlement document or settlement document item	None	Create a new settlement document
Forwarding settlement document or internal settlement document	Invoice Posted	Create a credit memo to reverse the settlement document or settlement document item	Create a credit memo to reverse the settlement document or settlement document item	None	Create a new settlement document
Credit memo	In Process/Ready for Posting	Cancel the existing credit memo	Cancel the existing credit memo	None	Create a new credit memo for the existing settlement document, reversing the settlement document or the settlement document item. Create a new settlement document.
Credit memo	In Process/Ready for Posting	Cancel the existing credit memo	Cancel the existing credit memo	None	Create a new credit memo for the existing settlement document, reversing the settlement document or the settlement document item. Create a new settlement document.
Credit memo	Invoiced Posted	Create a credit memo for the credit remaining amount in the forwarding order.	Create a credit memo for the credit remaining amount in the forwarding order.	None	Create a new settlement document
Forwarding settlement document, internal settlement document, or credit memo	Posting Started	Skip the change process for these settlement documents and report as errors	Skip the change process for these settlement documents and report as errors	Skip the change process for these settlement documents and report as errors	Reexecute the change process once the documents move out of the lifecycle status

i Note

If you specify a change strategy of **Delta Only** and the invoicing status of the forwarding order is **Over Invoiced** or **Not Up-to-Date**, the system processes the change as if you had a strategy of **Reverse and Repost**.

The system sets the invoicing status for the order to **Not Up-to-Date** when the charge in the order has a significant change in either a new agreement or calculation sheet being determined for the order, or due to the change of a charge that has already been settled in the order.

When you've enabled the setting **Settlement Change Process: Reverse and Repost** in Customizing for **Transportation Management** under **Basic Functions > General Settings > General Settings for TM** and the settled charge line increases, the system sets the invoicing status of the forwarding order as **Not Up-to-Date**. If the setting is disabled, the invoicing status is set to **Partially Invoiced**.

The invoicing status of the order is set to **Over Invoiced** if there's a net negative value change in the order charge to what was already settled, due to the addition of new charges that are negative in nature. In these cases, the system reverses the settlement documents that you posted earlier to update the follow-on documents.

In a collective settlement document, the system applies the change process only for the items that have changed in a forwarding order.

i Note

You can't post a forwarding settlement document with a net negative value. The credit memo, which is created automatically for the settlement documents to be reversed during the change process, is always created for the full value of the settlement document for the order that has changed. For a collective settlement, the credit memo is created for the full amount for the settlement document item corresponding to the order that has changed.

The new settlement documents can be collective settlement documents in the following situations:

- You enable an **Invoice Creation Method** of **Collective** when you run the background processing function **Creation and Transfer of Forwarding Settlement Documents** (program /SCMTMS/CFIR_CREATE_BATCH).
- You select the **Collective Settlement** checkbox in the relevant settlement profile.

i Note

In execution-based settlement scenarios, such as a buyer's consolidation, a route-based settlement, or a trailer-unit based settlement, the forwarding order doesn't contain the charge calculation details as the charges are only calculated in the forwarding settlement documents. Therefore, for these scenarios, you can't use change tracking and change management in the forwarding settlement process. For more information, see [Execution-Based Settlement](#).

Forwarding Settlement Document and Credit Memo Settings

You've enabled all the relevant settings for working with forwarding settlement documents and credit memos. For more information about credit memos, see [Credit Memos for Forwarding Orders](#).

The system determines the settlement document type as follows:

1. From **BAdI: Forwarding Settlement Document Type Determination**

For more information, see Customizing for **Transportation Management** under **Business Add-Ins (BAdIs) for Transportation Management > Settlement > Forwarding Settlement**.

2. From the forwarding settlement document type

For more information, see Customizing for [Transportation Management](#) under [Settlement](#) > [Forwarding Settlement](#) > [Define Forwarding Settlement Document Types](#).

3. From the default credit memo type

For more information, see Customizing for [Transportation Management](#) under [Settlement](#) > [Forwarding Settlement](#) > [Define Credit Memo Reason Codes and Types for Forwarding SDs](#).

If the system uses the credit memo type for a change process, you must specify a credit memo type in which you've enabled the following settings:

- [Influence Invoicing Status](#)
- [Automatically Enter Credit Amount](#)

More Information

Subject	More Information
Change management process for settlement documents	Automatic Change Management Process for Settlements
Example of an Automated Change Management Process in a Forwarding Settlement Scenario	Example of Automated Change Management in an FWSD Scenario
Background processing function to automatically change a forwarding settlement document and an invoice posting	Creation and Transfer of Forwarding Settlement Documents
Controlled Change Process for Freight and Forwarding Settlements	Controlled Change Process for Freight and Forwarding Settlements

Example of Automated Change Management in an FWSD Scenario

Use

1. You are a logistics service provider (LSP) and you work with a carrier to deliver your transportation requirements. Your carrier reports that they picked up the cargo at a customer location. When this is done, you create the forwarding settlement document to bill the customer.

However, you subsequently find out that the quantity of the cargo you get from your customer for transportation does not match the quantity requested by your customer originally. You subsequently update the forwarding order with the change in quantity. This results in a change to the charges in the forwarding order.

2. You have enabled the required Customizing settings for an automated change management process. For more information, see [Automatic Change Management for Forwarding Settlements](#).
3. You have enabled the background processing function [Creation of Forwarding Settlement Documents](#) (/SCMTMS/CFIR_CREATE_BATCH), with a [Creation Type](#) of [Include Changed FWOs](#) or [Only Changed FWOs](#). Alternatively, you go to the impacted forwarding order in the personal worklist or in the individual order and you create the forwarding settlement document.

In your business process, you keep a one-to-one relationship between your forwarding order in TM and your billing document in Sales and Distribution (SD). In the settlement profile that the system applies to the forwarding order and the customer, you have defined the change process strategy for forwarding settlement documents as [Reverse and Repost](#).

In the background, the system performs the following tasks:

- TM creates a credit memo with the amount equal to the amount in the existing forwarding settlement document. The credit memo cancels the existing forwarding settlement document.
- On posting the credit memo to SD, the system creates a credit memo in SD to cancel the existing billing document.
- TM creates a new forwarding settlement document for the changed forwarding order. The forwarding settlement document now has an amount that is equal to the amount in the forwarding order after the change.
- On posting the forwarding settlement document to SD, SD creates a new billing document.

4. Alternatively, in your business process, you don't keep a one-to-one relationship between your forwarding order in TM and your billing document in SD. You define a change process strategy for forwarding settlement documents as **Delta Only** in the relevant settlement profile. The system applies the settlement profile to the forwarding order and the customer.

You make a change to the forwarding order, and the invoicing status of the forwarding order changes to **Partially Invoiced**. In the background, the system performs the following tasks:

- TM creates a new forwarding settlement document only for the extra charge.
- On posting the forwarding settlement document to SD, SD creates a new billing document only for the extra charge.

Alternatively, you make a change to the forwarding order, and the invoicing status changes to **Over Invoiced** or **Not Up-to-Date**. The system automatically uses the **Reverse and Repost** change process strategy. In the background, the system performs the following tasks:

- TM creates a credit memo with the amount equal to the amount in the existing forwarding settlement document. This cancels the existing forwarding settlement document.
- On posting the credit memo to SD, SD uses the credit memo to cancel the existing billing document.
- TM creates a new forwarding settlement document for the changed forwarding order. The amount in the new settlement document is equal to the amount in the forwarding order after the change.
- On posting the forwarding settlement document to SD, SD creates a new billing document.

More Information

Subject	More Information
Change management process for forwarding settlements and internal settlements	Automatic Change Management for Forwarding Settlements

Automatic Change Management for Freight Settlements

Use

You're a requester of transportation services, such as a shipper or a logistics service provider (LSP). You create a freight order, freight booking, or service order in your Transportation Management (TM) system to request transportation services. An LSP can provide transportation services if you're a shipper. Alternatively, a carrier can provide transportation services if you're an LSP.

Sometimes, there can be a difference between what you expect to pay against your service provider expects to be paid. To keep good business relationships, you can work closely with your service provider to resolve these differences in a collaborative way. For more information, see [Dispute Management for Freight Settlements](#). When you resolve a dispute using the dispute management process, the system updates the freight order with the changed charge information.

Alternatively, you may not have a dispute management process in place with your service provider. Instead, your service provider communicates a discrepancy directly to your shipping department. You agree with the change, and you update the freight order directly with the changed information.

You use the background function [Creation of Freight Settlement Documents](#) (program /SCMTMS/SFIR_CREATE_BATCH) to create, change, and transfer the freight settlement documents to Materials Management (MM) (see [Creation and Transfer of Freight Settlement Documents](#)). The freight settlement document contains the changes from the freight order. Alternatively, you go to the impacted freight order from the personal worklist or the individual document, and you create the freight settlement document.

Change Management Process Settings

The system uses the automatic change management process that you specify in the relevant settlement profile in Customizing. For more information, see Customizing for [Transportation Management](#) under [►Settlement ► Define Settlement Profile ► Change Process FSD ►](#).

You can control the change process strategy for settled charge lines when there's an increase in the charge lines from the Customizing activity for [Transportation Management](#) under [►Basic Functions ► General Settings ► General Settings for TM ►](#). For more information, see [Controlled Change Process for Freight and Forwarding Settlements](#).

TM uses the following elements to determine what approach to take with the change:

- [Change Process FSD](#) setting in the settlement profile
- Default credit memo type

You must specify a default credit memo type in Customizing for [Transportation Management](#) under [►Settlement ► Freight Settlement ► Define Credit Memo Reason Codes and Types for Freight SDs ►](#). You must also specify the following settings for the default credit memo type:

- Influence invoicing status
- Automatically enter the credit amount
- Lifecycle status of the freight settlement document

You have a freight order for transporting your cargo that you agree with your carrier, for the freight cost of 2000 USD. Consider the following business scenarios for change:

Additional Detention Charge

Your carrier communicates to you that there are additional charges of USD 300 due to an unplanned delay at the unloading point. You verify this request from the carrier and agree to pay for the unplanned detention charge. In the relevant freight order, you change the transportation charge by adding the detention charge for USD 300. This takes the freight order charge to USD 2300. The freight order has an invoicing status of [Partially Invoiced](#).

Additional Discount due to Better Volume

Your purchasing department communicates that the carrier has agreed to an additional discount due to a better volume of transactions. Because of this, there's a downward revision of the overall freight charges of USD 200. You add the discount charge line in the freight order to make this change. The freight order charges are reduced to USD 1800. In this case, the freight order has an invoicing status of [Partially Invoiced](#).

Decrease in Rate for the Base Freight

Your purchasing department communicates that the carrier has agreed to a reduced rate for freight due to a better volume of transactions. Because of this, there's a downward revision of the overall freight charges for the freight order of USD 100. You change the basic freight charge to reflect this reduction. The freight order changes to USD 1900. In this case the freight order has an invoicing status of [Not Up-to-Date](#).

Increase in Rate for the Base Freight

Your purchasing department communicates that the carrier has increased rate for freight. Because of this, there's an upward revision of the overall freight charges for the freight order of USD 50. You change the basic freight charge to reflect this increment. The freight order changes to USD 2050. In this case, if the setting **Settlement Change Process: Reverse and Repost** in Customizing activity for **Transportation Management** under **Basic Functions > General Settings > General Settings for TM** is enabled the system sets the invoicing status of the freight order as **Not Up-to-Date**. If the setting is disabled, the invoicing status is set to **Partially Invoiced**.

You can select the following change process strategies:

- **Create New FSD for Delta Amount**

Lifecycle Status	System Action	Example
New/In Process/Ready for Posting	If the invoicing status of the freight order is Partially Invoiced , create a new freight settlement document for the remaining amount.	Additional Detention Charge The system creates a new freight settlement document for the remaining amount of USD 300. Additional Discount due to Better Volume The system cancels the freight settlement document for USD 2000 for the freight order. The system creates a new freight settlement document for the revised value of the freight order of USD 1800. Change in Rate for the Base Freight The system cancels the freight settlement document for USD 2000 for the freight order. The system creates a new freight settlement document for the revised value of the freight order of USD 1900.
Accruals Posted/Invoice Verified	◦ If the invoicing status of the freight order is Partially Invoiced, create a new freight settlement document for the remaining amount. ◦ If the invoicing status of the freight order is Over Invoiced, to reverse the amount, the system creates a credit memo for the freight order with reference to the freight settlement document or freight settlement document item in the case of a collective freight settlement document. You then create a new freight settlement document for the full value of the freight order charges.	◦ Example for Partially Invoiced Scenario Additional Detention Charge The system creates a new freight settlement document for the remaining amount of USD 300. On posting the freight settlement document to Material Management (MM), the system creates a new purchase order and a service entry sheet (SES) amount for USD 300. ◦ Example for Over-Invoiced Scenario Additional Discount due to Better Volume

Lifecycle Status	System Action	Example
	For an airfreight booking, if a charge correction advice document (CCAD) already exists for the freight booking, the system creates a CCAD.	The system creates a credit memo for USD 2000 with reference to the freight settlement document. If the settlement for this order is by a collective settlement document, the system refers to the freight settlement document item. This reverses the amount in the existing freight settlement document or freight settlement document item. On posting this credit memo to MM, the system reverses the posting of USD 2000. The system creates a new freight settlement document for USD 1800. On posting the new freight settlement document to MM, the system creates a new posting for USD 1800 Change in Rate for the Base Freight The system creates a credit memo for USD 2000 with reference to the freight settlement document. If the settlement for this order is by a collective settlement document, the system refers to the freight settlement document item. This reverses the amount in the existing freight settlement document or freight settlement document item. On posting this credit memo to MM, the system reverses the posting of USD 2000. TM creates a new freight settlement document for USD 1900. On posting the new freight settlement document to MM, the system creates a new posting for USD 1900. Air Freight Scenarios Where a CCAD Already Exists Example for a Partially Invoiced Scenario Additional Detention Charge

Lifecycle Status	System Action	Example
		The system creates a new CCAD for the remaining amount of USD 300. On posting the CCAD, the system reverses the posting of USD 2000 and creates a new posting for USD 2300. Example for Over-Invoiced Scenario Additional Discount due to Better Volume. The system creates a new CCAD for the remaining amount of USD -200. On posting the CCAD, the system reverses the posting of USD 2000 and creates a new posting for USD 1800. Change in Rate for the Base Freight The system creates a new CCAD for the remaining amount of USD -100. On posting the CCAD, the system reverses the posting of USD 2000 and creates a new posting for USD 1900.
Canceled/Reversed	Create a new freight settlement document with the updated amount.	Additional Detention Charge The system creates a new freight settlement document for the revised value of the freight order of USD 2300. Additional Discount due to Better Volume The system creates a new freight settlement document for the revised value of the freight order of USD 1800. Change in Rate for the Base Freight The system creates a new freight settlement document for the revised value of the freight order of USD 1900.

- **Reverse and Repost with New FSD**

Lifecycle Status	System Action	Example
New/In Process/Ready for Accruals	Cancel the freight settlement document or freight settlement document item and create a new freight settlement document.	Additional Detention Charge The system cancels the freight settlement document for USD 2000 for the freight order. It creates a new freight settlement document for the remaining amount of USD 2,300.

Lifecycle Status	System Action	Example
		Additional Discount due to Better Volume The system cancels the freight settlement document for USD 2000 for the freight order. It creates a new freight settlement document for the revised value of the freight order of USD 1800. Change in Rate for the Base Freight The system cancels the freight settlement document for USD 2000 for the freight order. It creates a new freight settlement document for the revised value of the freight order of USD 1900.
Accruals Posted/Invoice Verified	Regardless of whether the invoicing status of the freight order is Partially Invoiced, Over Invoiced, or Not Up-to-Date to reverse the amount the system creates a credit memo for the freight order with reference to the freight settlement document or freight settlement document item in case of a collective freight settlement document. It then creates a new freight settlement document for the full value of the freight order charges. In the case of an air freight booking, if a CCAD already exists for the freight booking, the system creates a CCAD.	Additional Detention Charge The system creates a credit memo for USD 2000 with reference to the freight settlement document. If the settlement for this order is by a collective settlement document, the system refers to the freight settlement document item. This reverses the amount in the existing freight settlement document or freight settlement document item. On posting this credit memo to MM, the system reverses the posting of USD 2000. TM creates a new freight settlement document for USD 2300. On posting the new freight settlement document to MM, the system creates a new posting for USD 2300. Additional Discount due to Better Volume The system creates a credit memo for USD 2000 with reference to the freight settlement document. If the settlement for this order is by a collective settlement document, the system refers to the freight settlement document item. This reverses the amount in the existing freight settlement document or freight settlement document item. On posting this credit memo to MM, the system reverses the posting of USD 2000. The system creates a new freight settlement document for USD 1800. On posting the new freight settlement document to MM, the system creates a new posting for USD 1800.

Lifecycle Status	System Action	Example
		Change in Rate for the Base Freight The system creates a credit memo for USD 2000 with reference to the freight settlement document. If the settlement for this order is by a collective settlement document, the system refers to the freight settlement document item. This reverses the amount in the existing freight settlement document or freight settlement document item. On posting this credit memo to MM, the system reverses the posting of USD 2000. The system creates a new freight settlement document for USD 1900. On posting the new freight settlement document to MM, the system creates a new posting for USD 1900 Air Freight Scenario when a CCAD Exists Additional Detention Charge The system creates a new CCAD for the remaining amount of USD 300. On posting the CCAD, the system would have reversed the posting of USD 2000 and created a new posting for USD 2300. Additional Discount due to Better Volume The system creates a new CCAD for the remaining amount of USD -200. On posting the CCAD, the system would have reversed the posting of USD 2000 and created a new posting for USD 1800. Change in Rate for the Base Freight The system creates a new CCAD for the remaining amount of USD -100. On posting the CCAD, the system would have reversed the posting of USD 2000 and created a new posting for USD 1900.
Canceled/Reversed	Create a new freight settlement document with the updated amount.	Additional Detention Charge The system creates a new freight settlement document for the revised amount of USD 2,300. Additional Discount due to Better Volume The system creates a new freight settlement document for the revised value of the freight order of USD 1800.

Lifecycle Status	System Action	Example
		Change in Rate for the Base Freight The system creates a new freight settlement document for the revised value of the freight order of USD 1900.

i Note

- You can run the **Creation of Freight Settlement Documents** background function to automatically make changes in freight settlement documents. For more information, see [Creation and Transfer of Freight Settlement Documents](#).
- You can create a freight settlement document either in the personal worklist or in the individual freight order.
- Regardless of the change process strategy, if the invoicing status of a freight order is **Not Up-to-Date**, the system automatically tries to apply the change process strategy of **Reverse and Repost with New FSD**.
- The system doesn't automatically execute the change process and reports an error when you change a freight order after you begin the freight settlement process, and the freight order has an existing credit memo that isn't enabled to influence the invoicing status.

You must make a manual change in the settlement process. You can change or cancel the relevant credit memo and create the freight settlement document again.

- Irrespective of the change process strategy, if a freight settlement document has a lifecycle status of **Posting Started**, **Cancellation Requested** or **Reversal Started** and you run the function **Create and Transfer Freight Settlement Documents**, the system reports an error.

You must wait until posting is complete and the system updates the lifecycle status in TM to either **Accruals Posted** or **In Process** before you take further action. If the system updates the status to **In Process**, errors have occurred during the posting of the settlement documents.

Charge Correction Advice Document

In an air freight scenario, you can create a CCAD to recover an overpaid amount from your carrier or pay an underpaid amount to your carrier.

You can only manually create a CCAD for air freight orders.

A CCAD contains the amount that has already been settled, the new negative amount to be corrected, and the final correct settlement amount for the freight order.

When posting the CCAD to MM, the system takes the following steps:

- Posts an SES return to reverse the amount you previously posted from TM
- Posts an SES with the revised freight order charge

More Information

Subject	More Information
Change management process for forwarding settlements	Automatic Change Management for Forwarding Settlements
Working with a dispute case	Dispute Management for Freight Settlements

Subject	More Information
Charge correction advice management process	Charge Correction Advice Management
Controlled Change Process for Freight and Forwarding Settlements	Controlled Change Process for Freight and Forwarding Settlements

Controlled Change Process for Freight and Forwarding Settlements

When you change any settled charge lines, the system updates the invoicing status of settled freight orders, freight bookings, service orders, or forwarding orders to reflect the change. After the invoicing status is set, the system uses the change process strategy that you've defined in the Customizing activity for **Transportation Management** under **Settlement** > **Define Settlement Profile** to process the change process for posting.

Depending on whether you've enabled or disabled the setting **Settlement Change Process: Reverse and Repost** in Customizing for **Transportation Management** under **Basic Functions** > **General Settings** > **General Settings for TM** the invoicing status of a freight order, freight booking, service order, or forwarding order is set as follows:

Scenario	Invoicing Status with settings disabled	Invoicing Status with settings enabled
Reduction or deletion of a settled charge or deactivating a charge line	Not Up-To-Date	Not Up-To-Date
Increment of a settled charge line	Partially Invoiced	Not Up-To-Date
Addition of new charge line or activating an inactive charge line	Partially Invoiced	Partially Invoiced
Changes in business partner or agreement	Not Up-To-Date	Not Up-To-Date

Related Information

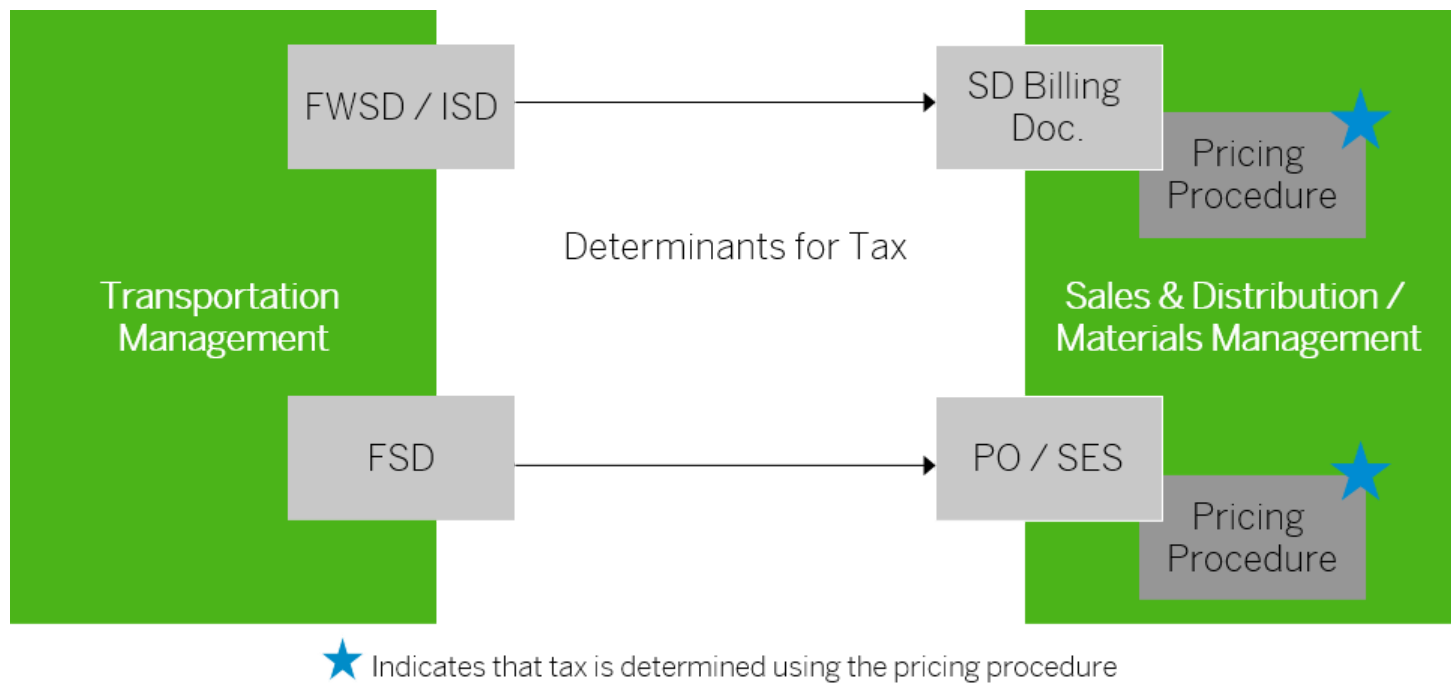
[Automatic Change Management for Freight Settlements](#)

Tax Determination

Use

Transportation Management (TM) enables information to be transferred to Materials Management (MM) and Sales and Distribution (SD), where it can be used to determine tax when settlement documents are posted. Taxes for forwarding settlement documents are calculated using the Sales and Distribution (SD) component, but taxes for freight settlement documents are determined using the Materials Management (MM) component.

You can set up the pricing procedures in the respective modules to establish rules based on your business requirements. TM can change the way in which tax is determined by transferring influencing factors or determinants to the SD and MM modules. For example, you post a settlement document for a freight order or freight booking to MM, or a forwarding order to SD. When you do so, the system also transfers the location country/region details from the order. The location country/region details are used to populate the country/region fields in the pricing communication structures. The communication structures are used when taxes are calculated during the pricing procedure in SD and MM.



Integration

The following tables contain a summary of the fields that are populated in the pricing communication structure, which can be used to determine the tax code.

Forwarding Settlement Integration

Communication Structure	Field in TM	Comment
KOMK - LAND1	Country/Region of the Customer	Country/region of the customer being billed
KOMK - ALAND	Country/Region of the Sales Org	Country/region of the sales organization
KOMK - SLAND1_A	Source Country/Region	FWO – Source country/region, stage source country/region, or item source country/region
KOMK - SLAND1_Z	Destination Country/Region	FWO – Destination country/region, stage destination country/region, or item destination country/region
KOMK - TAXK1 to TAXK9	BP Tax Group	Mapping of the tax group to the classification in SD from the tax data of the business partner
KOMP - TAXM1 to TAXM9	Charge Type Tax Group	Mapping of the tax group to the classification in SD from the charge type level that is set up in TM

Freight Settlement Integration

Communication Structure	Field in TM	Comment
KOMK - LAND1	None	Country/region of plant in MM-PO or country/region of company code when the plant is determined during the creation of the purchase order
KOMK - ALAND	Country/Region of the Carrier	Country/region of the carrier
KOMK - SLAND1_A	Source Country/Region	FO – Source country/region, stage source country/region, or item source country/region

Communication Structure	Field in TM	Comment
KOMK - SLAND1_Z	Destination Country/Region	FO – Destination country/region, stage destination country/region, or item destination country/region
KOMP - TAXIM	Material Tax Indicator	Assignment of the tax indicator to the charge type level

Prerequisites

Transportation Management

For Freight Settlement:

- You've specified whether the tax is calculated at charge type level for countries/regions.
 - For TM embedded in SAP S/4HANA, you specify this in the Customizing activity [Define Tax at Charge Type Level](#) in Customizing for [Transportation Management](#) under [►►Settlement ► Freight Settlement ► Integration for Settlement Posting ► Tax ►](#).
 - For integration with an external SAP S/4 HANA ERP system, you specify this in the Customizing under [►►Integration with other SAP Components ► Transportation Management ► Invoice Integration ► Invoicing ► Tax Setting for TM in SAP S/4HANA ►](#) in the SAP S/4 HANA ERP system.
- You've specified the tax indicators for the charge types for countries/regions.
 - For TM embedded in SAP S/4HANA, you specify this in the Customizing activity [Assign Tax Indicators for Charges](#) in Customizing for [Transportation Management](#) under [►►Settlement ► Freight Settlement ► Integration for Settlement Posting ► Tax ►](#).
 - For integration with an external SAP S/4 HANA ERP system, you specify this in the Customizing under [►►Integration with other SAP Components ► Transportation Management ► Invoice Integration ► Invoicing ► Tax Setting for TM in SAP S/4HANA ►](#) in the SAP S/4 HANA ERP system.

i Note

During freight settlement posting, the system prioritizes the Customizing settings in the TM for SAP S/4HANA system even though you've specified the Customizing settings in the SAP ERP system.

For Forwarding Settlement:

- You've specified whether the tax is calculated at charge type level for countries/regions in the Customizing activity [Define Tax at Charge Type Level](#) under [►►Transportation Management ► Settlement ► Forwarding Settlement ► Tax ►](#).
- You've specified the tax indicators for the charge types for countries/regions in the Customizing activity [Assign Tax Indicators for Charges](#) under [►►Transportation Management ► Settlement ► Forwarding Settlement ► Tax ►](#).

Features

On a country/region-specific basis, you can specify the tax influencers, such as the tax indicator and tax classification for the charge-type level, charge category level, or subcategory level. You can also specify whether tax is to be applied at charge level. These settings are transferred to pricing procedures in SD and MM, and is used when the forwarding settlement document or freight settlement document is posted.

More Information

Split Criteria in Settlement Document Creation

Use

When you create a collective forwarding settlement document or freight settlement document, the system attempts to combine the settlement for the forwarding orders or freight documents you have selected into a single forwarding settlement document or freight settlement document.

However, the split criteria arise when particular conditions are different between the orders the system is trying to combine. As a result, the system creates multiple settlement documents.

The system makes a split in no particular order of priority.

Features

Split Criteria for Forwarding Settlement Documents

The system uses the following split criteria when it attempts to create a collective forwarding settlement document:

- Sales organization
- Bill-to party
- Payer
- Sold-to party
- Payment term
- Credit segment
- Document currency
- Calculation level
- Stage, if you specify the **Stage Split** setting in the **Define Settlement Profile** Customizing activity. For more information, see Customizing for **Transportation Management** under **► Settlement ►**.
- Settlement group, where you group the charge types that you want to settle at the same time in your process. For more information, see [Flexible Creation of Forwarding Settlements](#).

Split Criteria for Freight Settlement Documents

The system uses the following split criteria when it attempts to create a collective freight settlement document:

- Procuring company code
- Purchasing organization
- Carrier or additional agreement party

- Invoicing party
- Payee
- Payment term
- Document currency
- Calculation level
- Stage, if you specify the [Stage Split](#) setting in the [Define Settlement Profile](#) Customizing activity.

Example

Forwarding Settlement

You have two forwarding orders that you want to combine into one forwarding settlement document.

FWO	Sales Organization	Sold-to Party / Bill-to Party / Payer	Payment Terms	Freight Cost (USD)
9100021	0001	Delicious chocolates	0001 – Due immediately	200
9100291	0001	Delicious chocolates	0002 – Net due in 15 days	300

You select the orders and select the combine option to create the collective settlement documents on the worklists screen. However, as the payment term in the order does not match, the system creates two forwarding settlement documents.

FWSD	Forwarding Order	Sales Organization	Sold-to Party / Bill-to Party / Payer	Terms of Payment	Freight Cost (USD)
9700021	9100021	0001	Delicious chocolates	0001 – Due immediately	200
9700022	9100291	0001	Delicious chocolates	0002 – Net due in 15 days	300

Freight Settlement

You have three freight orders that you want to combine into one freight settlement document.

Freight Order	Procuring Company Code	Purchasing Organization	Carrier / Invoicing Party / Payee	Payment Terms	Freight Cost (USD)
8100021	0001	0001	Fast Trucking Company	0001 – Due immediately	200
8100229	0001	0001	Fast Trucking Company	0001 – Due immediately	240
8100291	0002	0002	Fast Trucking Company	0001 – Due immediately	300

You select the orders and select the combine option to create the collective settlement documents on the worklists screen. However, as the company code and purchasing organization in the order do not match, the system creates two freight settlement documents.

Freight Settlement Document	Freight Order	Procuring Company Code	Purchasing Organization	Carrier / Invoicing Party / Payee	Terms of Payment	Freight Cost
900021	8100021 8100229	0001	0001	Fast Trucking Company	0001 – Due immediately	440
900022	8100291	0002	0002	Fast Trucking Company	0001 – Due immediately	300

Related Information

[Freight Settlement Integration using MM External Services Management Forwarding Settlement](#)

Carrier Collaboration

As a shipper, you can collaborate with carriers for various processes in Transportation Management (TM) in SAP S/4HANA, even if the carrier doesn't use TM.

Once you give your carrier access to the relevant TM features, the carrier can, for example, respond to your freight agreement requests for quotation (RFQs) with their proposed rates.

i Note

This documentation is aimed at shippers, and also at carriers or service provider representatives for carrier organizations who work with shippers.

Prerequisites

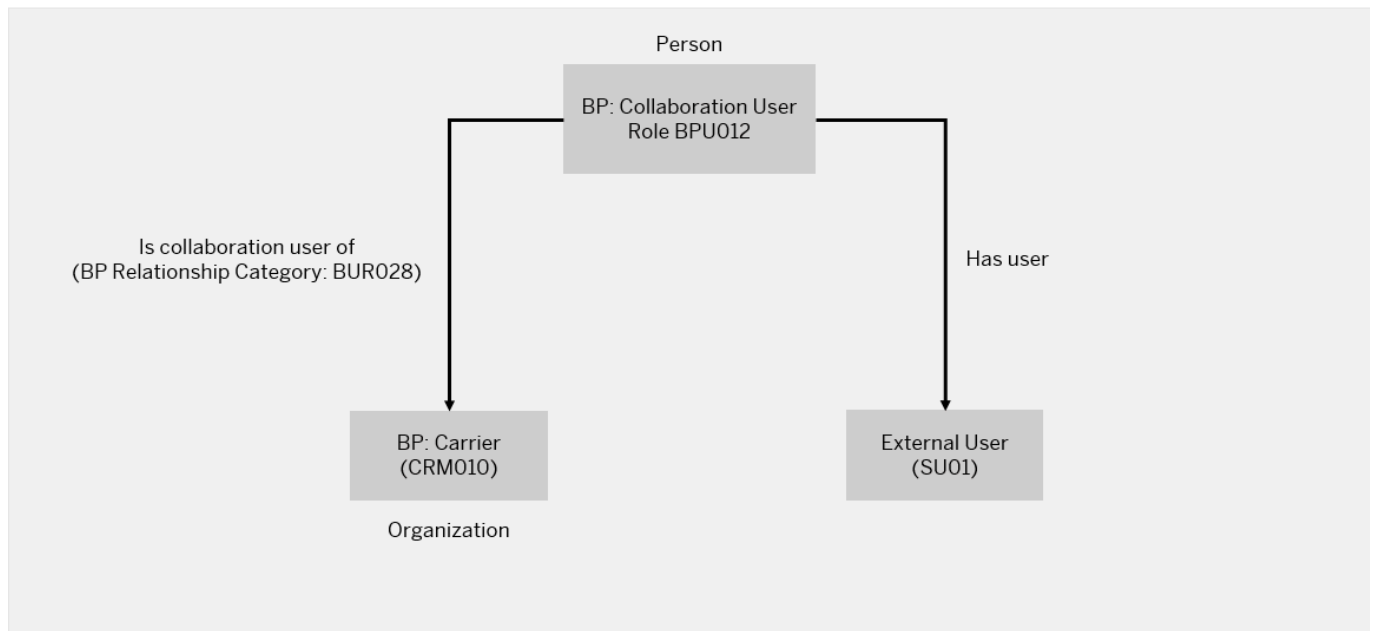
You've performed the following steps for each carrier that you want to collaborate with:

1. In the **Maintain Business Partner** (BP) transaction, you've created a business partner with the business partner role of **Carrier**.
2. In the **Maintain Collaboration Users** app, you've defined a collaboration user and specified that this collaboration user is for the business partner with the role of **Carrier**.

For more information, see [Maintain Collaboration Users \(F4911\)](#).

The following graphic describes the mapping of a carrier organization with an external user.

Mapping of Carrier Organization with External User



You've also performed the following steps for the relevant business document types:

- You've assigned an attachment schema that is configured to use the Attachment Service. For more information, see [Working with Attachments](#).

More Information

[Management of Quotations for Transportation Services](#)

[Strategic Freight Procurement](#)

Agreement Management

As a shipper, you can collaborate with carriers during your strategic freight procurement process in Transportation Management (TM) in SAP S/4HANA, even if the carrier doesn't use TM.

For example, you can request quotations for transportation services and carriers can respond with their proposed rates. You can then negotiate the best rates from your carriers.

More Information

[Management of Quotations for Transportation Services](#)

[Strategic Freight Procurement](#)

Management of Quotations for Transportation Services

As a shipper, you can request quotations for transportation services and negotiate the best rates during your strategic freight procurement process, even if the carrier doesn't use Transportation Management (TM) in SAP S/4HANA.

As the carrier or a service provider representative negotiating freight agreements with shippers, you can use the **Manage Freight Agreement RFQs** app to monitor your freight agreement requests for quotation (RFQs). You can submit your responses to the

shipper with your proposed rates for the transportation services.

As a carrier or a service provider representative, you can use the [Manage Freight Agreement RFQs](#) app to do the following:

- Monitor freight agreement RFQs by status, for example, freight agreement RFQs that are open for response and are waiting for carrier confirmation
- Use a spreadsheet to enter your rate details
- Use attachments and notes
- Respond to freight agreement RFQs
- Display freight agreement RFQs in a worklist sorted by response deadline
- Display freight agreement RFQs in a list that you can filter by different criteria
- Choose not to negotiate rates with the shipper by rejecting your participation in a freight agreement RFQ

i Note

If as a shipper, you decide that you don't want carriers to collaborate with you using the [Manage Freight Agreement RFQs](#) app, you can exchange freight agreement RFQ details using a spreadsheet instead. For more information, see [Responses from Carriers Using Spreadsheets](#).

Prerequisites

- The shipper has configured the prerequisites for collaboration with carriers (see [Carrier Collaboration](#)).
- The shipper has configured the prerequisites for strategic freight procurement (see [Strategic Freight Procurement](#)).
- The shipper has maintained carrier text types to enable carriers to add notes.

Procedure

1. In the [Create Freight Agreement RFQ Master](#) app, the shipper creates a freight agreement RFQ master to detail their transportation requirements (see [Freight Agreement RFQ Master](#)).
2. The shipper publishes the freight agreement RFQ master and the system automatically creates and sends individual freight agreement RFQs to the carriers.
3. The system notifies each carrier about their individual RFQ, basically asking the carrier to bid for the provision of future transportation services in a trade lane.
4. In the [Manage Freight Agreement RFQs](#) app, the carrier evaluates the freight agreement RFQ and responds to the shipper.

If the carrier does not want to bid for the transportation services, the carrier can choose to reject participating in the freight agreement RFQ.

If the carrier wants to participate in the bid, they can respond to the shipper with their rates.

If needed, the carrier can download the freight agreement RFQ in a spreadsheet, enter their rate details, and then upload the freight agreement RFQ spreadsheet back to the [Manage Freight Agreement RFQs](#) app.

i Note

The uploaded spreadsheet is also attached to the freight agreement RFQ as an attachment.

5. The shipper monitors and reviews the carrier responses for the freight agreement RFQ.

6. The shipper accepts or rejects the rates from the carrier.
7. The shipper requests a confirmation of the rates submitted by the carrier.
8. Once the rates are confirmed, the shipper creates or updates the freight agreement for the carrier.

More Information

[Requesting of Quotations for Transportation Services](#)

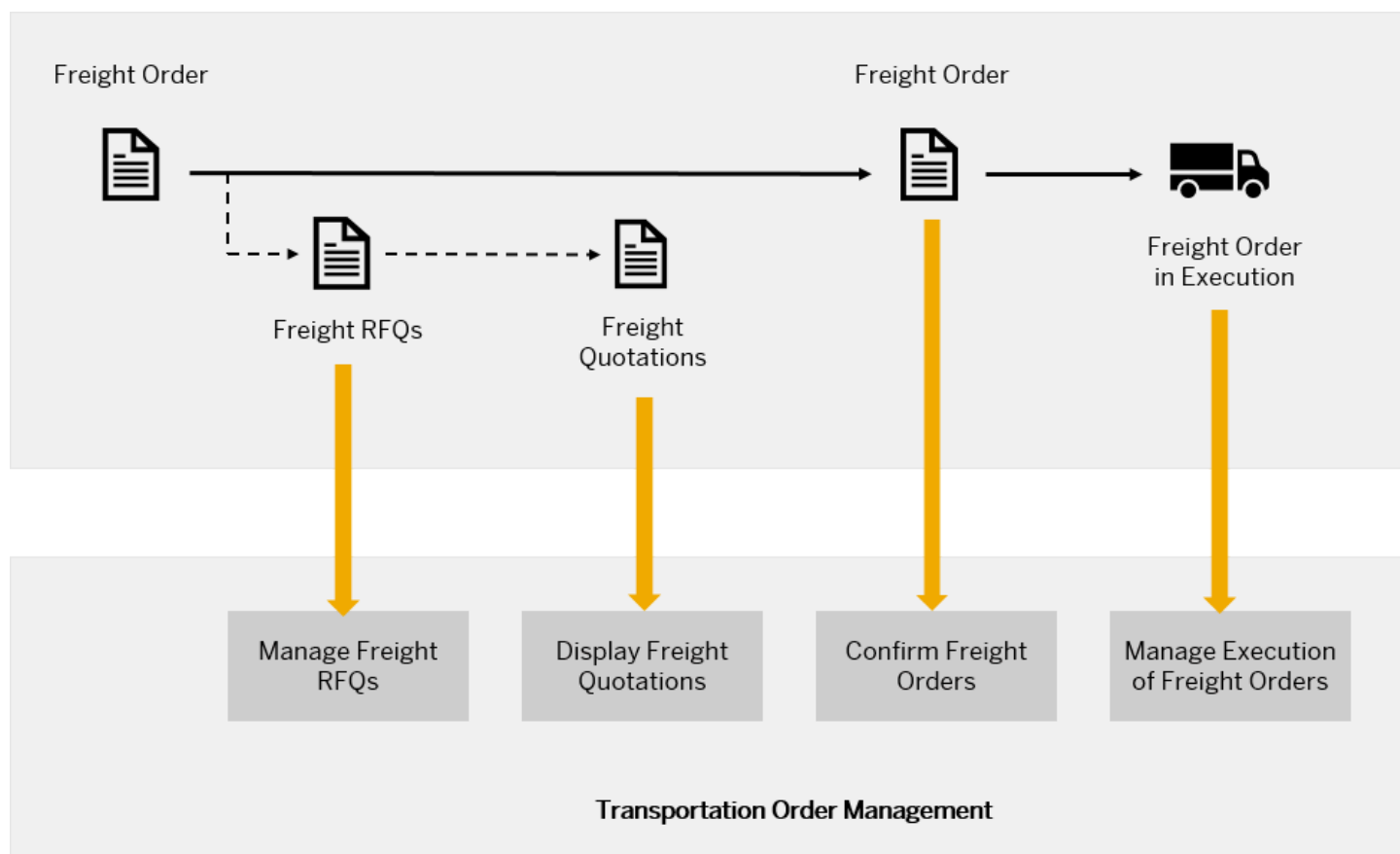
Order Management

As a carrier or a service provider representative, you can use the transportation order management apps to respond to freight tendering requests in the freight subcontracting process.

You have the following features available in transportation order management:

- Freight tendering
- Freight order confirmation
- Freight order execution

The following graphic gives you an overview of transportation order management:



Freight Tendering

As a carrier or a service provider representative, you can display the freight requests for quotation (RFQs) that you have received from a shipper. You can also display the freight quotations that you submitted to a shipper for a specific freight RFQ.

For more information, see [Freight Request for Quotation](#) and [Freight Quotation](#).

Freight Order Confirmation

As a carrier or a service provider representative, you can view, confirm, or reject freight orders that have been sent to you by a shipper. This helps you to plan your workload independently of the tendering process.

For more information, see [Freight Order Confirmation](#).

Freight Order Execution

As a carrier or a service provider representative, you can view and monitor the execution of freight orders that have been sent to you by a shipper. You can report events and report actual quantities of the items for the freight orders that you are executing.

For more information, see [Freight Order Execution](#).

Freight Request for Quotation

As a shipper, when you use a freight request for quotation (RFQ) tendering process to send out one or more freight requests for quotation (RFQs) to carriers, the carriers can use the transportation order management apps to respond to your freight tendering requests. For more information as a shipper, see [Freight Tendering](#) and [Tendering](#).

As a carrier or a service provider representative, you can use the **Manage Freight RFQs** app to manage and respond to the freight RFQs that you have received from the shipper.

You can use the following features in the **Manage Freight RFQs** app:

- On the app tile, view the number of freight RFQs that are due for a response from you and when they are due. The app tile also displays when the information was last retrieved from the system and allows you to refresh the information.
- Use the app to view all freight RFQs, and to list those that are open, closed, and responded to.

Open freight RFQs require a response from you. The freight RFQs that you respond to are sent to the shipper for evaluation. Closed freight RFQs are those that have been evaluated by the shipper and are either rejected or awaiting an award. Closed freight RFQs are also those whose response time has been exceeded or that were canceled.
- Perform a search on the freight RFQs by using the filter criteria.
- Choose a freight RFQ from the list to view details of the freight RFQ such as source and destination details, transportation details, cargo details, information about whether the cargo is considered dangerous goods, transportation stops, notes, and attachments.
- Choose a transportation stop in a freight RFQ to view the items that are to be loaded or unloaded at that stop. You can also view details of the cargo items.
- Perform the following actions in a freight RFQ:
 - Accept a freight RFQ without making any changes.
 - Accept a freight RFQ by proposing changes to the price, currency, or start and end dates.

i Note

This action is available to you if the shipper has configured the tendering settings to allow you to propose changes.

- Reject a freight RFQ and select a reason for rejection.
- A freight quotation is created when you accept a freight RFQ. The freight quotation is listed in the freight RFQ details.
- As a carrier, you can revise the quotation multiple times within the response deadline.

Prerequisites

The shipper has configured the prerequisites for collaboration with carriers (see [Carrier Collaboration](#)).

Freight Quotation

As a carrier or a service provider representative, when you accept freight RFQs, the system creates freight quotations and sends them to the shipper.

You can use the [Display Freight Quotations](#) app to view all existing freight quotations.

You can use the following features in the [Display Freight Quotations](#) app:

- On the app tile, view the number of freight quotations that have been awarded to you or rejected by the shipper. The app tile also displays when this information was last retrieved from the system and allows you to refresh this information.
- Use the app to view all freight quotations, and to list those that are pending, rejected, and awarded.

The pending freight quotations are those that are being evaluated by the shipper. The awarded freight quotations are those that have been evaluated by the shipper and have been awarded to you. The rejected freight quotations are those that have been evaluated by the shipper and have been rejected.

- Perform a search on the freight quotations by using the filter criteria.
- Choose a freight quotation from the list to view details of the freight quotation such as source and destination details, transportation details, cargo details, information about whether the cargo is considered dangerous goods, transportation stops, notes, and attachments.
- Choose a transportation stop in a freight quotation to view the cargo items that are to be loaded or unloaded at that stop. You can also view details of the cargo items.

Prerequisites

The shipper has configured the prerequisites for collaboration with carriers (see [Carrier Collaboration](#)).

Freight Order Confirmation

As a carrier or a service provider representative, you can use the [Confirm Freight Orders](#) app to manage and respond to freight orders sent to you by the shipper.

You can use the following features in the [Confirm Freight Orders](#) app:

- On the app tile, view the number of freight orders that require your confirmation.
- Use the app to view all the freight orders, and to list those that require your confirmation or have been confirmed or closed.

Closed freight orders are the freight orders that you have either canceled or completed.

- Perform a search on the freight orders by using the filter criteria.

- Choose a freight order from the list to view details of the freight order such as source and destination details, transportation details, cargo details, information about whether the cargo is considered dangerous goods, driver details, transportation stops, notes, and attachments.
- Choose a transportation stop in a freight order to view details of the stop and the items to be loaded or unloaded at that stop.
- Update driver and vehicle registration details for a freight order.
- Propose changes to the confirmed start date and time, confirmed end date and time, price limit, or currency by choosing [Confirm with Changes](#).

Prerequisites

The shipper has configured the prerequisites for collaboration with carriers (see [Carrier Collaboration](#)).

Freight Order Execution

As a carrier or a service provider representative, you can use the [Manage Execution of Freight Orders](#) app to monitor the execution of freight orders sent to you by a shipper.

You can use the following features in the [Manage Execution of Freight Orders](#) app:

- On the app tile, view the number of freight orders that you are currently executing.
- View all freight orders, and list those that are in execution or have been closed or blocked.

A closed freight order is a freight order that you have either canceled or completed.

A blocked freight order is a freight order that is currently blocked for execution by the shipper.

- Perform a search on the freight orders by using the filter criteria.
- From the list of freight orders, you report a planned event or an unplanned tour event to the shipper for a freight order in execution.

For a planned event, you select the event listed for a location. The reporting date and time is defaulted to the current date and time. Additionally, you can add a comment. For an unplanned event, you select a reason for the event. An example for a planned event could be [Arrival](#) or [Loading Begin](#), and an example of an unplanned event could be [Delay](#).

You cannot report events for a blocked freight order.

- Choose a freight order from the list to view details about the freight order such as transportation details, source and destination details, cargo details, information about whether the cargo is considered dangerous goods, transportation stops, tour events, notes, and attachments.
- Report a planned event for a freight order when viewing the details of a freight order.
- Choose a transportation stop in a freight order to view details of the stop, stop events, and details of the items to be loaded or unloaded at that stop.
- Report a planned or unplanned event in a transportation stop. When you report an event, you enter the date and time and add a comment. You also need to choose a reason if you report an unplanned event.
- Report the actual quantities of each item in a transportation stop.
- Under [Tour Events](#), add a new, unplanned event that affects the entire transportation, and also report the event.
- View documents or links attached by the shipper. As a carrier, you can upload documents or links.

Prerequisites

The shipper has configured the prerequisites for collaboration with carriers (see [Carrier Collaboration](#)).

Settlement Management

As a carrier or a representative of a service provider, you can use the settlement management apps to process freight orders for settlement.

You have the following features available in the settlement management:

- Manage Charges for Freight Orders
- Manage Freight Orders for Invoicing
- Manage Freight Invoices
- Display Credit Memos

Manage Charges for Freight Orders

As a carrier or a representative of a service provider, you can view, update, or add charges for a freight order. You can also view the details of a freight order such as stages, cargo etc.

For more information, see [Manage Charges for Freight Orders](#).

Manage Freight Orders for Invoicing

As a carrier or a representative of a service provider, you can use this app to create invoices for freight orders that you've received from shipper.

For more information, see [Manage Freight Orders for Invoicing](#).

Manage Freight Invoices

As a carrier or a representative of a service provider, you can use this app to manage invoices that you've submitted for a shipper.

For more information, see [Manage Freight Invoices](#).

Display Credit Memos

As a carrier or a representative of a service provider, you can use this app to display the credit memos that you've submitted to a shipper.

For more information, see [Display Credit Memos](#).

Manage Charges for Freight Orders

App ID: F7104

With this app, as a carrier or a logistics service provider, you can monitor the charges for the freight orders that you've received from the shipper.

Key Features

- On an app tile, view the number of freight orders that require your approval, freight orders that have new disputes that you haven't submitted to the shipper, and freight orders with disputes that you have submitted to the shipper.
- View all freight orders that you're currently executing, and list them by the status of the dispute.

A new dispute is a freight order that you're editing to update the charges but haven't submitted to the shipper.

A submitted dispute is a freight order for which you've updated the charges and submitted to the shipper for review.

A freight order for approval is a freight order for which the shipper has proposed charges and has sent to you for approval.

A freight order with no dispute or a resolved dispute is listed under **No Dispute**.

- Perform a search on the freight orders using the filter criteria.
- Choose a freight order from the list to view details about the freight order such as transportation details, source and destination details, cargo details, transportation stages, charges, notes, and attachments.
- Edit a freight order to perform the following actions:
 - Add a charge or update an existing charge.
 - Propose the rate amount for a new or an existing charge with or without a calculation base.
 - Propose the price per unit or quantity for a new or an existing charge with a calculation base.

i Note

- For charges having amounts in percentages and values, you must propose charges using two different fields **Proposed Rate Amount in %** and **Proposed Rate Amount**.
- You can't add a new charge that is a percentage charge type.
- You can only delete a charge line that you've added to the freight orders. You can add a charge line only for **No Dispute**, **For Approval**, and **New Dispute** freight orders. The charge lines that you've added are known as unplanned charges.

If you select multiple charge lines and try to delete them, the system deletes only the unplanned charges.

- Tax handling isn't supported.

- Submit a freight order with updated charges to the shipper for review.
- Discard a new dispute.
- Accept a freight order with proposed charges from the shipper.
- View or add documents or links attached to a dispute.

i Note

You can attach documents to a freight order only if the freight order has a dispute.

- View or add notes attached to a dispute.

You can add a note by entering the text and choosing  (*Submit*).

i Note

You can add notes to a freight order only if the freight order has a dispute.

- Personalize the charges table and group, sort, or filter the charges by a particular field.

You can also manage and save your personalized views of the table using variants.

Prerequisites

The shipper has configured the prerequisites for collaboration with carriers (see [Carrier Collaboration](#)).

Manage Freight Orders for Invoicing

App ID: F7732

With this app, as a carrier or a logistics service provider, you can create invoices for freight orders that you've received from a shipper.

Key Features

You can use this app to do the following:

- View all the freight orders based on their invoicing status.

For freight orders where you're added as an additional agreement party, you can't view those freight orders using this app.

Status	Description
Not Invoiced	Freight orders where invoices haven't been submitted
Partially Invoiced	Freight order where invoices need your approval for posting
Completely Invoiced	Freight order where invoices have been created and submitted

- Use various filter criteria, for example, [Source Location](#), [Destination Location](#), or [Means of Transport](#) to search freight orders
- View various details of a freight order, for example, [Net Amount in Document Currency](#), [Source Location](#), and [Destination Location](#)
- Create freight invoices for [Not Invoiced](#) or [Partially Invoiced](#) freight orders

Supported Device Types

- Desktop
- Tablet

Related Information

[Manage Freight Invoices](#)

Manage Freight Invoices

App ID: F7106

With this app, as a carrier or a logistics service provider, you can manage invoices that you've submitted to a shipper.

Key Features

You can use this app to do the following:

- On the app tile, view the number of invoices that you've created but not submitted and invoices that need your approval
- View all the invoices that you're currently processing and list them by their status:

Status	Description
Submitted	Invoices that are submitted by the shipper for further processing
Not Submitted	Invoices that you've created but not submitted for further processing
For Approval	Invoices that need your approval for changes to the charges proposed by the shipper
Rejected or Canceled	Invoices that are canceled by you or rejected by the shipper

- Use various filter criteria, for example, [Invoice Date](#), [Document Currency](#), or [Terms of Payment](#), to search invoices
- On the list page, view various invoice details, for example, [Invoice Date](#), [Gross Amount](#), [Proposed Gross Amount](#) and [Status](#)
- Select an invoice to view various details, for example, [Invoice Date](#), [Terms of Payment](#), and [Invoice Items](#)
- View invoice items with relevant freight order details such as source and destination location, and the date and time of departure and arrival

You can select the invoice item to view its transport details, cargo details, and the charges.

- You can also see if a charge line has a dispute or not. Disputed charge lines are marked in red and non-disputed ones are marked green.

Disputed charge lines are the ones which you've inserted or edited manually and the shipper needs to act upon them.

i Note

The charge lines that you've excluded from the invoice using the [Exclude](#) button doesn't have any color associated with it.

- View, upload, download, or delete an attachment
- View or add a note

Once you've added a note, you can submit it to the shipper.

Not Submitted

You can do the following with [Not Submitted](#) invoices:

- Edit the invoice reference and the invoice date

- Add a tax charge or a charge to an invoice item
- Discard invoices that you don't need. Discarding an invoice deletes it permanently from the system
- Delete the charge lines that you, as a carrier, have added to the invoice
- Edit the rate amount, price per unit, or quantity of an existing charge of an invoice item
- Exclude or include a charge line from the invoice item
- Submit the invoice to the shipper

After editing, you must save the invoice before submitting it to the shipper.

Submitted

You can do the following with **Submitted** invoices:

- Create a credit memo for the invoice

You can create a credit memo to reverse an invoice that has been processed by a shipper and post the credit memo to financial accounting for further processing. There can be only one credit memo for an invoice. When you create a credit memo, you can also attach a note for the shipper in the credit memo.

You can also add a note when you create a credit memo.

- Recall an invoice

You can recall an invoice only if the financial posting of an invoice isn't yet processed.

Successful recall of an invoice results in the following changes:

- The invoice goes back to the **Not Submitted** status
- You can resubmit the invoice with or without any changes
- Any existing disputes corresponding to the invoice are canceled.

For Approval

You can do the following with **For Approval** invoices:

- View an invoice, invoice item details, and the charges proposed by the shipper
- Edit an invoice or its item
- If the charges proposed by the shipper are acceptable, approve the shipper's proposal
- If the charges proposed by the shipper aren't acceptable, edit the charges and propose new charges
- Recall an invoice

Rejected

You can view the details of invoices that have been canceled by you or rejected by the shipper.

Supported Device Types

- Desktop
- Tablet

Related Information

[Display Credit Memos](#)

[Manage Freight Orders for Invoicing](#)

Display Credit Memos

App ID: F7502

With this app, a carrier or a logistics service provider can display the credit memos that they've submitted to a shipper. They can create a credit memo to reverse an invoice that has been processed by a shipper. They can post the credit memo to financial accounting for further processing. There can be only one credit memo for an invoice.

Key Features

You can use this app to do the following:

- On the app tile, view the number of credit memos that you've submitted to the shipper
- Use various filter criteria, for example, [Credit Memo Date](#) or [Terms of Payment](#), to search credit memos
- View the various credit memo details, for example, [Credit Memo Reference](#), [Gross Amount](#), and [Status](#)
- View credit memo items with the relevant freight order details such as source and destination location, and the date and time of departure and arrival

You can select the freight order to view its details and the charges that are applicable to the credit memo.

- Manage notes and attachments that are relevant for the credit memo

Note that you can upload, download, or delete an attachment, but you can only add a note and can't delete or edit the note.

Supported Device Types

- Desktop
- Tablet

Related Information

[Manage Freight Invoices](#)